

Civilization Sandbox — Game Development Blueprint

Authoritative Version 2.0: complete early-modern product scope, physical-world specification, simulation-grounded storytelling, coded-intelligence architecture, prototype program, and production contract

Persistent fictional human societies begin from limited early-modern settlements, physically construct towns and states, develop different solutions to geography and history, interact through economics, diplomacy and war, and leave inspectable chronicles and enduring scars while an unseen omniscient player may observe, direct, damage, restore or forcibly alter nearly any part of the world.

Document authority. Version 2.0 is the project-level source of truth for the north-star product, the complete early-modern Version 1.0 game, prototype sequence, simulation architecture, game-AI contracts, storytelling and history systems, engine decision, AI-assisted production workflow, testing, performance, persistence, risks and later-age expansion. Repository specifications and architecture decision records may refine implementation details but may not contradict confirmed decisions without an approved later revision.

Field	Value
Version	2.0
Baseline revised	Version 1.1, 156 pages; complete replacement rather than supplement
Revision scope	Integrates the 94-page authoritative revision prompt, 28 locked creator decisions, sequential historical-version strategy, rough-prototype program, AI Decision Laboratory, simulation-grounded storyteller, historical event graph, chronicles and retention architecture
Research verification date	July 17, 2026
Primary implementation target	Unity 6.3 LTS, C#, Entities/DOTS, Burst, Job System, Entities Graphics and URP; exact patch/package versions pinned at repository bootstrap
Version 1.0 product boundary	Complete fictional early-modern / roughly 1700s-equivalent civilization sandbox
Authoritative world format	Flat bounded 2.5D world with real 3D terrain, roads, bridges, buildings and infrastructure; camera-facing pixel-art people and mobile entities
Default player mode	Completely unrestricted; every power implemented in the current build is available immediately
Required top-level blueprint sections	87 plus technical subsections and appendices

Status legend

- CONFIRMED

Creator-locked requirement or governing constraint.
- RECOMMENDED

Selected engineering or production default; implement unless a measured phase gate disproves it.
- EXPERIMENTAL

Bounded spike required before production reliance.
- DEFERRED

Excluded from the current milestone or Version 1.0.
- ARCHITECTURE RESERVED

Current boundaries must permit the future capability without premature implementation.
- CREATOR DECISION REQUIRED

Primarily artistic, commercial, ethical, tonal or preference-driven.
- SUPERSEDED

Older decision explicitly replaced by this edition.

How to use this blueprint

The blueprint is a source of truth, not one implementation backlog. Sections 4-68 define product, simulation, storytelling, AI, engine and verification architecture. Sections 69-76 define the falsifiable prototype and release sequence. Sections 77-87 govern risk, completion, implementation tasks, assumptions, bibliography and technical contracts. A feature is not done because it compiles or appears once: it must satisfy authoritative behavior, decision contract, invariants, deterministic scenarios, performance, persistence, traceability, accessibility and creator-visible acceptance.

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1 Cover and document identity

This edition's cover, subtitle, version, date and authority identify it as the complete replacement source of truth. The cover thesis summarizes the locked product: a physically represented, persistent, early-modern civilization sandbox with unrestricted omniscient player intervention, sequential later-age expansion, explainable coded intelligence and simulation-grounded history.

The PDF presentation is authoritative for review and distribution. The Markdown, HTML, CSS, diagram sources and build scripts are authoritative revision assets and must remain versioned with it.

2 Document control

CONFIRMED Version 2.0 supersedes Versions 1.0 and 1.1. Conflicts are resolved in favor of creator-locked decisions in the final revision prompt, then the defining game identity, verified current evidence and the strongest documented engineering recommendation.

The document-control table in front matter records the implementation target, historical boundary, world presentation, default player mode and required structure. Repository ADRs may narrow implementation choices but cannot weaken confirmed decisions. Any later change to unrestricted powers, omniscience, physical city growth, early-modern Version 1.0 scope, decision-level control, storytelling authority or AI verification requires a blueprint revision or explicit creator-approved superseding ADR.

Every release stores the blueprint version, content-schema version, save version, engine revision, package lock and behavior-contract catalog hash in its build manifest.

3 Revision summary

Version 2.0 integrates the complete 94-page revision brief rather than adding an appendix. The major changes are:

- Version 1.0 is now explicitly a complete early-modern/1700s game; later ages are sequential major-version expansions.
- Development is separated into deterministic, scale, AI, visual, rough-settlement, divergence, vertical-slice and full-production gates.
- Simulation-grounded storytelling, Story Director profiles, event graph, chronicles, historical maps and retention tiers are core architecture and Version 1.0 scope.
- Historical event recording begins before complex systems so later causality is not reconstructed from incomplete state.
- The first 25 tasks and phase roadmap are reordered around repository governance, deterministic persistence, event history, AI Decision Laboratory and physical founding proof.
- Performance and risk controls now include event-graph queries, chronicle generation, historical map storage, narrative repetition and story-authority violations.

All prior locked decisions remain: unrestricted default powers, binding orders plus Force Outcome, full omniscience, bounded flat 2.5D world, camera-facing pixel billboards, modular 3D buildings, exterior-only interiors, society-dependent growth, no manual placement, households and inherited tendencies, behaviorally active traits, contextual speech, ecosystems, extinction/recovery and persistent landmark ruins.

4 Executive summary

4.1 Product thesis

CONFIRMED The project is a large-scale, real-time, emergent civilization sandbox. Persistent fictional people begin in vulnerable early-modern founding groups; select plausible sites; construct physical settlements; form households, workplaces, institutions, governments and civilizations; exchange goods and knowledge; fight, migrate, recover and disappear; and continue toward industrial, modern and radically divergent futures. The player is an unseen omniscient external force who may simply watch, directly determine civilization-level decisions, alter material conditions, or use Force Outcome powers that bypass ordinary feasibility.

The defining proposition is **visible causality becoming persistent history**. A place does not become a city because a population threshold flips an abstract state. People clear land, carry materials, establish paths, build structures, create markets, finance projects, extend roads, damage infrastructure, flee war, return to ruins, and reshape the landscape. At street and neighborhood zoom, people and structures physically express the same authoritative state that drives regional and civilizational outcomes. At world zoom, those local processes become trade networks, political orders, military fronts, technological paths, migration systems and historical scars.

The second differentiator is **causal divergence**. Two civilizations confronting the same problem should be capable of developing materially different solutions because their geography, deposits, ecology, institutions, urban form, ownership, knowledge carriers, political coalitions, labor skills, external contacts and previous accidents differ. Awareness, understanding, operation, repair, component manufacture, full manufacture, mass production and adoption are separate capability dimensions. A society may operate imported machinery it cannot reproduce, understand nuclear principles without possessing the industrial chain for a reactor, or retain rail infrastructure after losing the institutions that maintain it.

The third differentiator is **freedom without compulsory micromanagement**. The default mode is not balanced around points, unlocks, charges or cooldown currencies. All powers implemented in the current version are available immediately. Normal direct control is expressed through binding decisions—policy, priorities, diplomacy, mobilization and major projects—while autonomous actors determine exact sites, routes, staffing, finance and execution. Force Outcome is deliberately separate: it can create, destroy, relocate, restore or impose world state directly, but the resulting world must still simulate shortages, legitimacy, maintenance, social reaction and historical memory.

4.2 Selected production decision

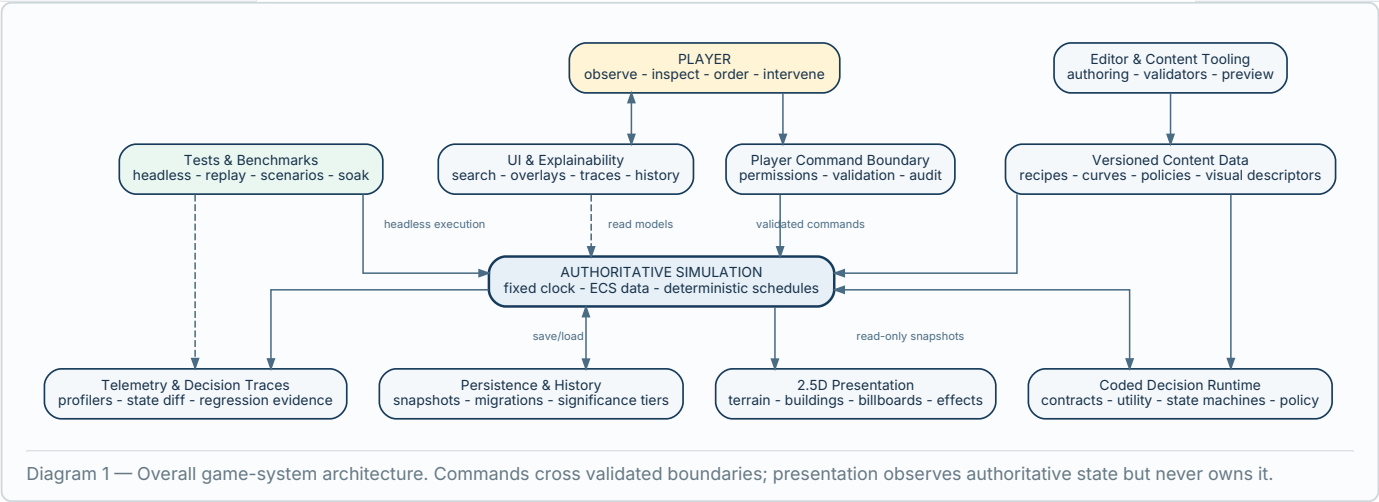
RECOMMENDED Use **Unity 6.3 LTS**, C#, Entities/DOTS, Burst, the Job System, Entities Graphics and URP as the initial production stack, subject to hard engine and architecture benchmarks before broad feature work. Unity 6.3 is the current long-term-support release and is supported through December 2027; Unity's current release model also offers update releases, but this project should remain on a pinned LTS revision unless a documented blocking defect requires change. [S-ENG-01](#) [S-ENG-02](#)

Unity remains the strongest project-specific compromise between data-oriented simulation, editor and asset tooling, terrain and 3D presentation, Windows development, Linux player builds, headless execution, test automation, profiling, package maturity and coding-agent operability. The recommendation is provisional rather than ideological. Phase 1 must demonstrate representative 10,000-person simulation, headless replay, deterministic scheduling, modular building rendering, camera-facing billboard density, save/load and clean-checkout agent operation. If Unity fails after two bounded, evidence-led optimization passes and the limiting factor is demonstrably engine/runtime-specific, a four-week Bevy comparison spike ports the headless benchmark and persistence harness—not the full game.

4.3 Architectural position

The authoritative simulation is fixed-step, camera-independent, headless-capable, seedable, batch-benchmarkable and isolated from presentation. Every person keeps a stable persistent identity, but persistence does not imply identical update frequency or visual existence at every zoom. Systems run on explicit cadences, due-time queues or state-change events. The renderer consumes read-only snapshots and may cluster, simplify or omit distant visuals; camera position, frame rate and visibility never alter authoritative decisions.

The simulation uses stable domain identifiers, integer or fixed-point authoritative quantities, explicit units, ordered event queues, keyed deterministic random streams, spatial partitions, hierarchical routes, inventory reservations, bounded utility decisions, state machines, policy rules and limited hierarchical planning. It does not promise universal cross-platform bitwise equality for rendering. It does require canonical replay equality for authoritative state on the supported reference build and a defined compatibility envelope.



4.4 First-playable definition

RECOMMENDED The first playable is a vertically complete comparative civilization simulation rather than a disposable survival prototype. It includes thin but real implementations of world generation, founding-site choice, persistent people, lightweight households, inherited baseline tendencies, behaviorally active traits, needs, health, occupations, physical construction, paths and roads, institutions, government, economy, logistics, culture, capability emergence, diplomacy, trade, physical conflict, omniscient inspection, decision traces, binding civilization orders, indirect intervention, Force Outcome, save/load and indefinite continuation.

The baseline quantities remain defensible and are retained unless a phase gate produces contradictory evidence. A Standard world is approximately 8 km × 8 km and begins with four groups of 40 people. A Large world is approximately 16 km × 16 km and is the 10,000-person benchmark. The first playable remains early-modern in content. It does not require nuclear weapons, skyscrapers, modern cities, advanced vehicles, complete ecosystems, full speech libraries or every form of government. Its schemas, route model, building modules, knowledge graph, save format and authority hierarchy must nevertheless support those later systems without replacing the authoritative simulation.

Table 1 — First-playable quantitative baseline

Scope item	Baseline quantity	Governing purpose
Commodity/resource classes	18	Food, water, construction, fuel, tools, weapons and logistics bottlenecks without microscopic item simulation
Building archetypes	16	Survival, housing, extraction, storage, production, exchange, services, governance and defense
Occupations	16	Legible labor allocation, education pressure and role specialization
Institution types	6	Household, workplace/guild, market, school, faith institution and government/council
Government structures	4	Distinct authority and ownership arrangements over one composable governance model
Cultural tendencies	12	Behavioral and institutional variation without fixed national personality labels
Technology domains	6	Agronomy, construction, materials/tools, transport/logistics, health/sanitation and institutions/information

Scope item	Baseline quantity	Governing purpose
Diplomatic actions	12	Contact through alliance, knowledge exchange, coercion, war and peace
Major player powers	24	Six inspection tools, six binding orders, six material interventions and six Force Outcomes
Primary UI workspaces	13	Inspection, comparison, overlays, causal tracing, intervention and history

4.5 Performance and stability gates

The first playable is not accepted because it opens, compiles or produces plausible screenshots. On the recommended reference tier it must maintain a 20 Hz authoritative base loop without unbounded backlog, target 60 frames per second at 1080p in ordinary settlement views, remain operable at 1×, 2×, 5× and 10×, simulate 10,000 persistent people in the Large benchmark, remain below a 6 GB working set, and preserve camera-independent results. A representative 100-year save should remain below 100 MB compressed, complete within 10 seconds with a main-thread snapshot hitch below 250 ms, and load within 15 seconds. These are initial engineering contracts, not marketing promises.

The release gate additionally requires a 24-hour wall-clock soak, a 500-game-year accelerated run, 100-year deterministic command-log replay, schema migration fixtures, interrupted-save recovery, corruption diagnostics, AI scenario suites, metamorphic tests, multi-seed behavioral distributions and trace-volume measurement. Retail tracing is selective and bounded; test and debug builds can enable fuller traces without making that overhead part of ordinary play.

4.6 Feasibility verdict

The north-star game is a studio-scale, multi-year simulation platform. It is not credible as one giant coding-agent prompt, a sequence of unreviewed scripts, or a content-first solo project. It is technically and production-feasible only as a staged program in which each phase creates an independently runnable product slice, every consequential behavior has a bounded contract, and all state changes are protected by tests, traces, migrations, performance evidence and rollback.

The largest technical risk is interaction complexity: routing, physical construction, inventories, institutional decisions, warfare, ecosystem change and long historical retention can multiply one another even when any single system appears affordable. The largest design risk is that sophisticated simulation feels arbitrary, visually unreadable or emotionally inert. The largest physical-world risk is that a convincing city consumes pathfinding, rendering and save budgets before it produces meaningful history. The largest AI-only production risk is architectural drift: coding agents can create duplicated concepts, hidden omniscience, nondeterministic shortcuts, unbounded searches, inconsistent scoring, save-incompatible state or locally plausible features that violate the decision hierarchy.

The mitigation is not a better master prompt. It is a repository harness: source-of-truth documents, behavior contracts, canonical decision primitives, deterministic fixtures, adversarial review, trace schemas, performance baselines, stable content IDs, migrations, small commits and creator acceptance based on observed behavior. OpenAI's current guidance for long-horizon Codex work similarly emphasizes repository-resident plans, scoped milestones, validation after each milestone, concise AGENTS.md routing and continuously updated documentation. [S-AI-01](#) [S-AI-02](#) [S-AI-03](#)

4.7 Immediate directive

After approval, create the repository and execute **Task 1: establish repository governance and a reproducible Windows bootstrap**. No gameplay feature precedes protected version control, pinned dependencies, command-line build/test scripts, the deterministic headless shell, `AGENTS.md`, the decision log, the behavior-contract template, the performance harness and a recoverable baseline build. The exact next technical milestone is the AI Decision Laboratory, not a giant civilization brain.

5 One-sentence game pitch

An endless, real-time, physically represented civilization sandbox where persistent fictional people autonomously build societies and cities, develop different technological answers to geography and history, trade and fight across generations, and can be fully inspected, directed, damaged, restored or forcibly reshaped by an omniscient unseen player.

6 Detailed player fantasy

6.1 Witness causality become history

The player opens a new bounded world and sees terrain, rivers, elevation, vegetation, exposed surface resources and several founding groups searching for viable places to survive. No society is pre-scripted as “the rail civilization,” “the military empire” or “the religious city.” Identity emerges from site choice, institutions, material success, trauma, law, trade, migration and conflict.

At close zoom the player can follow a named builder carrying lumber, a hungry child returning to a household, a healer entering a clinic, a laborer leaving a polluted workshop, a patrol moving along a road, or a refugee family arriving at temporary shelter. At neighborhood zoom, those actions form construction sites, queues, markets, schools, workshops, social tensions, speech bubbles and visible traffic. At regional zoom, they become roads, ports, stockpiles, shortages, migration corridors, front lines and power networks. At world zoom, the player sees states, alliances, trade dependence, extinct cultures, ruined cities and technological divergence.

The player’s agency has six complementary forms:

1. **Interpretive agency:** inspect why an outcome occurred rather than accept hidden dice.
2. **Counterfactual agency:** change one pressure and watch the world branch.
3. **Directorial agency:** issue binding decisions—war, peace, priorities, policy, colonization or projects—without controlling every worker.
4. **Material agency:** destroy housing, add resources, protect a region, damage transport or alter leaders and let consequences propagate.
5. **Force Outcome agency:** directly create, destroy, relocate, restore or impose a result that ordinary actors could not currently achieve.
6. **Custodial agency:** preserve a world for centuries, track lineages of institutions and technologies, revisit ruins and compare alternate save branches.

6.2 Example ten-minute session

The player loads a 140-year-old Standard world at 5× speed. A northern republic has stopped exporting tools to a southern agrarian monarchy. The logistics overlay shows a bottleneck at a mountain bridge; the causal panel reports that flood damage reduced capacity, a border incident increased perceived threat, and a metalworkers’ faction gained influence over export policy. The player pauses and opens the republic’s decision panel.

A normal **Binding Order** instructs the republic to restore trade and prioritize an alternate route. The order is accepted at the government layer, but the trace reports that the alternate corridor lacks surveyed land, bridge timber and funding. The civilization therefore creates a survey project, negotiates land access and shifts a public-works budget. The player accelerates time and sees crews, materials and path traffic appear physically.

Impatient, the player could instead use **Force Outcome** to complete the bridge instantly. The bridge would exist, but the game would not fabricate trained maintainers, security or approach roads. If the player also destroys nearby housing to create space, displaced households would seek shelter, prices would rise, political anger could increase and a permanent intervention record would connect those consequences to the forced redevelopment.

The player then switches to the civilization-knowledge perspective. The monarchy incorrectly believes the republic's army is massing nearby; the omniscient view shows that the observed movement is a construction convoy. A diplomatic order can propose verification or peace, but each party's institutions still interpret and implement it. The session has no required victory. The result—successful reconciliation, a nationalist backlash, a bridge collapse, a forced peace with lingering hostility, or an entirely new trade route—becomes history.

6.3 Play styles and rule sets

CONFIRMED DEFAULT

New worlds begin in **Unrestricted Sandbox**. Every player power implemented in that build is available immediately, without influence points, charges, balancing cooldown currencies, discovery unlocks, achievements, persistent progression or per-world progression. “All powers” means all powers implemented in that version; it does not require future-era powers to exist in the first playable.

Optional modes are later or secondary presets:

Mode	Restrictions	Intended experience
Unrestricted Sandbox	None beyond command validation and data integrity	Default WorldBox-like freedom: observe, order, create, destroy, restore and experiment
Observation Only	No authoritative state-changing commands	Watch, inspect, compare and archive history
Limited Influence	Selected powers, budgets or cooldowns	Optional constrained experimentation
Discovery Mode	Tools unlock through world knowledge or achievements	Optional progression-oriented play
Challenge Scenario	Authored restrictions and objectives	Bounded puzzles, survival or historical experiments
Custom Rules	Per-category toggles	Creator/player-defined combinations

Changing a world's rule set is allowed but produces a permanent “rules changed” history event. No optional progression system may silently restrict the default sandbox.

7 Design pillars

7.1 Persistent people, scalable causality

Every person retains a stable identity and authoritative life state: name, age, health, nutrition, injury, disease, skills, education, occupation, wealth, possessions, home, household, traits, temperament, beliefs, political attitudes, patriotism, needs, goals, action, decision trace, social ties, experiences, settlement and civilization membership. Persistence does not require every component to update every render frame. It requires that every outcome follows declared schedules and rules regardless of observation.

Design test: leaving a region unseen for ten game years and returning must produce the same authoritative checkpoint as keeping the camera there under the same seed and command log.

7.2 Physical historical continuity

Settlements, roads, farms, factories, ports, railways, military bases, power systems, neighborhoods, skyscrapers, ruins and environmental scars must physically exist. Abstract metrics may summarize capacity but may not replace the causal assets that provide it. Construction, destruction, occupation and rebuilding are visible processes tied to real footprints, materials, labor and routes.

Design test: a housing district destroyed by bombardment must produce visible ruins and traceable displacement, not merely reduce an aggregate housing score.

7.3 Civilizations solve problems differently

Technology is a capability network of problems, principles, procedures, components, tools, materials, institutions, skills, supply chains and applications. Civilizations can specialize, copy, improvise, stagnate, forget or develop incompatible variants. There is no universal global ladder.

Design test: resource- and institution-matched seed suites must produce statistically distinguishable transport, energy, military, settlement and economic patterns—not merely different research speeds.

7.4 Unrestricted player freedom with layered control

The default sandbox exposes all implemented powers. Normal direct control acts at the level of decisions and priorities; exact physical implementation remains the work of simulated actors. Force Outcome is explicit and separate. Indirect intervention changes material and social conditions and must propagate through the same systems as natural events.

Design test: a player can order housing expansion, destroy a food warehouse, impose peace or instantly restore a settlement, and each command has a distinct validation path, decision trace and downstream history.

7.5 Omniscience without information overload

Nothing authoritative is fundamentally hidden from the default player. Search, filters, overlays, drill-down panels, timelines, comparisons and causal traces organize information. A separate civilization-knowledge perspective shows what selected societies believe, including uncertainty and misinformation.

Design test: a player can inspect the true reason for a war declaration and compare it with the declaring government's incomplete information.

7.6 Society-dependent urban form

City growth emerges from households, businesses, institutions and governments acting under property, finance, culture, transport, security, pollution and planning rules. Different societies visibly produce different urban forms. The player directs priorities but does not manually place ordinary buildings, roads, lots, zones or districts.

Design test: communal, speculative private, authoritarian, religious and military settlements facing similar geography must create materially different layouts through shared primitives.

7.7 Explainable coded intelligence

Citizens and civilizations use bounded coded systems—utility scoring, state machines, needs, rules, policies, social influence, controlled randomness and limited hierarchical planning—not live language models. Every important decision emits a structured trace. Behavior is specified by contracts, tested in deterministic fixtures and reviewed adversarially.

Design test: every important selected action can show candidates, preconditions, observations, normalized scores, modifiers, randomness, controlling policy and invalidation reason.

7.8 Persistent scars, extinction and recovery

War, disaster, extraction, pollution, famine, migration and collapse leave persistent consequences. Species, states, cultures and populations can disappear in distinct ways. Recovery may occur from viable remnants. World-defining ruins and geographical scars survive compaction unless deliberately reclaimed, demolished, buried or restored.

Design test: centuries after a metropolis is destroyed, its street pattern, bridges, major towers, contamination and cultural significance remain inspectable.

7.9 Performance, migration and verification are product features

Headless execution, deterministic replay, save migration, profiling, scenario fixtures, trace schemas, regression ledgers and CI are not late polish. They are required from the first commit because the creator cannot safely compensate for unbounded code generation through manual inspection.

Design test: a coding agent can implement one behavior from a contract, produce a clean evidence pack and have another agent reproduce and challenge the result without conversational memory.

8 Explicit non-goals

The following are deliberate exclusions, not reduced ambition.

Table 2 — Explicit non-goals

Non-goal	Reason	Reconsideration point
Manual placement of ordinary buildings, roads, lots, zones or districts	Would convert the core into a city-builder and bypass society-dependent implementation	Never for normal direct control; Force Outcome may place or delete objects as a god power
Live LLM citizens or governments	Cost, latency, nondeterminism, privacy, localization and inability to reproduce 10,000 persistent minds	Optional offline flavor tooling only; never authoritative behavior
One giant civilization brain	Unbounded, untestable and authority-violating	Never; use layered contracts and bounded planners
Fully rendered building interiors	Multiplies pathfinding, assets and occlusion while adding little civilizational value	Exceptional future set pieces only, outside normal building simulation
Room-by-room skyscraper simulation	Hundreds of floors would consume state and rendering without proportional causal value	Never in the core model; internal capacity remains authoritative and abstracted
Detailed romance or genealogy simulator	Pulls scope toward a life simulator and creates an unbounded social graph	Optional family-history depth after household continuity is proven
Pairwise relationship score for every person	Quadratic state and low player value	Capped meaningful ties only
Realistic anatomical gore or prolonged suffering animation	Not required to communicate casualties and destruction	Maintain stylized low-detail pixel violence
Full projectile physics for large wars	Incompatible with high time speeds and physical war scale	Representative visual projectiles may depict authoritative aggregate fire events
Microscopic simulation of every grain, coin, nail and bullet	State and hauling explosion	Use physical batches; preserve unique artifacts and vehicles where valuable
Universal linear technology tree	Contradicts divergent capability development	Never
Spherical globe as the core world	Adds camera, terrain, streaming and routing risk before core proof	Distant planetary expansion with architectural extension

Non-goal	Reason	Reconsideration point
Multi-planet simulation in initial releases	Requires orbital, scale and logistics abstractions, not content alone	After modern/future systems stabilize
Camera-dependent authoritative simulation	Observation would change history	Never
Black-box machine learning for citizens, governments, diplomacy, technology, politics, economy or culture	Weak explainability, reproducibility and save stability	Experimental only for bounded crowd/combat/tuning tasks with fallbacks
Scripted off-screen outcomes	Violates persistence and player trust	Never; lower-frequency scheduling is allowed, fabrication is not
Perfect historical reenactment of a real year	The world is fictional and emergent	Separate historical rule sets may be future content
Mandatory victory or optimal strategy	The game is an endless sandbox	Optional challenge scenarios
Multiplayer in the first production cycle	Networking, determinism and moderation multiply every risk	After stable single-player saves and command logs
Unrestricted executable mod code in early access	Save security, determinism and support risk	Begin with validated data mods; sandboxed scripting later
Final art before engine, AI and settlement gates	Art would conceal architectural failure and create rewrite pressure	After physical-city-growth gate

9 Comparable-game analysis

9.1 Evidence standard

Comparable games are design references, not assumed implementation blueprints. Official descriptions, developer statements and public talks can verify visible goals or disclosed techniques. Proprietary internal architecture is treated as unknown unless a primary source states otherwise. “Adopt” below means a project-specific design inference, not a claim that the referenced game uses the same runtime.

Table 3 — Comparable-game synthesis

Reference	Verified or observable lesson	Adopt	Do not copy or infer
WorldBox	Immediate world observation and godlike interventions are central to its public identity S-GAME-01	Low-friction powers, autonomous history, inspectable civilizations and destructive freedom	Opaque progression as default; unknown internal pathfinding or save architecture
Dwarf Fortress	Interconnected systems and persistent history create stories from failure	Shared causal primitives, selective depth, ruins and historical continuity	Inaccessible information design or unbounded detail for its own sake
RimWorld	Readable motives and crises make interacting systems legible	Visible needs, traceable incidents, paced notifications	A storyteller that fabricates remote civilization outcomes
Songs of Syx	Settlement growth and collapse remain readable across large population scales	Explicit population tiers, selective aggregation, recovery after decline	Advertised population count as proof of a particular technique
Foundation	Organic, gridless growth and modular building language S-GAME-02	Desire paths, terrain-led form, modular architecture and constrained parcel geometry	Unbounded freeform geometry in the first playable
Manor Lords	Roads, plots and landscape create organic settlement form	Route-led parcels and visible land-use consequences	Manual building placement as the project's core loop
Workers & Resources: Soviet Republic	Logistics and infrastructure can be exposed at configurable depth	Physical flows, capacity overlays and optional complexity presets	Mandatory player micromanagement of every network
Cities: Skylines	Transport capacity and citizen movement create city-scale feedback	Hierarchical routes, congestion, access and land-use overlays	Per-person long-range path searches as the main simulation sink

Reference	Verified or observable lesson	Adopt	Do not copy or infer
The Sims	Needs and autonomy can create readable local behavior	Bounded utility, hysteresis, visible motives and contextual speech	Detailed romance, interiors and exhaustive social simulation
Crusader Kings / Europa Universalis / Victoria	Political, diplomatic and economic layers gain meaning over long horizons	Factions, legitimacy, institutions, treaties and strategy layers	Fixed nations, deterministic history or replacing people with statistical pops
Kenshi	Persistent recovery after defeat can be more interesting than game over	Wounds, displacement, rebuilding and changing world ownership	Assumptions about undisclosed implementation
Spore — Civilization Stage	Broad strategic silhouettes and divergent material culture are legible	Society-specific vehicles, structures and world-scale reading	Discrete stage transitions or customization replacing simulation

9.2 Synthesis

The strongest combined direction is **WorldBox interaction freedom + Dwarf Fortress historical causality + Foundation-like organic physical form + grand-strategy political layering + Workers & Resources logistics discipline**, unified by an original persistent-person, capability-network and omniscient-causal interface. Complexity belongs in authoritative causes, not in compulsory menus or repetitive manual placement. No single reference unifies all defining constraints; this project must validate its own combination through project-specific phase gates.

10 Research findings

10.1 Entity count is not the primary performance metric

Ten thousand compact person records are tractable in a data-oriented runtime. The expensive work is what each person causes: path requests, neighborhood scans, inventory reservations, structural changes, rendering, collision, speech, history and institutional feedback. Unity Entities is explicitly data-oriented; Burst and the Job System are intended for compiled parallel processing. [S-ENG-03](#) [S-ENG-04](#) [S-ENG-05](#)

Recommendation: budget work per simulated minute, candidate-set cardinality, cache invalidation, memory bandwidth and structural change—not merely “number of citizens.” Every system declares cadence, maximum candidate count, allocation policy and benchmark scenario.

10.2 Determinism must be designed above the engine

Multithreading, floating-point behavior, hash iteration, job completion order and platform differences can undermine replay. Counter- or key-based random streams allow independent reproducible samples that do not depend on a shared consumption order. [S-TECH-02](#)

Recommendation: stable ordering keys, integer or fixed-point authoritative accumulators, deterministic reductions, keyed RNG, canonical serialization, command logs and explicit compatibility envelopes. Promise canonical replay on supported builds, not universal identity for visual calculations.

10.3 Hierarchical routing is mandatory

Running long-range fine-grid A* for every person is incompatible with large cities and high speeds. Hierarchical pathfinding abstracts regions and refines locally; shared corridors or flow fields can serve groups with common destinations. [S-TECH-01](#) [S-TECH-03](#)

Recommendation: strategic region graph → settlement/road graph → short local traversal and steering. Cache commute and shipment corridors; invalidate affected segments rather than clearing all routes.

10.4 Utility AI works only when bounded and explainable

Utility scoring is useful for contextual choice, but unrestricted candidates, arbitrary curves and per-feature scoring frameworks create instability and performance risk. [S-TECH-04](#)

Recommendation: role- and state-specific candidates; explicit preconditions; normalized canonical curves; commitment inertia; deterministic tie-breaking; controlled keyed randomness; structured traces; per-decision budgets. Utility selects among legal possibilities; it does not grant authority or knowledge.

10.5 Semantic saves are required

Memory dumps couple saves to transient engine layout. Versioned semantic schemas with stable field IDs, reserved removed keys, explicit migrations and checksums are required for a long-lived simulation. Message-Pack-CSharp and LZ4 are suitable implementation tools; Protocol Buffers compatibility rules illustrate the underlying discipline. [S-TECH-05](#) [S-TECH-06](#) [S-TECH-07](#)

Recommendation: versioned domain chunks, human-readable manifest, stable IDs, content hashes, migration history, atomic replacement and previous-good backup. Landmark significance and visible scars are explicit save semantics, not incidental mesh persistence.

10.6 Property and metamorphic tests are essential for behavior

Example tests catch known outcomes; property tests check invariants over broad generated inputs; metamorphic tests validate directional relationships when an exact answer is not stable. QuickCheck established the property-based pattern, and metamorphic testing addresses systems where a perfect oracle is unavailable. [S-TECH-11](#) [S-TECH-12](#)

Recommendation: all important decisions require deterministic scenarios, invariants, metamorphic relations and—where controlled randomness matters—multi-seed distribution tests. “Looks intelligent in one video” is not evidence.

10.7 Explainability is a production dependency

A civilization simulation can be correct yet feel random. Causal UI cannot be reliably reconstructed from final state because the losing candidates, stale beliefs, policy constraints and random adjustment have disappeared.

Recommendation: important decisions emit structured traces during execution. Retail retains bounded recent traces and promotes major causes to history. Debug and test modes can record complete candidate data. A decision system that cannot explain itself is incomplete.

10.8 Organic cities need constrained generation

Terrain-led streets and modular construction create convincing form, but unconstrained freeform geometry increases collision, navigation, serialization, occlusion and asset validation costs.

Recommendation: validated footprint families, parcel-aligned modules, finite road splines, entrance rules, construction envelopes, skyline modules and culture/era palettes. Organic does not mean arbitrary.

10.9 Camera-facing billboards reduce asset cost but create semantic risk

A single camera-facing person sprite eliminates four- or eight-direction animation sets, but combat direction, tool use and crowd readability can become ambiguous.

Recommendation: encode direction through motion vectors, lean, equipment and weapon orientation, projectiles, shadows, facing indicators, pose states, lane alignment and selected-entity outlines. The prototype must test melee, ranged combat, work, carrying, injury and vehicle boarding before art expansion.

10.10 Omniscience is an information-design problem, not a simulation shortcut

Full player knowledge does not mean governments know the truth. The simulation must preserve per-civilization knowledge while the UI exposes authoritative state to the player.

Recommendation: every query supports an observer perspective: omniscient ground truth or selected civilization belief. Knowledge filters belong in coded decision boundaries, not UI-only masking.

10.11 Long-horizon coding-agent work needs a repository harness

Official OpenAI guidance emphasizes durable plans, small scoped diffs, repository instructions, milestone validation and living documentation. [S-AI-01](#) [S-AI-02](#) [S-AI-03](#) [S-AI-04](#)

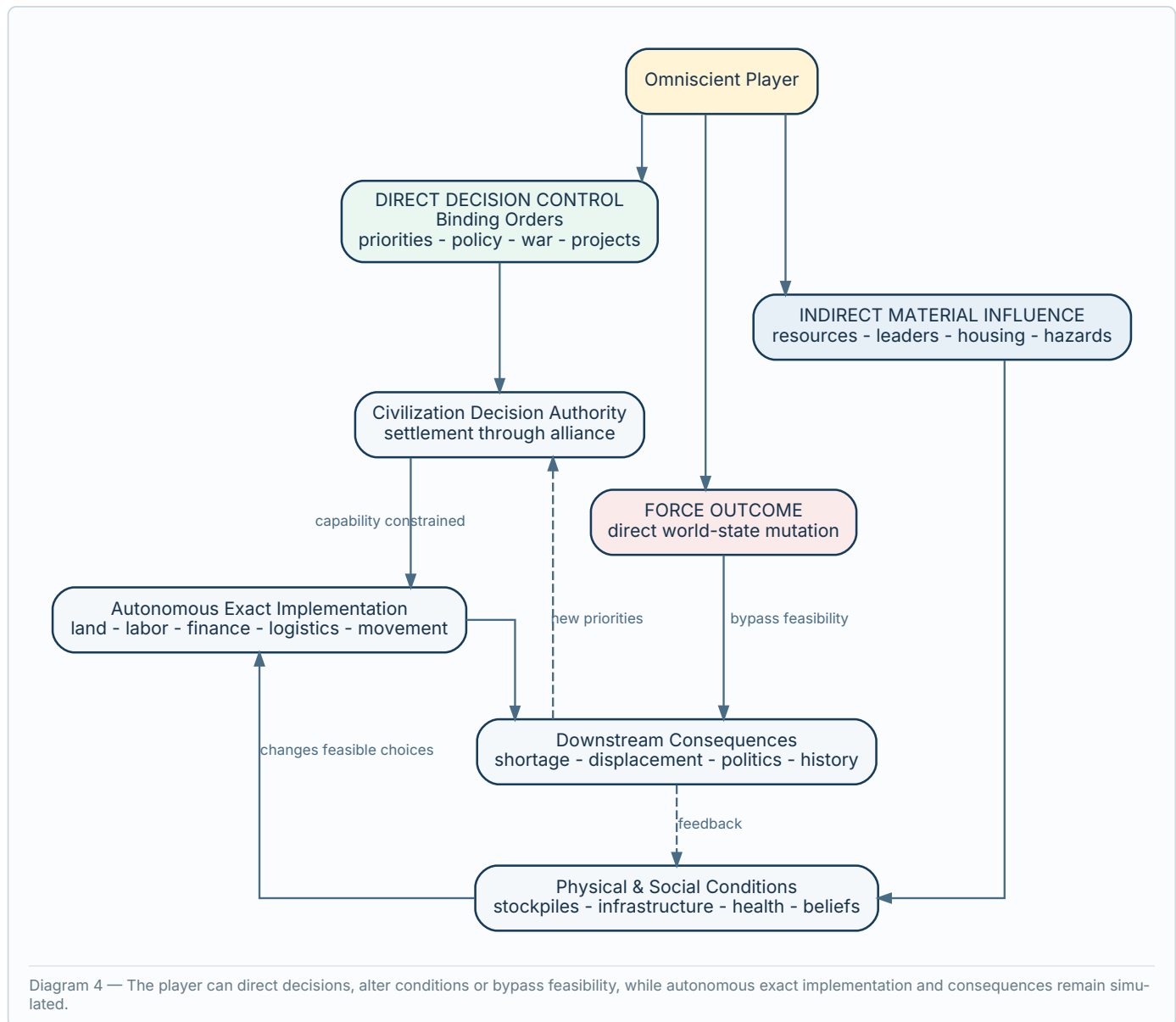
Recommendation: `AGENTS.md` routes to focused specifications; behavior contracts define observable results; CI runs headless evidence; independent reviewers search for invalid assumptions; every behavioral defect becomes a fixture. The repository, not conversational memory, is authoritative.

10.12 Engine selection remains a benchmark decision

Unity's LTS, ECS, testing and editor ecosystem make it the strongest initial choice, but no document can prove the combined load of 10,000 persistent people, physical cities, overlays, billboards, saves and AI-only development.

Recommendation: retain Unity 6.3 LTS provisionally and design Phase 1 to disprove it quickly. Do not build months of content before the representative benchmark.

11 Core gameplay loop



11.1 Moment-to-moment loop

- 1. Observe:** run at pause, 1×, 2×, 5× or 10×; watch people, construction, transport, politics, conflict and disasters.
- 2. Detect change:** notifications and overlays expose shortages, breakthroughs, migration, policy change, war, extinction and unusual behavior.
- 3. Inspect:** select a person, household, workplace, institution, building, route, settlement, civilization, species, capability or event.
- 4. Explain:** read the causal chain, decision trace, blockers, confidence, observer perspective and intervention ancestry.
- 5. Choose stance:** remain passive; issue a binding order; alter material conditions; or use Force Outcome.
- 6. Watch implementation:** simulated actors acquire land, materials, labor, finance, authority, transport and knowledge—or fail visibly.

7. **Compare:** follow second-order consequences, contrast societies, or load a branch save for a counterfactual experiment.

11.2 Simulation-side loop

The game continuously senses authoritative local and institutional state; validates current commitments; allocates labor, goods and services; moves people and shipments; executes construction and production; updates health, prices, legitimacy, attitudes and knowledge; evaluates diplomacy and military operations; publishes presentation snapshots and traces; records important history; and compacts only detail that no longer carries current or landmark significance.

11.3 What the player is not required to do

The player does not place every farm, draw every road, zone every district, hire every worker, set every price, choose every technology, move every squad or manage every production queue. Direct orders operate at the appropriate decision layer: "expand housing," "build a railway connection," "mobilize," "prioritize education," "colonize this region" or "seek peace." Households, firms, planners, engineers and military organizations select exact implementation.

12 Long-session gameplay loop

A long session spans decades or centuries and has no mandatory end. Attention cycles through incidents, local episodes, settlement arcs, civilization eras and historical ages. The player may follow a person for a day, a city for fifty years, a religion through schism and extinction, or a ruined metropolis across centuries of scavenging and myth.

Scale	Typical duration	Player attention	Authoritative outputs
Incident	Minutes	Injury, speech, invention attempt, local fire, route failure	Goals, traces, casualties, repairs
Local episode	Days to months	Harvest, construction program, election, raid, outbreak	Prices, housing, legitimacy, migration, capability progress
Settlement arc	Years	Growth, specialization, redevelopment, decline	Built form, institutions, class structure, neighborhood change
Civilization era	Decades	Industrial path, expansion, alliance system, political reform	Infrastructure networks, knowledge portfolio, borders, military doctrine
Historical age	Centuries	Collapse, recovery, extinction, global order	Ruins, successor states, rediscovery, environmental scars
Player experiment	Any	Counterfactual orders and interventions	Divergent save branches and intervention ancestry

The durable loop is:

follow a pattern → form an expectation → intervene or abstain → watch implementation → inspect second-order consequences → compare history → choose a new pattern

Tracking lists support people, households, settlements, civilizations, factions, religions, technologies, wars, artifacts, species, ruins and player experiments. The player may name an era or bookmark a world state without changing simulation authority.

13 Version and age strategy

CONFIRMED The project is developed one historical possibility space at a time. The complete north-star architecture remains visible, but production does not implement thin 1700s, industrial, electrical, mechanized, atomic, information-age, future and space systems simultaneously. Each major version must be a coherent game rather than an unstable cross-era technology demo.

13.1 Version 1.0 is the complete early-modern game

Version 1.0 is centered on a fictional roughly 1700s-equivalent early-modern stage. It must be enjoyable and commercially coherent even if no later age is ever produced. It therefore requires deep support for settlement founding, farms, hunting, fishing, forestry, mines, workshops, crafts, markets, roads, ports, canals where justified, fortifications, neighborhoods, households, wealth, education, health, culture, religion, governments, factions, diplomacy, colonization, migration, rebellion, conquest, occupation, muskets, artillery, cavalry, ships, militias, standing armies, supply lines, historical chronicles and unrestricted player intervention.

Version 1.0 does **not** include token implementations of electricity, automobiles, tanks, aircraft, nuclear weapons, computers, spaceflight or multi-planetary civilization. Those systems require new infrastructure, precision, energy, communications, routing, urban, military and presentation capabilities. Their interfaces may be reserved; their gameplay must wait.

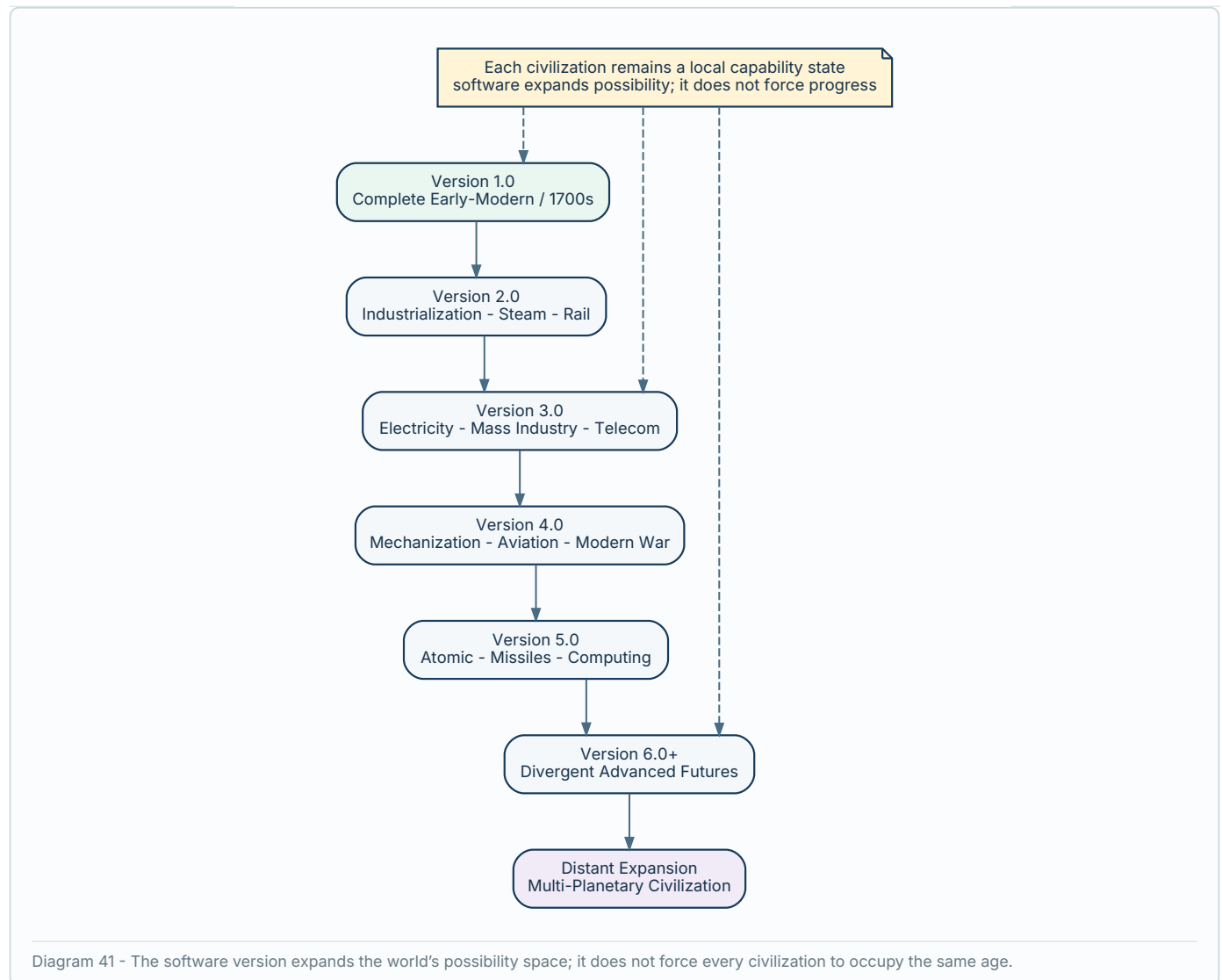
13.2 Minor versions deepen the same age

Version 1.x releases add early-modern content and quality: additional occupations, governments, doctrines, resources, regional styles, technologies, trade patterns, religious institutions, military organization, UI, modding, accessibility, performance and historical presentation. A minor release does not introduce an entire later age merely because an isolated asset is easy to add.

13.3 Later major versions expand what is possible

Table 18 - Sequential historical possibility spaces

Major version family	New possibility space	Required architectural expansion
1.0	Early-modern / 1700s	Complete causal foundation: people, settlements, state, trade, war, knowledge, story and persistence
2.0	Industrialization, steam, factories and rail	Machine production, coal energy, rail logistics, industrial labor, pollution and dense urban growth
3.0	Electricity, mass industry and telecommunications	Power grids, communications, precision manufacture, larger services and multi-center cities
4.0	Mechanized transport, aviation and modern warfare	Fuel networks, automobiles, aircraft, mechanized armies, high-speed logistics and new tactical abstractions
5.0	Atomic energy, missiles, computing and information systems	Radiation, strategic weapons, computing, electronic communication and global surveillance/coordination
6.0+	Divergent advanced futures	Multiple energy, automation, biotechnology and social-development paths proven by later research
Distant expansion	Multi-planetary civilization	Orbital spatial model, interplanetary logistics, celestial resources and scale transitions



13.4 Ages are local capability states, not one global variable

The simulation must not use a single authoritative `CurrentAge` flag to determine what every civilization can do. A civilization is represented by its scientific understanding, practical knowledge, tooling, materials, energy sources, component production, repair ability, institutions, infrastructure, manufacturing precision, military adoption and civilian adoption. UI labels such as “Early Industrial” are summaries over those states.

Civilizations may advance unevenly, specialize, regress, import equipment they cannot reproduce, retain obsolete infrastructure, skip familiar historical routes, reject technologies, combine old and new systems or lose the institutions needed to maintain prior achievements. This unevenness is central to the game’s identity.

13.5 Architecture-reservation rule

Use the rule **design the doorway now; do not build the room yet**. Version 1.0 schemas should permit later energy sources, transport classes, manufacturing precision, communications, utilities, advanced medicine, pollution, new military doctrines and additional building categories. Reservation means stable identifiers, extensible interfaces, explicit units and migration-safe data. It does not mean implementing dormant later-age simulation branches or balancing future content before the early-modern game works.

14 Player influence model

14.1 Governing rule: unrestricted by default

SUPERSEDED Version 1.0 recommended per-world discovery progression as the default. Version 2.0 replaces that recommendation throughout.

CONFIRMED Unrestricted Sandbox is the default initial mode. Every state-changing tool implemented in the current version is available immediately. No influence currency, charges, balancing cooldowns, achievement gates, discovery gates, profile progression, world progression or resource cost is required. Optional restricted modes may reuse the same command architecture later, but the default must prioritize freedom and experimentation.

14.2 Three authoritative intervention classes

Table 4 — Player intervention classes

Class	Authority	Examples	Result semantics
Binding Order	Changes a decision owned by a settlement, region, government, civilization or alliance	Development priority, housing, factory, mobilization, war, peace, education, research, migration, colonization	Decision is obligatory but execution remains constrained by real capability
Indirect Material Intervention	Alters physical, biological, social or informational conditions	Destroy housing, contaminate water, add resources, kill a leader, introduce belief, protect habitat	Normal systems propagate the changed condition
Force Outcome	Bypasses ordinary feasibility and directly mutates authoritative state	Instantly create a building, relocate a population, restore a species, impose a government, complete an impossible project	Mutation must preserve data invariants, produce aftermath and enter permanent history

Observation, search, pause, comparison, overlays and tracking do not mutate state and are not treated as powers with costs.

14.3 Binding direct orders

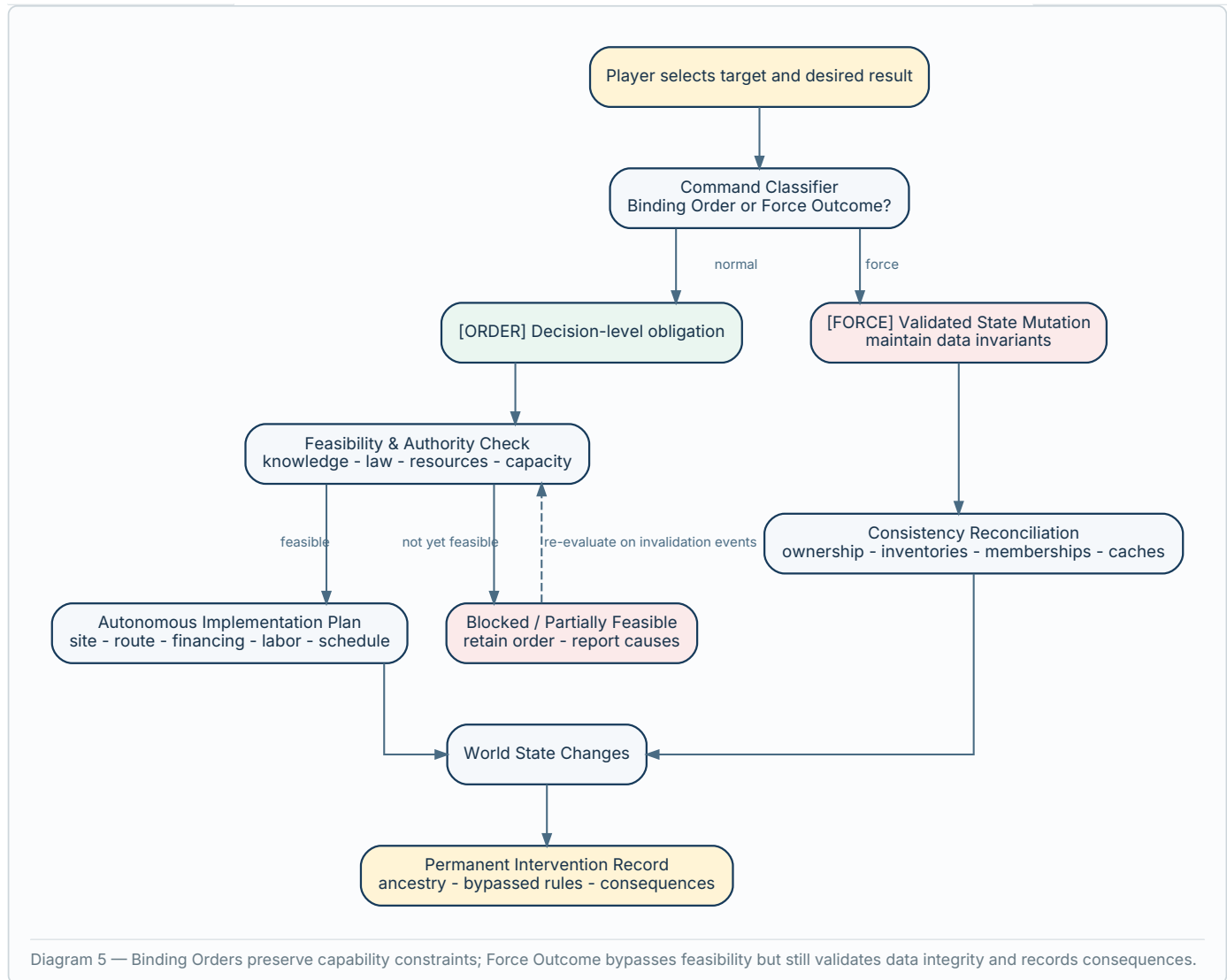
The player may directly decide or order matters belonging to settlements, towns, cities, regions, governments, countries, civilizations and alliances. Valid categories include development, economy, military mobilization, war, peace, diplomacy, expansion, colonization, industrial policy, education, research, infrastructure, government policy, resource allocation, migration and major projects.

An order binds the receiving decision layer. It does not teleport prerequisites or micromanage lower layers. An order to build a factory requires a project definition, legal authority, site candidates, land rights, funding, labor, materials, transport, knowledge, utilities and maintenance. An invasion order may create mobilization and operational plans yet fail because readiness, supply, morale, transport or political stability are insufficient. A peace order ends official war but does not erase insurgency, claims, casualties, hatred, occupation or ruined infrastructure.

Every active order stores issuer, target layer, objective, priority, start time, expiration or cancellation rule, feasibility state, active projects, blocked reasons, controlling policy, progress and intervention ancestry. The player can inspect whether the civilization is obeying, what it is doing, and what prevents completion.

14.4 Force Outcome

Force Outcome is an explicit category rather than a hidden “cheat” path through normal commands. It can create or destroy structures, force a government change, end war, relocate people, create deposits, complete projects, restore species or civilizations and impose diplomatic relationships.



Force mutations pass through a consistency-reconciliation layer. Creating a factory must establish a stable building ID, footprint, owner, capacity, damage state, inventories and route entrances; it may still have zero workers and no supply. Relocating a population updates people, households, homes, memberships, claims, routes and migration history; the destination may have no food or shelter. Restoring a civilization creates plausible continuity claims and surviving institutions, not an exact frozen copy unless the player explicitly restores a saved snapshot.

14.5 Indirect control through conditions

The player may destroy factories, farms, food, shelter, housing, roads, railways, bridges, ports, power and communications; kill or alter leaders; create shortages; add or remove deposits; contaminate land or water; create disasters; protect areas; reveal resources; alter health; or introduce artifacts, technologies, diseases, beliefs and ideologies.

These interventions must not stop at aggregate modifiers. Destroyed housing changes physical capacity and tenancy, creating displacement, overcrowding, homelessness, migration, anger, disease, poverty, rebuilding demand, crime opportunity and political instability. The exact outcome varies with household wealth, mutual aid, law, government capacity, available land, weather, discrimination and nearby transport.

14.6 First-playable power set: 24

Observation and omniscient inspection — 6

1. Inspect/follow any stable entity and open its causal trace.
2. Search by name, occupation, settlement, trait, belief, status, event or resource.
3. Bookmark entities, locations, events and save branches.
4. Toggle omniscient and selected-civilization knowledge perspectives.
5. Compose and compare overlays.
6. Open intervention history and causal ancestry.

Binding orders — 6

1. Set settlement development priority.
2. Order a major physical project category such as housing, road, port, fortification or public institution.
3. Set civilization economic, industrial or education priority.
4. Set research/capability priority.
5. Order diplomacy, war, peace, mobilization or demobilization.
6. Order migration, colonization, resettlement or strategic expansion.

Indirect material interventions — 6

1. Add, remove, reveal or conceal a resource deposit.
2. Modify local fertility, rainfall, water quality or pollution.
3. Damage or destroy selected buildings, stockpiles or infrastructure.
4. Start or suppress fire, flood, storm or epidemic pressure.
5. Alter a person's health, trait, skill, reputation or leadership status.
6. Seed knowledge, an artifact, specialist, belief or ideology carrier.

Force Outcomes — 6

1. Instantly create, complete, repair or destroy a physical structure.
2. Relocate or create a population group with reconciled households.
3. Force government, law, leadership or border state.
4. Force a diplomatic relationship, treaty, war termination or peace.
5. Restore an extinct species, destroyed settlement or civilizational continuity claimant.
6. Set an authoritative project, inventory, environmental or capability outcome beyond current feasibility.

14.7 Safeguards and auditability

Every state-changing command records world time, mode, target, parameters, prior state summary, validation, bypassed rules, resulting state, affected stable IDs and causal ancestry. Normal orders and Force Outcomes are visually distinct in history. Undo is implemented as another recorded command or save-branch operation, not silent deletion of history.

The UI warns about mass displacement, unique knowledge-carrier loss, treaty invalidation, protected-area damage, extinction and save-scale consequences, but warnings do not restrict the unrestricted sandbox. Command validation protects data integrity, not balance.

15 Player omniscience

15.1 Full authoritative inspection

CONFIRMED The default player is fully omniscient. Hidden diplomacy, forces, espionage, uncontacted groups, artifacts, underground deposits, exact environmental state, internal priorities and true decision causes are inspectable. Interface organization may collapse detail, but no authoritative category is fundamentally inaccessible.

Omniscience does not modify AI. Governments and people use their own knowledge stores. The player can deliberately switch to a civilization-knowledge perspective to understand miscalculation.

The player can inspect individual, household, settlement, government, military, diplomatic, economic, technological, environmental and hidden-state data through search, drill-down panels, causal traces and comparison views. Section 46 specifies overlay composition and civilization-knowledge perspectives.

16 World generation

16.1 World representation

CONFIRMED The core world is a flat, bounded 2.5D terrain. It has real 3D elevation, rivers, roads, bridges, railways, buildings and infrastructure. The camera uses an angled top-down view. A spherical globe is not part of the core game; planetary and multi-planet systems require a future topology extension.

World data uses three spatial resolutions:

Layer	Baseline resolution	Authoritative purpose
Strategic raster	16 m cells	Elevation sample, slope, water, soil, biome, movement cost, land cover, deposits and claims
Region/save chunk	256 m × 256 m	Cache and save partition, scheduling, presentation streaming and spatial indexing
Local settlement/nav layer	2–4 m adaptive cells plus graph	Footprints, entrances, parcels, short routes and construction access where development exists

The strategic raster is not the visible terrain mesh. Presentation creates terrain geometry and LOD from authoritative fields. Roads and rivers are graph features with geometry, capacity and state, not decorative textures.

16.2 World presets

Table 5 — First-playable world presets

Preset	Player-facing size	Internal padded size	Starting groups	Starting people	Intended mature band
Small	4 km × 4 km	4,096 m square	2	80	1,500–3,000
Standard	8 km × 8 km	8,192 m square	4	160	4,000–7,000
Large	16 km × 16 km	16,384 m square	6	240	8,000–10,000 baseline; higher experimental

These sizes preserve the Version 1.0 rationale: Standard supports several watersheds, travel times and polities while remaining inspectable; Large exercises the 10,000-person performance contract without pretending to represent a full planet.

16.3 Generation pipeline

1. Create a 128-bit seed, generation-version identifier and topology record.
2. Generate macro elevation with broad landforms, ridges and erosion-informed modifiers.
3. Compute drainage, watersheds, rivers, floodplains, lakes and coastlines.
4. Derive temperature, moisture, seasonality, elevation effects and rain shadows.
5. Derive soil, vegetation productivity, habitat, surface resources and carrying capacities.
6. Place concealed geological deposits with quantity, grade, depth, accessibility and extraction cost.
7. Create hazard fields for flood, fire, drought, storm, slope and disease-vector pressure.
8. Validate ecological and route diversity.
9. Identify broad candidate settlement regions.
10. Allow each founding group to select among candidates using its own knowledge, urgency, traits and controlled randomness.

16.4 Founding-site behavior contract

A founding group observes only knowledge available to it. It scores a bounded candidate set rather than searching every cell. The default minimal-survival profile weights water, food, arable land, timber, shelter, route access, security, resource opportunity and group preference, then subtracts hazard and claim conflict. Refugees, traders, military expeditions and colonists later use different profiles.

A plausible baseline normalized score is:

```
site = 0.22 water + 0.18 food + 0.14 arable + 0.08 timber + 0.06 shelter + 0.10 route + 0.08 security + 0.06 opportunity
+ 0.08 preference - hazards - claim conflict
```

The coefficients are policy data, not universal truth. The contract requires preconditions, candidate cap, knowledge filter, deterministic tie-breaking, keyed variation, trace output and invalidation when a site becomes impossible.

16.5 Terrain validity

A Standard seed should contain at least one navigable river, a secondary watershed, four ecological/resource regions, several viable founding sites, fresh water within a practical walking route, timber access, route corridors between regions, nonuniform hazards and concealed mineral diversity. The validator rejects worlds that lack viable interactions. It does not eliminate difficult or failed sites; it eliminates structurally unusable worlds.

16.6 Persistent ruins and terrain history

World generation establishes the base, but the world remains mutable. Extraction creates pits, spoil, roads and pollution. Rivers may be bridged, dammed or redirected by later systems. Urbanization changes parcels and terrain cover. War creates craters and ruins. Save chunks retain current physical state and historical-significance metadata so compaction cannot recreate a pristine generated map over centuries of change.

17 Time model

17.1 Authoritative clock

Store world time as a signed 64-bit count of simulation seconds from world epoch. Calendar conversion is a view. Content receives typed `WorldTime` and `Duration` values to prevent accidental mixing of wall time, frame time and game time. Money, inventories and long-lived accumulators use integer or fixed-point values with explicit overflow guards.

17.2 Fixed scheduling and speed

The base simulation loop runs at 20 fixed wall ticks per second. At 1× each tick advances three game seconds, producing one game hour per real minute. At 2×, 5× and 10× the game-time delta changes, but the system does not multiply every person's decision count by the speed factor. Due-time systems consume elapsed game time; discrete events retain canonical ordering.

Control	Game seconds per 50 ms wall tick	Approximate rate
Pause	0	Stopped
1×	3	1 game hour / real minute
2×	6	2 game hours / real minute
5×	15	5 game hours / real minute
10×	30	10 game hours / real minute

Table 6 — System cadence baseline

System	Cadence	Notes
Local movement and encounters	10–20 Hz	Route progress, local avoidance, hazard contact; no global replanning
Active combat sectors	10 Hz or event steps	Squad exchange and morale; visuals may interpolate at 60 Hz
Immediate hazards	2–10 Hz or event	Fire, drowning, acute wounds, collapse
Needs and individual goal review	Every 1–5 game minutes, staggered	Urgency can schedule earlier reconsideration
Work, hauling and construction	Every 5 game minutes or job event	Batched workplace processing
Household consumption and budget	Hourly/daily	Critical hunger bypasses normal delay
Market and price update	Every 6 game hours	Essential allocation can run earlier
Health and demography	Daily plus events	Births, deaths and acute outcomes are event-driven
Government review	Weekly/monthly plus crisis	Policy, budget and major projects
Diplomacy	Event-driven plus weekly	Sparse relevant relationships only
Culture and medium-rate attitudes	Monthly	Rolling lived conditions and institutional messaging
Ecology and resources	Daily/monthly/seasonal	Extraction events run at workplace cadence

17.3 Deterministic ordering

Event queues sort by due time, priority, stable target ID and sequence key. Randomness is addressed by world seed, stream family, stable entity ID, decision ID and occurrence counter. Daily and monthly work is staggered by stable-ID bucket to avoid “midnight spikes.” Continuous rates use rational fixed-point accumulators so larger time deltas do not create drift.

17.4 Long-run controls

A scheduler watchdog records backlog, overdue events, system duration and deferred work. If 10× cannot keep up, the UI reports “simulation limited by CPU” and advances at the maximum sustainable rate. It may lower presentation density or defer nonurgent UI queries, but it may not silently drop consumption, casualties, decisions or save integrity.

18 Person simulation

18.1 Persistent person record

Every person is a durable domain entity identified independently of Unity’s transient ECS entity handle. Components are decomposed by access pattern so systems process only relevant data. The minimum authoritative record contains:

Component group	First-playable fields	Primary update source
Identity	Stable ID, generated name, birth time, age, origin, alive/dead state	Creation, birth, death
Body and health	Vitality, nutrition reserve, injury, disease exposure, chronic burden, disability tags	Health, hazards, care
Baseline aptitudes	Physical health tendency, strength, cognitive aptitude, disease susceptibility, fertility and longevity tendencies	Controlled inheritance and variation
Behavioral traits	Temper threshold, courage, discipline, empathy, ambition, curiosity, sociability, aggression, caution, resilience and other bounded tendencies	Inheritance, creation, rare slow change
Skills and education	Domain skills, literacy, numeracy, apprenticeship, credentials, practical exposure	Work, schooling, mentoring, injury, age
Work	Occupation, workplace, role, schedule, compensation/entitlement, commitment	Labor and workplace systems
Material life	Household, home, possessions, liquidity, ownership shares, debt and obligations	Household, economy and property systems
Culture and belief	Culture exposure, language tags, faith/sect, conviction, ritual participation	Social exposure, institutions and events
Politics	Patriotism, issue attitudes, loyalty, trust, hostility, grievance, efficacy and faction support	Lived conditions, institutions, propaganda and events
Needs and fast state	Nutrition, rest, shelter, safety, care, belonging, autonomy, status, purpose, mood, fear, anger and exhaustion	Periodic needs and event reactions
Position and intent	Authoritative location, route progress, current action, goal, target and invalidation conditions	Movement and decision systems
Experience and ties	Major experiences, capped persistent social ties, military exposure and knowledge-carrier roles	Event systems
Appearance descriptors	Age band, culture, era, occupation, wealth, class, equipment, religion where relevant, health, injury and exposure	Derived authoritative presentation record

Names are generated from validated culture-specific phoneme and pattern data, never from a live model. The canonical name remains stable; social display names may change. Visual descriptors are simulation-facing tags, not direct sprite or mesh references.

18.2 Life course

Birth is evaluated at household and community level from reproductive-age population, health, nutrition, housing, cultural norms, child survival expectations, fertility tendency, law and economic security. The simulation does not model sexual activity. Children consume resources, receive guardianship, join households, inherit baseline tendencies and culture exposure, and may attend school or perform age-appropriate work according to law and circumstance.

Adulthood eligibility for occupations depends on age, capacity, skills, education, local demand, law, social access and reachable workplaces. Occupation selection is a bounded decision balancing compensation or entitlement, safety, commute, status, aptitude, household need, belief, discrimination and employer choice.

Aging and death respond to age, nutrition, disease, pollution, work, housing, climate, injuries and care. Medicine reduces specific risks; it does not add a universal lifespan bonus. Death creates vacancies, inheritance, household disruption, grief/cohesion effects, knowledge-carrier loss, succession and historical events where significant.

18.3 Health model

Health is causal but bounded. It separates current vitality, nutrition reserve, acute conditions, chronic burden, active disease exposures, care access and functional limitation. First-playable content includes hunger and malnutrition, injury, respiratory or contamination burden, waterborne disease pressure and one generic communicable-disease family with data-driven variants. Full epidemiology, detailed genetics and organ-level simulation are deferred.

A daily transition uses recent summaries from home, workplace and travel rather than querying raw environmental cells per person. Acute events schedule earlier processing. Health determines work capacity, movement, combat vulnerability, fertility, care demand, appearance state and mortality without collapsing the person into one "health percentage."

18.4 Needs and wellbeing

Needs are dimensions with level, trend, safe threshold, preference weight and social interpretation. The first playable tracks nutrition, rest, shelter/thermal comfort, safety, care, belonging, autonomy, status/respect and purpose/meaning. Wellbeing is a rolling multidimensional summary, not a single hidden truth.

The same destroyed home can therefore produce different outcomes. A wealthy household rebuilds; a poor household becomes displaced; a cohesive neighborhood organizes mutual aid; an excluded household radicalizes; a patriotic person volunteers after blaming a foreign attack; another patriotic person opposes the government for failing the country. These differences must emerge from resources, attitudes, traits, ties, institutions and evidence rather than arbitrary personality flavor.

18.5 Individual decision sequence

1. Validate whether the current commitment remains legal, possible and useful.
2. Pull a small role- and state-specific candidate set.
3. Read knowledge-filtered observations and cached local opportunities.
4. Reject candidates whose preconditions fail.
5. Evaluate canonical normalized considerations.
6. Apply traits, fast state, medium-rate attitudes, law, higher-layer priorities and social context.
7. Apply commitment inertia, cooldown, hysteresis and change penalty.
8. Apply a bounded keyed random adjustment where the contract permits.
9. Select through stable tie-breaking or deterministic weighted choice.
10. Reserve required unique resources, commit and schedule reconsideration.

11. Emit a structured decision trace.

No personal decision may scan the world, use undiscovered information or perform work owned by a higher layer. A person can decide to seek employment; the person cannot declare national war. A government can authorize mobilization; it cannot directly choose each citizen's footstep.

18.6 Person acceptance criteria

- 10,000 people retain stable identity through save/load and 500 game years of births and deaths without ID reuse.
- Displayed health, job, home, politics, goals and current action link to authoritative causes.
- No personal decision exceeds its candidate or spatial-query budget.
- Camera position does not change canonical replay checksums.
- At least 95% of working-age, non-incapacitated people in a stable Standard scenario have a valid work, care, study, household, travel or search activity rather than unexplained idleness.
- Unique resource reservations prevent two people from consuming or carrying the same item/batch entitlement simultaneously.
- Appearance descriptors update consistently after occupation, injury, displacement and equipment changes.

19 Households

19.1 Lightweight household

CONFIRMED The household is a lightweight co-residence, support and economic unit. It exists because dependents, housing, food, pooled money, migration, illness, unemployment, inheritance and survival cannot remain coherent if people are isolated records. It is not a romance or genealogy game.

A household stores members, responsible adults or guardians, dependents, home, tenancy/ownership, shared inventory, cash or entitlement policy, possessions, debts, inheritance rule, migration commitment, shared hazards and joint survival priorities. Partner links exist only where required by active family law. Parent links are retained for inheritance and continuity, not an exhaustive relationship drama graph.

19.2 Household decisions

Households decide among shared food allocation, emergency rationing, shelter repair, rent or tax payment, savings, care support, school participation, household production, relocation, splitting, joining, adoption/guardianship and inheritance. These decisions use shared needs and rules; they do not override individual emergency actions. A starving member may seek food even while a household's planned purchase is delayed.

Household migration is joint where practical. A move reserves destination housing or temporary shelter when available, assigns transport and possessions, and creates a shared commitment. War, eviction, disaster or Force Outcome can break the normal process and create separated members, missing dependents or fragmented possessions, all of which become explicit recovery tasks.

20 Inheritance

20.1 Controlled trait inheritance

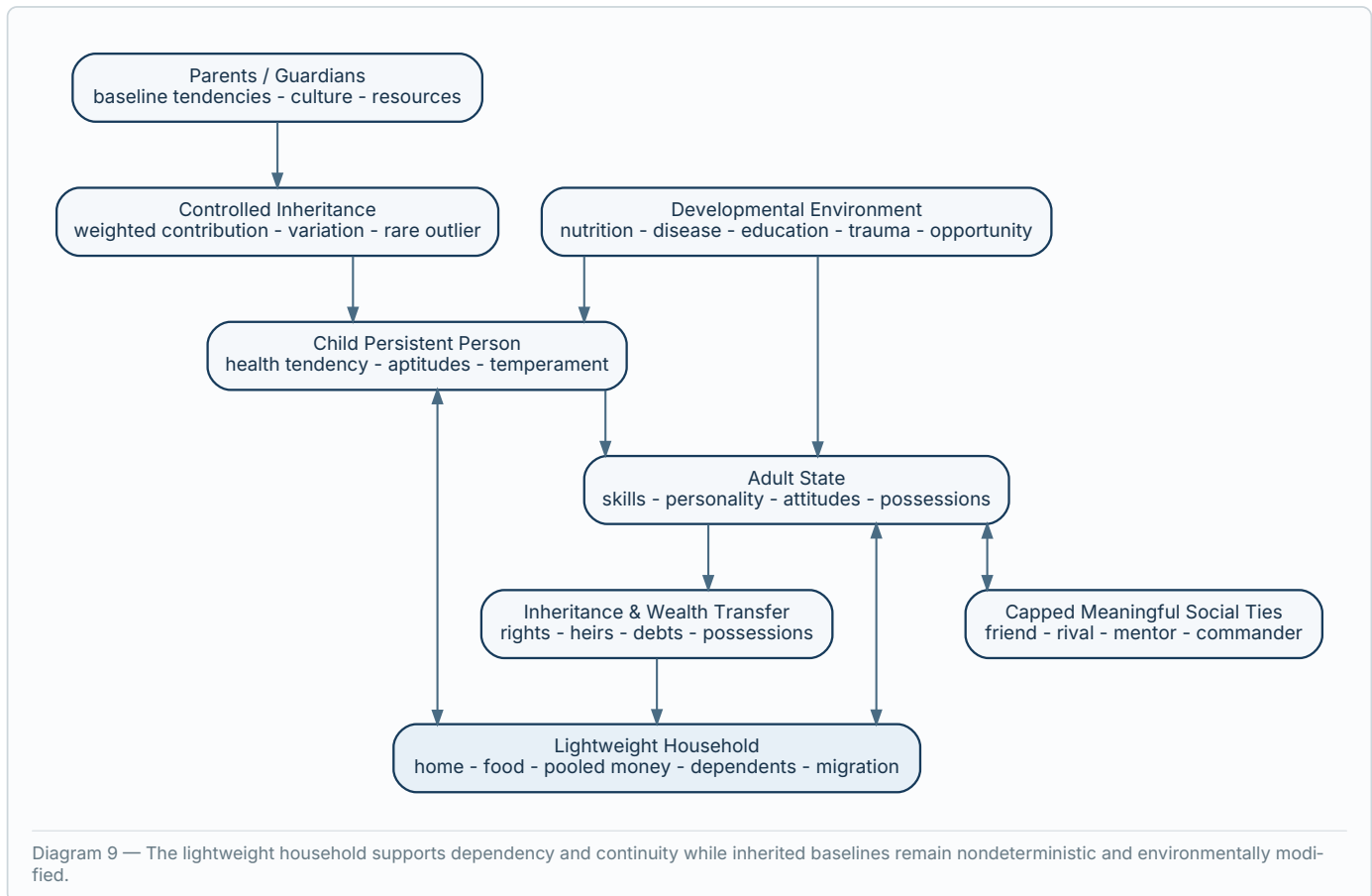
Children inherit a controlled mixture of parental baseline tendencies and aptitudes. Inheritance may affect health tendency, strength, disease susceptibility, cognitive aptitude, temperament, anger threshold, risk tolerance, sociability, fertility and longevity. The model uses weighted parental contribution, controlled variation, rare outliers and environmental modification.

Political values, patriotism, religion, class attitudes and ideology are not genetic. They are learned through household experience, education, propaganda, institutions, war, culture, legitimacy and personal treatment. A child may inherit a lower temper threshold while learning political restraint from education and law.

A recommended implementation represents inheritable quantitative baselines as bounded fixed values. For each field:

```
child = clamp(weightA × parentA + weightB × parentB + population prior + keyed variation + rare outlier)
```

Weights and variation are content-defined by trait family; no player-facing eugenic “quality” score is created. Environment and development then modify expressed state without rewriting the original baseline.



20.2 Inheritance and transfer

Death transfers property according to law, household rule, testament where supported, creditors, state claim and available heirs. Knowledge does not transfer automatically: plans, books, tools and mentorship can survive, but tacit skill may die with the carrier. A household can inherit debt, disputed land, an unfinished building or a politically significant artifact.

The save model retains minimal parent and heir references when historically or legally relevant. Ordinary remote kinship can be compacted to tombstone references; world-defining lineages, succession claims and preserved institutions retain explicit evidence.

20.3 Non-goals and limits

- No exhaustive family tree computation for every UI query.
- No detailed romance, attraction or dating system.
- No pairwise affection score among all people.

- No deterministic hereditary ideology.
- No thousands of genetic loci.
- No automatic household unity: members can leave, disagree or be separated when systems justify it.

21 Behavioral traits

21.1 Traits are coded modifiers, not labels

CONFIRMED Traits directly modify utility scores, thresholds, goal selection, risk evaluation, reaction speed, social behavior, military behavior, politics, migration, work, innovation, rebellion, crime and cooperation. They must have contextual advantages and disadvantages. “Good trait” and “bad trait” categories are prohibited.

Table 7 — Behavioral trait semantics

Trait	Primary coded effects	Contextual strengths	Contextual costs
Temper threshold	Anger response curve, grievance salience, reaction latency	Rapid resistance to abuse or threat	Escalation, conflict, impulsive retaliation
Courage	Fear threshold, willingness to enter danger	Defense, rescue, exploration	Excess risk, casualties
Discipline	Commitment persistence, compliance, training efficiency	Reliable work and military cohesion	Rigidity, slow adaptation
Empathy	Weight on others’ harm, aid and civilian restraint	Mutual aid, legitimacy, care	Personal sacrifice, reluctance in harsh tradeoffs
Ambition	Status and leadership utility, project pursuit	Innovation, political organization	Rivalry, corruption opportunity
Curiosity	Exploration, experimentation and learning utility	Discovery, adaptation	Resource-consuming failed experiments
Aggression	Coercive and confrontational action utility	Deterrence, decisive combat	Crime, escalation, diplomatic cost
Loyalty	Commitment to people, institutions or identity	Cohesion, resistance	Entrenchment, factional conflict
Conformity	Response to norms and institutions	Coordination, rapid diffusion	Suppression of novelty
Independence	Resistance to pressure and authority	Reform, entrepreneurship	Coordination cost, insubordination
Sociability	Social contact and influence frequency	Information spread, networks	Exposure to rumor or disease
Caution	Risk penalty and evidence requirement	Safety, stable planning	Missed opportunity, slow response
Resilience	Recovery curve after trauma and failure	Long-term adaptation	Can prolong untenable commitments if not separated from judgment

21.2 Temper example

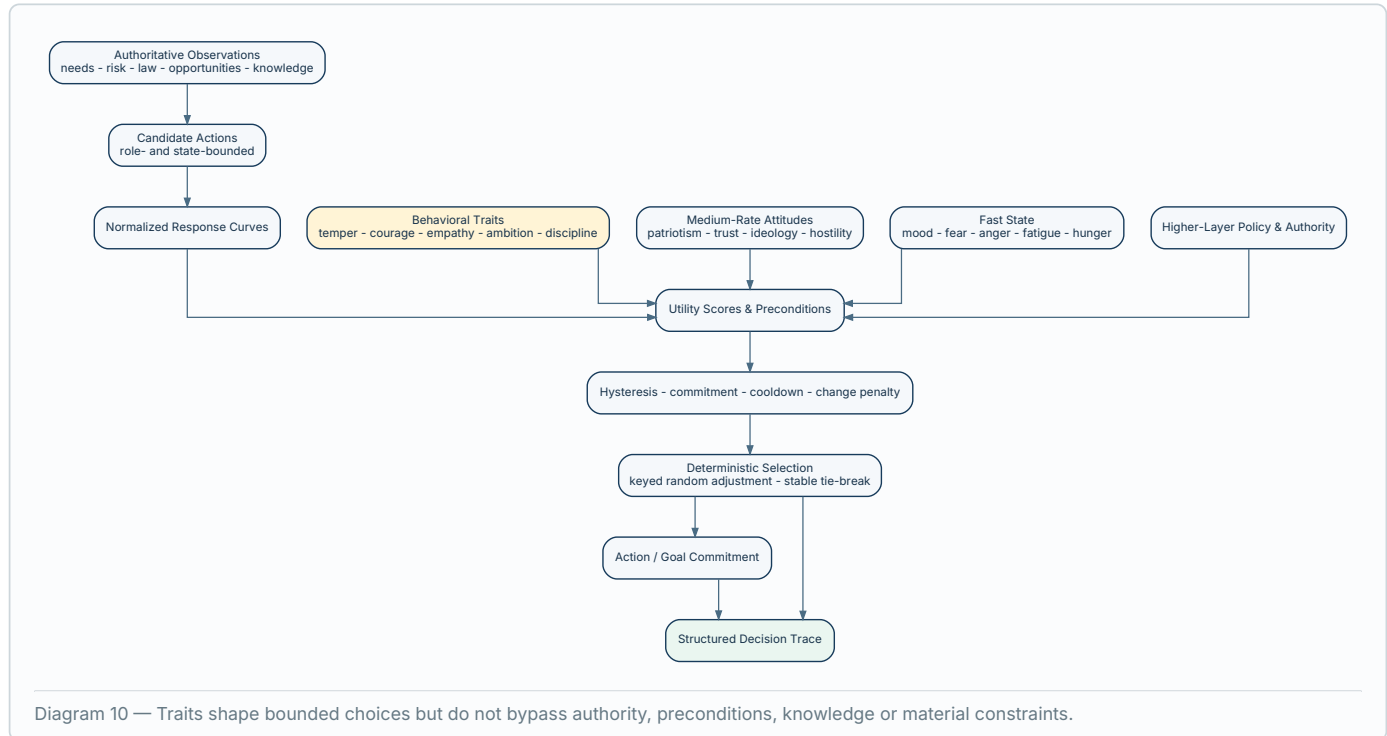
A low temper threshold raises anger faster under insults, shortages, unfair treatment, property destruction, family harm, repression, workplace abuse or military defeat. It does not guarantee violence. Violence remains a candidate that must pass opportunity, law, fear, discipline, social support, expected consequence and weapon-access checks. The trace should show “anger +0.31 from family harm” separately from “attack rejected: high arrest risk and household evacuation priority.”

21.3 Patriotism is learned and dynamic

Patriotism is a medium-rate attitude, not an inheritable personality trait. It develops through family, education, propaganda, war, government legitimacy, culture, religion, shared threat, victory, defeat and personal experience. It modifies willingness to volunteer, accept sacrifice, defend institutions, support projects, resist occupation, remain during crisis or donate resources.

Patriotism is not blind obedience. A highly patriotic person who believes the government betrayed the country may organize opposition, support a coup, join resistance or refuse an unjust order. The attitude has a target and interpretation: homeland, people, constitution, ruler, revolution or civilizational identity may compete.

21.4 Trait-to-decision pipeline



Trait effects are declared in validated policy data and executed by canonical evaluators. A trait may adjust a response curve, threshold, candidate eligibility, commitment duration or random-variation amplitude. It may not inject arbitrary code through content data.

21.5 Trait acceptance criteria

- Removing a trait changes only declared modifiers and does not corrupt current commitments.
- Increasing courage cannot make an impossible action feasible; it can only alter selection among feasible actions.
- Patriotism responds to education, legitimacy, war and personal experience, and cannot be inherited as a genetic value.
- Major experience can shift slow personality only within configured rates and evidence thresholds.
- Trait-driven behavior is visible in traces and statistically measurable across seed suites.
- No trait guarantees crime, rebellion, heroism, loyalty or innovation.

22 Personality change

| 22.1 Three rates of change

Fast-changing state includes mood, immediate fear, anger, exhaustion, hunger and short-term morale. It changes over minutes or days and directly affects urgent choices.

Medium-changing attitudes include patriotism, political loyalty, religious commitment, trust, hostility, ideology, attitudes toward outsiders and support for war. They change through repeated evidence, institutions and major events over weeks to years.

Slow-changing personality includes temperament, risk tolerance, sociability, discipline, empathy, ambition and core behavioral tendencies. It changes only gradually through major repeated experiences such as combat, trauma, prosperity, education, betrayal, cooperation, displacement, long poverty or institutional stability.

The person must remain recognizable over time. A single bad day does not rewrite personality; a decade of war may.

23 Social ties

| 23.1 Lightweight social ties

Each person retains a small capped set of meaningful persistent ties: household member, friend, rival, mentor, student, commander, subordinate, trusted colleague, political ally or personal enemy. Default cap: 8 persistent non-household ties, with temporary local contacts held in short-lived buffers. Historical figures may receive a higher cap only through explicit significance policy.

Ties support loyalty, betrayal, revenge, mentorship, influence, household aid, military cohesion and political organization. They are not a central relationship game. Social influence otherwise operates through cohorts—household, workplace, neighborhood, institution, unit, faction and culture.

24 Speech bubbles and social information

| 24.1 Purpose

Contextual speech bubbles make local state legible and create human-scale stories. They are not full conversations and do not use live LLM generation. Most lines reflect actual state; selected conversation events also spread information, rumor, attitudes, religion, propaganda, fear, knowledge, reputation or misinformation.

| 24.2 Content model

Lines come from authored pools, pregenerated validated libraries and grammar-safe templates. Each line has localization key, context tags, culture/era/occupation/mood/event tags, required observations, forbidden states, salience, cooldown, speaker constraints and optional simulation effect. Substitutions are typed—person name, place, commodity, policy, relative time—and must pass grammar for every locale.

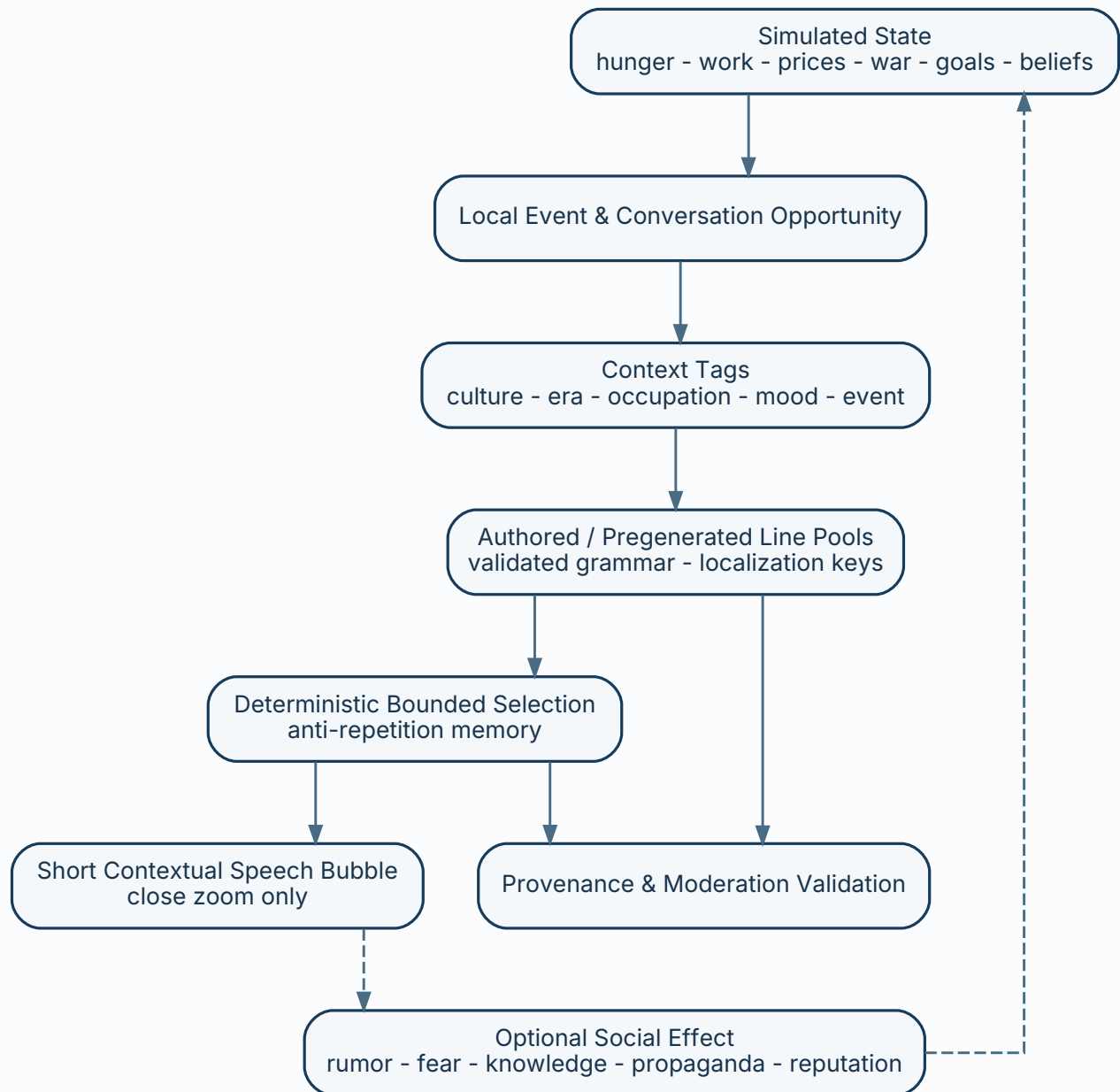


Diagram 33 — Speech content is authored and validated; only declared conversation events alter social information.

24.3 State-reflective examples

Hunger: "The grain stall is empty again."

Work: "The road crew still needs stone."

War: "My brother never came back from the pass."

Housing: "Three families are sleeping in the hall."

Policy: "The council raised the market tax."

Patriotism: "The government failed us, but this is still our home."

Migration: "There is work across the river."

Technology: "The new pump saves half the hauling."

Rumor: "They say the northern army has crossed the ridge."

The line selector must not reveal hidden information the speaker lacks. A rumor can be false; the bubble indicates heard belief, not authoritative truth.

24.4 Social transmission

A conversation event requires proximity or shared institution, communication capacity, available time and willingness. It carries a compact payload: claim, confidence, source reputation, emotional charge and decay. Receivers update belief or attitude according to trust, prior belief, evidence and traits. The renderer may show a bubble without simulating transmission; only events tagged with effects do so.

24.5 Repetition control

The system tracks recent line family per person, location and global view. It rotates synonyms and context combinations, suppresses low-value repetition and uses aggregated crowd chatter at high density. First playable requires a thin library sufficient to validate architecture; thousands of localized lines are deferred content.

24.6 Speech acceptance criteria

- No line states information the speaker cannot know unless explicitly tagged rumor/misinformation.
- Templates produce grammar-safe output for test locales.
- Bubble density remains within frame and clutter budgets.
- Simulation-effect conversations emit a trace and bounded social update.
- Repetition metrics remain below configured family frequency in a 10-hour observed-settlement test.

25 Settlement simulation

25.1 Settlement as a socio-spatial system

A settlement is a persistent organization of people, parcels, buildings, roads, workplaces, institutions, stockpiles, services, authority and recognized territory. Its class is descriptive, not a promotion flag. Population creates demand and capacity but does not alone turn a village into a city.

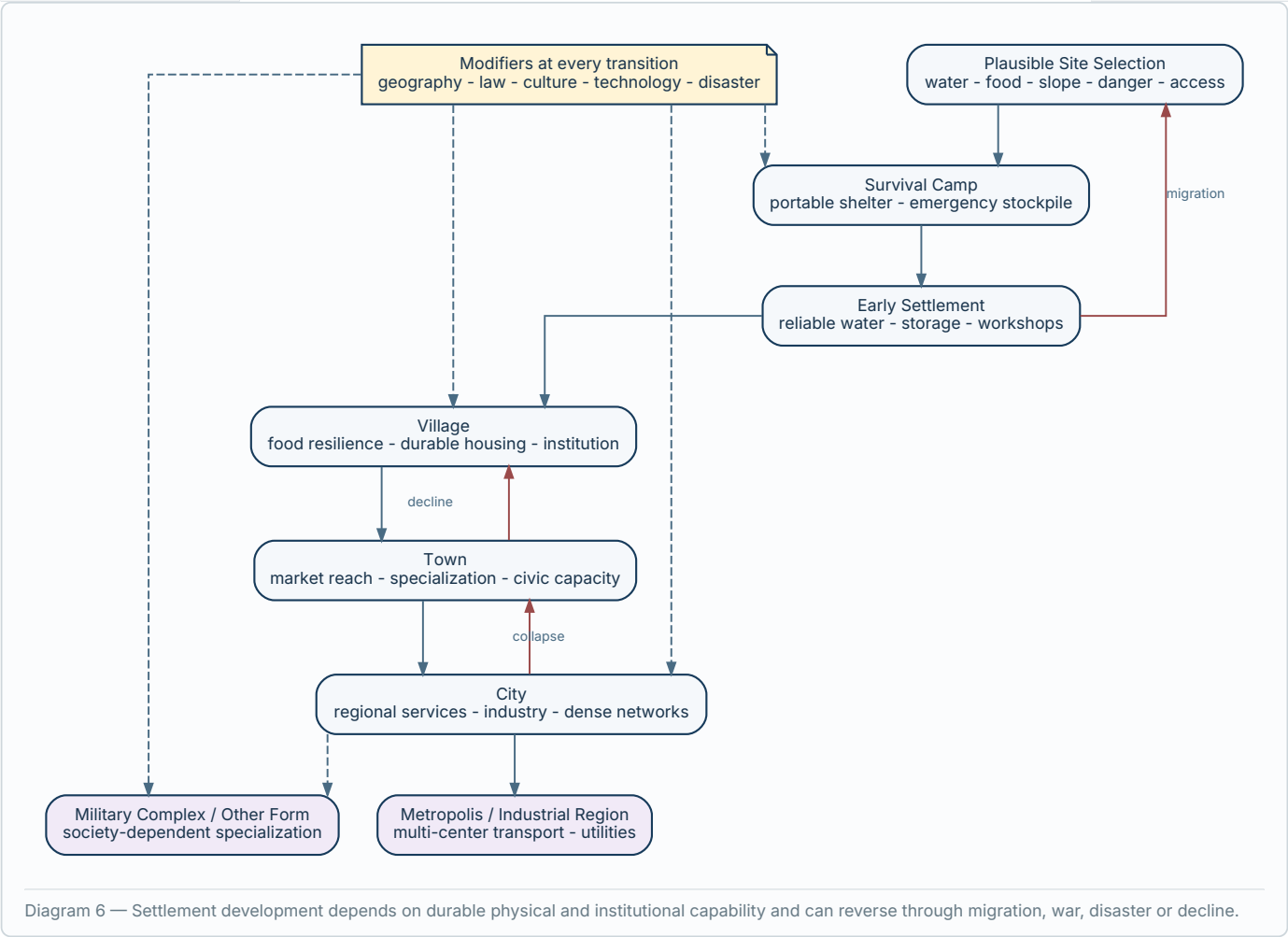


Table 8 — Settlement development states

State	Required capability	Typical physical form	Reversion causes
Survival camp	Water, temporary shelter, emergency food plan, recognized group	Portable shelters and common stockpile	Death, exhaustion, forced migration
Early settlement	Recurring food, durable shelter, storage, leadership	Homes, work yards, paths, first shared building	Harvest failure, disease, conflict
Village	Multi-season food resilience, specialized work, institution, maintained routes	Farms, workshops, market place, neighborhood clusters	Depopulation, service loss
Town	Surplus exchange, civic capacity, route node, multiple institutions	Denser center, public works, specialized districts	Trade collapse, fiscal failure, war
City	Regional services, diversified economy, administrative reach, infrastructure	Multiple centers, formal roads, industry, utilities	Prolonged systemic collapse
Metropolis / industrial region	Linked centers, mass transport, utilities, regional governance	Continuous urban region and skyline	Deferred content; architecture reserved
Military complex / other form	Society-specific strategic specialization	Bases, fortifications, logistics and controlled access	Mission loss, supply collapse, political change

25.2 Founding and early growth

A founding group selects a site under the behavior contract, creates a charter, designates a common stockpile and assigns provisional roles. People physically clear, haul, build and finish structures. Construction consumes commodity batches, labor time and tool capacity. Visual phases correspond to authoritative progress.

The settlement expands around water, farms, paths, workplaces, markets, defense and existing structures. Households and businesses choose locations according to ownership, access, profit, employment, safety, wealth, transport, culture, social ties, pollution, law, discrimination and available land. Governments and institutions later plan public works, districts, bases, new towns, rail corridors, utilities and redevelopment.

25.3 Projects, not direct mutations

A settlement planner creates bounded project candidates, priority weights, reservations and authorization requests. It does not place every citizen or mutate personal inventory. A housing project may be initiated by households, a landlord, a communal body, a business, a religious institution or government according to law. The selected actor obtains land rights, reserves inputs, recruits labor and schedules construction.

The player can order “expand housing” or “build a port.” The settlement selects exact location and design under its own policy. Force Outcome is the only normal player path that directly places or completes the object.

25.4 Services and spatial capacity

Water, food distribution, shelter, storage, health, education, sanitation, safety, administration, market access and defense derive from actual buildings, staff, stockpiles and routes. Capacity is spatial: a clinic across a destroyed bridge is not available. Overlays expose nominal capacity, reachable capacity, utilization and bottleneck.

25.5 Land and rights

Parcels are created from roads, natural boundaries, claims and survey. Rights are bundles: occupancy, use, extraction, construction, transfer, inheritance, taxation, exclusion and requisition. Law maps those rights to communal, household, private, elite, company, government, religious, military or occupied control. Mixed systems coexist.

25.6 Damage, abandonment and recovery

Buildings degrade through age, fire, flood, war, neglect, pollution and extraction damage. Damage reduces function and raises accident risk. Repair competes with new construction. Abandoned buildings may become shelter, salvage, hazard, heritage, religious site, military position or knowledge artifact. Destruction leaves parcels, foundations, routes, contamination and history rather than deleting the place.

25.7 Settlement acceptance criteria

- At least 80% of valid baseline seeds produce one surviving village by year 15 without intervention; failures remain explainable.
- At least two founding groups develop measurably different land use or specialization by year 25.
- Every completed structure has attributable deliveries, person-hours and authority.
- Roads emerge from traffic and projects, not population thresholds.
- No settlement planner teleports labor or goods.
- Damage, salvage and rebuilding preserve stable identities and causal history.
- Normal player orders never bypass settlement siting and construction logic.

26 Physical city growth

26.1 Society-dependent hybrid growth

CONFIRMED City development combines organic household and business location with institutional and government planning. Early growth follows water, food, paths, work, ports, markets, resources and defense. Later governments may build planned roads, districts, public housing, industrial zones, bases, forts, administrative centers, new towns, railways, highways, ports, utilities and redevelopment.

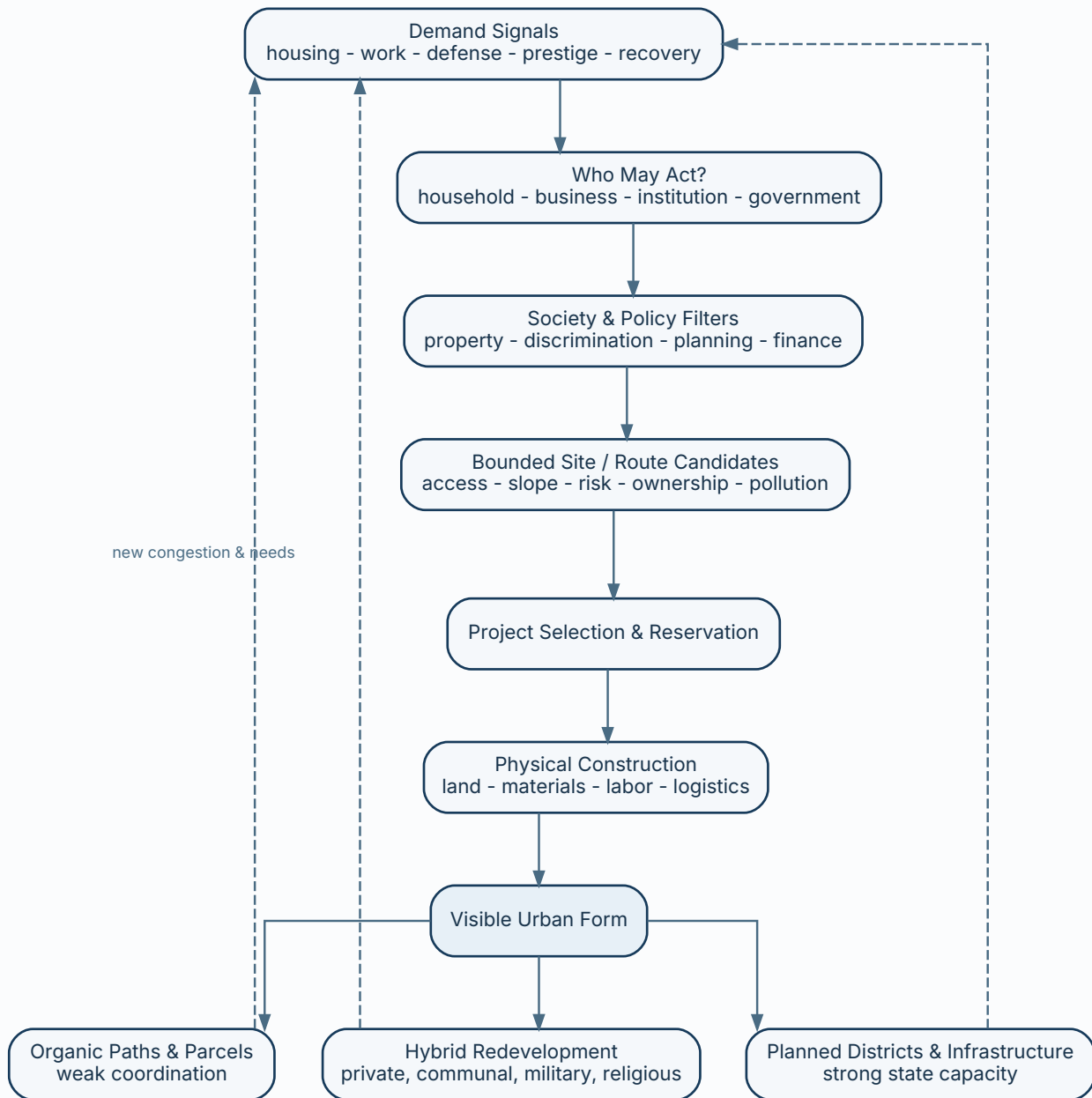


Diagram 7 — Urban form emerges from shared physical constraints filtered by ownership, government, culture and institutional capacity.

Weak government may produce sprawl and unreliable services. An authoritarian government may demolish neighborhoods for monumental routes. A communal society may prioritize compact shared facilities. Private development may create commercial centers, speculative subdivisions, industrial corridors and wealth segregation. Military settlements prioritize defense, logistics, controlled access and sightlines. Religious settlements center sacred institutions. Colonial systems emphasize extraction and transport. Post-disaster cities rebuild around surviving infrastructure and scarred land.

26.2 Location and route selection

Building and infrastructure candidate generation is bounded. Candidate parcels must satisfy footprint, entrance and slope constraints. Scores then consider route access, hauling distance, water, sanitation, fire and flood, land rights, input and customer access, pollution, security, culture, neighborhood ties, expected expansion, finance and demolition cost.

Road construction follows a parallel process. Travel creates a decaying heat field. Sufficient repeated movement creates a desire path. Households, businesses, governments and militaries may formalize that path or propose a strategic alternative. The route becomes an authoritative graph edge with geometry, surface, width, capacity, condition, owner, maintainer, toll/access, seasonal reliability and upgrade lineage.

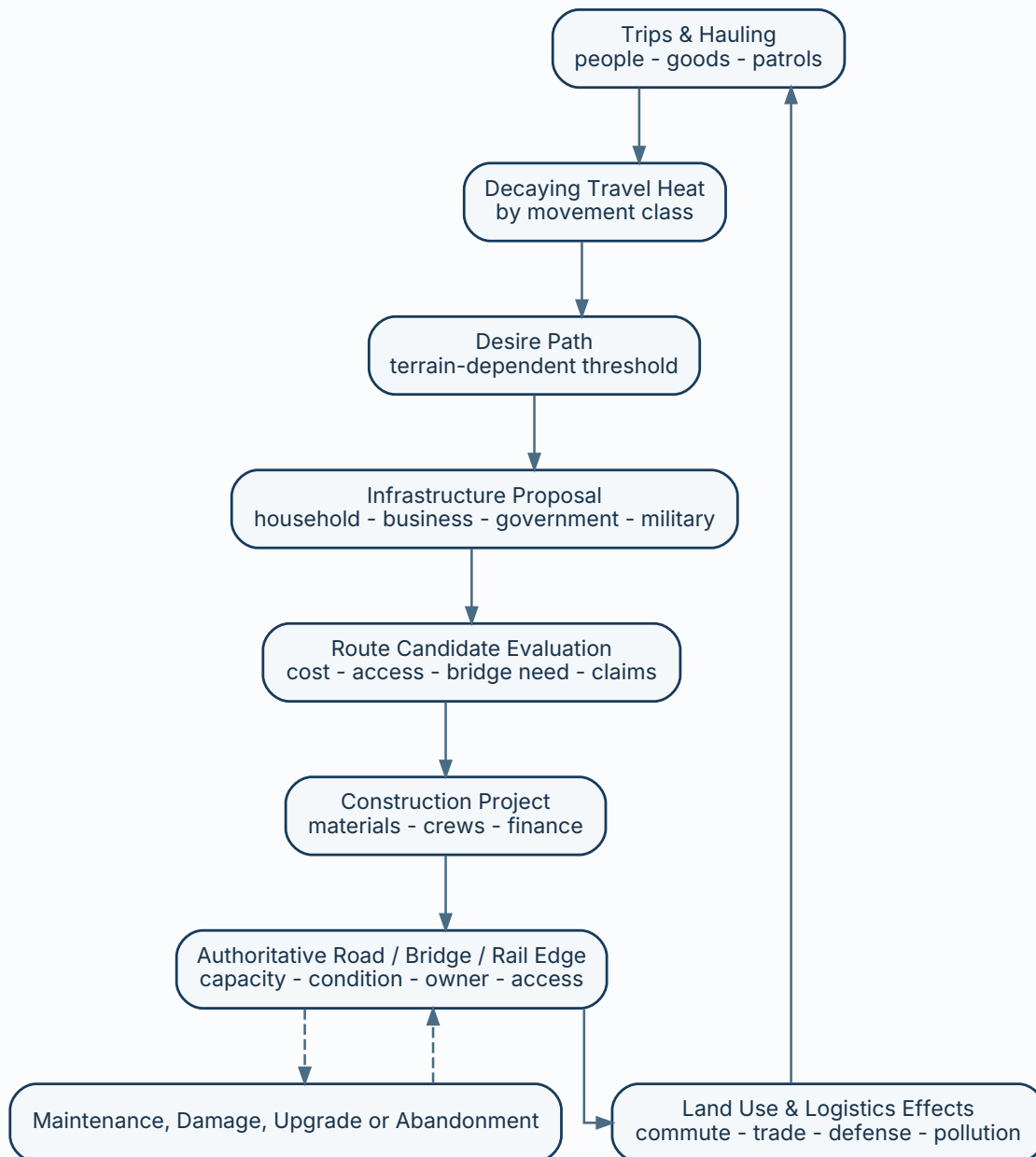


Diagram 8 — Repeated movement can create paths; authority and investment convert them into roads, bridges, railways and other durable infrastructure.

26.3 Neighborhoods and districts

Neighborhoods are derived stable clusters based on travel graph, density, land value, culture, wealth, work access, services, safety, pollution and exclusion. Boundaries evolve gradually. Formal districts may overlay neighborhoods where government planning or law creates them, but they do not replace the physical graph.

Household location utility includes affordability, tenure, commute, kin/cohort proximity, cultural and faith access, school, market, safety, pollution, crowding, law and discrimination. Businesses consider labor, customers, inputs, transport, land cost, utilities and externalities. Government projects consider strategic and political value in addition to technical feasibility.

26.4 Physical construction sequence

Every project progresses through proposal, authority, site/route selection, land acquisition, design variant, finance, reservation, site preparation, foundation, structure, utilities/equipment, commissioning and occupation. Work packages can overlap where the contract allows. Visual construction uses the same stage identifiers. Cancelling a project releases reservations, leaves partial work or salvage, and records sunk cost.

26.5 No manual placement

The player cannot normally draw a road or place a building. Direct decision UI chooses objective, scope, priority, constraints and target region. A "rail connection between A and B" order can include acceptable cost, protected areas and deadline, but route and engineering remain simulated decisions. A Force Outcome tool may place a selected route or structure directly; the resulting object still requires maintenance, operation and supporting networks.

26.6 Growth into skyscrapers and industrial regions

ARCHITECTURE RESERVED Later cities can assemble apartment blocks, factories, power plants and skyscrapers from modules. A skyscraper comprises foundation, podium, repeated floor modules, mechanical sections, roof, antenna and façade variations. Its interior is not rendered; capacity, occupants, employment, utilities, equipment, damage and evacuation remain authoritative building state.

The road graph generalizes to rail, highways, ports and utilities. The city model must not be replaced when new eras arrive. Later content adds route types, building modules, power and mass-transport constraints, not a separate "modern city" simulation.

26.7 City-growth go/no-go criteria

Before heavy art investment, a prototype must demonstrate:

- visible camp-to-village growth using real deliveries and labor;
- two society presets producing different layouts from the same terrain;
- at least 1,000 visible building instances with LOD and selection;
- 5,000 moving billboards in a city stress scene with acceptable frame budget;
- road/parcel changes causing deterministic route and land-use effects;
- destruction of a neighborhood causing displacement and visible rebuilding;
- save/load preserving footprints, construction stages, ruins and route graph;
- no camera-dependent authoritative behavior.

27 Roads and infrastructure

CONFIRMED Roads, bridges, ports, canals, later railways and utilities are physical authoritative assets rather than invisible bonuses. They occupy space, have owners and access rules, possess capacity and condition, affect land values and city form, and can be damaged, blocked, upgraded, repaired or abandoned.

27.1 Desire paths and formal projects

Repeated travel accumulates decaying movement-class heat. Plausible desire paths may improve through local action, but a durable road, bridge, canal, port or fort road becomes an explicit project with route candidates, land rights, finance, materials, labor and maintenance responsibility. A path cannot become a paved network edge merely because a texture darkened.

27.2 Hierarchical network

Infrastructure participates in the strategic region graph, settlement road graph and local entrance/steering layer. Long trips reuse route corridors; local movement resolves only near active entities. Damage invalidates affected edges and caches rather than forcing global recomputation. Military and civilian access, tolls, border controls, weather, bridge limits and transport modes modify route cost and capacity.

27.3 Urban and logistical effects

New access changes parcels, commute, market reach, extraction viability, defense, pollution and development demand. Those consequences feed back into project priorities. A new port may create warehouses and neighborhoods; a destroyed bridge may create shortages, refugee congestion, military isolation and political pressure.

27.4 Presentation and inspection

The normal view shows paths, surfaces, intersections, bridges, construction, damage, traffic and blockages. Overlays expose capacity, ownership, access, condition, congestion, shipments and military control. Distant rendering simplifies geometry but retains route topology and major damage.

27.5 Acceptance criteria

- Repeated travel can create a path candidate without fabricating a completed road.
- A blocked route produces traceable detours, shortages and project pressure.
- Network edits invalidate bounded caches.
- People visibly enter and leave building entrances through valid local routes.
- A save/load checkpoint preserves route, project, reservation and traversal state.
- Road and bridge stress scenarios remain within pathfinding and rendering budgets.

28 Civilization simulation

28.1 Definition

A civilization is a recognized political-cultural network that may contain settlements, regions, governments, institutions and minority cultures. It is not synonymous with one ethnicity, state, government or city. Its record contains membership and citizenship rules, settlements, claims, government capacity, treasury, obligations, public inventories, cultural composition, capability portfolio, diplomatic relationships, military organization, strategic priorities, law, property regime and historical summaries.

28.2 Layered authority

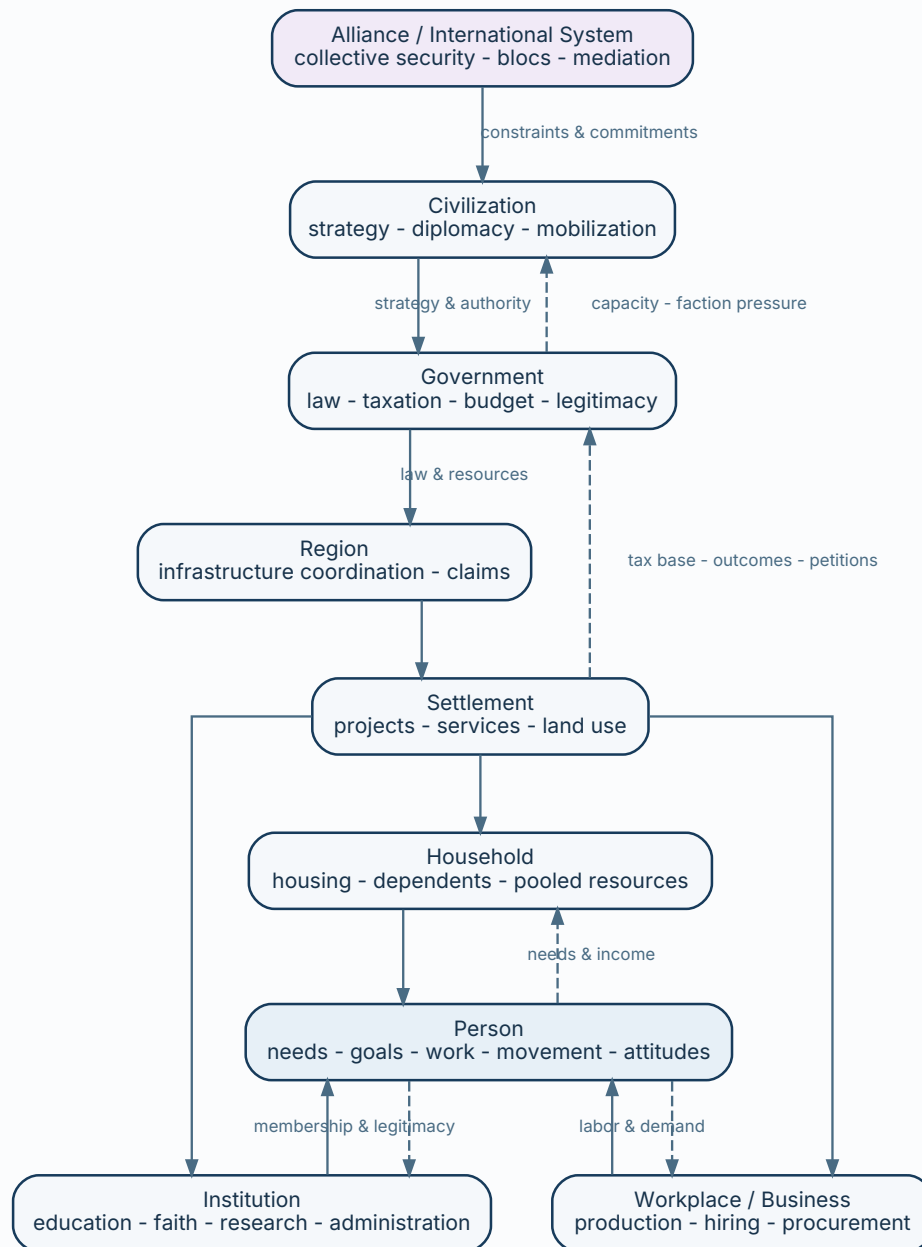


Diagram 3 — Higher layers coordinate and constrain; lower layers supply labor, demand, support and outcomes. No layer micromanages decisions below its authority.

The authority model is enforced in code and behavior contracts. A civilization chooses strategic posture and major programs; government authorizes law, budgets, taxation, policing, mobilization and diplomacy; regions coordinate infrastructure and claims; settlements choose projects and services; workplaces produce and hire; households allocate shared resources; people act locally. Higher-level policy may constrain or prioritize lower decisions but cannot call lower-level movement or inventory mutation APIs directly.

28.3 Strategic planning

Civilization planning is event-driven plus periodic review. It maintains a small portfolio of goals—food security, route access, defense, legitimacy, revenue, knowledge, health, territorial claim, alliance, integration or recovery. Each goal has urgency, expected value, cost, risk, political support, authority requirement, confidence and active projects. It selects among bounded plan templates rather than searching arbitrary future action sequences.

Direct player orders enter at this layer when appropriate. A binding colonization order creates an obligatory strategic goal. The civilization still selects expedition, target candidates, membership, supply and route. If all feasible candidates fail, the order remains blocked and its trace explains why.

28.4 Formation, merger, fracture and extinction

A civilization forms through durable mutual recognition, shared authority or treaty structure, and membership identity. It may be a city-state or multi-settlement polity. Merger inherits institutions, debts, treaties, claims, knowledge and autonomy; it does not instantly homogenize culture or law.

Separatism depends on material benefit, legitimacy, representation, culture, coercion, security, organization, leadership, resources and viable alternatives. State collapse occurs when authority, fiscal capacity, control, legitimacy and services fall below recoverable thresholds across regions. People and settlements persist under successors, occupiers, institutions, militias or no recognized authority.

Extinction is split into state extinction, cultural extinction and population extinction. A country may disappear while its people and identity survive. A culture may disappear through assimilation even when descendants remain. Population extinction means no surviving members. Recovery can occur through descendants, refugees, rebels, exiles, preserved institutions, surviving settlements and restoration movements.

28.5 Civilizational metrics

The UI reports population resilience, food and energy security, productive capacity, logistics reach, treasury and credit, military readiness, capability depth, institutional capacity, legitimacy, faction balance, diplomatic network, cultural influence and environmental condition. A composite convenience index may be offered for comparison but never drives AI directly.

28.6 Civilization acceptance criteria

- Identical starting groups under different geography and government produce measurable divergence.
- Higher layers never issue personal movement or consumption commands.
- Binding player orders are represented as explicit strategic obligations with blockers and progress.
- State collapse preserves people, assets, cultures and successor possibilities.
- Restoration from remnants produces continuity claims and real capacity limits, not an exact duplicate.
- Strategic reviews remain within candidate and time budgets.

29 Culture and religion

29.1 Culture as carried tendencies

Culture is a changing set of overlapping tendencies carried by people, households and institutions. It is not a civilization-wide static label. The first-playable tendency set remains: militarism/civilian restraint, expansionism/territorial restraint, trade openness/autarky, scientific openness/traditionalism, religious/secular authority, centralization/local autonomy, individual/communal property, hierarchy/egalitarian preference, risk/precaution, outsider acceptance/exclusion, education/apprenticeship preference and stewardship/extraction.

These are not moral ratings. Centralization can coordinate infrastructure and magnify information failure. Communal property can protect access and suppress individual investment in some contexts. Trade openness can enrich a society and expose it to blockade.

29.2 Cultural change

Monthly updates use rolling summaries of household and workplace experience, war, occupation, displacement, prosperity, inequality, education, communication, institutions, propaganda, law, migration, technology success, disaster, foreign contact and player interventions that people can observe. Change is cohort-sensitive and path-dependent. Institutions and older cohorts create inertia; crises can create discontinuity.

Culture changes utility weights, policy support, location choice, institutional behavior, speech pools and visual descriptors. It never forces identical action.

29.3 Religion and belief institutions

Religion uses validated doctrine modules, practices, organization, sacred places or objects, membership and institutional capacity—not generated scripture. Doctrine may address authority, ritual, charity, education, violence, conversion, family, property, outsiders, environment and succession. Religions form when a belief community gains sustained identity, carriers and institutions. Sects form through doctrinal or political divergence. “Cult” describes a small high-commitment or restricted organization; it is not an automatic evil modifier.

Faith institutions can provide belonging, charity, education, legitimacy and trans-settlement networks. They can also create exclusion, conflict or resistance depending on doctrine, conditions and government response. No supernatural effect occurs unless a world rule or player power explicitly creates it.

29.4 Patriotism, identity and government opposition

Patriotism can attach to place, people, constitution, ruler, religion, revolution or civilizational myth. Education, propaganda, war and legitimacy alter it. A patriotic person may oppose a government believed to have betrayed the country. The political model therefore separates support for polity, government, law, leader and current policy.

29.5 Cultural extinction and revival

A culture becomes locally extinct when no active carriers remain in a region; functionally extinct when only passive records or isolated speakers remain; and globally extinct when no viable carriers, institutions or recovery mechanism remain. Revival may use descendants, records, ritual sites, language material, exiles and political movements. A revived culture is historically linked but can differ from its predecessor.

30 Government and politics

30.1 Composable governance

Government is a composition of executive selection, deliberative body, franchise, administrative reach, fiscal system, property defaults, rights and policing, succession/removal, regional structure, military authority and legitimacy sources. This supports monarchies, councils, oligarchies, democracies, dictatorships, military rule, religious rule, company settlements, communal systems and hybrids without a moral hierarchy or separate engine per label.

The first playable retains four presets over shared components:

Preset	Authority pattern	Public works / property tendency	Main instability risk
Communal assembly	Eligible adults and rotating delegates	Communal parcels and labor contribution	Slow response, deadlock

Preset	Authority pattern	Public works / property tendency	Main instability risk
Council with selected/elected head	Council budget; head executes	Mixed communal and household rights	Elite capture, legitimacy dispute
Hereditary/chiefly rule	Lineage leader with advisors	Ruler/elite grants and customary use	Succession crisis, extraction abuse
Appointed/autocratic magistracy	Concentrated executive and administrators	State allocation and requisition	Coup, coercion cost, information failure

30.2 Legitimacy and support

Legitimacy is a vector: popular support, elite support, institutional compliance, legal/traditional legitimacy, performance legitimacy, coercive control and external recognition. A low-popularity state may remain stable through elites and coercion but becomes costly and shock-sensitive. A popular government may fail if administration cannot execute policy.

30.3 Factions

Factions form around material interest, workplace, land, military, faith, region, culture, policy and leadership. A faction has support weights, active members, organizers, resources, institutions, grievances, program, risk tolerance and methods. People may weakly support several factions while active membership remains limited.

30.4 Policy and enforcement

Policies are versioned definitions with authority requirement, administrative cost, target, enforcement mechanism and expected effects. Taxes, land access, school funding, tariff, conscription, rationing, sanitation, religious establishment/toleration and policing priorities alter actual transactions, projects, services and legal decisions. A policy does not apply an immediate civilization-wide bonus; incomplete enforcement and resistance remain visible.

30.5 Coups, reform and rebellion

Political change proceeds through grievance, salience, organization, coalition, petitions or elections, strikes or noncompliance, government response, and only then coup or rebellion if force, opportunity and expected support exist. Unhappiness cannot spawn an army. Rebellion uses persistent people, weapons, supply, leadership and the warfare system.

Revolution changes authority, property, institutions and obligations. It does not guarantee liberation, repression, prosperity or collapse. Post-change legitimacy and capacity must be rebuilt.

30.6 Political feedback

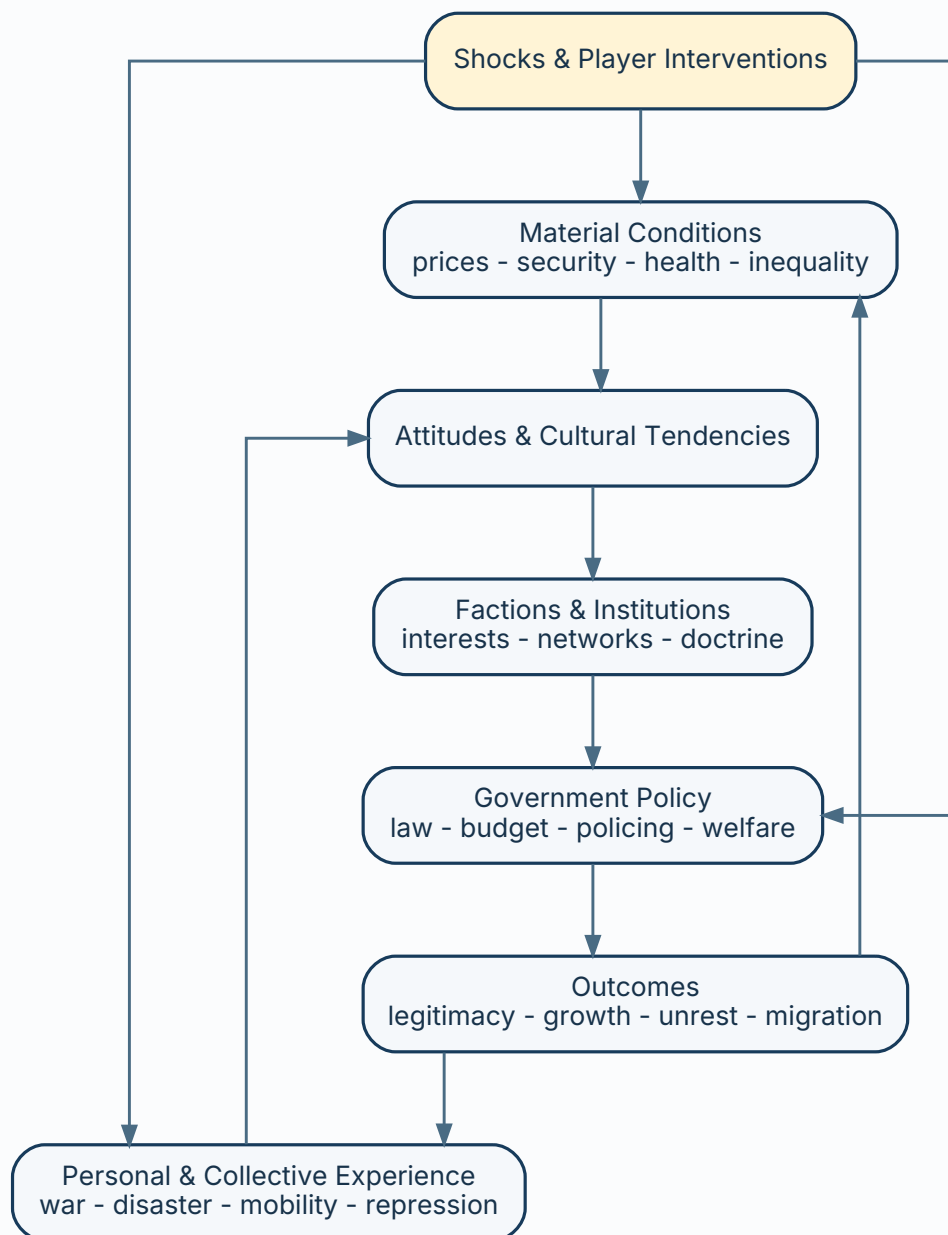


Diagram 18 — Politics and culture emerge from repeated material and institutional feedback rather than static ideology bonuses.

30.7 Direct player government control

Binding orders can set government policy, budget priority, mobilization, diplomacy, infrastructure, migration, research and strategic projects. The government must still execute through legal authority and administrative capacity. Force Outcome may replace a government or impose a law instantly; that creates succession, legitimacy, enforcement and resistance consequences rather than automatic compliance.

30.8 Political acceptance criteria

- Presets produce different authorization, land allocation, revenue and succession through shared components.

- Policy effects are attributable to enforcement and compliance.
- Leadership transitions survive save/load and replay.
- Rebellion requires actors, organization, resources and opportunity.
- Patriotism can support either government defense or opposition depending on perceived national interest.
- Forced government change produces coherent membership, law and legitimacy state and a permanent intervention record.

31 Economy

31.1 Selected abstraction

RECOMMENDED The economy uses **physical commodity batches plus institution-mediated allocation, periodic markets and compact ledgers**. Location, transport, ownership, shortage and production remain causal; individual coins and microscopic items do not.

Pure stockpile command simulation cannot express mixed ownership, wages, credit or inequality. Full item-and-coin simulation would multiply state and hauling. Pure equilibrium models would fabricate local availability and weaken off-screen causality. The hybrid accepts reconciliation and damping complexity because it best fits the vision.

31.2 Commodity representation

Commodities are fungible classes stored in integer quantities. Quality, age, spoilage, provenance and ownership are represented where they affect decisions. Inventories belong to households, buildings, workplaces, shipments, governments, military units and selected people. Soil fertility, land, pollution, water flow, route capacity and labor are resources but not ordinary commodity stacks.

Table 9 — First-playable commodity and economic content

Class	Type	Core use
Fresh water	Essential flow/batch	Drinking, food, sanitation, production
Staple food	Perishable	Calories, seed conversion, rationing
Protein food	Perishable	Nutrition quality and trade
Seed stock	Seasonal capital input	Planting and recovery
Timber	Raw	Construction, tools and fuel conversion
Fuelwood	Fuel	Heating, cooking and workshops
Stone	Raw	Durable construction and roads
Clay	Raw	Brick, pottery and sanitation
Fiber	Raw	Textiles, rope and sacks
Iron ore	Finite raw	Metal production
Coal	Finite fuel/raw	Smithing efficiency and later powered industry
Medicinal inputs	Renewable/harvested	Treatment and sanitation
Lumber	Processed	Efficient construction
Masonry/brick	Processed	Durable buildings and infrastructure
Tools	Durable capital good	Productivity and capability gates
Construction goods	Processed bundle	Fittings, fasteners and prepared parts

Class	Type	Core use
Textiles/consumer goods	Household good	Clothing, status, trade and wellbeing
Weapons/ammunition	Durable/consumable bundle	Hunting, defense and war

31.3 Production and workplaces

A workplace stores recipes/capabilities, equipment, capacity, labor roles, inventories, maintenance, quality, ownership and operating policy. Recipes transform reserved inputs, labor time and equipment wear into outputs, by-products, pollution and waste. Production occurs in lots; the game does not simulate each hammer strike.

A lot starts only when inputs, legal access, tools, workers and storage are available. Progress derives from skill, health, equipment, infrastructure and interruptions. Finished output enters a real inventory. The decision trace distinguishes "planned," "reserved," "in progress," "blocked," "completed" and "failed."

31.4 Labor

People supply labor through employment, household production, communal obligation, state assignment, military service or self-employment. Jobs declare role, skill, schedule, compensation/entitlement, safety, duration and location. Matching is local or regional: the system prefilters by travel reach, eligibility, law and skill before scoring.

Labor scarcity can raise compensation, delay harvest, attract migrants, encourage mechanization, create coercive pressure or cancel projects. Traits affect willingness and persistence but cannot substitute for qualification or physical access.

31.5 Money, prices and credit

Societies may use barter, shares, dues, ration entitlements or currency. Currency balances are signed 64-bit minor units; physical transport of money is abstracted until secure monetary logistics becomes strategically relevant. Markets update local prices from available stock, purchase demand, incoming shipments, reserve policy, recent clearing, market power and expectations. Daily change limits and hysteresis prevent uncontrolled oscillation, with crisis rules explicitly allowing larger movements.

The first playable supports simple debts: principal, interest, schedule, collateral/guarantor and default. Banking, securities, central banking and complex money supply are deferred. Merchant and government credit can finance construction or trade, creating real risk and unequal access.

31.6 Ownership and accounting

Assets have legal owners and controllers: person, household, business, institution, government, communal body, military or occupier. Ledgers track relevant revenue, cost, wages/entitlements, taxes, debt and maintenance. Full double-entry accounting per person is unnecessary, but all transfers must reconcile quantity and ownership. Debug builds run conservation checks by commodity and declared source/sink.

31.7 Allocation mechanisms

The active political economy selects among household self-provision, communal stores, markets, business procurement, government purchase/requisition, military allocation, charity and illicit exchange. These mechanisms share inventories, rights and reservations. A government can order industrial priority, but exact procurement and labor still follow its institutions and law.

31.8 Economic acceptance criteria

- Commodity conservation holds except declared production, consumption, decay, extraction, loss and intervention.
- Closing a route produces location-specific stock and price effects after causal travel delay.

- Markets cannot purchase unavailable goods or teleport inventory.
- Different property/government arrangements produce different allocation and city form using shared primitives.
- Price and labor systems converge under stable benchmark conditions.
- Poverty, insolvency, rationing and failure have inspectable ledgers and causes.
- Force Outcome inventory changes create explicit sources/sinks and history rather than breaking conservation diagnostics.

32 Resources and logistics

| 32.1 Resource fields and knowledge

Natural resources have quantity/reserve, grade, accessibility, depth, extraction difficulty, renewal/finite state, seasonality, transport access, pollution profile, strategic importance, ownership/claim and discovery state. Existence is global; knowledge is held independently by people and institutions.

A civilization's knowledge progresses through unknown, suspected, surveyed, proven, developed, depleted/abandoned and rediscovered. The omniscient player can inspect true deposits. The civilization-knowledge overlay shows belief, confidence, map age and source.

| 32.2 Hybrid visualization

Surface resources are visible: forests, crops, livestock, fish activity, exposed stone, water, mines, quarries, wells and extraction facilities. Underground and continuous fields—coal, oil, uranium, ore grade, soil quality, water table and pollution concentration—use overlays. Extraction visibly changes terrain, infrastructure, labor, pollution, settlement growth and transport.

32.3 Logistics architecture

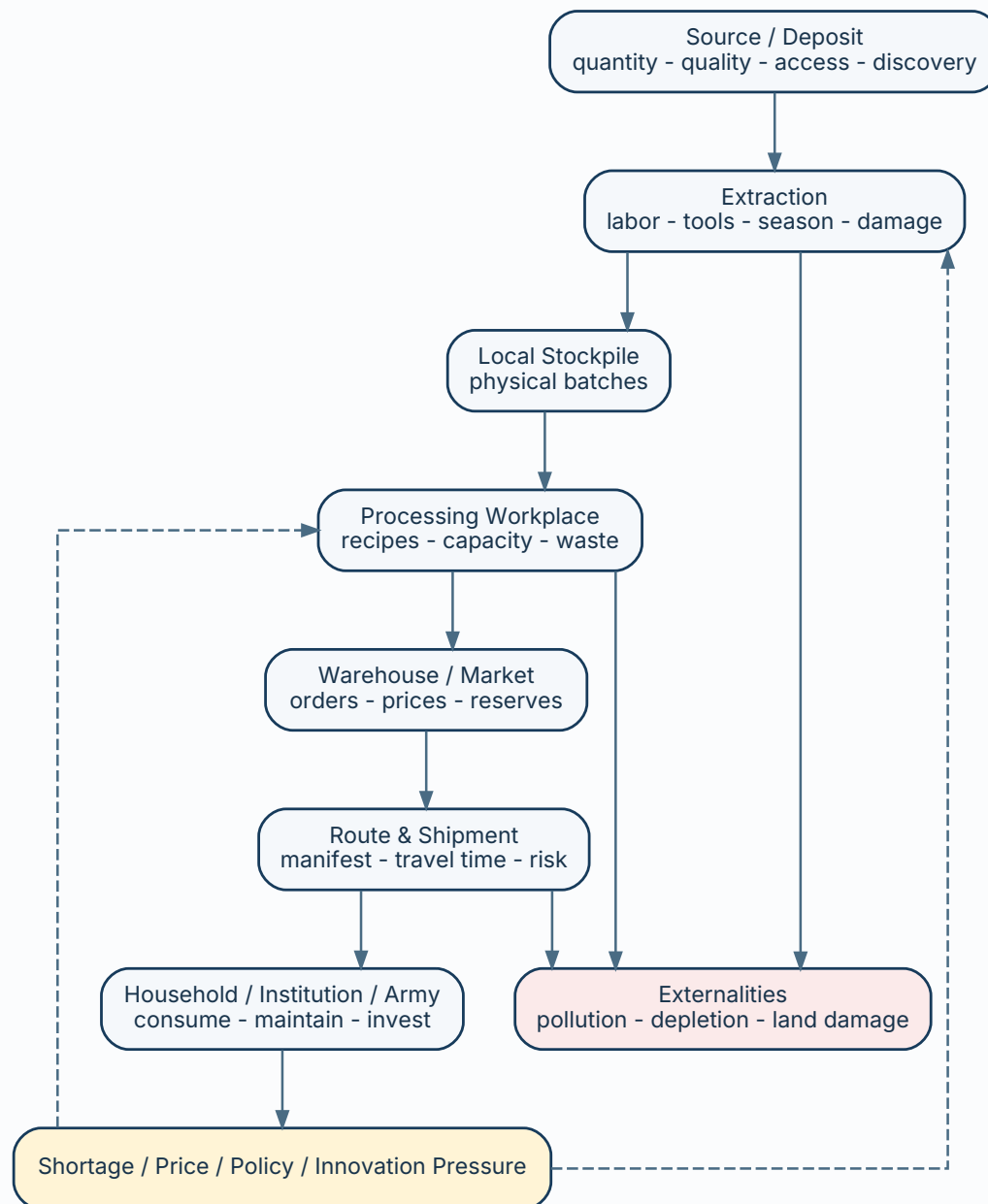


Diagram 15 — Goods remain physically located and move through inventories, processing, storage and cargo manifests.

A shipment has stable ID, owner, cargo manifest, origin, destination, corridor, transport capacity, departure/arrival estimate, risk, escort and current segment. People, animals or vehicles may perform movement, but planning is group-level. The hierarchy is building access → settlement road graph → regional graph → strategic corridor.

First-playable transport modes are porter, pack animal, handcart/wagon, river boat/ferry and organized convoy. Vehicles are durable equipment with presentation instances; they do not use rigidbody traffic physics. Later rail, road vehicles, ships and aircraft extend the same shipment and route model.

32.4 Reservations and priorities

Inventories support reservations so plans cannot spend the same batch twice. Priority classes are emergency survival, committed maintenance, contractual, public project, commercial and discretionary. Reservations expire or release on cancellation. Military requisition may override lower priority but creates ownership, market, legitimacy and political consequences.

32.5 Renewable and finite resources

Water, wild food, timber, medicinal plants, pasture and soil productivity renew regionally according to climate, extraction, pollution and land use. The ecosystem does not track every plant. Overharvest reduces future yield and can change erosion or habitat.

Stone, clay, iron, coal and later oil/uranium are deposits. Extraction consumes reserves and can increase cost as accessible material is exhausted. Abandoned works retain infrastructure, hazards, scrap and records. Depletion changes settlement economics rather than deleting an icon.

32.6 Resource-flow overlays

The player can inspect production, storage, transport, consumption, waste, loss, theft, shortage, export and import. Selecting food can show farms, harvest forecasts, granaries, markets, households, government reserves, armies and routes. Flow widths encode quantity; line style encodes status; labels and patterns supplement color.

32.7 Logistics acceptance criteria

- Every cross-settlement transfer has origin inventory, manifest, route, travel time and destination.
- Route failure cannot duplicate or silently delete goods.
- Large benchmarks support thousands of concurrent shipments without per-cargo long-range path searches.
- Deposits and stockpiles expose true state to the player but filtered belief to AI.
- A destroyed bridge invalidates affected corridors and propagates shortage, price, policy and migration pressure.
- A forced resource deposit reconciles claims, geology overlay, extraction candidates and intervention history.

33 Technology and knowledge

33.1 Capability network

CONFIRMED There is no universal linear technology tree. The authoritative model is a directed capability hyper-graph. Nodes represent problems, principles, procedures, components, tools, materials, institutions, skills and applications. Requirements may have alternative branches.

“Mechanized water pumping” is not one unlock. A variant can require pressure understanding, seals, cylinders, valves, heat source, workshop, maintenance skill, finance and demand. Another society may solve the same drainage problem with channels, animal power or wind pumps.

33.2 Knowledge states

Table 10 — Capability and knowledge dimensions

State	Meaning
Awareness	The outcome may be possible

State	Meaning
Conceptual understanding	Relevant principles are understood
Practical procedure	People can perform a method
Plans/records	Transferable instructions or measurements exist
Prototype	A local experimental instance exists
Operational	Existing instances can be used
Repair	Failures can be diagnosed and repaired
Component manufacture	Key parts can be produced
Full manufacture	Complete instances can be built
Mass production	Repeated scaled output is supported
Institutional adoption	Organizations depend on it
Civilian adoption	Households/businesses use it
Military adoption	Armed forces standardize it

These are dimensions rather than a mandatory ladder. A civilization may operate a captured engine without understanding it, or understand a principle while lacking industrial tolerance.

33.3 First-playable domains

The first playable retains six domains with 8–12 validated capability definitions each: agronomy; construction/civil engineering; materials/tools; transport/logistics; health/sanitation; and institutions/information. No society must acquire all capabilities. Every definition requires visible effects, prerequisites, blockers, trace terms and test scenarios.

33.4 Invention lifecycle

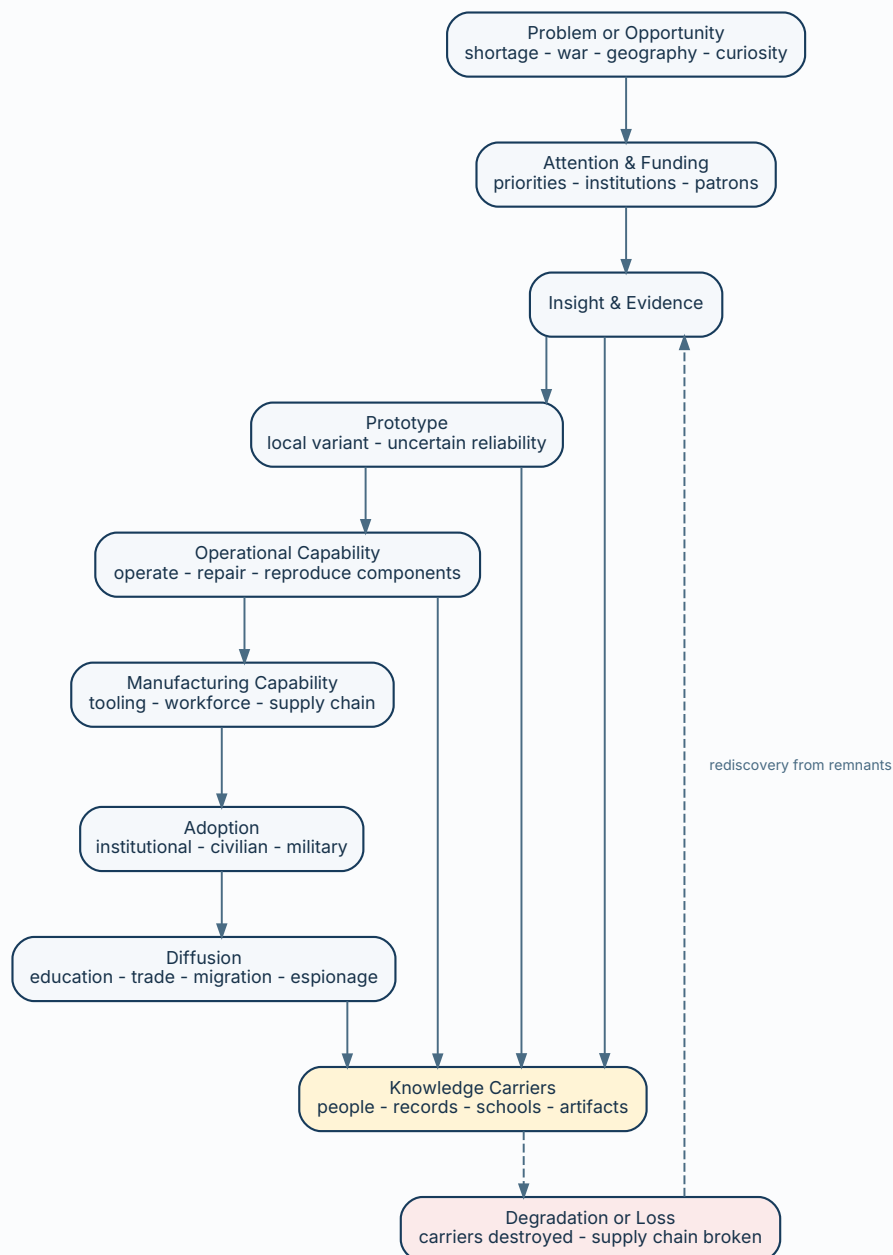


Diagram 16 — Knowledge develops through evidence and carriers, may diffuse imperfectly, and can be lost when institutions and production collapse.

Problems and opportunities create attention. Institutions, workplaces, patrons, governments or individuals form projects. Contributors are selected by aptitude, education, experience, access, health and interest. Experiments consume time, tools and materials and produce evidence, failures, spillovers or prototypes. Breakthroughs select only feasible variants from available components and knowledge. Adoption is a separate decision influenced by cost, reliability, law, culture, infrastructure, incumbent interests and demand.

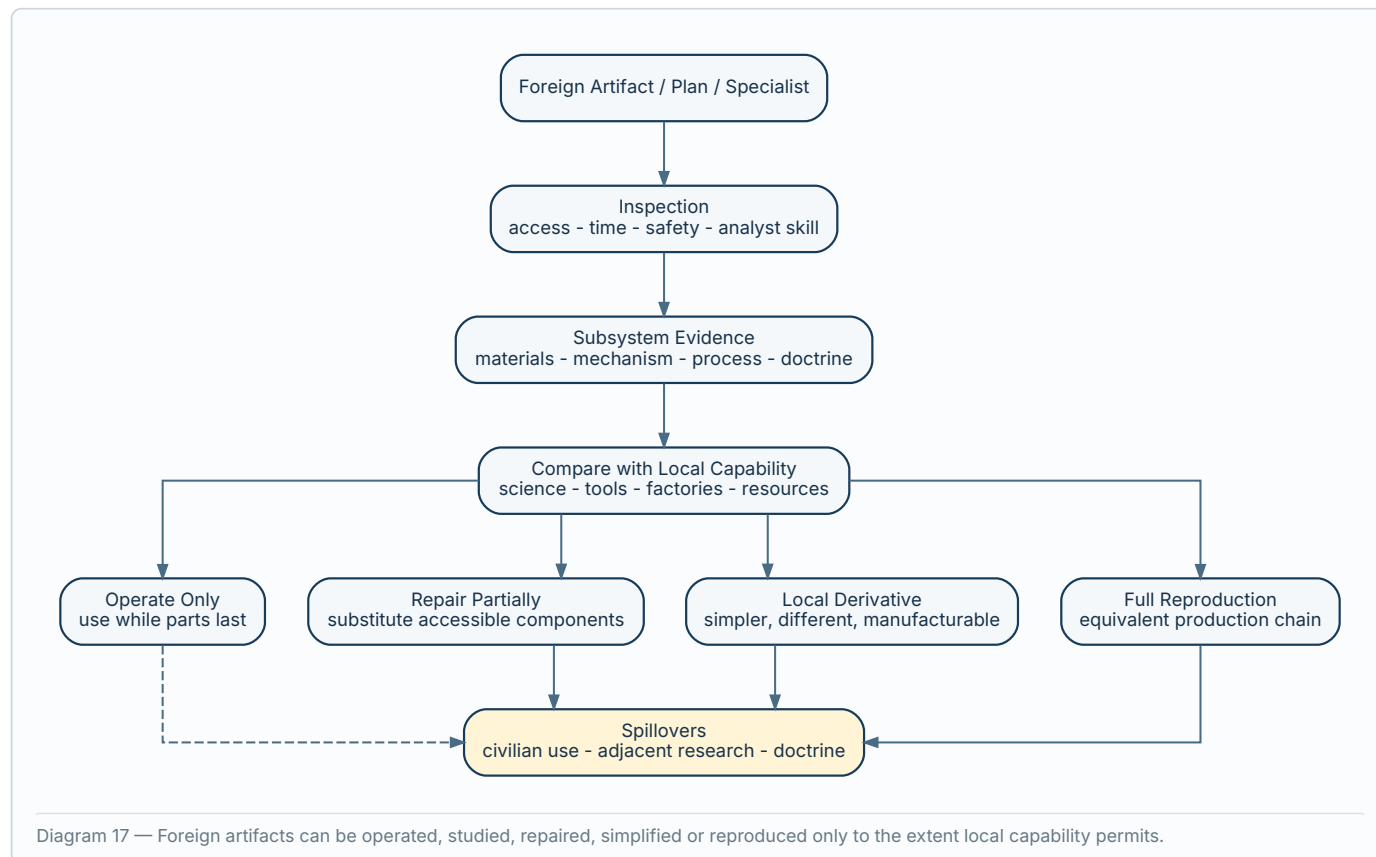
Randomness affects timing and choice among feasible variants. It cannot conjure uranium enrichment, precision machine tools or mass production without supporting capability.

33.5 Technology behavior contracts

Important technology decisions each have separate contracts: recognize pressure, fund investigation, select contributors, choose experiment, test prototype, adopt variant, standardize, copy, conceal, abandon and rediscover. Scenario tests include an inventor with conceptual knowledge but no manufacturing base; an artifact without suitable materials; a school destroyed during adoption; and a government order to pursue nuclear power before prerequisites exist.

The expected result of the last scenario is not “research failed” as one opaque state. The trace identifies missing scientific understanding, uranium knowledge, specialists, process equipment, safety institutions, electrical infrastructure and political authorization. The binding order remains active or is explicitly cancelled.

33.6 Reverse engineering



An artifact exposes subsystems. Analysts generate evidence according to access, condition, safety, tools, time and prior knowledge. Evidence may reveal principles, repair methods, components, substitute recipes, tolerances or adjacent applications. Full reproduction requires local materials, processes, tools, skills and supply chains. A surviving future vehicle may transform doctrine without granting infinite copies.

33.7 Diffusion and loss

Knowledge spreads through apprenticeships, schools, records, products, migration, trade, espionage, conquest and independent rediscovery. Adoption can be blocked by cost, secrecy, law, infrastructure or opposition. Knowledge decays when carriers die or leave, records burn, practice stops, schools close, factories are destroyed or supplies disappear. Awareness may survive after manufacture is lost.

33.8 Technology acceptance criteria

- Paired seed suites produce statistically different solution portfolios under different resources and institutions.

- Discovery never grants manufacture or adoption automatically.
- Every blocked capability can identify missing requirement states.
- Destroying carriers and production assets can cause partial knowledge loss.
- Reverse engineering produces bounded evidence or local variants.
- Failed projects can produce documented spillovers.
- A player-seeded artifact enters the same evidence and adoption systems unless Force Outcome explicitly grants a later state.

34 Diplomacy

| 34.1 Relationship model

Diplomacy operates among recognized political actors: settlements, governments, civilizations, alliances, insurgent authorities and exceptional institutions where permitted. A relationship contains contact, recognition, communication, claims, treaties, trade dependence, threat by domain, commitment trust, affinity/friction, grievances, power asymmetry, faction preferences, active incidents and war goals. Pairwise records are sparse and created only for contact or strategic relevance.

| 34.2 Believed information

AI actors use believed state with confidence and age, never omniscient truth. Contact can arise through exploration, migration, trade, refugees, military movement, player intervention or shared institutions. Recognition progresses from unknown to rumor, encounter, mapped, diplomatic recognition and institutional connection. Misinformation and stale maps enable miscalculation without cheating.

| 34.3 First-playable actions

The twelve baseline actions remain: establish contact/recognition; recognize/dispute claim; open/restrict/terminate trade; offer aid; negotiate migration/refugee access; sign non-aggression; form defensive alliance; exchange knowledge/specialists; demand tribute/concession; conduct espionage; declare war with goals; and negotiate armistice/peace/reparations/borders.

| 34.4 Decision process

Governments evaluate diplomacy on incidents, material threat changes, treaty review and periodic strategy. A bounded feasible proposal set is scored on goal value, security/economic benefit, legal compatibility, credibility, domestic support, capacity, military balance, information confidence, escalation, leader tendencies and controlled randomness. Both parties evaluate proposals; authority or communication can fail.

| 34.5 Treaties

Treaties are executable state with parties, clauses, review, obligations, verification, breach and consequence. Clauses can regulate tariffs, access, aid, defensive triggers, borders, prisoners, technology transfer and demilitarized areas. Breach creates evidence and reaction, not automatic war.

| 34.6 Player control

A binding diplomatic order obligates the selected authority to seek or accept a relation within legal and operational limits. A forced peace immediately terminates official hostilities but leaves claims, resentment, insurgency, prisoners, damage and displaced people. A forced alliance establishes the relationship yet may have low legitimacy or unreliable compliance.

34.7 Acceptance criteria

- All wars have traceable goals or player command.
- AI uses believed information with confidence.
- Treaties create enforceable obligations and recorded breaches.
- Refugees remain persistent people carrying skills and knowledge.
- Three-actor scenarios produce alliance or balance behavior unavailable in two-actor tests.
- Forced diplomatic outcomes retain downstream political consequences.

35 Warfare

35.1 Design position

Warfare is physical and visible. People fight in the world; soldiers move along routes; fortifications, fires, explosions, retreat, occupation, damage, casualties and refugees appear at the corresponding locations. The authoritative model must scale without per-projectile physics or unrestricted tactical cognition.

RECOMMENDED Use strategic operations + route movement + sector encounters + squad-level resolution. Persistent people remain unit members; combat evaluates squads or formations and allocates wounds, deaths, capture and experience back to individuals.

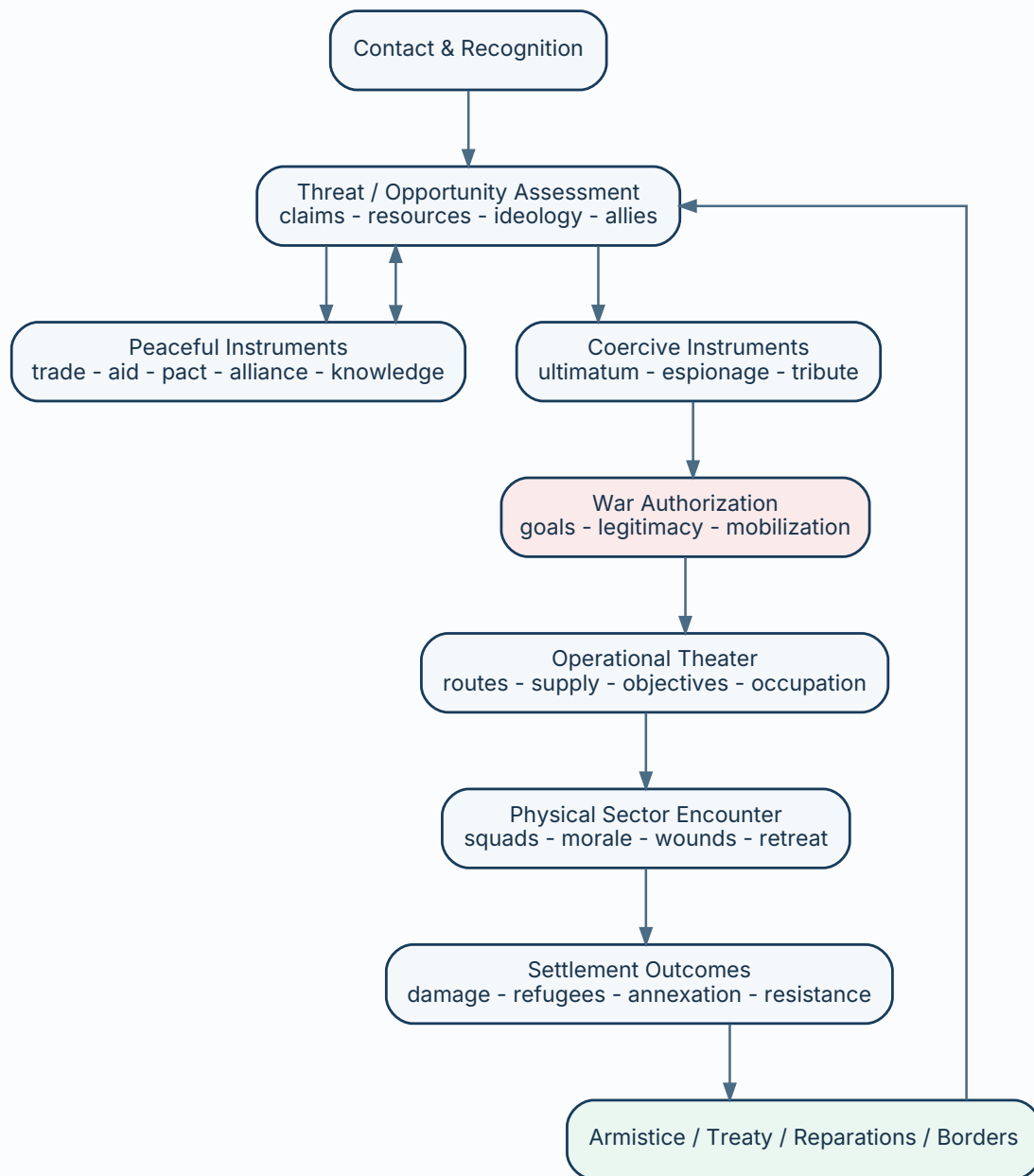


Diagram 19 — Political goals authorize operations; routes, supply and physical encounters produce occupation, destruction and settlement outcomes.

35.2 Military organization

A force has recruitment policy, unit hierarchy, persistent commanders, training, readiness, equipment manifests, supply demand, morale, cohesion, doctrine, pay/entitlement, mobilization and legal authority. The first playable supports militia and small standing guards. Squads contain approximately 8–20 people; task groups combine squads for route movement and operations.

Recruitment removes people from civilian work. Mobilization therefore affects harvest, production, households and politics. Soldiers retain beliefs, home, wealth and ties. Wounds, capture, desertion, trauma, promotion and death persist after demobilization.

35.3 Operations and supply

Governments authorize goals such as raid, defend claim, seize route, occupy settlement, compel tribute, aid ally or suppress rebellion. Military planners select bounded plan templates from objectives, route access, supply, enemy estimates, season, morale and policy. Armies use the same strategic graph as trade and migration.

Food, ammunition, equipment, medical supplies, replacements and transport are physical. Foraging and requisition can supplement supply but damage local inventories, ownership and legitimacy. A force without a valid supply path loses readiness and may retreat, disperse or forage.

35.4 Encounter resolution

Active sectors evaluate ready combatants, equipment, training, doctrine, terrain, fortification, command, morale, fatigue, ammunition, surprise, intelligence, weather, retreat routes, civilian density and rules of engagement. The model emits pressure, morale change, equipment loss, wounds, deaths, capture, retreat and position. Representative projectiles and explosions visualize aggregate fire; presentation cannot change outcomes.

Casualties are assigned to persistent people using exposure, role, equipment, health, cover and keyed random streams. Directional readability for camera-facing billboards uses weapon orientation, projectile direction, facing indicator, movement lean, shadows, poses and selection outlines.

35.5 Occupation, civilians and rebuilding

Occupation controls parcels, routes, government buildings and public inventory. Civil authorities may collaborate, resist or collapse. Occupiers choose requisition, protection, integration, autonomy, repression or withdrawal. Civilians flee, shelter, resist, collaborate, scavenge or return according to conditions and traits.

War damages physical infrastructure and creates ruins, mines or contamination in later eras. Rebuilding consumes real labor and material. Peace changes official relations and claims but does not repair damage or erase hostility.

35.6 Violence presentation

CONFIRMED Violence is visible but stylized. Pixel-art impacts, injuries, bodies/casualties, explosions, fire, fleeing civilians and ruined infrastructure communicate events. Blood and damage remain low-detail. No realistic anatomical gore, prolonged close-up suffering or high-fidelity wound animation is required. Simulation severity can remain serious even when visual presentation is restrained.

35.7 Future nuclear effects

DEFERRED Nuclear weapons are not required in the first playable. Later implementation should model a layered gameplay footprint: blast damage, thermal ignition, prompt casualties, infrastructure failure, fallout/contamination, displacement, long health burden, environmental effects, political response, doctrine, deterrence, cultural trauma and persistent ruins. The model should remain bounded and game-relevant rather than becoming a weapon-effects laboratory.

35.8 Warfare acceptance criteria

- Battles occur at physical sectors with visible movement and destruction.
- Off-screen and on-screen encounters use identical authority.
- Supply failure changes readiness and decisions.
- Mobilization has measurable civilian labor and household effects.
- The same seed and command log reproduces the authoritative encounter.
- Soldiers return with persistent wounds, experience and political effects.
- Peace terms correspond to goals and control.
- Camera-facing sprites remain directionally readable in prototype combat.

36 Environment

| 36.1 Environmental state

The environment is a set of spatial and temporal fields: elevation, slope, geology, water, soil fertility, moisture, erosion, salinity, vegetation biomass, timber yield, temperature, precipitation, storm exposure, fire fuel, drought stress, pollution, habitat, wildlife capacity, flood risk and long-term climate forcing. Region systems publish summarized exposure to homes, workplaces, routes and settlements.

| 36.2 Pollution and climate

Extraction and production emit air, water and soil loads. Pollution moves through bounded local and downstream models, affecting health, fertility, fish/wild food, land value, migration and politics. Remediation consumes labor and materials; natural decay depends on pollutant type.

The architecture stores monthly normals and forcing variables so industrial systems can later modify greenhouse, aerosol and land-cover effects. The first playable uses natural variability and local land-use feedback. It does not claim real-world predictive climate fidelity.

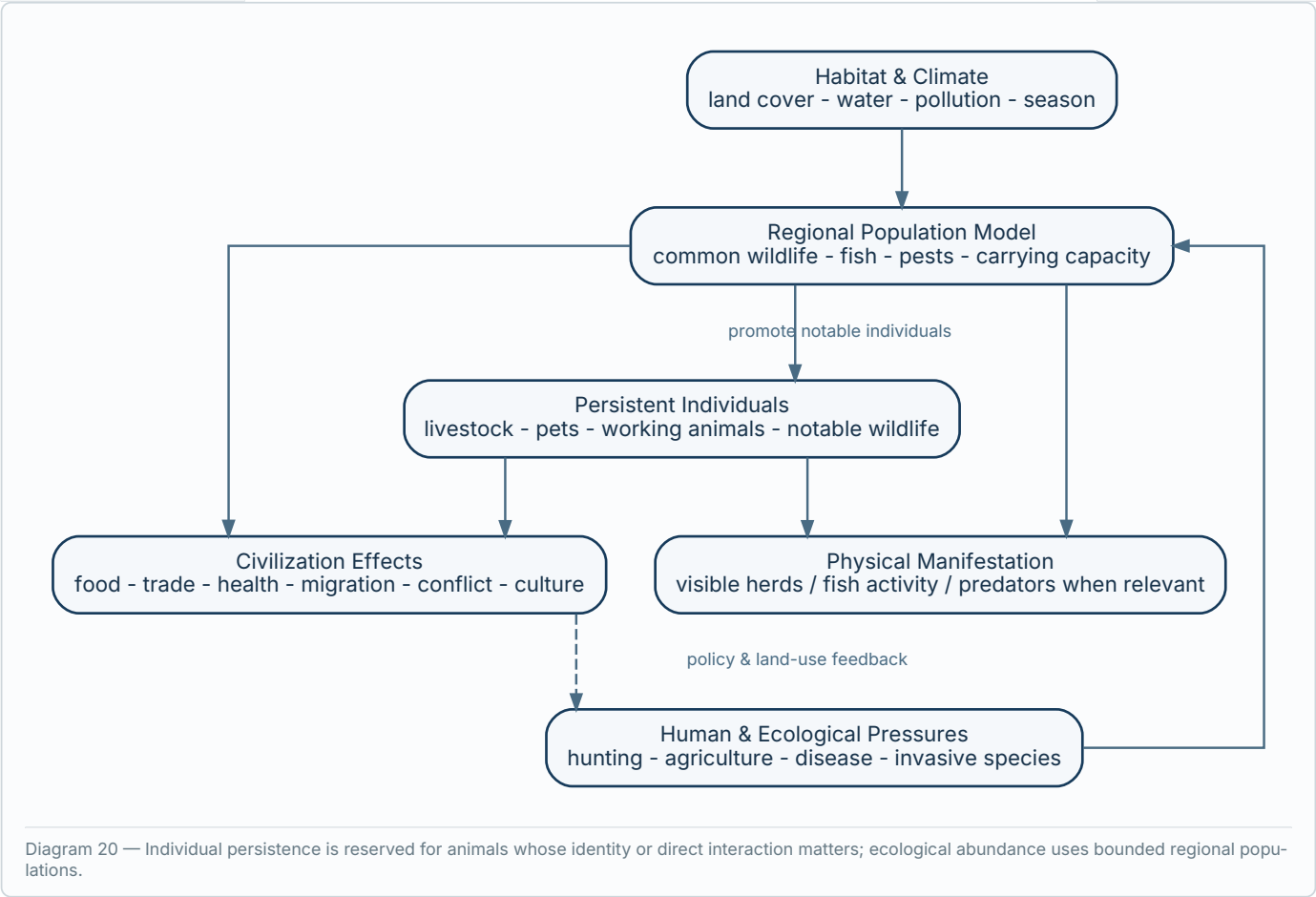
| 36.3 Environment acceptance criteria

- Wildlife and resource populations respond directionally to habitat, harvest and pollution.
- Removing habitat does not increase carrying capacity without another modeled cause.
- Local extinction and recolonization remain distinct from global extinction.
- Restored species require habitat and food after Force Outcome.
- Environmental overlays distinguish authoritative fields from civilization knowledge.
- Old damage remains in save state and affects future land use.

37 Ecosystems

| 37.1 Hybrid ecosystem

CONFIRMED Persistent individual animals are used where identity matters: livestock, working animals, pets, rare or notable wildlife and historically important animals. Common wildlife, fish, herds, insects, pests and small animals use regional population models. Physical animals appear when relevant, but the save does not retain every fish or insect.



Hunting, fishing, overharvest, habitat loss, pollution, domestication, disease, invasive species, predator pressure, food, climate and migration interact. Ecological changes affect nutrition, trade, health, settlement, culture, technology, politics and conflict.

38 Extinction and recovery

38.1 Extinction states

Species can become locally extinct, functionally extinct or globally extinct. Global biological extinction is permanent under normal simulation when no viable population or preserved recovery mechanism remains. Viable remnants may survive in isolated habitat, captivity, farms, preserves, hidden regions or laboratories.

Human civilization disappearance is separated into state extinction, cultural extinction and population extinction. A state may vanish while its people and culture survive. Cultural extinction occurs when language, identity, religion, customs and institutions no longer have viable carriers. Population extinction means no surviving members.

38.2 Recovery from remnants

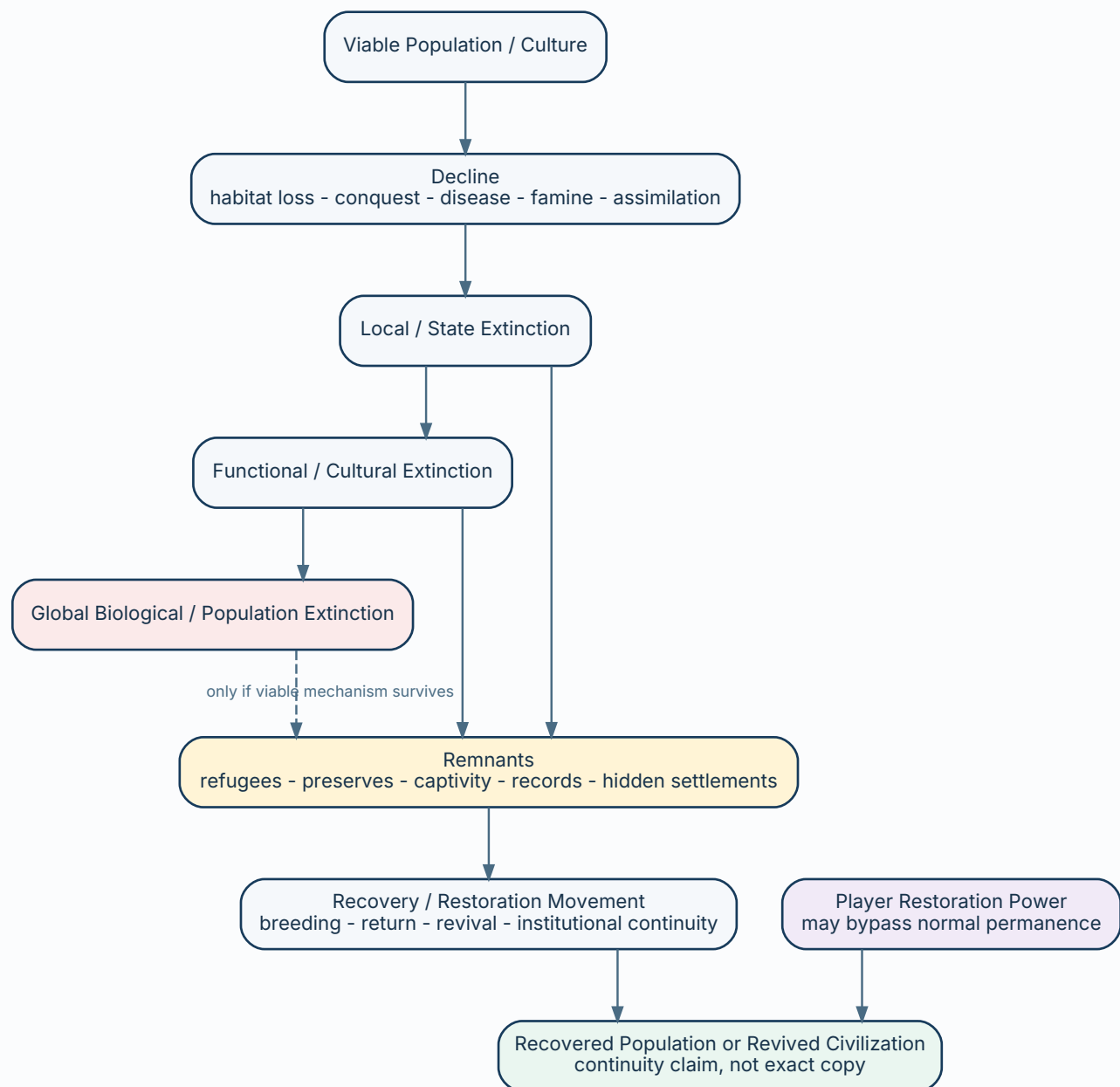


Diagram 21 — Recovery depends on viable remnants, preserved knowledge or explicit player/advanced technological intervention.

Species recover through breeding populations, habitat restoration, captivity or scientifically plausible preservation/biotechnology in advanced eras. Civilizations return through descendants, refugees, rebels, exiles, institutions, surviving settlements, cultural revival and restoration movements. The revived entity may claim continuity but is not an exact copy.

The unrestricted player may restore extinct species or civilizations. Force restoration creates a viable defined population and authoritative history. It does not automatically provide habitat, food, institutions, immunity or political recognition.

39 Disasters

39.1 Disaster lifecycle

A disaster has cause, preconditions, forecast signals where plausible, footprint, intensity over time, affected channels, response opportunities, direct effects, secondary effects, recovery and historical summary. Flood follows hydrology; fire follows fuel, weather and access; disease follows exposure and sanitation. The simulation does not choose "destroy civilization" as an arbitrary result.

First-playable hazards are seasonal heat/cold, drought, river flood, wildfire/settlement fire, severe storm, contamination/disease pressure, soil degradation and extraction pollution. Earthquake, tsunami, tornado, volcano, sea-level change and advanced industrial climate effects are later expansions using the same footprint and damage interfaces.

39.2 Response and recovery

Settlements and governments may warn, evacuate, fight fire, ration, quarantine, relocate, repair, aid and construct protection. Culture, trust and institutions affect compliance. Recovery consumes materials, labor, money, housing and route capacity. External aid travels physically.

39.3 Player-created disasters

The unrestricted player can invoke plausible hazards, amplified events or direct destruction. All are recorded. A direct destructive power may have no natural cause but must still propagate fire, casualties, infrastructure failure, displacement and response. The game does not hide the intervention as natural history.

40 Persistent ruins

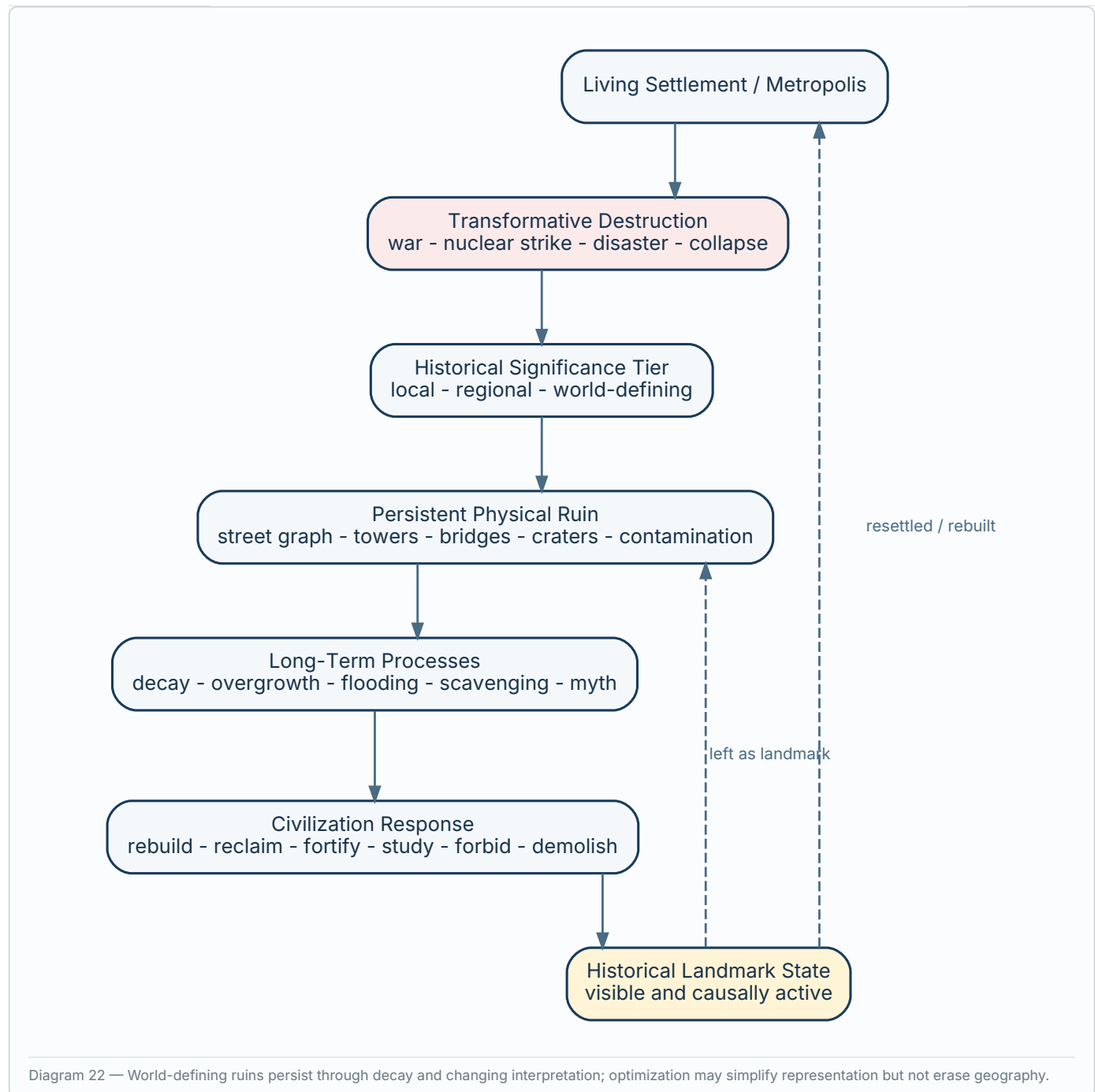
40.1 Historical-significance tiers

Every damaged or abandoned asset receives a significance tier based on event importance, scale, uniqueness, references, player bookmark, cultural meaning and geographic effect:

Tier	Examples	Compaction rule
0 Routine	Minor debris, ordinary failed construction	May merge into regional summary after cleanup/decay
1 Local	Notable house, small battlefield, neighborhood fire	Retain while locally referenced; simplify geometry later
2 Regional	Fort, major bridge, industrial accident, town ruin	Permanent identity; LOD geometry may simplify
3 World-defining	Destroyed metropolis, nuclear scar, civilization-ending battlefield, unique monument	Permanent physical geography and canonical history unless deliberately transformed

40.2 Landmark ruins

CONFIRMED A metropolis destroyed by war, nuclear strike, catastrophic disaster or civilizational collapse remains physically visible for the life of the world if it is not resettled, reclaimed, demolished, rebuilt, deliberately cleaned or naturally buried. Street pattern, districts, towers, industrial sites, bridges, railways, highways, craters, government complexes, bases and monuments remain recognizable.



Ruins can decay, overgrow, flood, remain radioactive, be scavenged, fortified, studied, mythologized, visited, prohibited or reused. They influence religion, politics, diplomacy, technology, migration and culture. A ruin inspector shows original identity, surviving assets, damage layers, contamination, later use and historical references.

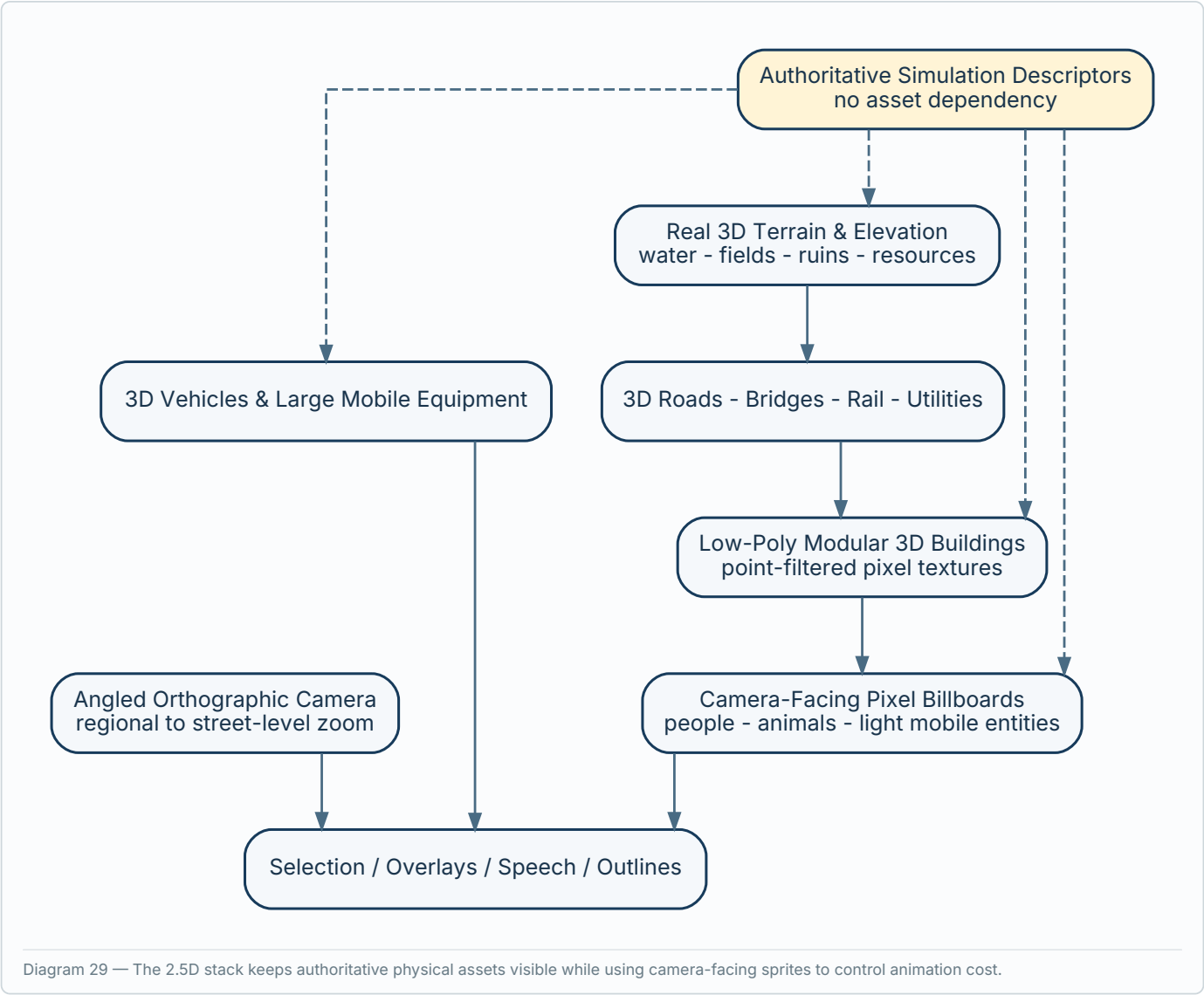
40.3 Scar acceptance criteria

- A Tier 3 metropolis remains recognizable after centuries and save compaction.
- Compaction never deletes structures cited by canonical history without an explicit transformation event.
- Cleanup, demolition and rebuilding are physical projects.
- Contamination and exclusion alter later pathfinding, land value and policy.
- Player restoration preserves old damage history even when structures return.

41 Visual representation

41.1 Locked direction

CONFIRMED Use a flat bounded 2.5D world with an angled top-down camera, real 3D terrain and elevation, real 3D roads, bridges, railways, buildings and infrastructure, and camera-facing pixel-art billboards for people and many mobile entities. The aesthetic is detailed, readable, approximately 32-bit-era pixel art: stylized, cohesive across eras, procedurally variable and compatible with large counts. It is not strict console emulation, photorealism or extremely abstract 8-bit art.



41.2 Visual density

The world shows fields, crops, orchards, forests, rivers, rocks, fences, utility poles, rail infrastructure, industrial clutter, warehouses, road structures, defenses, ruins, pollution, debris, ports, markets, mines, wells, livestock and farms. Density scales by zoom, region, settlement type, era, wealth, land use and performance budget. Decorative clutter is presentation-only and must not obscure authoritative entrances, routes or hazards.

41.3 Levels of detail

Zoom band	People	Built world	Information
Close / street	Individual billboards, equipment, speech and action states	Full modules, entrances, construction and damage	Names, needs, goals, traces, local hazards
Neighborhood	Simplified billboards and selective labels	Full footprints, local roads, service traffic	Housing, work, ownership, service overlays
Settlement	Instanced billboards; density clustering where needed	Simplified modules and skyline	Projects, shortages, flows and alerts
Region	Aggregate movement plus tracked individuals	Terrain, roads, sites and ruins	Trade, migration, claims and war
World	Ordinary people omitted; tracked pins	Political/resource/environment map	Diplomacy, eras, extinction and comparisons

Visual omission does not change authoritative presence. Selected and tracked people remain locatable at any zoom.

41.4 Effects and violence

Weather, smoke, fire, construction particles, combat effects and blood/damage use pooled screen-space budgets and distance LOD. Authoritative damage comes from simulation events; effects depict it. Accessibility settings reduce flash, shake, blood, weather opacity and motion without changing state.

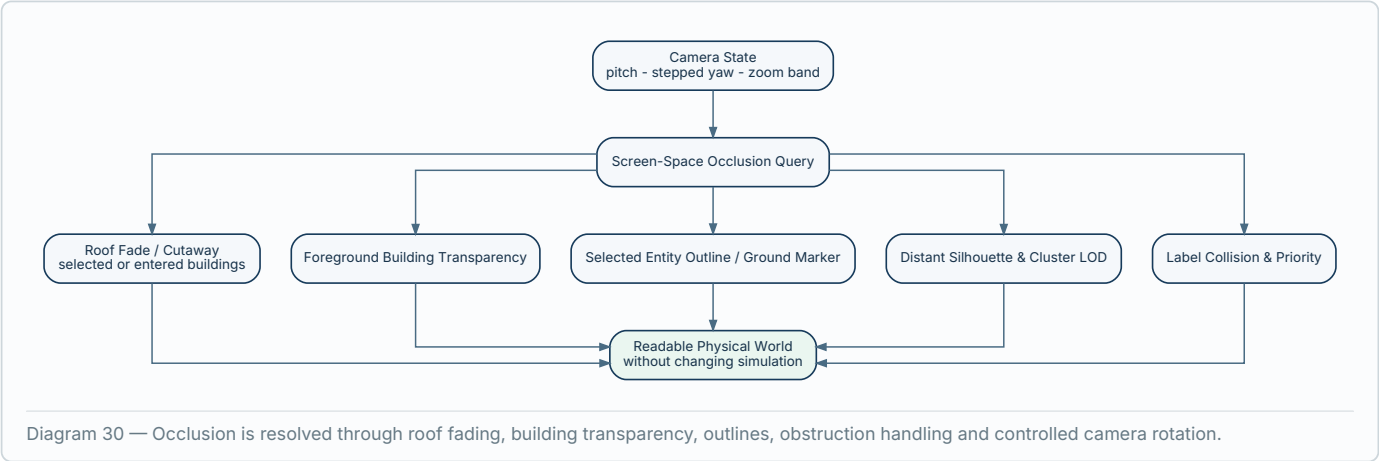
42 Camera and controls

42.1 Camera

Use an orthographic or near-orthographic 3D camera at approximately 40 degrees downward pitch. Smooth pan and continuous zoom support regional overview through individual inspection. Yaw uses limited stepped rotation—initially 90-degree increments, with 45-degree evaluation—so billboards and building entrances remain readable. Free rotation is experimental, not the default.

The camera supports keyboard/edge pan, middle-drag, pointer-centered zoom, rotate, jump to alert/bookmark/settlement, follow entity or shipment, previous-location history, UI-hidden cinematic pan and overlay framing.

42.2 Occlusion



Buildings between camera and selection fade or become transparent within bounded rules. Roofs fade for selected occupants. Selected entities receive outlines or silhouettes. Foreground vegetation and clutter simplify. Distant silhouettes replace detail. Occlusion behavior is presentation-only.

42.3 Selection and input

Click selects the highest-priority visible entity; repeated click cycles candidates. A context stack lists person, building, parcel, route, resource, settlement and effect. Drag performs a region query, not tactical unit selection by default. Search and overlays select hidden or remote entities.

UI view actions and simulation commands are separate. Commands validate target, authority, type, parameters and data integrity. The unrestricted sandbox bypasses balance restrictions, not structural validation.

42.4 Accessibility baseline

Full keyboard remapping, scalable UI, pause-and-inspect, high-contrast and color-vision-safe overlays, pattern/label alternatives, reduced motion, configurable effects, adjustable speech-bubble density, notification filters and readable minimum text are required. Controller support is deferred until dense inspection UI stabilizes; code must not hard-code mouse input.

43 Character presentation

43.1 Camera-facing billboards

CONFIRMED People use camera-facing pixel-art billboards rather than complete four- or eight-direction sprite sets. This reduces animation and asset burden and permits procedural layering. Billboards remain vertically aligned to the world, scale consistently and receive terrain-following shadows.

Direction and action are communicated through movement vector, lean, pose, shadow offset, equipment orientation, weapon/tool orientation, projectiles, effect origin, facing indicators and short specialized states. The sprite itself faces the camera; the semantic action does not.

43.2 Procedural appearance

Characters are assembled from validated layers or modular parts: body/age band, skin/hair variation, culture/era clothing, occupation layer, wealth/class layer, equipment, military role, religion where visually relevant, health/injury, environmental exposure and refugee/work-condition markers. Layer combinations are validated for silhouette, palette, clipping and animation compatibility.

The simulation emits descriptors; presentation resolves them to assets. Changing art does not migrate authoritative state. Thousands of unique-looking people emerge from combinations rather than individually authored characters.

43.3 Action readability contracts

Melee combat: weapon orientation, lunge/brace pose, target line, impact effect, shadow shift and temporary facing marker.

Ranged combat: aiming pose, weapon angle, muzzle/projectile origin, target direction and reload/cooldown cue.

Construction/work: tool layer, repeated pose, material prop, worksite effect and progress indicator.

Carrying: visible bundle/cart attachment, altered lean and reduced movement speed.

Vehicle use: boarding cue, hidden or seated occupant state, vehicle ownership/manifest and disembark point.

Panic: accelerated animation, fear icon when selected, erratic-but-authoritative steering and crowd direction.

Injury: posture, reduced pace, bandage/damage layer and health marker.

Crowds: lane alignment, spacing, density fade, cohort markers and selected-person outline.

43.4 Animation vocabulary

The first playable uses idle, walk, run/panic, carry, work, construct, melee, aim/fire, reload, hurt, incapacitated/dead, enter/exit and seated/vehicle states. A state can have two to four frames plus procedural bob/lean. The objective is semantic clarity, not cinematic performance.

43.5 Speech bubbles

At close zoom, a bounded number of bubbles appear using priority, screen-space budget and cooldown. Bubble tails attach to the billboard's authoritative world position. Selected and tracked people can override normal density limits. Speech content is specified in Section 32.

43.6 Character presentation benchmark

A representative city scene must display 5,000 moving people, 200 close-detail billboards, 40 active speech bubbles and 100 combatants without violating the GPU/CPU presentation budget. The benchmark tests camera rotation, slopes, occlusion, crowding and mixed cultural/equipment layers.

44 Building presentation

44.1 Modular low-poly 3D with pixel textures

CONFIRMED Buildings use low-poly modular 3D geometry with pixel-art textures. Every authoritative instance has physical footprint, height, entrances, construction, damage, repair, abandonment, demolition, ownership, capacity, function, era, culture, quality, materials and infrastructure requirements.

Visual kits cover homes, farms, workshops, markets, warehouses, schools, faith/government buildings, factories, rail, bridges, ports, bases, fortifications, apartments, skyscrapers, power facilities and future structures. First-playable content is smaller, but the assembly contract is common.

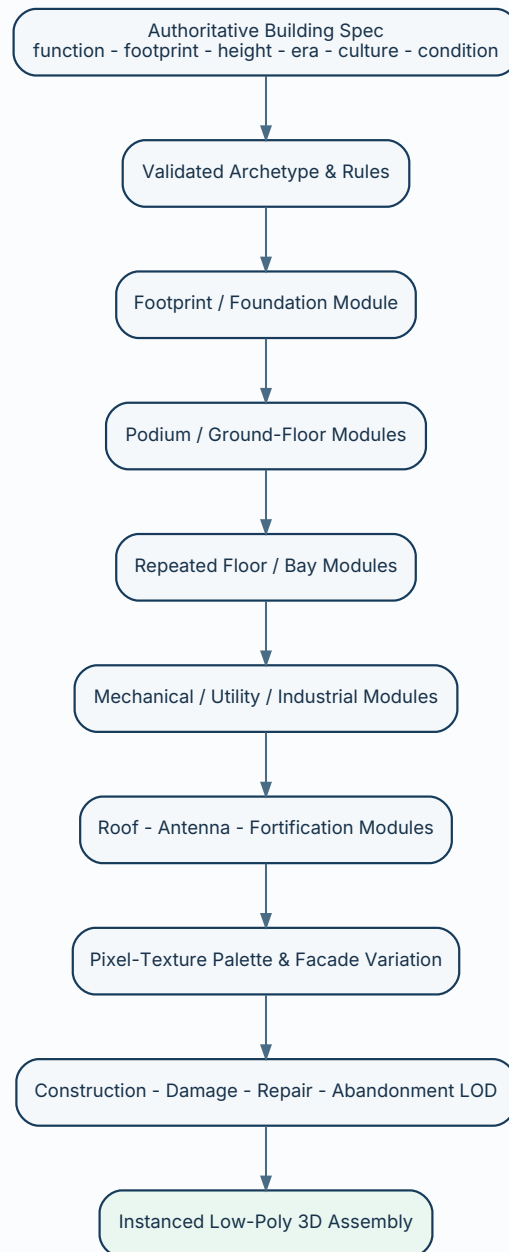


Diagram 31 — Buildings are assembled from authored, validated modules; procedural variation does not invent arbitrary geometry.

44.2 Exterior-only interiors

People enter and leave through real entrances. Internal movement is abstracted through building state: occupants, capacity, inventory, production, employment, health, injury, equipment, storage, utilities, damage and functional modules. The game does not render rooms, cutaways or hundreds of skyscraper floors.

When a person enters, authoritative location becomes “inside building ID / functional module” plus entry time and activity. Presentation may hide the billboard or show a roof-fade marker. Evacuation, fire and collapse use building capacity and exits without rendering interior paths.

44.3 Construction and damage states

Visual stages map to authoritative work packages: site preparation, foundation, frame, enclosure, equipment, commissioning. Damage layers include superficial, functional, structural, fire, contamination and collapse. Repair restores specific modules. Demolition and salvage create debris and material recovery.

44.4 Skyline and era continuity

Cities develop recognizable skylines through height rules, density, land value, technology, law and society. Buildings do not globally reskin at an era threshold. Older structures remain, retrofit, decay or are replaced. Skyscrapers use foundation, podium, repeated floor modules, mechanical sections, roof structures, antennas and façade variants. Their actual capacity and utilities remain authoritative.

44.5 Infrastructure presentation

Roads, bridges, rail, ports, utilities and defenses are spline or modular systems tied to authoritative edges and nodes. Surface, width, capacity, condition, owner, access and damage determine visual state. Distant LOD preserves route readability and landmark form. Ruined bridges and rail corridors remain recognizable after use ends.

44.6 Validation

Every module kit passes footprint/entrance validation, occlusion test, texture density standard, LOD generation, construction/damage completeness, culture/era compatibility, selection collider and save-stable descriptor mapping. No content generator may output an unvalidated mesh directly into production.

45 User interface and explainability

45.1 Information architecture

The interface has a world canvas, contextual inspector and global workspaces. The canvas shows physical state, selection and intervention cursor. The inspector drills from selected entity into needs, ownership, inventory, goals, traces and links. Global workspaces provide time, search, alerts, overlays, comparisons, history, orders, Force Outcome, saves and world rules.

The UI reads immutable query models. It never edits ECS state directly.

45.2 Primary workspaces: 13

1. World creation and rules.
2. Main HUD, time, alerts and bookmarks.
3. Person inspector.
4. Household/home inspector.
5. Settlement/city inspector.
6. Workplace/institution/building inspector.
7. Civilization/government inspector.
8. Technology/capability portfolio.
9. Economy, inventories, prices and logistics.
10. Diplomacy and treaty workspace.
11. War and operations workspace.
12. History, ruins, extinction and comparison.
13. Player powers, binding orders, Force Outcome and intervention log.

45.3 Decision traces and “Why?”

Every consequential decision trace includes entity, decision ID, tick, candidates, preconditions, observations, score terms, modifiers, random adjustment, rejected candidates, selected action, expected effect, invalidation reason, controlling policy and relevant higher priority.

The UI exposes three depths:

- **One-line cause:** “Mara chose the eastern granary because hunger was critical and the western bridge is closed.”
- **Factor panel:** normalized contributors, blockers, dates, source entities and beliefs.
- **Technical trace:** contract version, candidate records, RNG key, thresholds, policy IDs and performance duration.

Retail players may hide technical depth; it remains available because the player is omniscient. Test builds can compare full traces.

45.4 Order progress UI

Binding orders show authority owner, feasibility, active projects, blockers, expected next review, resources reserved and reasons for deviations. The UI distinguishes “refused” (invalid authority or explicit lawful resistance in an optional restricted mode), “accepted,” “blocked,” “in progress,” “partially complete,” “completed” and “cancelled.” In the unrestricted default, a valid decision-level order is binding; resistance appears in implementation consequences, not refusal to register the order.

Force Outcome uses a separate visual language and previews affected records, reconciliation operations and likely immediate consequences. The preview is informative, not a balance restriction.

45.5 Inspectors

The person inspector shows identity, health, nutrition, injury, disease, skills, education, work, wealth, possessions, home, household, traits, personality, beliefs, religion, attitudes, patriotism, happiness dimensions, needs, goals, action, decision scores, ties, experiences and memberships. Every summary links to source entities.

Settlement and civilization inspectors show population, production, consumption, food, resources, employment, housing, infrastructure, institutions, plans, military, diplomacy, technology, manufacturing, tensions, movements, culture, factions, trade, logistics, ownership and wealth distribution. Bottleneck panels rank constrained systems with evidence.

45.6 History and ruins

The timeline combines canonical events, graphs, interventions and visible landmarks. Filters cover people, settlement, civilization, government, culture, species, war, technology, economy, disaster, migration and ruin. Selecting a ruined metropolis opens original city identity, damage event, surviving geometry, later scavenging, contamination, myths and current claims.

45.7 Notifications

Notifications are configurable by severity and topic. Critical categories include save integrity, world-rule change, Tier 3 ruin threat, civilization collapse, extinction, unique knowledge-carrier loss, mass displacement, war declaration, Force Outcome reconciliation failure and performance degradation. Routine events aggregate into summaries.

45.8 Explainability acceptance criteria

- Every major technology, war, migration wave, revolt, collapse, extinction and price shock has a nonempty explanation.

- Factors link to current state or retained evidence.
- Randomness is quantified rather than used as “random event” explanation.
- Player intervention ancestry is visible where causal propagation retained it.
- A playtester can explain why a civilization cannot manufacture a selected artifact without reading code.
- UI can compare ground truth with selected civilization belief.

46 Information overlays

46.1 Overlay architecture

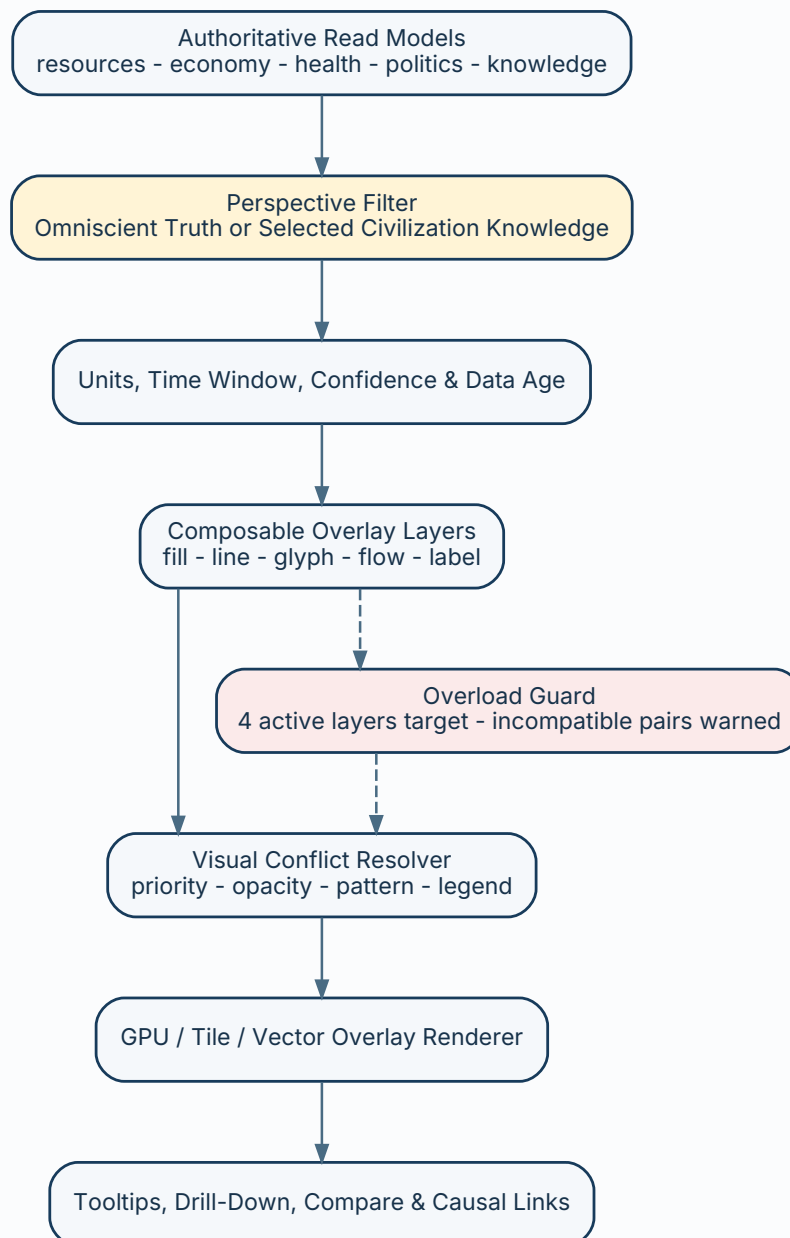


Diagram 32 — Overlays are query products with explicit observer perspective, confidence, units, legend and composition rules.

Required overlay categories are deposits, food production, stockpiles, hunger, water, fertility, timber, wildlife, pollution, ownership, trade, flows, infrastructure, power, transport, military control, borders, population, wealth, health, disease, housing, employment, construction, technology, culture, religion, political attitudes and civilization knowledge.

46.2 Composition and overload control

Overlays are composable where semantics remain clear. A base layer, one primary thematic layer and up to two secondary line/marker layers is the default. Incompatible combinations produce a warning or automatic legend split. Opacity, pattern, labels, contour, arrows, icon and line style encode meaning; color never carries meaning alone.

Every overlay declares unit, time, aggregation, data age, observer perspective, uncertainty, source and update cadence. Hover reveals exact values; selection opens the authoritative source.

46.3 Knowledge perspective

The perspective selector offers Omniscient, selected civilization, selected government, selected institution and selected person where useful. Believed overlays show confidence and staleness. Unknown areas remain unknown only within that perspective. A compare mode shows “true versus believed” difference for deposits, forces, routes, disease, population and diplomacy.

46.4 Performance

Overlay data is tiled, cached and incrementally updated. Query generation is separate from simulation and can defer nonurgent requests. Dense arrows and labels use screen-space budgets. Retail overlays do not enable full decision tracing globally; they request bounded summaries.

47 Save and history model

47.1 Save architecture

RECOMMENDED Saves are versioned domain snapshots plus compact history in an atomic container. Use a human-readable manifest and MessagePack-CSharp chunks with LZ4, stable numeric keys, checksums and explicit migrations. Never serialize raw ECS memory, transient handles, scene objects or presentation caches.

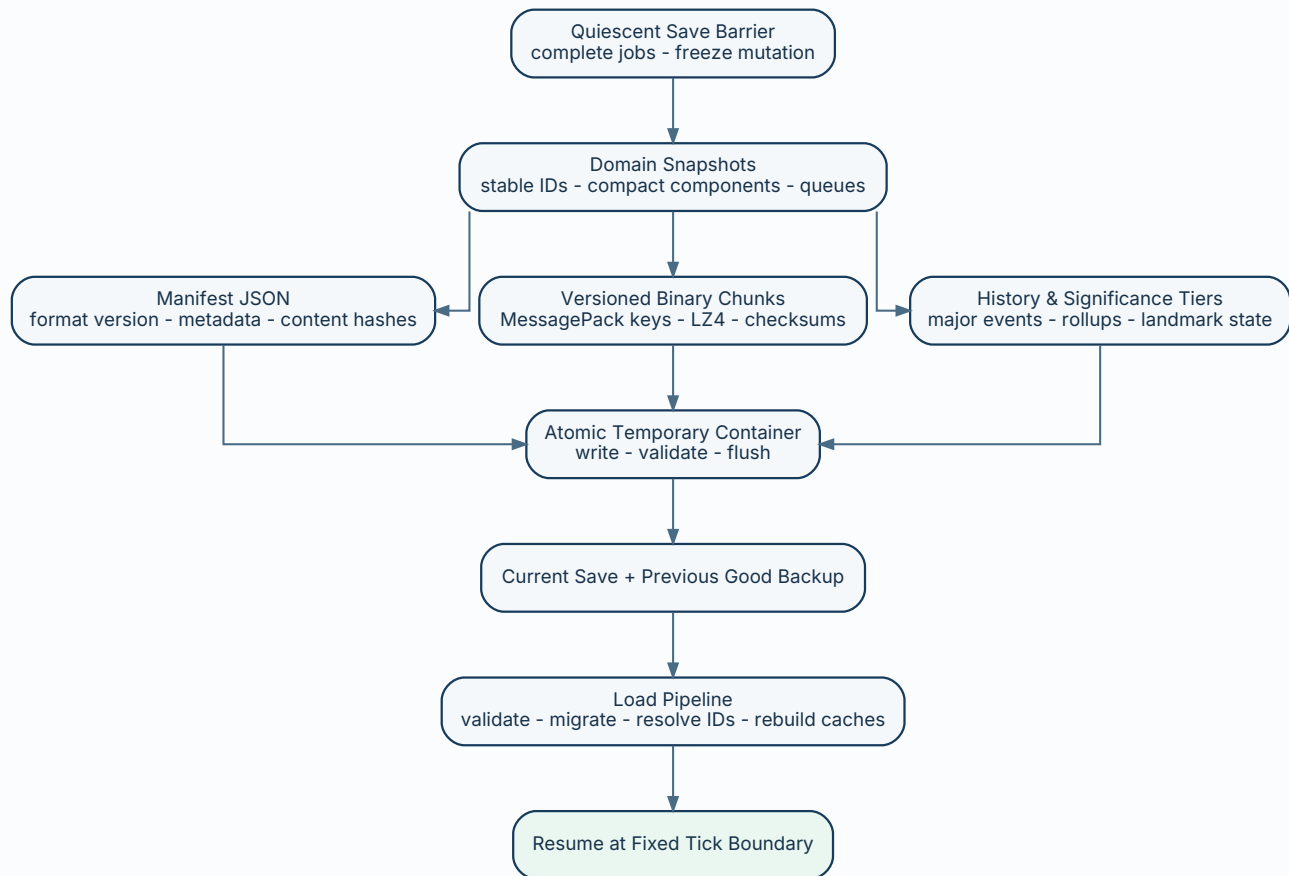


Diagram 23 — Saves cross a quiescent barrier, serialize semantic domain state, validate, replace atomically and rebuild derived caches on load.

A representative container contains manifest, world fields, people, organizations, buildings/routes, economy, knowledge, politics/diplomacy/war, ecosystems, history, landmark geometry/state, interventions, checksums and preview.

47.2 Save process

1. Record request and slot.
2. Reach a quiescent fixed-tick boundary after scheduled jobs.
3. Capture staged/copy-on-write domain snapshots and resume when safe.
4. Serialize chunks on worker threads where supported.
5. Run structural, cross-reference, count, invariant and checksum validation.
6. Write temporary container, flush and atomically replace current save.
7. Retain previous-good backup and rotated autosaves.
8. Report duration, size, compaction and schema.

No autosave overwrites the only good copy. Force Outcome does not bypass save invariants.

47.3 Load and migration

Load validates checksums and manifest, rejects unsupported future formats, backs up old saves, applies sequential pure migrations, deserializes domain records, resolves stable IDs, reports optional missing content, rebuilds ECS and caches, runs invariants and resumes at the stored tick. Every migration has input/output fixtures and rollback notes. Removed numeric keys remain reserved.

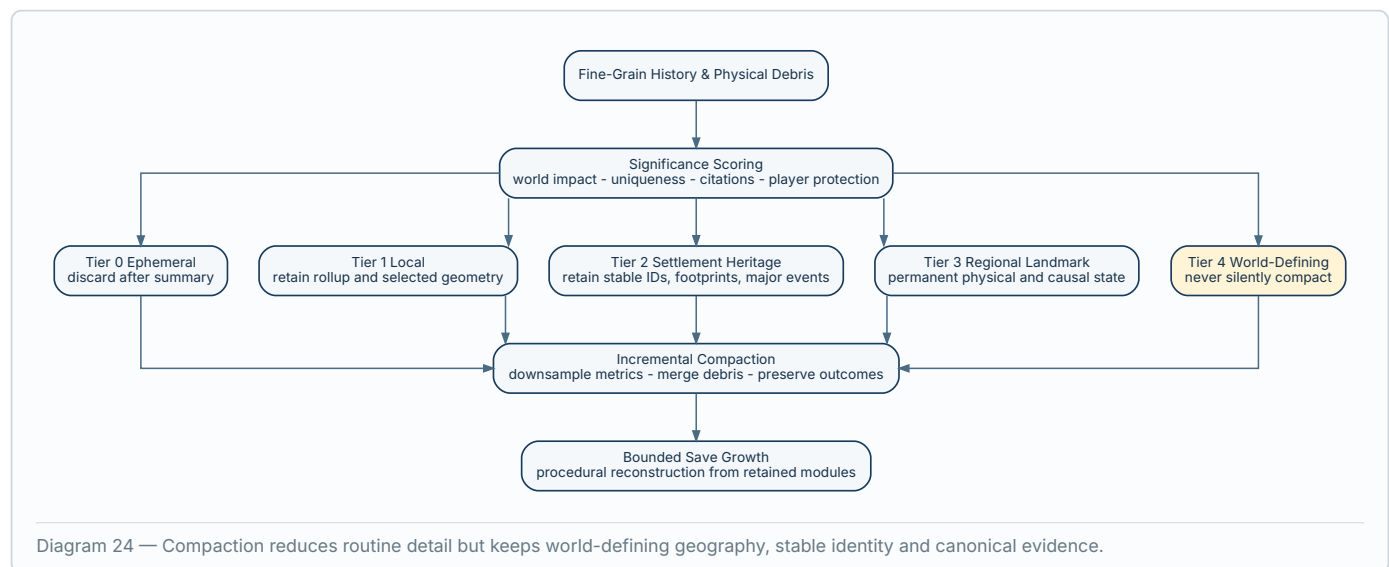
47.4 History tiers

Canonical major events—founding, war, treaty, government change, invention, extinction, disaster, Tier 3 ruin and player intervention—are permanent. Domain rollups retain prices, population, health, culture, capability, environment and territory at decreasing cadence. Ephemeral traces and routine path/transaction events use bounded buffers and summaries.

Recommended retention:

- Person routine actions: 30 game days full, two years monthly summaries, then annual life summaries.
- Settlement economy/health: one year daily, twenty years monthly, then annual.
- Civilization metrics: monthly for twenty years, annual thereafter.
- Important decisions: latest 32 per person and 256 per major organization, with promotion for major events.
- Intervention records and Tier 3 landmark evidence: permanent.

47.5 Landmark preservation and compaction



Physical ruin state is not merely an old event. Current footprints, surviving structural modules, route geometry, contamination and land restrictions remain authoritative. Compaction may replace many ordinary debris entities with a region debris field or merge repeated façade modules, but it must preserve the recognizable city form and stable landmark IDs.

A landmark preservation manifest records original city boundary, street graph checksum, protected structures, damage layers, significance reason and dependent historical references. Any cleanup or demolition creates an explicit transformation event.

47.6 Stable identity

`StableEntityId` combines immutable world identity with a monotonic local ID or equivalent nonreused representation. Tombstones retain minimum identity and destruction/death data when referenced by history, lineage, treaties, knowledge or claims. Presentation and ECS handles may be rebuilt.

47.7 Save acceptance criteria

- Round-trip yields the same canonical checksum after cache rebuild.
- Interrupted writes leave previous or new valid save, never partial replacement.
- Every supported schema fixture migrates automatically.
- Representative 100-year Large save remains below 100 MB under baseline retention.

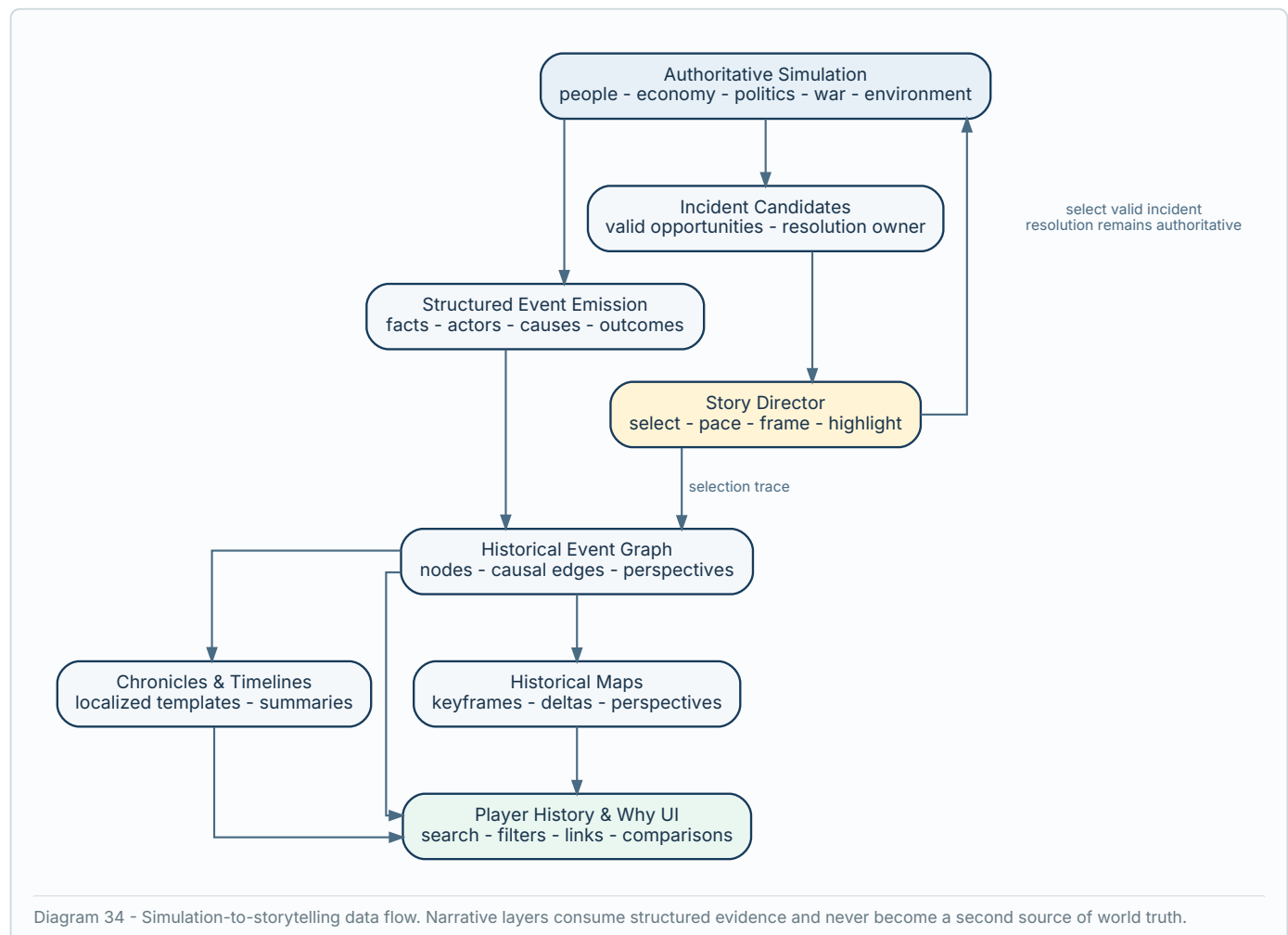
- Snapshot hitch <250 ms, save <10 s and load <15 s on reference hardware.
- Corruption identifies affected chunk and offers backup/recovery.
- Tier 3 ruins remain recognizable and historically linked after aggressive compaction.

48 Emergent storytelling architecture

CONFIRMED A simulation-grounded storyteller and historical-chronicle layer is a core Version 1.0 feature. Its purpose is to make causality legible, pace attention and preserve memorable history without inventing outcomes that the simulation did not produce.

The governing rule is strict:

The authoritative simulation determines what is possible and what actually happens. Story systems may select, pace, frame, summarize and highlight valid events; they may not silently override simulation truth.



48.1 Three cooperating layers

1. **Authoritative simulation:** produces crop failures, shortages, discoveries, construction, migrations, coups, rebellions, wars, extinctions, recoveries, player interventions and physical remnants through ordinary systems.

2. **Story direction:** evaluates eligible incident opportunities, manages pacing and prominence, and chooses which valid developments deserve attention.
3. **Historical narrative:** records event nodes, causal links, participants, locations, outcomes and later consequences, then renders chronicles through authored templates and procedural grammar.

Live LLM generation is not used. Narrative text uses structured events, localization keys, authored templates, grammar-safe substitutions, culture and era tags, generated proper names, measured facts and causal links.

48.2 Structured event contract

Every promotable event declares an event type, authoritative time span, location, participants, affected systems, direct causes, contributing conditions, immediate state changes, player-intervention ancestry, evidence references, historical significance and presentation keys. Events are immutable facts; later interpretation is stored separately.

A food shortage, for example, is not merely a sentence. It references stockpile decline, damaged farms, blocked routes, consumption demand, ration policy, affected households, prices, deaths, migration and the later projects or political changes it caused. A chronicle may summarize that graph, but it cannot replace it.

48.3 Scope and non-goals

The storyteller does not predetermine battle winners, discover technologies, create religions, kill leaders, collapse settlements or spawn resources by narrative fiat. It may identify a valid opportunity for a festival, relief effort, diplomatic mission, investigation or crisis response when the responsible systems can actually resolve it. It may delay a low-priority notification or space unrelated disasters within configured bounds; it may not make off-screen societies fake outcomes because the camera moved away.

48.4 Acceptance criteria

- Pure Simulation mode produces identical authoritative checksums to Chronicler mode for the same command log except where both select from explicitly story-directable, simulation-valid opportunity sets.
- Every story-selected incident has a decision trace, preconditions, cooldown and resolution owner.
- Every generated sentence can be traced to structured facts and localization keys.
- Narrative presentation never exposes information unavailable to its selected perspective unless the omniscient player view is active.
- A 500-year run retains readable high-significance history without unbounded event or prose growth.

49 Story Director

The Story Director is a deterministic policy layer over valid incident candidates. It exists to manage attention and pacing, not to act as an adversarial deity unless the selected profile explicitly requests stronger pressure.

49.1 Candidate generation

Simulation systems publish **incident candidates**, not finished outcomes. A candidate includes preconditions, required actors, required resources and knowledge, exclusions, target set, severity range, cooldown, repetition key, resolution system, expected notification class, historical record requirements and performance budget.

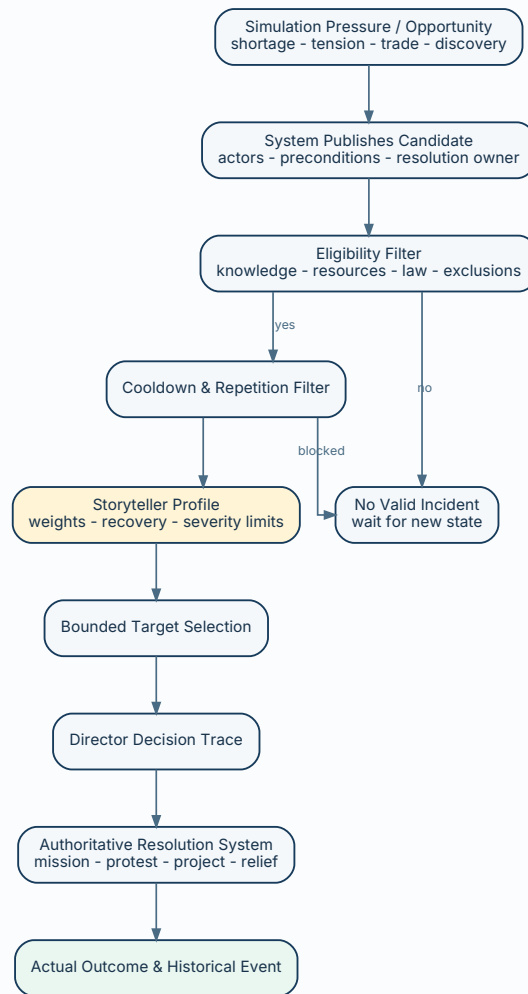


Diagram 35 - Incident-candidate selection. The director chooses only among feasible opportunities and hands resolution back to authoritative systems.

Examples include a government commissioning a survey after resource pressure, a faction calling a protest after legitimacy falls, a settlement holding a memorial after a major loss, merchants proposing a mission when price differences and safe routes exist, or a relief institution organizing aid after displacement. The director may increase the presentation priority of a naturally forming war crisis; it cannot decree that a city loses the war.

49.2 Authority boundary

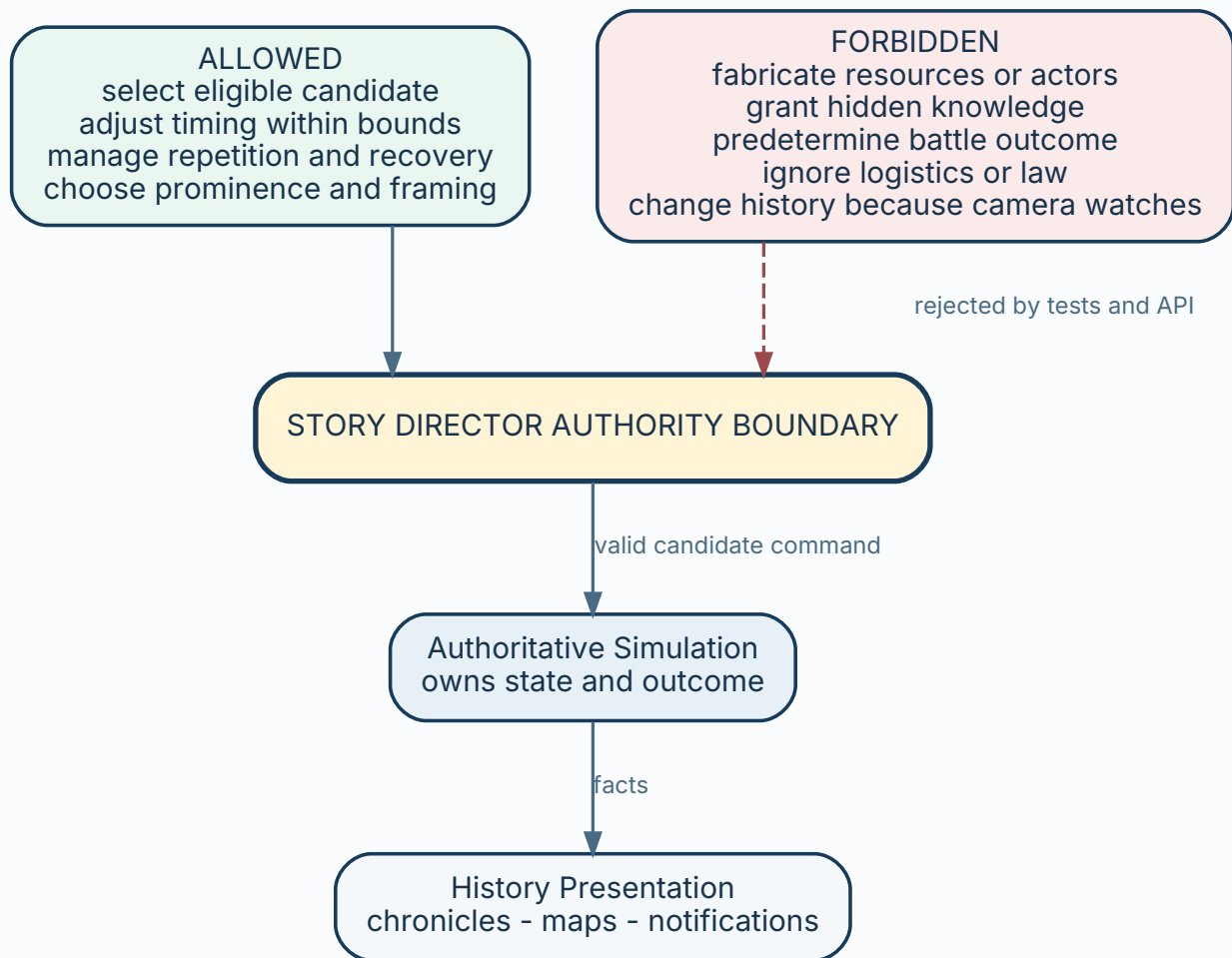


Diagram 36 - The director may select, pace, frame and notify. It may not fabricate resources, knowledge, actors or outcomes.

The director may:

- rank eligible candidates;
- adjust timing within declared bounds;
- suppress immediate repetition;
- protect recovery windows;
- highlight neglected civilizations or under-observed systems;
- prefer event chains connected to existing tensions;
- escalate unresolved pressure by selecting stronger **eligible** responses;
- choose notification prominence and chronicle framing.

It may not:

- bypass incident preconditions;
- create hidden knowledge;
- ignore logistics, law or geography;
- alter a battle result after selection;
- target a civilization merely because the camera is watching;

- violate deterministic replay;
- exceed profile and world-rule limits.

49.3 Pacing state

Pacing state stores recent incident categories, severity load, recovery debt, targeted actors, unresolved chains, notification density and profile parameters. It is versioned, saved and replayed. Pacing uses bounded rolling windows rather than an opaque global “drama score.” Different regions and civilizations may have independent pressure histories so a quiet society is not forced into crisis because another region is already chaotic.

49.4 Director decision trace

Each selection trace records candidate pool size, rejected preconditions, repetition and cooldown filters, profile multipliers, recovery constraints, chosen target, deterministic random contribution, selected candidate, alternative candidates and the authoritative system responsible for resolution. The player can inspect a simplified “Why was this highlighted?” explanation; debug mode exposes full scoring.

49.5 Incident-definition quality gate

An incident is incomplete until it has positive, negative and boundary scenarios; a no-valid-target case; save/load continuity; repetition tests; perspective rules; chronicle keys; performance measurements; and a documented answer to “what real system resolves this?”

50 Storyteller profiles

Storyteller profiles are saved world rules. They change candidate weighting, pacing and notification policy within the authority boundary; they do not replace the simulation.

Table 19 - Storyteller profiles

Profile	Governing behavior	Intended player experience
Pure Simulation	Minimal selection influence; only essential notifications and naturally triggered incidents	Benchmarking, testing and players who want the least narrative shaping
Chronicler	Light pacing, recovery protection, meaningful chain preference and clear summaries	Recommended default; readable history without an adversarial game master
Adversarial	Higher weight for valid pressure, shorter recovery windows and stronger escalation	Harsh worlds where institutions are repeatedly tested
Chaotic	Favors unusual but feasible combinations, reversals and cross-system chains	Surprising alternate history without impossible events
Custom	Per-category controls for disaster, disease, war, politics, migration, invention, positive events and notifications	Creator/player-defined experiments

50.1 Chronicler default

Chronicler may space unrelated catastrophes, allow recovery after severe losses, highlight significant people and places, surface long causal chains and reduce notification noise. It should not protect favored civilizations, guarantee satisfying arcs or convert failure into a hidden comeback bonus. The world can still become calm, catastrophic, unfair or extinct when the underlying systems produce those conditions.

50.2 Custom controls

Custom settings include disaster opportunity, disease opportunity, political instability, war pressure, economic volatility, religious movement, migration, invention opportunity, positive civic incidents, maximum simultaneous severe chains, recovery duration, notification aggregation and chronicle detail. Each control has a documented effect range and cannot authorize impossible candidates.

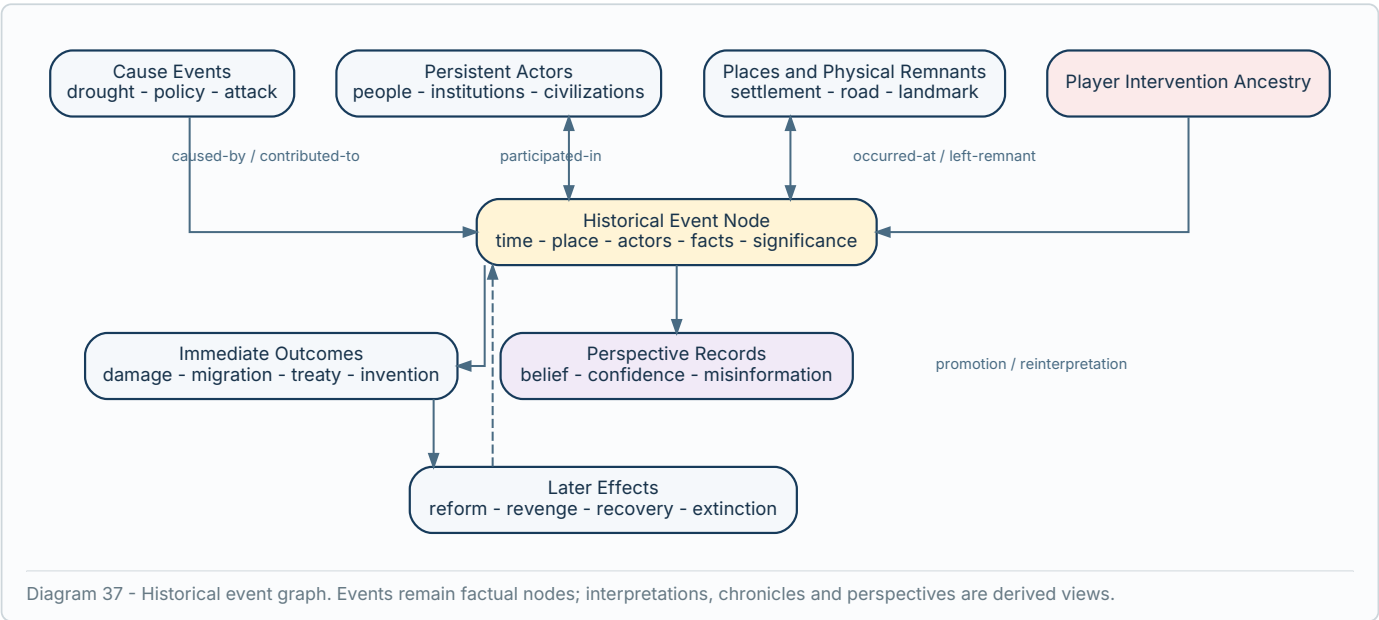
Changing profile or parameters in an existing world creates a permanent history event and stores the prior configuration. Replaying a command log includes those changes.

50.3 Profile verification

Paired-seed tests compare profiles over hundreds of seeds. They must show expected differences in **selection and presentation** while preserving invariants, authority boundaries and camera independence. Pure Simulation is the reference for detecting unwanted narrative intervention.

51 Historical event graph

The Historical Event Graph is the durable causal index of world history. It connects structured events to persistent people, households, institutions, settlements, civilizations, capabilities, physical remnants and player interventions.



51.1 Node and edge model

Event nodes carry stable IDs, type, start/end time, spatial bounds, participants, authoritative facts, significance, presentation keys and source-system version. Edges are typed: caused-by, contributed-to, responded-to, escalated, ended, damaged, displaced, discovered, copied, inherited, memorialized, rebuilt-from and player-intervention ancestry.

Entities do not store an ever-growing unbounded list of every event. The graph maintains indexed significant associations, while local ambient events compact into summaries. A ruined fort can reference the siege that damaged it; a technology variant can reference the artifact, teachers and workshops that enabled adoption; a government can reference the coup and coalition that created it.

51.2 Event chains

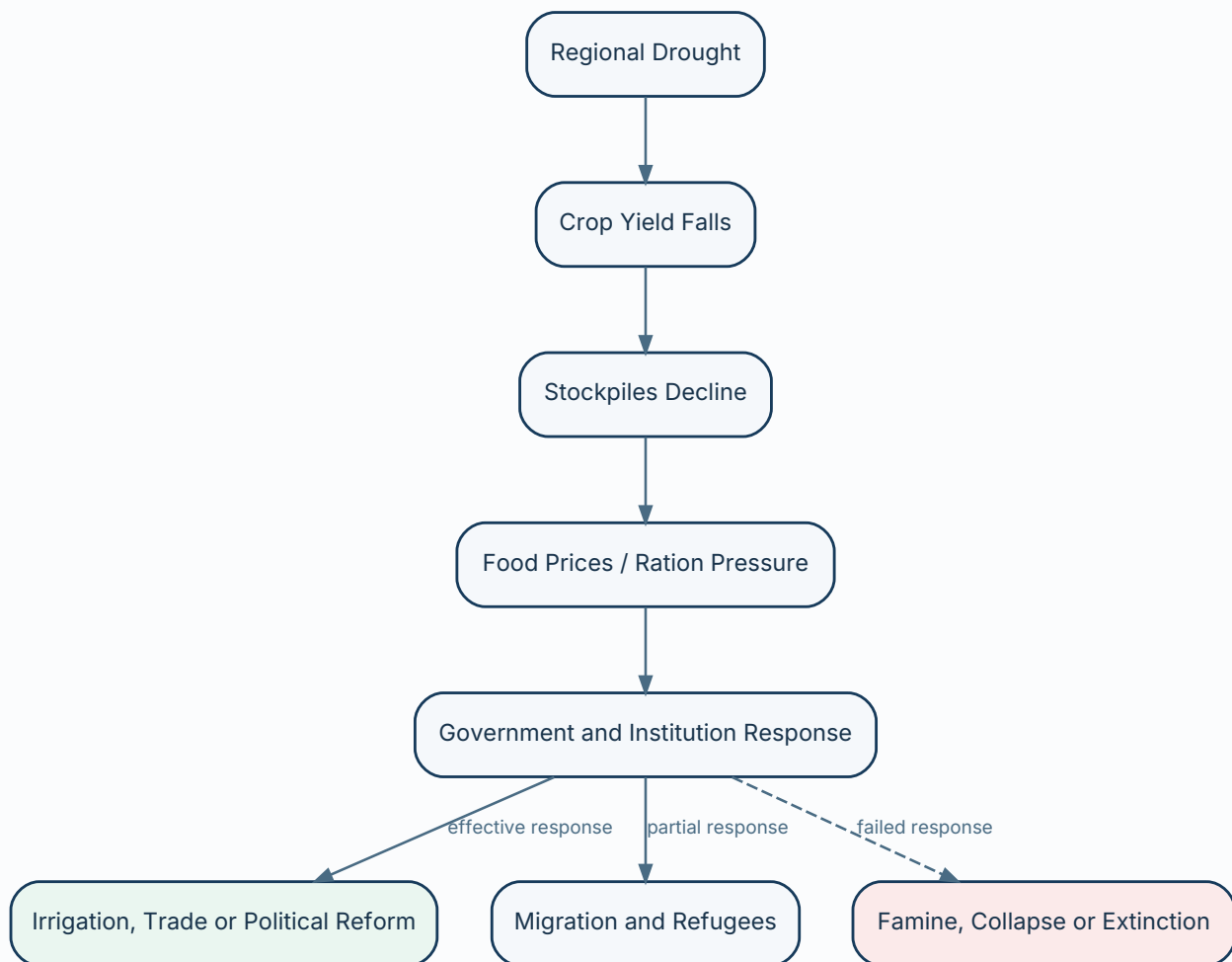


Diagram 38 - Example event chain. A drought becomes history through concrete intermediate systems rather than one scripted narrative event.

Chains are not fixed scripts. A drought may reduce yields, raise prices, trigger ration policy, strengthen mutual aid, provoke protest, encourage imports, cause migration or be absorbed by reserves. The graph records whichever path actually occurred. Query tools can show immediate causes, deeper contributing conditions, downstream effects and counterfactual levers without pretending that one cause explains every outcome.

51.3 Perspective and uncertainty

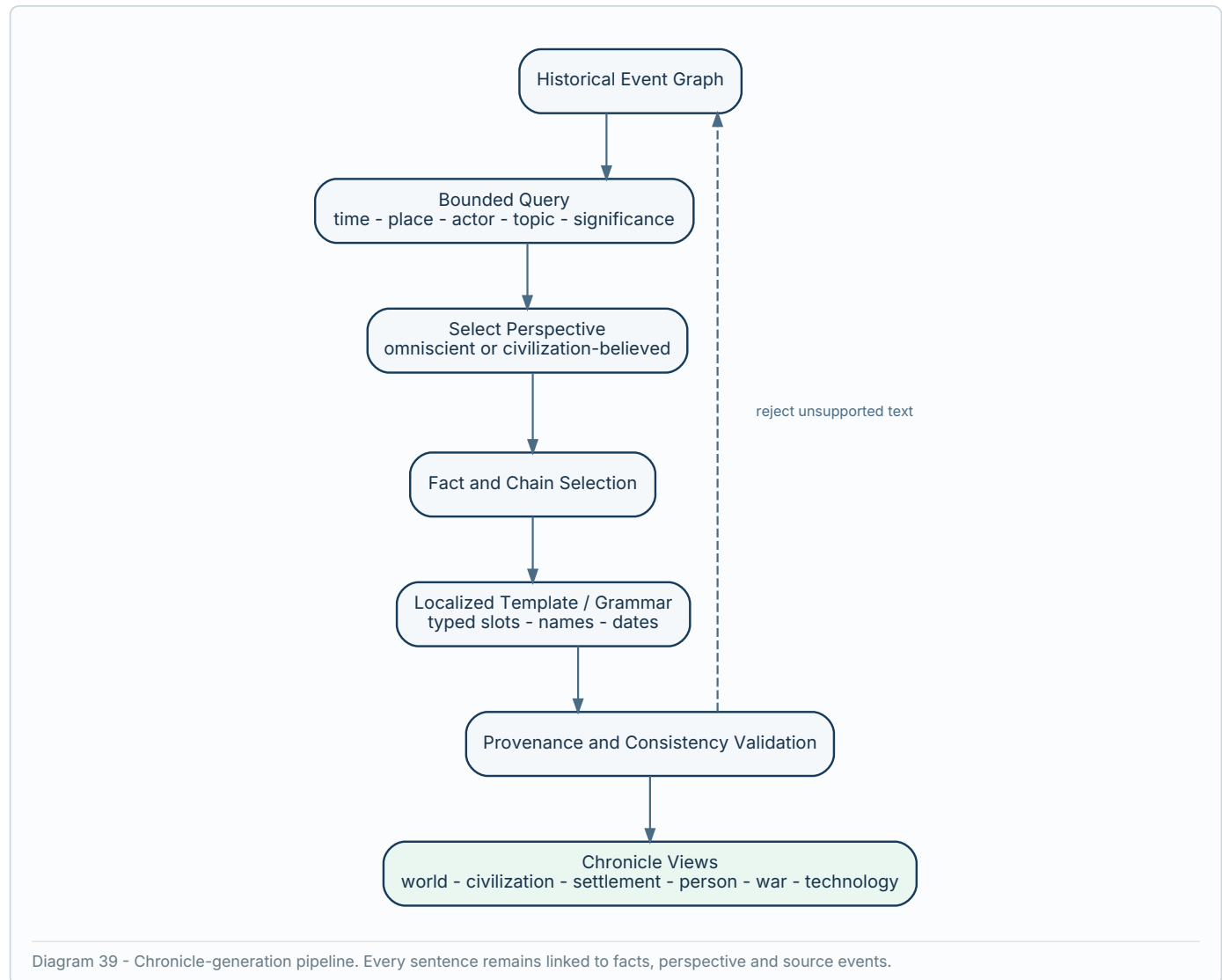
Authoritative event facts are separate from what contemporaries knew. A civilization may record a convoy as an invasion, a disease as divine punishment or a technology as foreign theft. Perspective records carry observer, evidence, belief confidence, misinformation ancestry and later corrections. The omniscient player can compare the factual graph with one or more historical perspectives.

51.4 Graph performance and integrity

The graph is chunked by time and region, indexed by entity and significance, and serialized independently from transient decision traces. Referential integrity tests ensure that compaction never deletes a node still required by a landmark, chronicle, capability lineage or intervention record. Queries have explicit depth, time and result caps.

52 Chronicles and historical presentation

Chronicles turn the event graph into readable historical products. They are generated from structured facts through authored templates and procedural grammar; they are not live free-form LLM output.



52.1 World timeline

The world timeline merges major events, trend graphs, territory changes, extinctions, inventions, wars, disasters, migrations, governmental transformations and player interventions. Filters limit time, geography, civilization, person, topic and significance. Period labels are generated from clusters of transitions and may be renamed by the player without changing history.

52.2 Civilization chronicle

A civilization chronicle presents origins, settlements, governments, institutions, wars, treaties, trade dependencies, migrations, cultural changes, capability lineages, collapses, successor states and revivals. It distinguishes state continuity, cultural continuity and population continuity.

52.3 Settlement history

Settlement history follows founding, named districts, major projects, fires, sieges, population changes, economic roles, rebuilding, abandonment and ruin. Physical landmarks link directly to the event entries that created or transformed them.

52.4 Person life summary

A person summary includes household origins, education, occupations, migrations, service, inventions, leadership, relationships of real significance, injuries, beliefs, major decisions and death. Historical importance is derived from event participation and consequences, not a permanent generic stat assigned at birth.

52.5 War and technology histories

War histories include causes, declarations, belligerents, objectives, operations, supply failures, civilian effects, occupation, peace terms and long-term consequences. Technology lineages show awareness, prototypes, teachers, artifacts, reverse engineering, local variants, manufacturing, adoption, loss and rediscovery.

52.6 Player intervention history

Every state-changing command is visually distinguished from natural events. The record includes time, target, command class, prior state, requested result, validation, bypassed rules, immediate mutation and linked downstream consequences. The player can therefore inspect how a forced bridge, restored species or imposed peace altered later history.

52.7 Text quality and localization

Templates use complete clauses, plural/gender/number-safe variables, locale-aware dates and typed fact slots rather than arbitrary concatenation. Pseudo-localization and missing-key tests run in CI. Unsupported prose is a defect: if a sentence cannot cite event facts, it is not emitted.

53 Historical maps

Historical maps let the player inspect prior territorial, demographic, ecological, infrastructure and settlement conditions without rewinding authoritative time.

53.1 Snapshot strategy

The save stores periodic coarse historical map keyframes plus event-driven deltas for world-defining changes: settlement founding, border transfer, major road/port construction, district destruction, coastline or river alteration, abandonment and landmark formation. The default cadence becomes coarser with age; high-significance periods receive denser keyframes.

Historical views may show:

- borders, claims and occupation;
- settlements and population density;
- roads, ports, canals and later railways;
- land use, forests and major ecological change;
- resources known to a selected civilization at that time;
- wars, migrations and trade routes;
- disaster footprints and persistent ruins;
- governments, cultures, religions and capability geography.

53.2 No authoritative rewind

Selecting a historical date changes only the read model. The live simulation remains paused or continues according to the player's chosen control; returning to an earlier authoritative state requires loading a prior save. A historical map may interpolate for presentation, but the UI labels interpolated or summarized data rather than implying exact archived positions for every person.

53.3 Perspective maps

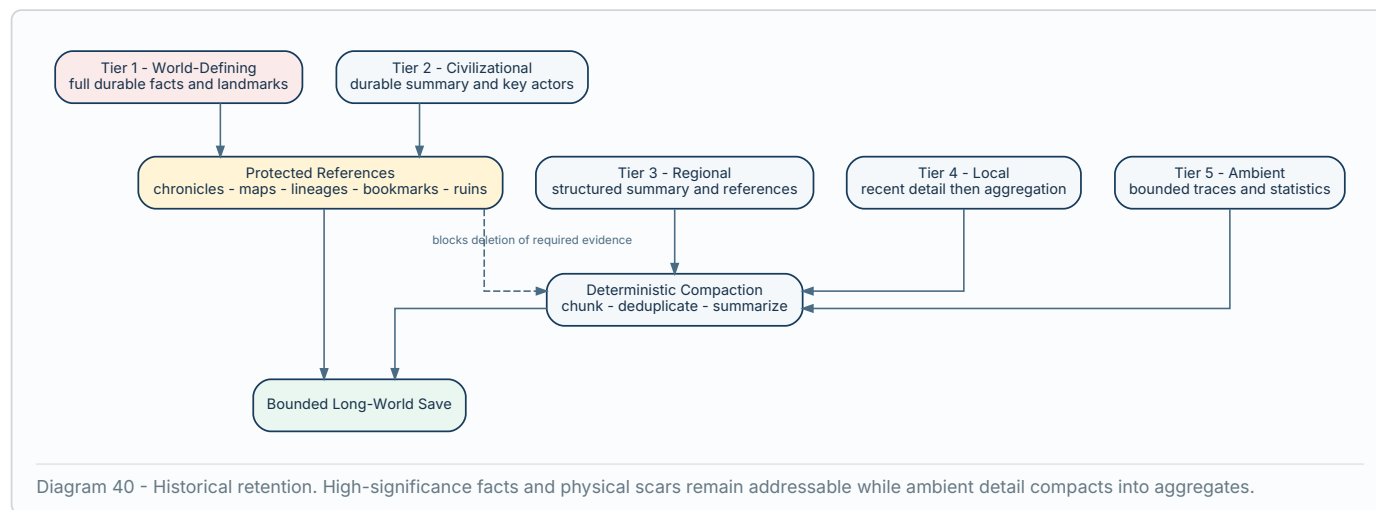
The omniscient map presents reconstructed authoritative state. Perspective maps show what a selected civilization recorded or believed, with uncertainty, missing regions, obsolete surveys and misinformation. Later discoveries may annotate that an earlier map was wrong without rewriting the original perspective.

53.4 Storage and acceptance

Map storage has an explicit bytes-per-century budget. Keyframes are chunked, compressed and deduplicated against current terrain and landmark data. A 500-year Large-world save must remain within the global save contract, and opening a historical century view must not stall the main thread beyond the UI query budget.

54 Historical retention

Historical retention preserves causality and world identity while bounding save growth. It uses significance tiers shared by the event graph, landmarks, chronicles and historical maps.



54.1 Retention tiers

- **Tier 1 - World-defining:** global extinctions, civilization-ending events, world wars, foundational inventions, major player interventions and landmark ruins. Preserve full factual node, essential edges, named participants and physical linkage indefinitely.
- **Tier 2 - Civilizational:** state formation, coups, major treaties, capital relocation, cultural extinction/revival and strategic infrastructure. Preserve durable summaries and important participants.
- **Tier 3 - Regional:** settlement founding, severe disasters, important projects, battles and migration waves. Preserve structured summaries and relevant landmark references.
- **Tier 4 - Local:** routine elections, workplace incidents, minor construction and household events. Retain while recent or referenced, then aggregate.
- **Tier 5 - Ambient:** ordinary transactions, movement, speech and low-consequence decisions. Keep only bounded traces, counters and statistical summaries.

54.2 Promotion and demotion

Significance is not assigned only at creation. A small event can be promoted when later consequences make it important—for example, a minor workshop fire that destroys the last knowledge records. Demotion may remove raw participant lists or detailed timestamps after references expire, but it cannot delete facts required by a surviving chronicle, lineage, landmark or player bookmark.

54.3 Physical ruin linkage

Landmark manifests preserve footprint, skyline envelope, material/damage classes, major structures, street graph and contamination or environmental scar. Fine debris can compact into deterministic ruin tiles or instanced clusters. The result must still look like the same destroyed city centuries later. Reclamation, rebuilding, flooding, burial or deliberate demolition creates new events and may transform the landmark rather than silently erase it.

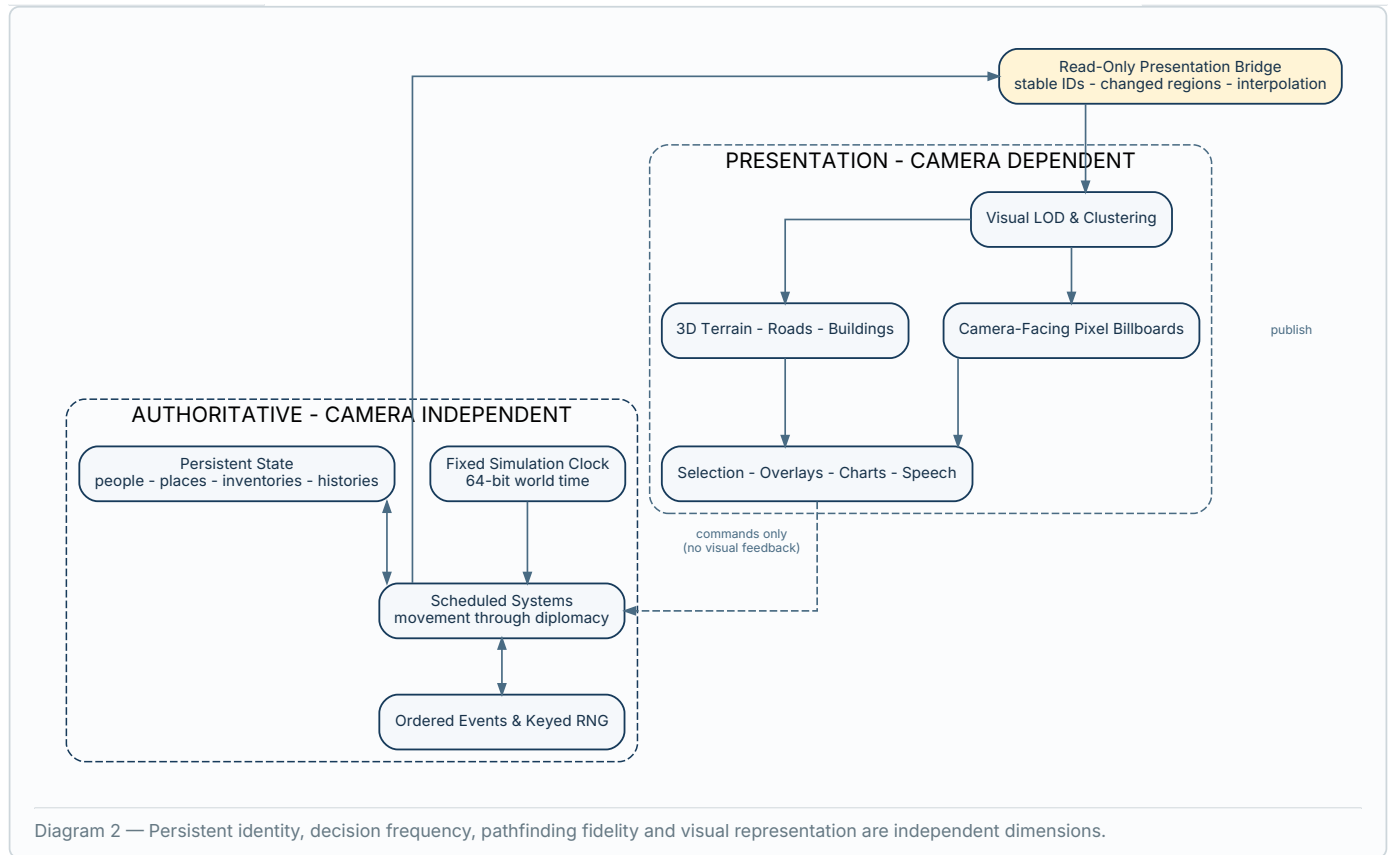
54.4 Retention verification

- Compaction is deterministic and migration-safe.
- Protected event and landmark IDs remain stable.
- All surviving graph edges resolve or produce an explicit diagnostic.
- Chronicle sentences remain supported after compaction.
- Historical maps retain world-defining geography.
- Save-size slopes are measured at 100, 250 and 500 years.
- A no-compaction debug run and a compacted run converge on authoritative current state.

55 Performance architecture

55.1 Philosophy

Performance comes from bounded work, data layout, scheduling, spatial locality, caching and variable cadence. Optimization may aggregate interchangeable goods, share routes, defer decisions and reduce visual density; it may not fabricate remote outcomes, delete people, ignore consumption or change behavior because the camera moved.



55.2 Persistent identity versus update detail

A person can remain persistent while needs update every few game minutes, work every five minutes, political attitudes monthly and visual position only while rendered. Long travel uses route progress and boundary events rather than one movement decision per second. Inside buildings, internal activity is authoritative without rendered paths.

The specification distinguishes:

- **Persistent identity:** stable person/object record across world life.
- **Authoritative state:** current causal facts required for outcomes.
- **Decision frequency:** when an entity reevaluates choices.
- **Visual representation:** whether/how it is drawn.
- **Pathfinding fidelity:** strategic corridor, road route or local steering detail.
- **Historical retention:** how much past evidence remains.

55.3 ECS and data layout

Components are small, unmanaged and grouped by access pattern. Large variable data uses buffers or indexed stores, not managed object graphs in hot loops. Structural changes are queued at barriers. Shared components are used cautiously. Stable ID maps rebuild on load. Authoring bakes immutable runtime definitions. Hot paths are Burst-compatible and allocation-free.

55.4 Spatial indexing and pathfinding

Maintain chunk membership, hashed local grids, settlement parcel/road graphs, regional routes, resource indexes, opportunity indexes and dirty-region sets. Every spatial query declares radius/count caps. Path requests are prioritized and cached. Commuters and shipments reuse corridors until graph versions change. Evacuation and military groups may use shared fields.

55.5 Scheduling and degradation

Each system declares read/write sets, cadence, dependency order, maximum entities/events, deferral behavior, deterministic reduction, profiler marker, budget and backlog fallback. When overloaded, the game can reduce clutter, shadows, speech bubbles, overlay resolution and UI refresh; cap requested speed; and defer nonurgent planning. It cannot skip authoritative due events silently.

55.6 Tracing cost

Decision tracing has three modes:

Mode	Retention	Intended use
Retail bounded	Selected/important decisions plus compact recent rings	Player “Why?” and low overhead
Debug selective	Full traces for chosen decision IDs, layers, regions or entities	Feature development and bug reproduction
Laboratory full	Complete candidate/precondition/score records for scenario fixtures	Automated behavioral comparison

Trace emission uses preallocated buffers, interned IDs and typed numeric terms. Full tracing is excluded from normal performance budgets unless explicitly enabled.

55.7 Telemetry

Benchmarks record p50/p95/p99 duration, job utilization, counts, chunk occupancy, structural changes, path requests/cache hit, query candidates, event backlog, allocations, memory, render batches, visible instances, CPU/GPU time, overlay cost, speech cost, save chunks, trace volume and checksum. Regressions compare against committed baselines.

55.8 Long-running worlds

All first-playable Large authoritative state remains in memory under 6 GB. Presentation assets and terrain meshes stream. Domain chunks permit later cold-storage or partition strategies but remote civilizations do not become fabricated summaries. Long-run soak tests monitor memory slope, ID growth, event queues, landmark growth, ecosystem stability, trace buffers and save size.

56 Engine comparison

56.1 Evaluation criteria

The comparison is project-specific: 10,000 persistent people; large visible cities; camera-facing sprites on 3D terrain; modular 3D buildings; roads and future skyscrapers; headless simulation; deterministic scheduling; multithreading; pathfinding; procedural construction; save migration; Windows development; Linux player builds; coding-agent compatibility; automated tests; editor integration; maintainability and licensing.

Criterion	Weight	Unity 6.3 + DOTS	Bevy	Unreal + MassEntity	Godot + custom ECS/ native
Data-oriented simulation and multithreading	18	5	5	5	3
Editor, terrain, UI and asset authoring	15	5	2	5	4
Coding-agent workflow and code legibility	12	5	4	3	4
Headless tests and automation	10	5	5	4	4

Criterion	Weight	Unity 6.3 + DOTS	Bevy	Unreal + MassEntity	Godot + custom ECS/ native
Pixel billboards on 3D terrain	8	5	3	3	4
Long-term ecosystem/maintainability	10	4	3	3	4
Windows development / Linux player	7	5	5	4	5
Save/content/mod foundation	7	4	4	4	4
Licensing/control	5	3	5	3	5
AI-only implementation risk (5 lower risk)	8	4	2	2	3
Weighted score / 5	100	4.65	3.76	3.82	3.86

Scores are engineering recommendations, not universal engine rankings.

56.2 Unity 6.3 LTS

Unity 6.3 is the current LTS and is supported through December 2027. Entities provides a data-oriented ECS; Burst and Jobs provide compiled parallel workloads; Entities Graphics supports entity-oriented rendering; Unity supplies testing, performance tools, batch/headless operation, terrain, UI, profiling and mature asset workflows. [S-ENG-01](#) [S-ENG-03](#) [S-ENG-04](#) [S-ENG-05](#)

Risks are DOTS complexity, mixed GameObject/ECS paradigms, package migration, nonautomatic determinism, licensing changes and agents defaulting to conventional MonoBehaviours. Pinning, assembly rules, examples and benchmark gates mitigate them.

56.3 Bevy

Bevy is ECS-first, open source and code-centric, which is attractive for headless simulation and agent-readable architecture. Its official repository continues to describe the engine as early-stage, with missing features/documentation and breaking API changes on a rapid cadence. [S-ENG-12](#) [S-ENG-13](#) The project would assume more editor, terrain, UI, asset and profiling infrastructure. Bevy is the best technical fallback if Unity's runtime—not project architecture—fails.

56.4 Unreal MassEntity

Unreal offers high-end world/editor tooling and MassEntity for large data-oriented populations. Current Unreal releases continue to develop Mass, but C++, reflection/build tooling, Blueprints, binary assets and engine scale increase coding-agent context and review burden. Its rendering advantage exceeds this project's needs. [S-ENG-14](#) [S-ENG-15](#)

56.5 Godot with custom ECS/native extension

Godot is open source, lightweight, text-friendly and supports headless execution. Current 4.7.1 maintenance release and official 4.7 documentation remain active. Large-scale ECS simulation would largely be custom C#, GDExtension or native code, and thread-safe API constraints require care. [S-ENG-16](#) [S-ENG-17](#) It is viable for a smaller target but medium-high risk for this combination.

56.6 Conclusion

No current evidence compels a change from Unity. The strongest decision remains Unity provisionally, with Bevy as an isolated benchmark fallback. Unreal and Godot are credible alternatives but increase project-specific AI-only development risk.

57 Final engine recommendation

57.1 Decision

RECOMMENDED Use Unity 6.3 LTS on Windows with C#, Entities/DOTS, Burst, Job System, Entities Graphics, URP, Unity Test Framework, Performance Testing API and command-line/headless scripts. Pin exact editor revision and packages. Use GameObjects/MonoBehaviours only for authoring, camera, UI shells, pooled effects and presentation adapters. Authoritative simulation packages may not reference rendering, UI, scene, camera, audio or editor assemblies.

57.2 Version policy

Commit package manifests and locks; disable automatic upgrades; upgrade only in planned compatibility windows on dedicated branches; require release-note review, migration plan, save fixtures, replay, benchmark comparison and rollback. Licensing and service terms are reviewed before commercial milestones. Unity's current pricing eligibility must be verified at those dates rather than assumed from this blueprint. [S-ENG-18](#)

57.3 Engine go/no-go benchmark

Unity is GO only if the Phase 1 representative benchmark demonstrates:

1. 10,000 persistent people with representative components and schedules.
2. Four to six settlements, workplaces, inventories, shipments, politics, diplomacy, ecosystems and active encounters.
3. 20 Hz base loop with p95 authoritative tick ≤ 25 ms at 1 $\times$ and ≤ 45 ms under 10 $\times$ stress, without growing backlog.
4. 60 FPS ordinary settlement view and ≥ 30 FPS stress floor with maximum prototype density.
5. Working set ≤ 6 GB and no sustained managed allocations in hot loops.
6. 100-year deterministic replay equality on the reference build.
7. Save ≤ 100 MB, save ≤ 10 s and load ≤ 15 s.
8. 5,000 camera-facing billboards, 1,000 modular buildings, overlays and selective speech within render budget.
9. A coding agent can implement a scoped change, build, test, benchmark and provide evidence without manual code repair.

Failure triggers at most two measured optimization iterations. If project architecture is the cause, fix it in Unity. If an engine/runtime constraint remains and a Bevy port plausibly removes it, run the four-week fallback spike before production expansion.

58 Game-AI architecture

58.1 Governing principles

The game's people and civilizations are not powered by live language models. They use coded and data-driven systems whose behavior is specified, bounded, deterministic, explainable, testable, benchmarkable, reproducible, regressible and reviewable. "Make intelligent AI" is not an acceptable implementation task.

The repository maintains one authoritative AI runtime and one catalog of versioned decision contracts. New behavior normally adds validated action definitions, policies, curves, preconditions and modifiers. It does not introduce another scoring framework, random API, blackboard, global singleton or hidden planner.

The golden principles are:

1. **Authority before choice.** A layer may consider only decisions it owns.
2. **Knowledge before utility.** Observations are filtered to what the actor can know.
3. **Preconditions before scoring.** Impossible actions do not win through weights.
4. **Bounded candidates.** Every decision declares maximum actions and query cardinality.
5. **Canonical curves and units.** Scores are comparable and inspectable.
6. **Commitment before churn.** Hysteresis, cooldown and invalidation prevent thrashing.
7. **Keyed randomness only.** Scheduling order cannot alter samples.
8. **Reservations before execution.** Unique resources and capacity cannot be double-spent.
9. **Trace every consequential choice.** Losing candidates and blockers remain inspectable.
10. **Tests before tuning.** Weights are changed against scenarios and distributions.
11. **Data varies policy; code executes policy.** Content does not inject arbitrary logic.
12. **Regression becomes knowledge.** Every observed incoherent behavior becomes a fixture.

58.2 Canonical AI primitives

The runtime contains one approved implementation of:

- utility normalization and score aggregation;
- linear, inverse, logistic, threshold, smoothstep and piecewise response curves;
- candidate enumeration with hard caps;
- preconditions and reason-coded failures;
- unique and divisible reservations;
- cooldowns and minimum commitment duration;
- hysteresis and change penalties;
- deterministic tie-breaking and weighted choice;
- keyed random stream access;
- goal invalidation and replanning events;
- cadence scheduling and stable bucket staggering;
- trace emission and bounded retention;
- knowledge filtering and confidence/staleness;
- layer-authority validation;
- policy loading, schema validation and versioning;
- performance markers and per-decision counters.

These primitives live in the smallest authoritative core package. Feature packages may compose them but not fork them. Any extension requires an architecture decision record, conformance tests and migration impact analysis.

58.3 Utility and selection semantics

All utility considerations normalize into a documented range, normally $[0,1]$ before weighting. The contract declares aggregation: weighted sum, multiplicative gate, minimum, maximum or limited hybrid. Hidden clamps are prohibited. Negative utility terms use explicit semantics rather than arbitrary subtraction from unrelated scales.

Candidate selection defaults to highest score after stable tie-break. Deterministic weighted choice is permitted where diversity is valuable and the contract declares random amplitude. Keyed randomness is never allowed to rescue a failed precondition. Random contribution appears in the trace as a separate term.

Hysteresis compares the current commitment with alternatives and imposes a switch penalty. Minimum duration prevents repeated reevaluation. Urgent invalidation—fire, destroyed target, critical hunger, direct order—can bypass ordinary delay. Cooldowns prevent repeated failed proposals but cannot hide permanent impossibility.

58.4 Knowledge filtering

Every decision receives an `ObservationContext` scoped to actor, layer, time and knowledge authority. It contains authoritative local facts the actor can directly sense, stored beliefs, institutional reports, maps, rumors and confidence. The context is assembled by a canonical knowledge service; decision code may not query unrestricted world stores.

The omniscient UI reads authoritative data separately. Unit tests deliberately supply false/stale observations to ensure decisions do not “cheat.” A camera, overlay or player inspector cannot write observations into AI state unless a player power explicitly reveals information.

58.5 Policy data versus execution

Policy variation is primarily data. The stable runtime loads action definitions, curves, weights, preconditions, cultural modifiers, government modifiers, trait modifiers, technology modifiers, institution rules, cadence and trace keys. Schemas validate units, ranges, references and cycles.

Data may choose among approved evaluators. It may not embed arbitrary scripts, call Unity objects or access mutable stores. New evaluator code requires a bounded task and tests. This separation permits later content packs, balance changes and society variation without rewriting the runtime.

58.6 Planning boundaries

Limited planning is justified for settlement projects, military operations, logistics and technology programs. Planners select among authored plan templates with bounded steps, costs and preconditions. They do not run open-ended search over arbitrary world states.

A plan is an explicit state machine: proposed → authorized → reserving → active → blocked → completed/cancelled. Every step belongs to one authority layer and creates lower-layer demand rather than direct mutation. Planners have search depth, candidate and runtime budgets.

58.7 Machine learning scope

CONFIRMED Machine learning is not the default implementation for governments, diplomacy, technology, settlement planning, citizen brains, politics, economy or culture.

EXPERIMENTAL ML may be explored for crowd avoidance, bounded local combat policies, heuristic comparison, offline parameter tuning or adversarial test generation. A learned policy cannot become authoritative without deterministic inference strategy, explainability, save/version stability, performance proof, reproducible training, versioned model artifacts, fixed input schema, fallback coded behavior and an architecture decision record.

58.8 AI completion criteria

A decision is complete only when it has an approved behavior contract, implementation using canonical primitives, trace schema conformance, deterministic scenarios, properties, metamorphic tests, multi-seed tests where needed, performance measurement, documentation, decision catalog entry and adversarial review. “The agents look smarter” is not a completion criterion.

59 Decision hierarchy

59.1 Required hierarchy and technique assignment

Table 11 — Decision layers, authority and primary techniques

Layer	Owned decisions	Primary techniques	Explicit exclusions
Immediate individual reaction	Flee, brace, interrupt, seek immediate safety	State machines, event reactions, thresholds	No strategic planning
Individual goals	Eat, rest, work, seek care, migrate, protest, fight	Bounded utility selection, commitment	No global search; no government policy
Household	Shared food, housing, dependents, budget, relocation	Shared needs, rules, reservations, bounded utility	No workplace production control
Workplace/business	Production lots, hiring, procurement, offers, maintenance	Demand signals, queues, utility allocation, policies	No personal movement control
Institution	Service access, curriculum, doctrine, research project, appointment	Policy rules, constrained goal selection	No national diplomacy
Settlement	Public projects, services, stockpiles, land-use coordination, defense	Utility-ranked projects, limited plan templates	No personal task micromanagement
Region	Route, survey, resource coordination and claims	Graph analysis, project portfolio	No settlement-internal placement
Government	Law, budget, policing, mobilization, diplomacy authorization	Policy state machine, strategic utility, coalition constraints	No tactical unit movement
Civilization	Long-term priorities, integration, external posture, major programs	Coarse hierarchical planning, bounded portfolio	No direct lower-layer execution
Alliance	Collective defense, sanctions, mediation, bloc policy	Relationship state, proposal evaluation, member commitments	No member-government internal policy
International system	Contact opportunities, balance pressures, treaty network effects	Sparse graph events and bounded evaluations	No omniscient "world director" decisions
Technology	Project, experiment, prototype, adoption, diffusion	Capability graph, opportunity detection, bounded experimentation	No global unlock tree

Authority validation occurs before candidate enumeration. A higher layer communicates through goals, policies, budgets, constraints, projects and commands. It cannot invoke lower-layer mutation methods. Lower layers publish demand, support, capacity and outcomes rather than requesting that the higher layer choose every action.

60 AI Behavior Contracts

Every consequential coded decision is implemented against a versioned AI Behavior Contract. The contract is machine-readable where practical and reviewable as Markdown. Coding agents implement a specified bounded decision; they do not invent an entire mind from a broad feature request.

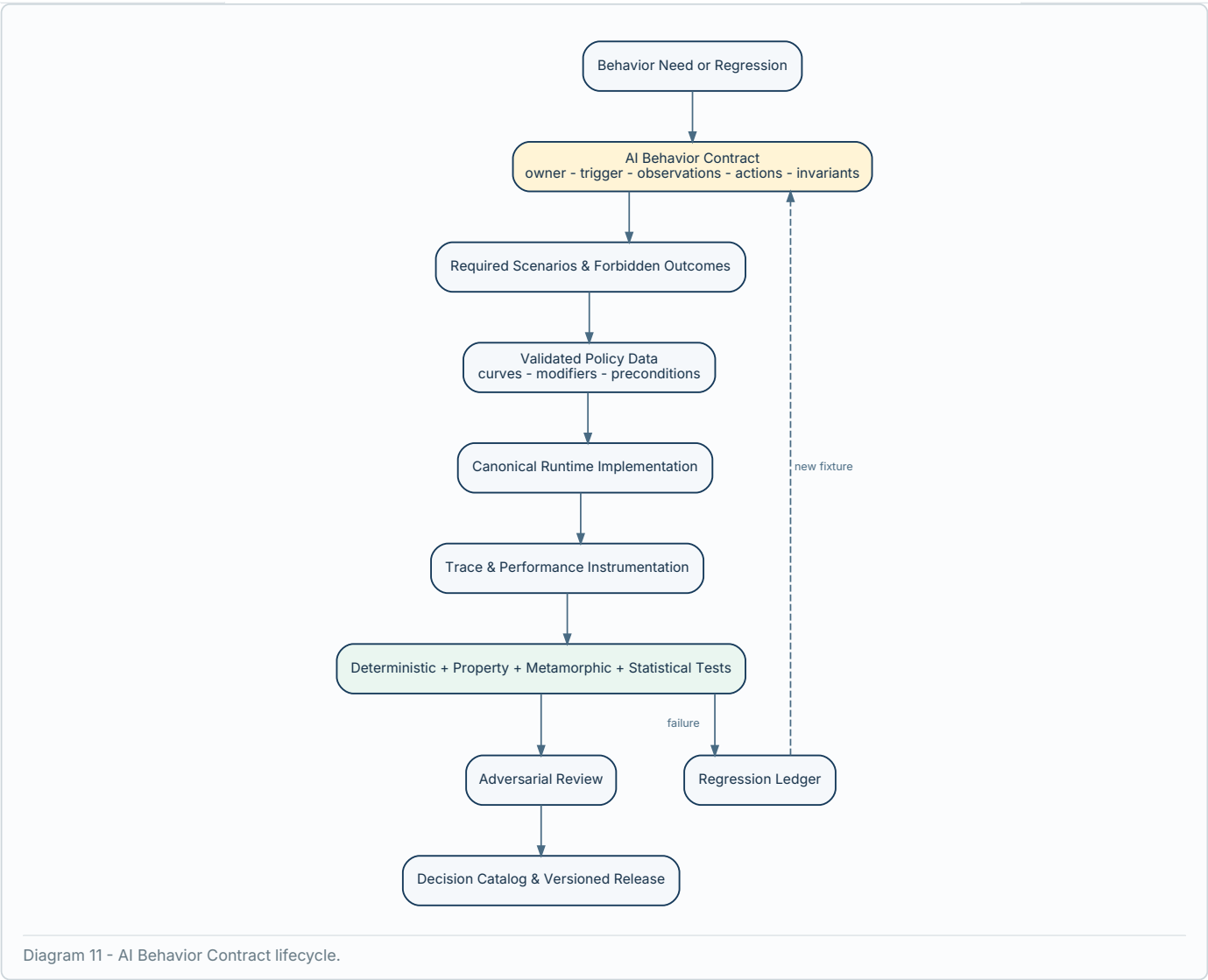


Table 12 - Required behavior-contract fields

Field group	Required content
Identity and authority	Decision ID/version, owning layer, purpose, explicit non-goals
Scheduling	Trigger, cadence, reconsideration, invalidation events
Information	Observation context, hidden information, knowledge age/confidence
Choice	Candidate actions, candidate/query caps, preconditions, response curves and utility terms
Stability	Reservations, commitment, cooldown, hysteresis and change penalties
Determinism	Keyed random stream, maximum random effect, tie-breaking and occurrence counter
Effects	State changes, commands to lower layers, failure/blocked outcomes
Evidence	Trace schema, invariants, scenarios, metamorphic relations, statistical tests and performance budget
Persistence	Save fields, migration behavior and backward compatibility

60.1 Contract review rule

A contract is approved before implementation. The implementer may propose refinements, but changing observations, authority, action set or randomness requires contract review. Runtime code must reject out-of-contract access through API boundaries and analyzers where possible.

60.2 Example

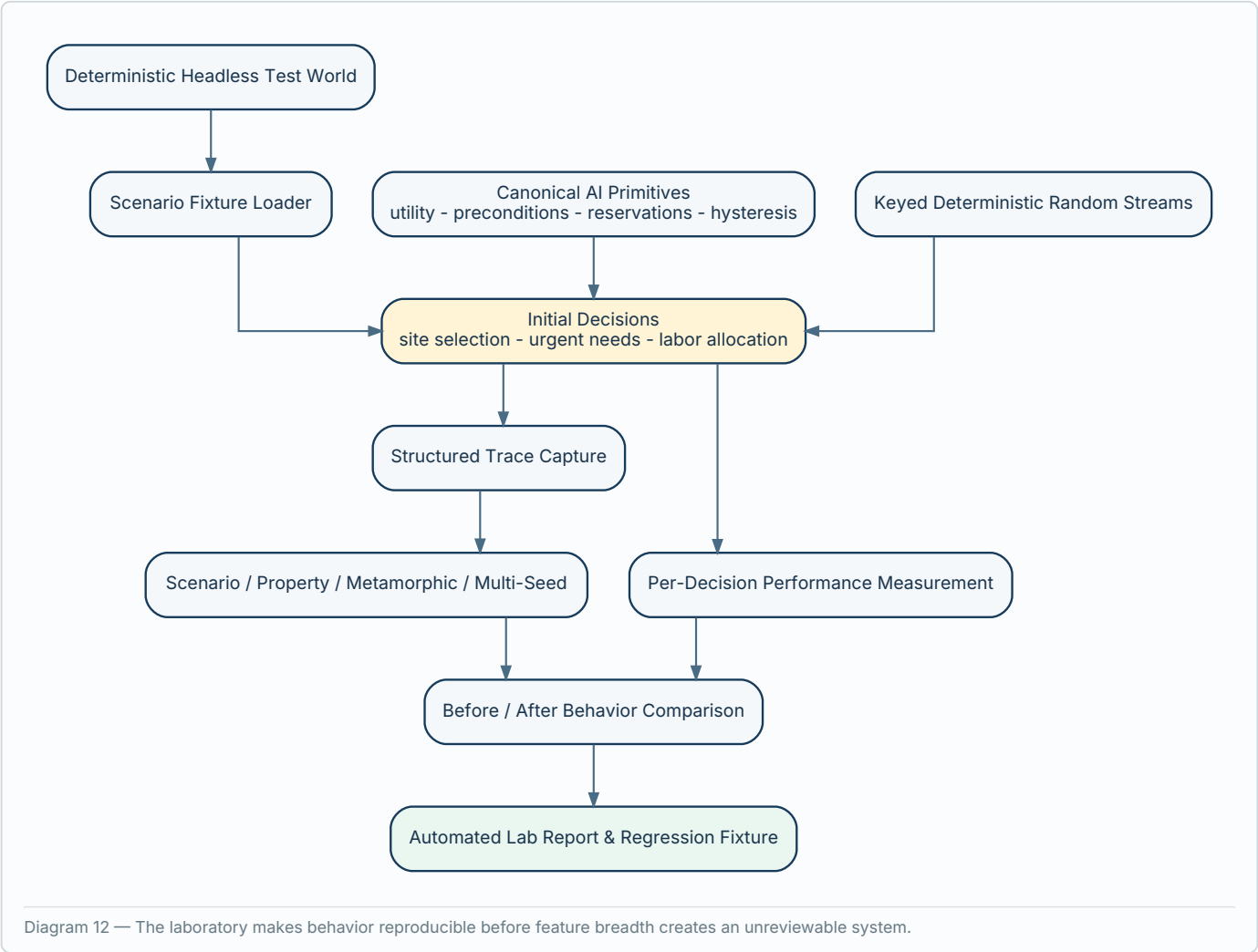
The canonical founding-site contract considers a bounded set of known candidates, reads water, food, soil, timber, hazard, security and travel observations available to the group, excludes undiscovered deposits, uses stable tie-breaking and emits all normalized scores. Its tests include reachable water, attractive flood risk, stale knowledge, unreachable candidates, camera independence and multi-seed survival distributions.

The complete template and example are retained in the technical appendices.

61 AI Decision Laboratory

61.1 Purpose and phase position

CONFIRMED The **AI Decision Laboratory** is an explicit early milestone before broad civilization behavior. It proves the primitives and evaluation method on a small number of decisions. The project must not begin with one giant civilization brain or dozens of independently authored utility systems.



61.2 Required components

The laboratory contains:

1. Deterministic headless test world.
2. Canonical bounded utility engine.
3. Precondition framework.
4. Reservation system.
5. Cooldowns.
6. Hysteresis and change penalties.
7. Goal invalidation.
8. Keyed deterministic random streams.
9. Structured decision tracing.
10. Scenario fixture loading.
11. Property-test support.
12. Metamorphic-test support.
13. Statistical multi-seed execution.
14. Before/after behavior comparison.
15. Per-decision performance measurement.
16. Regression fixture capture.
17. Automated HTML/JSON report generation.

The lab runs as a command-line program and CI job independent of the game renderer. It can optionally render a small diagnostic world, but the headless state and trace are authoritative.

61.3 First three decisions

61.3.1 Founding-group site selection

The contract owns a founding-group decision. Trigger: world generated or current site invalidated. Observations: group knowledge of candidate regions, water, food, arable land, timber, hazards, travel and claims. Candidate count: no more than 16 broad regions and 8 final sites. Actions: select site, continue search, emergency camp. Non-goals: permanent town layout, road construction or omniscient deposit use.

Required scenarios include viable river site, attractive site with hidden flood risk, no nearby timber, known hostile claim, exhausted refugees and two equivalent sites with deterministic tie-breaking. Metamorphic relation: adding reachable fresh water to a candidate must not reduce its water term or overall utility absent an explicitly modeled downside.

61.3.2 Individual urgent-needs selection

The contract owns immediate person goals. Trigger: scheduled reconsideration or urgent state change. Candidates: continue commitment, eat, drink, rest, seek shelter, seek care, flee, help dependent, idle. Preconditions ensure food exists and is accessible, care has capacity, and routes are known.

Scenarios include starving person beside reachable food, food across impassable river, food reserved by another, person carrying food home to a dependent, burning building, and direct higher-layer work priority that must yield to critical hunger. Property: selected action satisfies preconditions. Metamorphic relation: adding accessible food cannot increase starvation duration under equivalent conditions.

61.3.3 Settlement labor allocation

The contract owns project/job allocation, not personal movement. Inputs: job demand, eligible labor pool, skills, schedules, law, wages/entitlements, distance and strategic priorities. Actions: open role, raise compensation/entitlement, reassign permitted labor, defer project, request migration/training or cancel. Reservations prevent double assignment.

Scenarios include harvest versus construction competition, worker lacking tools, two jobs for one worker, conscription shock, public order to prioritize housing, and legal exclusion. Metamorphic relation: increasing a job's compensation cannot reduce its attractiveness unless another declared factor changes.

61.4 Scenario fixture format

A fixture is versioned data:

```
scenario_id: AI-NEED-001
world_seed: 0042...
simulation_version: 1.1-lab
initial_state_fixture: starving_person_reachable_food
commands: []
run_until: PT10M_GAME
allowed_actions: [EatFood, MoveToFood, ContinueCriticalCarry]
forbidden_actions: [Sleep, Wander, StartDistantJob]
expected_tendencies:
  - selected_action in [EatFood, MoveToFood]
  - starvation_level decreases by deadline
required_trace_terms:
  - hunger_urgency
  - food_accessibility
invariants:
  - inventory_nonnegative
  - selected_preconditions_true
performance_budget_us: 50
```

Fixtures use stable content IDs and schema versions. They can include checkpoints, tolerated ranges and multi-seed expectations. A failure produces state diff and trace diff.

61.5 Before/after comparison

For every AI change, the lab runs the affected fixture set on baseline and candidate builds. Reports show selected actions, score deltas, precondition changes, trace term changes, timing, allocation, success rates and distribution shifts. Intended differences require task-contract explanation. Unrelated decision changes are regressions unless explicitly accepted.

61.6 Performance measurement

Every decision contract declares maximum candidates, observations, spatial queries, allocations and p95 duration. Laboratory benchmarks use representative batch sizes, not one entity. They report per-decision and per-1,000-decision cost. Full trace overhead is measured separately from retail bounded tracing.

Initial laboratory targets on reference hardware:

Decision	Batch	p95 evaluation target	Allocations
Urgent person needs	10,000 evaluations	≤50 μs per decision; ≤8 ms total parallel job target	0 steady-state
Site selection	100 groups × 16 candidates	≤2 ms per group headless	0 after fixture load
Labor allocation	100 settlements × 64 jobs × bounded workers	≤5 ms per settlement review	0 steady-state

These targets are provisional and may be revised only with measured evidence and updated budgets.

61.7 AI Decision Laboratory go/no-go

The project does not proceed to broad person and government behavior until:

- all three decisions pass deterministic scenario suites;
- property and metamorphic tests run in CI;
- multi-seed reports show diversity without forbidden outcomes;
- save/load during a decision preserves subsequent choices;
- camera and rendering absence make no difference;
- canonical traces reproduce and compare cleanly;
- performance fits budget with 10,000-person batches;
- an implementer and adversarial reviewer independently approve evidence;
- at least ten deliberately seeded defects are detected by the harness.

Failure means redesign the primitives and fixtures, not tune more weights.

62 Behavioral evaluation

62.1 Evaluation dimensions

Behavior is evaluated for legality, coherence, responsiveness, diversity, stability, explainability, efficiency and exploit resistance. A decision may be legal but incoherent; coherent but too slow; diverse but unstable; explainable but exploitable. Reports keep these dimensions separate.

Dimension	Core question	Evidence
Legality	Did the actor have authority, knowledge and prerequisites?	Preconditions, authority checks, invariants
Coherence	Is the choice plausible from observations and goals?	Scenario expectations and trace review
Responsiveness	Does behavior change in the expected direction?	Metamorphic tests
Diversity	Do equivalent actors sometimes choose different feasible paths?	Multi-seed distributions
Stability	Does behavior avoid loops, oscillation and thrashing?	Long-run trace patterns and hysteresis metrics
Explainability	Can the choice be reconstructed from trace?	Trace completeness and UI tests
Efficiency	Is candidate/query/time cost bounded?	Performance reports
Exploit resistance	Can weights or commands produce degenerate behavior?	Adversarial review and fuzzing

62.2 Adversarial review checklist

The reviewer actively searches for:

- hidden omniscience or access to player overlays;
- unbounded candidate enumeration or world scans;
- shared mutable randomness;
- layer-authority violations;
- repeated cancel/reselect loops;
- resources reserved without release;

- invalid state after target destruction;
- weights that dominate every context;
- missing negative/edge scenarios;
- performance dependent on camera;
- save incompatibility or behavior-version drift;
- Force Outcome paths that bypass reconciliation;
- traces that omit losing candidates or controlling policy;
- culture or traits acting as deterministic stereotypes;
- “magic” aggregate modifiers without physical cause.

The reviewer may add tests but does not silently rewrite the implementation. Findings return to the implementer with reproducible fixtures.

62.3 AI regression ledger

Every observed stupid, exploitable, impossible or incoherent behavior becomes:

1. a saved deterministic fixture;
2. a named regression case;
3. an automated test;
4. a documented root cause;
5. a permanent CI requirement where practical.

The ledger is searchable by decision, layer, symptom, root cause and fixed version. It is project-specific behavioral knowledge available to every future agent session.

62.4 Tuning discipline

Weights are not tuned directly in production until scenario and distribution baselines exist. A tuning change names intended outcomes, affected contexts and unacceptable side effects. Automated search or offline optimization may suggest parameters, but humans/agents review trace semantics and keep values within documented ranges. Tuning never changes hard preconditions.

The tuning guide distinguishes:

- **shape tuning:** response curve and thresholds;
- **importance tuning:** weights among comparable considerations;
- **inertia tuning:** switch cost and commitment duration;
- **diversity tuning:** bounded random amplitude;
- **policy tuning:** culture/government/trait modifiers;
- **cadence tuning:** evaluation frequency and invalidation.

62.5 Behavioral dashboards

CI reports track action frequency, rejection reasons, average candidate count, p95 decision time, switch rate, idle rate, reservation failures, blocked order duration, technology portfolio diversity, migration response, rebellion incidence, casualty distribution, extinction/recovery and trace completeness. Changes outside tolerance fail or require explicit baseline approval.

62.6 Behavior release gate

A major behavior version can ship only when:

- all required fixtures pass;

- no unexplained unrelated distribution drift exists;
- traces remain schema-valid and readable;
- decision performance meets budget;
- save/load preserves active commitments and counters;
- adversarial review is resolved;
- regression ledger is updated;
- player-facing "Why?" text has localization keys;
- the creator accepts at least one representative playable scenario.

63 Decision tracing

Every important decision emits a structured trace suitable for tests, debugging and player explanation. A decision system that cannot explain its selected action and rejected alternatives is incomplete.

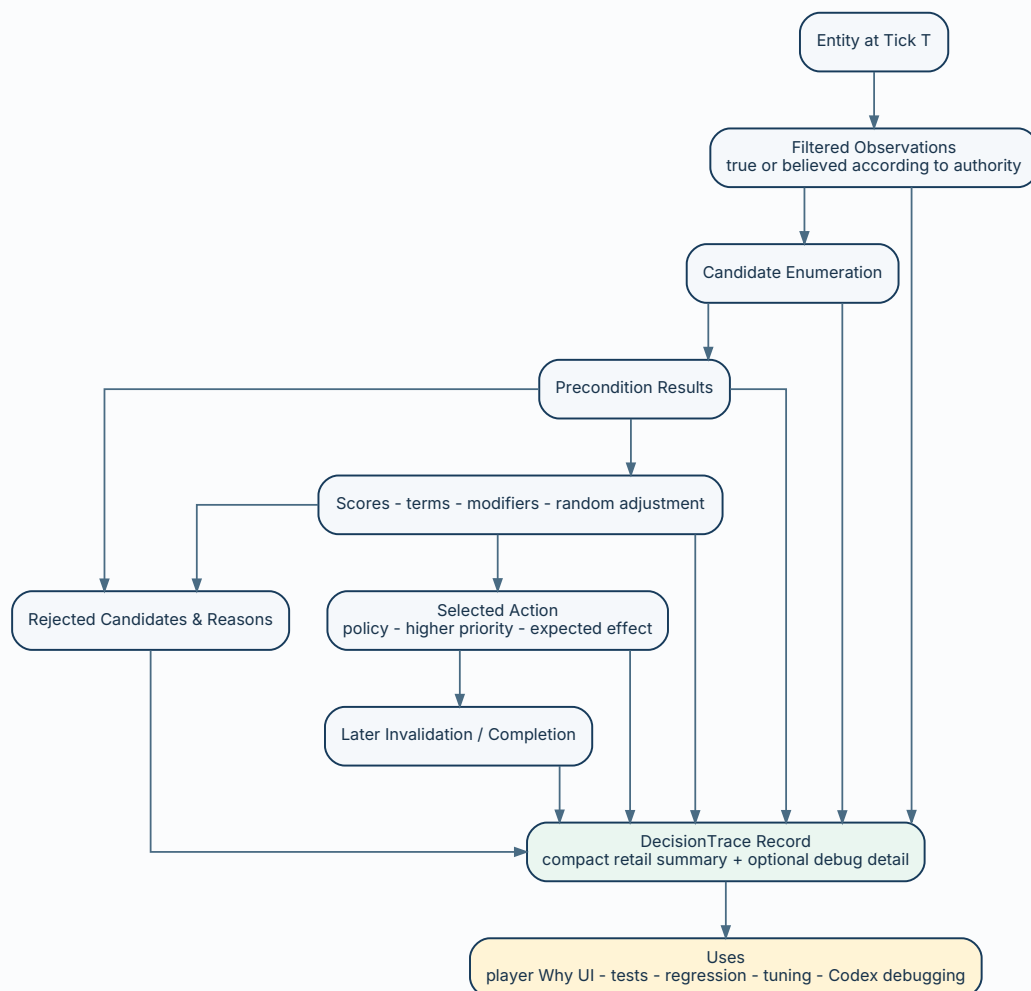


Diagram 13 - Decision tracing connects implementation evidence to the player-facing Why interface.

63.1 Required trace data

A trace records stable actor ID, world tick, decision/contract version, observation context, knowledge perspective, candidate list, precondition results, normalized terms, trait/attitude/policy modifiers, random key and bounded adjustment, cooldown/inertia, selected action, reservations, expected effect, controlling higher-layer priority and later invalidation or completion reason.

63.2 Trace modes

- **Retail normal:** outcome codes and compact top factors for selected/bookmarked entities and important events.
- **Retail inspection:** on-demand candidate and factor detail for a bounded recent window.
- **Test/debug:** complete candidate, random and invariant evidence with explicit memory/time budgets.

Full traces are not retained for every person forever. Important outcomes promote a causal summary to the event graph; low-significance traces expire or aggregate.

63.3 UI mapping

The UI exposes a one-line cause, ranked factor panel and technical trace. It distinguishes authoritative facts from believed information, player-order pressure from autonomous preference and direct Force Outcome ancestry from normal causality.

63.4 Trace acceptance

Traces must be deterministic, schema-versioned, localization-safe, cheap enough in configured modes, preserved across save/load for active decisions and validated against the actual state change. A trace that claims a factor not read by the decision is a test failure.

64 AI regression strategy

64.1 Adversarial review checklist

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- adversarial review is resolved;
- regression ledger is updated;
- player-facing "Why?" text has localization keys;
- the creator accepts at least one representative playable scenario.

65 AI/Codex development workflow

65.1 Operating principle

Coding agents write essentially all production code, but the repository—not chat history—is authoritative. The creator acts as game director, product owner, playtester and acceptance authority. Automated evidence and independent review compensate for limited code-level oversight.

Official OpenAI guidance for long-horizon Codex work recommends durable plans, scoped milestones, repository instructions, validation after each milestone and current documentation. [S-AI-01](#) [S-AI-02](#) [S-AI-03](#) [S-AI-05](#)

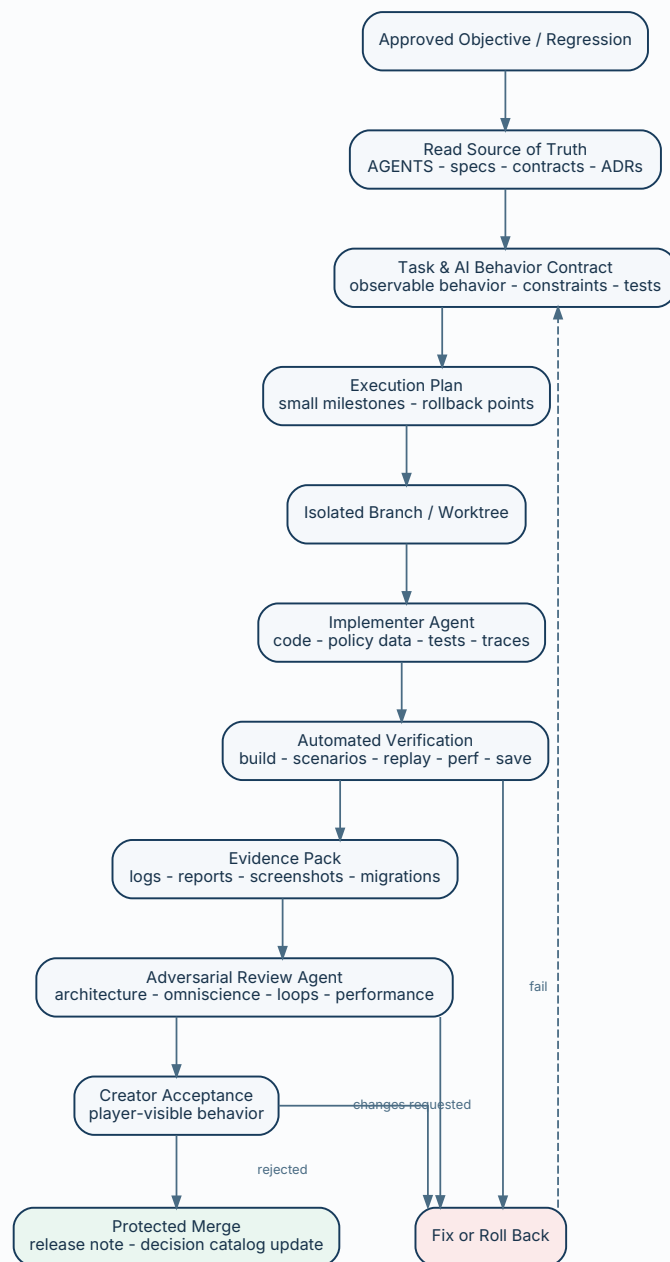


Diagram 25 — No consequential implementation is accepted without reproducible evidence and independent review.

65.2 Agent roles

Planner reads specifications and produces task contract and small milestone plan.

Implementer changes code/content and local tests within allowed files.

Verification agent runs clean builds, tests, replay, save and performance without modifying production code.

Adversarial reviewer searches for hidden omniscience, loops, unbounded queries, nondeterminism, authority violations, missing scenarios, exploitable weights, save issues and regressions.

Migration/documentation auditor validates schema, content IDs, behavior version and source-of-truth updates.

Creator evaluates player-visible behavior and preference decisions.

One high-risk agent session may not plan, implement and self-certify the same change.

65.3 Task contract

Every task defines title/ID, objective, player-visible behavior, authoritative inputs, outputs/state changes, affected packages, constraints/invariants, acceptance criteria, required tests, save/migration impact, UI/explainability, trace/telemetry, performance, explicit non-goals, allowed files, destructive disclosure, evidence and roll-back.

"Improve citizen AI," "make city growth," "add politics" and similar objectives are prohibited. A feature decomposes into contracts such as "Settlement selects one housing project from at most 12 candidates using HousingProject_v1 policy and emits trace schema 2."

65.4 Behavior-specific implementer package

A major behavior PR contains:

- approved behavior contract;
- scenarios including negative and boundary cases;
- policy data and schema validation;
- implementation using canonical primitives;
- trace fields and player-readable summary keys;
- deterministic replay checkpoint;
- property and metamorphic tests;
- statistical report where randomness exists;
- before/after performance and behavior report;
- save compatibility or migration;
- decision catalog and regression ledger update;
- adversarial review response.

65.5 Risk tiers

Tier	Examples	Required evidence
Low	Copy, isolated UI, test-only	Normal PR checks and review
Medium	New component, content definition, bounded UI	Task contract, tests, screenshots, local performance
High	Decision contract/runtime, scheduler, save schema, world generation, economy invariant	ADR, scenario suite, replay, migration fixtures, full perf, independent review
Critical/destructive	Engine/dependency migration, history rewrite, save-breaking change, repository rewrite	Explicit creator approval, backup/tag, roll-back rehearsal and two independent reviews

65.6 Branch and commit discipline

`main` is protected. One branch/worktree per coherent task. Commits are small, passing and reference task IDs. Generated/binary changes are isolated. Force push on shared branches is prohibited without explicit approval. Release candidates are tagged before migrations or dependency upgrades. Build/test commands must work from a clean checkout without manual editor clicks.

65.7 Evidence pack

Completion evidence includes build/test logs, changed tests, replay result, save result, migration output, benchmark comparison, profiler extract, trace report, screenshots/video for visible behavior, failures encountered, documentation, review findings and rollback commands. Evidence is stored as CI artifacts and summarized in the PR.

65.8 Architectural-drift control

Assembly dependency tests reject forbidden references. New managers, singletons, service locators, random APIs or scoring frameworks require ADRs and default to rejection. A weekly architecture-entropy job reviews dependency graph, TODOs, warnings, allocations, flaky tests, stale docs, behavior catalog consistency and unexplained checksum changes.

65.9 Destructive operations

Before deleting or replacing code, content, saves or history, the agent states what changes, why preservation is impossible, affected branches/saves/dependencies, backup/tag, migration/rollback and recovery tests. Creator approval is mandatory for critical operations.

66 Repository architecture

66.1 Top-level layout

```
/
AGENTS.md
README.md
ProjectSettings/
Packages/
Assets/                # authoring and presentation shell
GamePackages/
  com.project.simulation.core/
  com.project.ai.runtime/
  com.project.world/
  com.project.people/
  com.project.households/
  com.project.settlements/
  com.project.economy/
  com.project.knowledge/
  com.project.politics/
  com.project.diplomacywar/
  com.project.ecosystems/
  com.project.playerpowers/
  com.project.persistence/
  com.project.presentation/
  com.project.ui/
  com.project.content/
  com.project.telemetry/
  com.project.tooling/
Tests/
  Unit/
  Property/
  Metamorphic/
  Scenarios/
  Statistical/
  Integration/
```

```

Replay/
Persistence/
Performance/
Soak/
Fixtures/
Baselines/
Reports/
docs/
  GAME_VISION.md
  GAME_DESIGN.md
  SIMULATION_SPEC.md
  TECHNICAL_ARCHITECTURE.md
  DATA_MODEL.md
  SAVE_FORMAT.md
  PERFORMANCE_BUDGETS.md
  TEST_STRATEGY.md
  MILESTONES.md
  DECISIONS.md
  CONTENT_GUIDE.md
  UI_UX_SPEC.md
  RELEASE_PROCESS.md
ai/
  AI_ARCHITECTURE.md
  AI_GOLDEN_PRINCIPLES.md
  DECISION_LAYER_AUTHORITY.md
  AI_BEHAVIOR_CONTRACT_TEMPLATE.md
  AI_DECISION_CATALOG.md
  AI_EVALUATION_STRATEGY.md
  AI_TRACE_SCHEMA.md
  AI_TUNING_GUIDE.md
  AI_REGRESSION_LEDGER.md
decisions/ADR-NNN-title.md
plans/TASK-NNN-title.md
debt/
Build/
  bootstrap.ps1
  build.ps1
  test.ps1
  test-ai.ps1
  benchmark.ps1
  package.ps1
  verify-clean-checkout.ps1
Config/
  analyzer.ruleset
  benchmark-reference.json
  content-schema/
.github/workflows/

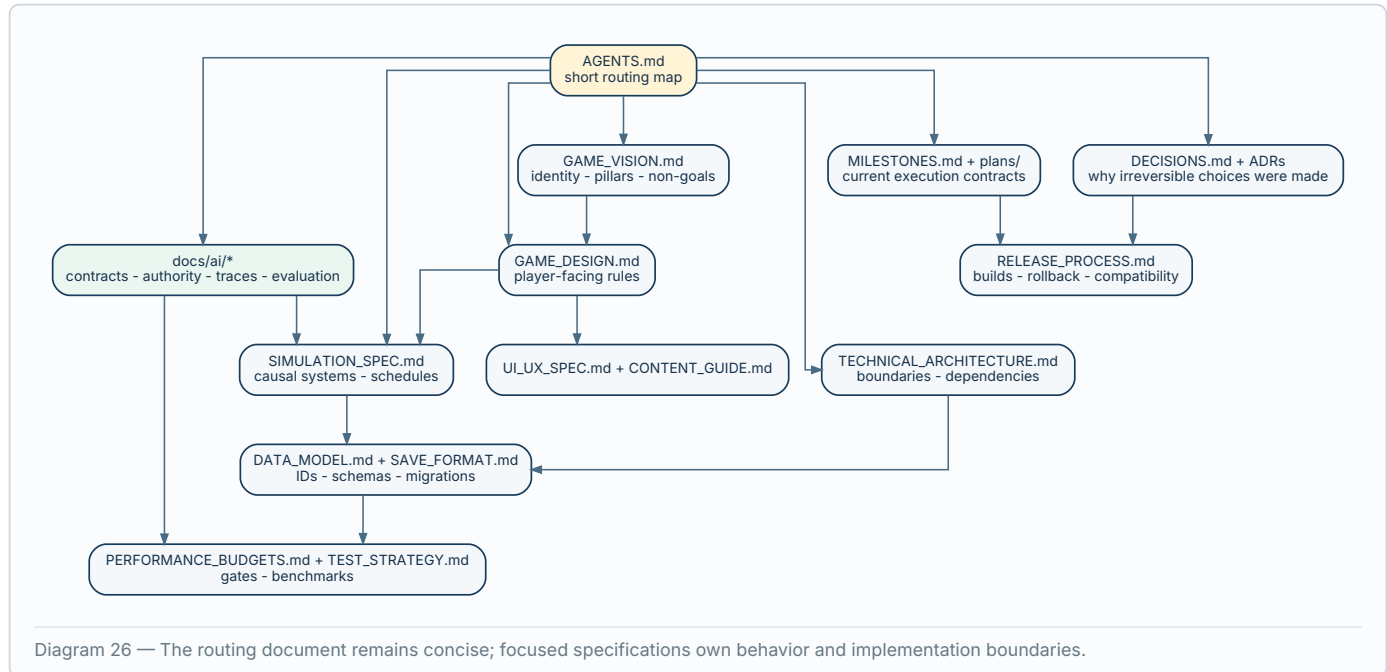
```

66.2 Dependency direction

Content definitions feed authoritative domain packages. Shared simulation and AI primitives are minimal foundations. Domain packages expose read models and command contracts to presentation/UI. Persistence serializes semantic DTOs. Telemetry observes declared interfaces. Circular references are forbidden.

Authoritative packages cannot reference `UnityEngine.UI`, rendering, camera, scene, audio or editor assemblies. Presentation may read snapshots and send commands but cannot access mutable stores. Persistence cannot serialize presentation. Editor tooling compiles only in editor assemblies.

66.3 Documentation hierarchy



`AGENTS.md` is a short map to build commands, safety rules and relevant documents. It does not duplicate the blueprint. `GAME_VISION` owns identity and non-goals; `GAME_DESIGN` owns player-facing rules; `SIMULATION_SPEC` owns causal behavior and cadence; AI documents own decision semantics; `TECHNICAL_ARCHITECTURE` owns boundaries; `DATA_MODEL` and `SAVE_FORMAT` own state; performance and test docs own gates; milestones and plans own current work; ADRs own rationale.

Every code change cites the documents it implements. Behavior changes update the authoritative document in the same PR. Contradictory documents fail documentation lint.

66.4 Decision catalog

`AI_DECISION_CATALOG.md` contains one row per decision ID: layer, owner package, contract version, trigger, cadence, candidate cap, knowledge context, trace schema, scenarios, performance budget, policy files, first release and status. CI checks that implemented decision IDs exist and retired IDs remain reserved.

66.5 Content architecture

Content is immutable, namespaced, schema-validated and versioned. Definitions reference stable IDs only. Validation checks missing references, prohibited cycles, units, ranges, localization keys, presentation mapping, AI evaluator compatibility and save metadata. Data mods later use the same schemas; early releases do not execute arbitrary code.

66.6 Scenario and regression storage

Each behavior scenario has human-readable source, compact binary fixture if needed, expected report and golden trace summary. Regression cases are named after observed behavior, not ticket numbers alone: `AI-REG-0047-worker-crosses-impassable-river`. The ledger records symptom, root cause, fix, affected contracts and permanent CI suite.

67 Testing strategy

67.1 Test pipeline

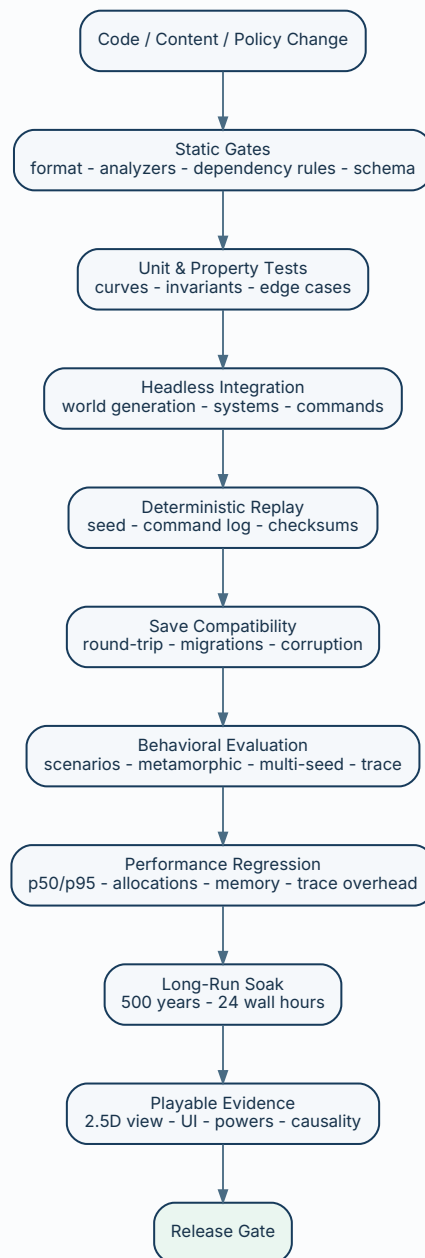


Diagram 27 — Verification escalates from pure rules to long-running worlds and creator-visible acceptance.

67.2 Static and architecture tests

Warnings-as-errors, formatting, analyzers, forbidden namespaces/dependencies, Burst compatibility, content schema, stable ID references, unauthorized RNG access, missing behavior contracts and documentation links run on every PR. Hot-path allocation analyzers and package graph checks are protected status checks.

67.3 Unit and property tests

Pure calculations cover fixed-point arithmetic, inventory conservation, price damping, utility curves, preconditions, reservations, trait modifiers, health transitions, resource renewal, knowledge state, treaty clauses, casualty allocation, time conversion, IDs and significance tiers. Property tests generate broad inputs to prove invariants such as nonnegative inventory, no reused IDs and valid selected-action preconditions.

67.4 Deterministic scenario tests

Controlled worlds define initial state, allowed actions, expected tendencies, forbidden actions, timing and trace terms. Required examples include:

- starving person beside reachable food;
- food behind an impassable river;
- worker without required tools;
- two jobs competing for one worker;
- village with timber but no stone;
- threatened government;
- inventor with knowledge but no manufacturing base;
- housing destroyed with displacement capacity elsewhere;
- forced factory with no workers or inputs;
- peace order with unresolved occupation.

Scenarios run headless from clean process and through save/load boundaries.

67.5 Invariant and property categories

- An agent cannot consume nonexistent food.
- An agent cannot use information outside its observation context.
- Two actors cannot reserve one unique item simultaneously.
- A civilization cannot manufacture a component it cannot produce.
- Camera position cannot affect state.
- Save/load cannot change subsequent deterministic choices.
- Actions cannot create negative inventory.
- Every selected action satisfies all hard preconditions.
- Force Outcome produces declared sources/sinks and valid references.
- Compaction cannot delete protected landmark identity.

67.6 Metamorphic tests

Metamorphic relations validate directional behavior:

- accessible food should not increase starvation under equivalent conditions;
- better roads should not increase travel time without another modeled cause;
- higher wages should not reduce job attractiveness absent an explicit counterfactor;
- higher perceived threat should not lower defensive priority without a competing reason;
- destroying a school should not increase education capacity;
- uranium alone should not create nuclear production;
- destroying housing should increase displacement pressure;
- removing habitat should not increase wildlife carrying capacity;
- increasing trace retention must not alter authoritative decisions;

- changing camera zoom must not alter any checksum.

Relations allow exceptions only when the trace names the competing modeled cause.

67.7 Statistical multi-seed tests

Randomized systems are evaluated across seed suites. Required distributions include founding survival, river transport preference, scarcity-driven migration, repression and rebellion risk, diversity among identical societies, geography-driven technology divergence, combat casualty ranges, speech repetition and ecosystem recovery. Tests use confidence intervals and trend thresholds rather than requiring exact seed outcomes.

67.8 Replay and persistence

A replay fixture contains world seed, generation version, content manifest, initial state, command log, expected checkpoints and checksums. Mismatches produce domain/component/trace diffs. Persistence tests cover round-trip, every migration, interrupted write, corruption, unknown optional fields, missing content, old landmarks and decision commitments in progress.

67.9 Performance and soak

Protected benchmarks measure simulation, decisions, routing, rendering, overlays, speech, tracing, saves and landmark growth. Soak suites include 24 wall hours, 500 game years, repeated save/load, high-speed operation, war/disaster churn and Force Outcome fuzzing. Memory and save-size slopes must be bounded.

67.10 Playable evidence

Automated correctness is necessary but insufficient. Creator-visible acceptance captures video or deterministic replay of site founding, city growth, direct orders, Force Outcome, omniscient inspection, war, ruins and recovery. UI screenshots include trace panels and overlays. The creator approves behavior, not merely metrics.

68 Performance budgets

68.1 Reference tiers

Budgets are engineering contracts measured on committed reference hardware. Final commercial minimum/recommended specifications remain a later creator/product decision. Initial reference tier: modern 8-core desktop CPU, 16 GB RAM, midrange discrete GPU, SSD, Windows development build; Linux player build is checked separately.

Table 13 — Initial runtime and persistence budgets

Budget	Target	Stress floor / failure trigger
Authoritative base loop	20 Hz; p95 ≤25 ms at 1×	No growing backlog; p95 ≤45 ms under 10× stress
Ordinary render	60 FPS at 1080p	≥30 FPS maximum prototype density
Persistent people	10,000 Large baseline	15,000 experimental only after evidence
Visible billboards	5,000 moving; 200 close-detail	Reduce visual LOD before simulation work
Modular buildings	10,000 instances world; 1,000 high-detail view	Instancing/LOD and geometry budget enforced
Active speech bubbles	40 default screen cap	80 debug stress; pooled and collision-managed
Active tactical sectors	8 ordinary, 16 stress	Coalesce operations, never merge people
Path requests	≤500 new long-range/tick stress; ≥80% corridor cache in mature city	Queue must remain bounded

Budget	Target	Stress floor / failure trigger
Individual decision evaluation	p95 $\leq 50 \mu\text{s}$ typical urgent-needs decision	Candidate cap violation fails
Major organization decision	p95 $\leq 5 \text{ ms}$	Review cadence defers nonurgent work
Retail trace volume	$\leq 64 \text{ MB}$ rolling world budget default	Promote/summarize before growth
Debug full traces	Scenario-scoped; $\leq 2\times$ decision runtime	Not part of retail budget
Overlay generation	$\leq 4 \text{ ms}$ CPU ordinary; $\leq 8 \text{ ms}$ stress	Reduce resolution/refresh, not state
Working set	$\leq 6 \text{ GB}$ Large benchmark	Memory slope $\leq 1\%$ per 100 game years excluding expected state growth
Compressed save	$\leq 100 \text{ MB}$ representative 100-year Large	Landmark growth tracked separately
Snapshot hitch	$< 250 \text{ ms}$ main thread	Staged snapshot redesign required
Total save/load	$\leq 10 \text{ s}$ / $\leq 15 \text{ s}$	Failure blocks vertical slice
Soak	24 wall hours and 500 game years	No crash, corruption, unbounded queue or leak

68.2 System allocation at 20 Hz

A provisional 50 ms wall-tick budget is not spent uniformly. Fast movement/encounters may consume 8–12 ms; due decision jobs 5–8 ms; construction/economy/logistics 5–8 ms; environment/health 2–5 ms; politics/technology/diplomacy 1–4 ms averaged; snapshot/telemetry 2–4 ms; and 10–15 ms remains headroom and scheduling variance. Slow systems run staggered and may use separate jobs.

Budgets are revised only from profiler evidence. A system exceeding its allocation must reduce candidate work, cadence or data movement before requesting a larger global share.

68.3 Rendering budgets

Billboards use instancing/indirect rendering where practical, atlas batching, distance animation LOD and screen-space culling. Buildings use modular mesh instancing, occlusion, HLOD and pixel-texture atlases. Effects and speech have pooled caps. Selected/tracked entities receive priority.

The camera stress scene includes dense city, slope, mixed building heights, fire, combat, overlays and speech. Rendering may degrade gracefully; authoritative outcomes remain unchanged.

68.4 Ruin and history budgets

Tier 3 ruin geometry has a world budget based on protected blocks/modules, not unlimited debris. The compactor may collapse repeated ordinary elements while retaining streets, landmark silhouettes and damage fields. Save telemetry records bytes by significance tier and projects 500-year growth. A world cannot silently demote a landmark to meet size.

68.5 Speed budgets

At 1 $\times$ the simulation targets full 20 Hz with headroom. At 2 $\times$ and 5 $\times$ due-time systems process larger deltas and more events but preserve ordering. At 10 $\times$, the engine may cap actual advancement if CPU-limited and clearly report it. Benchmark acceptance requires useful 10 $\times$ operation, not necessarily exact 10 real-time acceleration in the largest worst-case war.

68.6 Regression governance

Protected benchmarks compare p50/p95/p99, allocations, memory, save size, candidate counts, cache hit and trace volume. Default failure thresholds are +5% p95 on protected hot paths, +10% memory/save without explained feature state, any new steady allocation, or any growing backlog. Baseline updates require evidence and reviewer approval.

69 Prototype Charter

The Prototype Charter converts the blueprint into falsifiable engineering experiments. It is approved before implementation tickets and updated only through an architecture decision record.

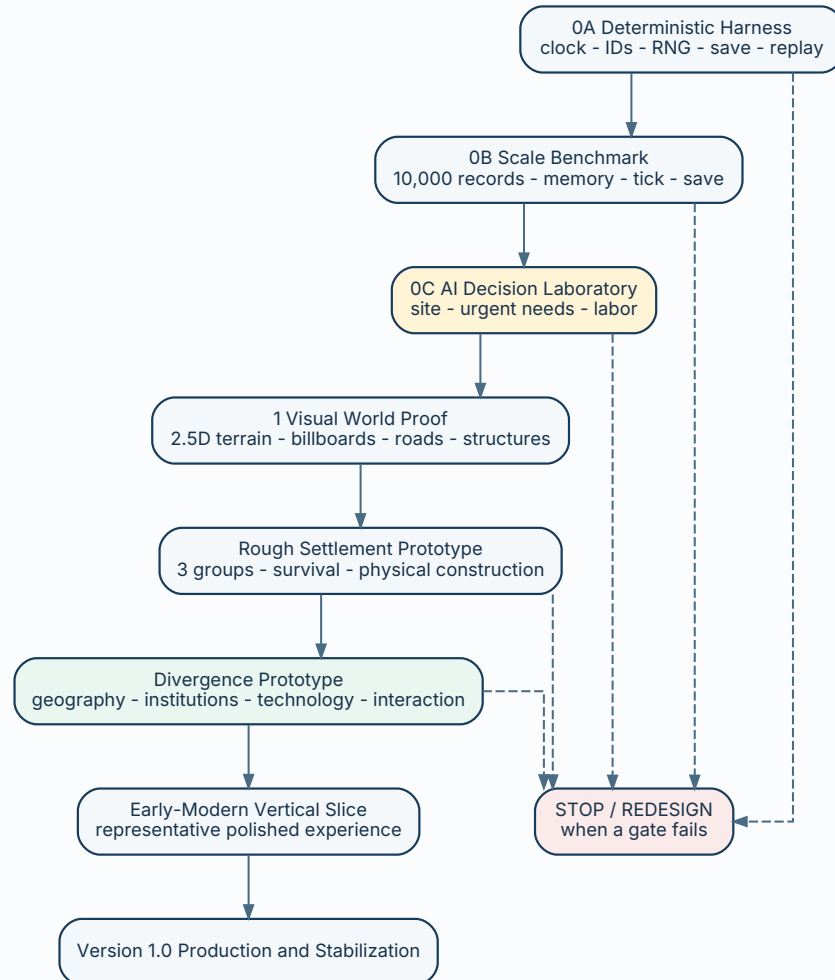


Diagram 42 - Prototype roadmap. Each gate proves one risk class before the next layer receives production scope.

69.1 Charter questions

The prototype program must answer, in order:

1. Can the project produce deterministic, headless, migration-safe world state?
2. Can 10,000 persistent person records exist within memory, save and scheduling budgets?
3. Can coding agents implement bounded, explainable decisions that survive behavioral tests?
4. Can the locked 2.5D stack display terrain, roads, modular buildings and billboards readably?
5. Can autonomous groups physically found and sustain settlements?
6. Do geography, resources and institutions create measurable civilization divergence?
7. Can diplomacy, conflict, player powers, chronicles and omniscient UI form a coherent representative experience?

69.2 Prototype rules

Prototypes use placeholder art, versioned fixtures and production-intent boundaries. They may discard presentation code only when explicitly marked; authoritative data, identity, persistence, test and decision contracts should be production-grade because later systems depend on them. Every prototype has a time box, measurable pass/fail criteria, captured evidence and a stop/redesign outcome.

No prototype is accepted because it looks promising once. It must run from a clean checkout, repeat under fixed seeds, save/load, expose traces, meet performance budgets and survive negative scenarios. Failed gates pause breadth. The response is profiling and bounded redesign, not adding more content.

69.3 Required pre-production package

Before Prototype 0A, create the Technical Architecture Specification, Prototype Game Design Specification, data model, test/benchmark plan, placeholder presentation specification, dependency graph, epic breakdown, implementation tickets, risk spikes and repository governance documents. The blueprint remains the product source of truth; these documents translate it into executable work.

70 Rough prototype specification

The rough playable prototype proves physical settlement formation. It is deliberately smaller than the vertical slice and much smaller than Version 1.0.

70.1 Recommended envelope

- One medium test region approximately 4 km x 4 km.
- Three founding groups of 40 people each, with configurable 30-60-person stress variants.
- Three contrasting viable environments: river/floodplain, forest/highland and dry or resource-poor frontier.
- A simplified four-season weather model.
- Eight occupations: farmer, gatherer, builder, woodworker, miner, hauler, hunter/fisher and coordinator.
- Six core resource classes: food, timber, stone, basic metal, tools and construction materials.
- Eight building archetypes: temporary shelter, house, storage, workshop, farm structure, extraction structure, meeting structure and defensive structure.
- One lightweight council/government pattern and one market/allocation mechanism.

70.2 Required systems

Survival includes hunger, nutrition, health, shelter, exposure, injury, death and thin population continuity. Work includes job demand, skill, tools, travel and reservations. Settlements choose sites, allocate labor, gather and store resources, create projects, build physically, form paths, improve selected roads, manage housing, respond to shortages and migrate or collapse.

The player can inspect every person and stockpile; add or remove food/resources; alter one trait; damage a building; issue settlement priorities; use one representative binding order and one representative Force Outcome; pause; and run at 1x, 2x, 5x and 10x.

70.3 Pass criteria

- Each group chooses a plausible reachable site using only its knowledge.
- At least two of three viable groups survive five game years in the reference seed suite without scripted aid.
- Shelter, food, storage and work emerge through physical projects and hauling.
- Collapse is explainable through traces, stockpiles, health and routes.
- Destroying housing produces visible ruins, displacement and rebuilding pressure.

- Save/load at arbitrary checkpoints preserves future checksums.
- The simulation can continue indefinitely after the evaluation window.
- The representative system remains within rough-prototype tick, memory and rendering budgets.

This gate does not require diplomacy, complex religion, naval warfare, full technology, polished chronicles, final art or large cities.

71 Divergence prototype

The divergence prototype is the main design go/no-go gate. It asks whether the simulation produces interesting, explainable societies rather than several copies of the same settlement optimizer.

71.1 Added systems

Add education, one school or apprenticeship institution, one faith/cultural institution, leadership, property/ownership variation, several cultural tendencies, capability opportunities, adoption, trade, borders, diplomacy, limited physical conflict, historical event recording, Chronicler profile and incident candidates. Technology domains are agriculture, construction, transport, metalworking and military equipment, using the same capability-state model intended for Version 1.0.

71.2 Controlled comparison design

Run matched worlds in which only defined environmental or institutional variables differ. Metrics include settlement form, road geometry, occupation distribution, food resilience, trade dependence, fortification, capability portfolio, adoption speed, government priorities and conflict behavior. Qualitative player review accompanies statistics; divergence must be understandable and visually apparent, not merely a hidden numeric distance.

71.3 Pass criteria

- Geography, resources and institutions each produce statistically measurable differences across fixed seed suites.
- At least three civilizations contact, trade, negotiate and sometimes fight through real systems.
- No universal technology ladder or one dominant settlement form appears in the test envelope.
- Players can inspect why one society selected a different project, policy or capability.
- Historical events and chronicles remain consistent with authoritative outcomes.
- Long runs remain stable; save/load and replay preserve results.
- Behavior contracts meet scenario, invariant, metamorphic and multi-seed gates.

Failure means redesigning the causal model, not expanding content. Heavy art, advanced government breadth and later-age production remain blocked until this gate passes.

72 Vertical-slice specification

72.1 Purpose

The first playable proves the complete defining proposition in one indefinitely runnable world:

Several persistent fictional human groups begin with limited early-modern capability, autonomously choose plausible settlement sites, form households and institutions, physically build and connect settlements, develop different social and technological solutions, exchange goods and knowledge, conduct diplomacy and physical conflict, and continue through growth, decline and recovery while an omniscient player may inspect every cause, issue binding civilization decisions, alter material conditions, or use Force Outcome.

It is not a scripted scenario, disconnected technical demo or content-complete game. “Complete” means the thin implementation of every defining pillar works together, saves, explains itself, scales to the declared benchmark and can continue indefinitely.

72.2 Scope-classification rule

Every feature is classified as one of four states:

- **Thin implementation required now:** player-visible and causally complete at minimal content depth.
- **Architecture Reserved:** schemas and boundaries must support later depth without replacement.
- **Deferred content:** more variants using the same architecture.
- **Future architectural expansion:** explicitly requires a new subsystem or scale abstraction.

Table 14 — First-playable feature classification

Feature	Thin implementation required now	Architecture Reserved	Deferred content	Future architectural expansion
Persistent people	10,000 identities, health, skills, traits, goals, appearance descriptors	More needs, disability, professions	Richer trait/appearance/content	None for core persistence
Households	Housing, pooled food/money, dependents, inheritance, joint migration	More family law	Romance/genealogy UI	None required
Social ties	Capped household/friend/mentor/rival/command ties	More tie types	Story-focused events	Large social-network research not planned
Settlements	Camp→village/town thin path, physical buildings and roads	City/metropolis, utilities, re-development	More archetypes and styles	Mass transport/power networks in industrial expansion
Buildings	16 archetypes, modular 3D, construction/damage/repair	Apartments, factories, skyscrapers, future structures	More modules/palettes	Exceptional interiors only if separately approved
Economy	18 commodities, workplaces, labor, markets, reservations	Finance, utilities, modern capital	More recipes and goods	Complex banking/money systems
Logistics	Physical batches, shipments, foot/wagon/river routes	Rail, trucks, ships, aircraft	More vehicles and hubs	Orbital logistics
Technology	Six domains, capability graph, invention/adoption/loss	Industrial through future networks	More nodes/variants	Some future science systems
Politics	Four government presets, factions, legitimacy, policy	Complex constitutions, parties, courts	More presets/policies	International organizations at depth
Diplomacy	Twelve actions, treaties, believed information	Blocs, sanctions, federations	More clauses and incidents	Global institutions
Warfare	Militia/guards, squads, physical sectors, supply, occupation	Artillery, vehicles, naval/air, nuclear effects	More weapons/doctrine	Strategic air/naval/orbital layers
Ecosystem	Regional wildlife/fish, livestock/rare animals, habitat and extinction	More species, invasives, biotechnology	Species content	Full planetary biosphere or multi-planet ecology
Speech	Tagged authored templates, limited social transmission	Large libraries/localization	Thousands of lines	No live LLM architecture
Player powers	24 tools across inspection/orders/material/Force	More powers by era	Optional restriction modes	None for core command model
Omniscience	True state, search, inspectors, overlays, knowledge perspective	More comparison and analytics	More overlay presets	None required
Save/history	Versioned chunks, interventions, significance tiers, landmark ruins	Larger worlds and mod manifests	More export tools	Cold-storage/partitioning beyond first-playable scale

Feature	Thin implementation required now	Architecture Reserved	Deferred content	Future architectural expansion
Presentation	2.5D terrain, billboards, modular buildings, LOD, occlusion	Dense modern skylines, advanced vehicles	Final art/animation/audio	Spherical/planetary rendering
AI development	Decision Lab, contracts, traces, scenario/property/metamorphic/statistical tests	More planners and evaluation	More behavior content	Learned policies only under separate experimental gate

72.3 World presets

The world sizes remain:

- **Small:** 4,096 m square, two groups, 80 starting people, intended 1,500–3,000 mature population.
- **Standard:** 8,192 m square, four groups, 160 starting people, intended 4,000–7,000 mature population.
- **Large:** 16,384 m square, six groups, 240 starting people, 8,000–10,000 benchmark population.

The strategic raster uses 16 m cells; local settlement navigation is adaptive at approximately 2–4 m plus route graph. Standard worlds must contain at least four ecologically distinct resource regions and multiple viable founding choices. Large is a scale gate, not the default art target.

72.4 Founding group: 40 people

Forty remains the recommended group baseline. It provides labor diversity, several households, dependents, potential teachers/healers/leaders, demographic continuity and meaningful risk without making a bad site immediately irreversible. A default group should contain approximately 24 adults, 8 adolescents and 8 children, adjusted by culture/rules and seed. At least six household units and no single indispensable specialist are recommended.

Starting traits and aptitudes are correlated by family only within controlled ranges. Starting culture and political attitudes are not identical. Founding groups possess limited maps and cannot see hidden deposits.

72.5 Starting supplies

Each group begins with bounded emergency stocks sufficient to survive initial site search but not a full year without production. Recommended per-group baseline:

Supply	Quantity band	Purpose
Staple/protein food	5,000–6,000 person-days equivalent combined	Search, first planting and construction transition
Seed stock	Enough for 12–16 ha first planting	Recovery and agriculture
Water containers and reserve	3–5 days	Travel safety, not permanent settlement
Timber/lumber/construction bundle	8–12 small structures equivalent	Emergency shelters and storage
Tools	30–40 mixed hand-tool sets	Agriculture, woodcraft, construction, extraction
Weapons/ammunition	Hunting and limited militia bundles	Defense and hunting, not prolonged war
Textiles/consumer goods	Basic clothing and repair reserve	Health and trade
Medicinal inputs	40-person emergency kit	Injury/disease transition
Pack animals or carts	Preset-dependent small number	Demonstrate transport capacity
Records/artifacts	Founding knowledge carriers	Literacy, recipes, law and culture continuity

Exact units live in content data and are balanced through survival seed suites. Shortages must be possible.

72.6 Starting knowledge

Groups begin with practical early-modern capabilities in shelter, farming, food storage, basic tools, water, sanitation, simple road/path work, household governance and local defense. They do not begin with every variant or full manufacturing chain. Knowledge is carried by people, records, tools and institutions. Loss during founding can materially change development.

Each group receives one or two modest specializations—river transport, masonry, animal husbandry, metal-work, medicine, record keeping or communal organization—to encourage divergence without deterministic identity.

72.7 Content quantities

The baseline content counts in Table 1 remain exact targets for the vertical slice. Additional detail:

Six institution types: household, workplace/guild, market, school, faith institution, government/council. Health and military functions may be building/workplace roles rather than separate institution categories in the thin implementation.

Sixteen occupations: farmer, herder/fisher, woodcutter, miner/quarrier, hauler, builder/carpenter, smith/tool-maker, trader, teacher/scribe, healer, administrator, faith worker, guard/soldier, craft researcher/inventor, domestic/care worker and general laborer.

Sixteen building archetypes: temporary shelter, dwelling, water point, farm support, livestock enclosure, store-house, lumber yard, quarry/mine works, builder workshop, smithy, market, school/scriptorium, meeting/government hall, faith hall, clinic/healer house and guard post/barracks.

Each archetype has at least two validated footprint variants, three material palettes and construction/damage states by vertical-slice completion. Fields and extraction sites use lighter geometry but remain authoritative work sites.

72.8 Society-dependent planning in thin form

The first playable must show at least three planning influences:

1. **Household/communal organic growth** around water, work and paths.
2. **Private or elite location choice** sensitive to access, land rights and wealth.
3. **Government public project** that reserves land and creates a planned road, defense or service building.

The same world fixture runs under at least two government/property presets and produces different building placement and access patterns. The player can order housing or a road connection but cannot select the exact footprint in normal mode.

72.9 Technology divergence proof

By year 50 in the comparative seed suite, at least two surviving societies must demonstrate different capability portfolios and visible solutions. Examples include masonry versus timber flood adaptation, wagon-road logistics versus river transport, centralized school records versus apprenticeship networks, or charcoal efficiency versus coal adoption. Divergence is statistical across seeds, not forced in every run.

The first playable does not require steam engines, railways or electricity. It must demonstrate that later railways can enter the existing route and project model.

72.10 Government and politics scope

Each founding settlement begins with one of four composable governance presets. It can collect revenue/dues, authorize projects, establish land rights, select/succeed leadership, fund one or more institutions, respond to crisis and generate faction pressure. Peaceful reform and coercive breakdown both appear in seed suites without scripts.

Patriotism and political loyalty are separate. At least one scenario must show patriotic opposition to a failing government and another patriotic defense of the same government under external threat.

72.11 Diplomacy and trade scope

Groups discover one another through exploration, migration, rumor or trade. They establish contact, exchange or restrict goods, form simple treaties, share/refuse knowledge, negotiate migration, create alliances, declare war and make peace. Trade uses real shipments and route travel. AI actors use believed information; the player can inspect truth.

72.12 Physical warfare scope

The thin warfare system includes militia/guard recruitment, squads, task groups, melee/hunting/limited black-powder weapon bundles, training, morale, fatigue, ammunition, food, command, raids, route interception, settlement assault, retreat, surrender, prisoners, basic fortification, wounds, death, displacement, property damage, occupation, annexation/reparations and rebellion.

Presentation shows people moving, fighting, firing, fleeing and becoming casualties; buildings and roads can be damaged. Tactical player movement control is excluded. Nuclear weapons, vehicles, naval and air combat are deferred.

72.13 Omniscient UI scope

The player can inspect every person and major organization, search stable entities, toggle required overlays, open true deposits, see secret diplomacy/forces, compare civilization knowledge, view decision scores, inspect binding-order progress and read intervention ancestry. The thin UI may use utilitarian visuals; information completeness and drill-down links are required.

72.14 Player control scope

The 24 powers in Section 14 are implemented. Binding orders are capability-constrained. Force Outcome supports direct structure, population, government, diplomacy, restoration and outcome mutation. At least one acceptance scenario demonstrates each category and its downstream consequences.

72.15 Speech scope

The first playable includes at least 600 unique source lines/templates across hunger, work, prices, war, housing, politics, religion, family, injury, fear, patriotism, migration, goals, local events and rumors. Only a subset needs final literary quality. The architecture must prove tags, localization keys, repetition control and optional social effects.

72.16 Ecosystem and extinction scope

Thin regional populations cover common wildlife, fish and habitat; livestock are persistent individuals or small groups. One species can become locally and globally extinct in controlled scenarios. Recovery from remnants and player restoration are implemented. Full food webs and many species are deferred.

Civilization state, cultural and population extinction are distinguishable. One regression fixture preserves a culture after state destruction; another destroys the last population.

72.17 Save and indefinite simulation

The world can save/load at any phase, including active construction, decision commitments, shipment, war, disaster, Force Outcome reconciliation and extinction recovery. A 500-year accelerated run remains coherent. "Indefinite" means no programmed terminal year or unavoidable overflow; it does not promise unbounded hardware-free growth.

72.18 First-playable performance contract

The first playable meets Section 68 budgets on reference hardware. Standard must feel responsive at 60 FPS ordinary view. Large must sustain 10,000 people and useful 10× operation. Full lab tracing is not enabled in retail performance tests. The camera-independence benchmark compares observed and unobserved runs.

72.19 Completion condition

The first playable is complete only when:

- every defining system is thin but connected;
- acceptance criteria in Section 74 pass;
- the AI Decision Laboratory gate remains green;
- engine, settlement and physical-city benchmarks are GO;
- a creator can play for several hours without encountering unexplained dead systems;
- save migration from at least two internal schema versions works;
- known limitations and deferred content are documented;
- no architecture-reserved feature requires replacing existing authority, persistence or physical-world state.

73 Version 1.0 scope

Version 1.0 is a complete, standalone early-modern civilization sandbox. It is larger and deeper than the vertical slice; it is not the technical prototype, rough settlement prototype or divergence proof.

73.1 Product scope

Version 1.0 supports several world sizes, persistent people and households, mature villages/towns/period-appropriate cities, society-dependent urban form, agriculture, resource extraction, crafts, workshops, markets, money and allocation systems, roads, ports and canals where appropriate, ships, fortifications, political factions, religion, education, technology divergence, trade, colonization, migration, rebellion, diplomacy, occupation, muskets, artillery, cavalry, naval operations, supply lines, unrestricted player powers, full omniscience, decision-level control, chronicles, historical maps, persistent ruins and indefinite simulation.

73.2 Content depth target

The exact launch counts remain subject to production evidence, but Version 1.0 should deepen the vertical-slice baselines rather than multiply systems indiscriminately. Content additions must reuse validated primitives and include data validation, localization keys, behavior scenarios, performance budgets and save compatibility. A new government or religion is not a label bundle; it must change authority, institutions, property, project selection, legitimacy and lived outcomes through shared systems.

73.3 Explicit Version 1.0 exclusions

No electricity grid, automobile network, tank, aircraft, nuclear weapon, computer network, global satellite system or multi-planetary simulation is required. No manual building placement, fully rendered interiors, live-LLM citizen dialogue, exhaustive genealogy, realistic gore, universal per-projectile army physics or black-box machine-learning civilization brain is added. Skyscrapers remain later-age architecture-reserved content.

73.4 Release principle

Version 1.0 must be worth playing if the project ends there. Later ages are expansions to a stable game, not excuses to leave early-modern politics, economy, warfare, UI, storytelling or performance unfinished.

74 Version 1.0 acceptance criteria

Version 1.0 is accepted only when the early-modern game satisfies the following objective product gates in supported world presets.

74.1 Autonomous society

Civilizations found viable settlements, maintain population continuity, organize households, allocate work, construct visible urban form, govern, trade, migrate, rebel, conquer, fragment, recover and sometimes become extinct without player micromanagement. Traits, attitudes, institutions and history measurably alter decisions while respecting authority and knowledge limits.

74.2 Physical and historical continuity

People, resources, roads, ports, fortifications, armies, buildings, ruins and recovery exist physically. Wars, disasters and player interventions produce displacement, damage, political reactions and persistent event/landmark links. Major ruins survive compaction and remain recognizable.

74.3 Early-modern completeness

Agriculture, crafts, markets, taxation or alternative allocation, property variation, education, religion, government, diplomacy, colonization, early-modern technology, muskets, artillery, cavalry, ships, fortifications and logistics form a coherent possibility space. Multiple viable development paths exist.

74.4 Player experience

The unrestricted default exposes all implemented powers. The player can inspect every authoritative layer, compare true and civilization-believed information, issue binding settlement/civilization decisions, alter material conditions, use Force Outcome, understand progress and inspect causal traces. Observation-only and constrained modes may be optional.

74.5 Story and information

Chronicler produces accurate, non-repetitive, localized chronicles from structured events. Historical maps, timelines, person/settlement/civilization histories, technology lineages and intervention records are queryable. The Story Director never produces impossible outcomes or camera-dependent history.

74.6 Technical release gate

All supported presets meet frame, tick, memory, save, load and long-run budgets on their declared hardware tiers. A Large benchmark supports 10,000 persistent people. Deterministic replay, migrations, atomic saves, corruption recovery, CI, architecture tests, scenario suites, statistical behavior evaluation, 24-hour soak and 500-year accelerated runs pass. Accessibility, onboarding, localization, crash diagnostics and release packaging meet the approved product plan.

74.7 Modularity for later ages

Content IDs, capability states, energy/transport interfaces, building modules, route classes, historical schemas and save migrations demonstrate at least one bounded extension fixture without modifying authoritative core assumptions. This proves the doorway exists without implementing the later room.

75 Phase-by-phase roadmap

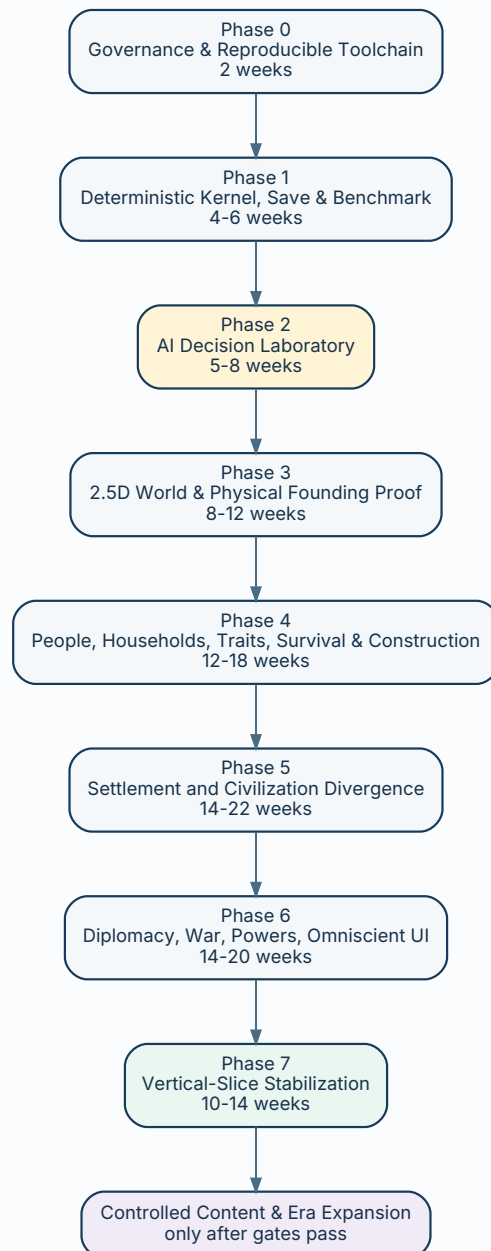


Diagram 28 — The roadmap treats the blueprint as a sequence of measured gates, not one implementation milestone.

75.1 Phase 0 — Governance and reproducible toolchain (2–3 weeks)

Objective: create a repository that coding agents can safely modify and independent sessions can reproduce.

Deliverables: protected Git repository; pinned Unity revision/packages; PowerShell bootstrap/build/test scripts; Windows clean-checkout verification; Linux player smoke build; assembly boundaries; source-of-truth documents; `docs/ai` set; task/ADR templates; analyzer and content-schema skeleton; CI artifact retention; baseline performance machine record.

Gate: a fresh Windows account clones, bootstraps, builds, runs one headless test and packages evidence without manual repair. A deliberate forbidden dependency and unauthorized RNG call fail CI.

Do not build: terrain, citizens, buildings or art.

75.2 Phase 1 — Deterministic simulation core and engine proof (5–7 weeks)

Objective: prove fixed scheduling, stable IDs, keyed randomness, canonical checksums, semantic saves, headless execution, spatial indexes and representative 10,000-entity throughput.

Deliverables: world clock; due queues; stable ID registry; fixed-point primitives; random streams; canonical state diff; versioned persistence skeleton; synthetic people/workplaces/shipments; hierarchical route spike; presentation bridge; 5,000 billboard and 1,000 building stress scene; telemetry.

Gate: engine benchmark in Section 57 passes or bounded optimization/fallback process begins.

Do not build: real civilization AI, politics, advanced rendering or final assets.

75.3 Phase 2 — AI Decision Laboratory (5–8 weeks)

Objective: validate the behavior-contract runtime before feature breadth.

Deliverables: canonical utility engine, preconditions, reservations, cooldowns, hysteresis, invalidation, knowledge contexts, trace schema, scenario loader, property/metamorphic/statistical runners, before/after reports and the first three decisions.

Gate: Section 61 go/no-go passes, including deliberate defect detection.

Do not build: giant settlement/government planners or large policy catalogs.

75.4 Phase 3 — World, people, households and survival (8–12 weeks)

Objective: produce autonomous physical founding groups that survive or fail for traceable reasons.

Deliverables: world generation; hydrology/resources/hazards; site choice; persistent people; households; inheritance; traits and three-rate state; health/needs; occupations; starting inventory; local movement; minimal visual billboards; omniscient inspectors; save/load through births/deaths.

Gate: settlement-formation precursor: groups choose sites, build temporary shelter, plant/forage, store food, care for dependents and survive seed suites; all decisions traced and tested.

75.5 Phase 4 — Physical settlement, economy and logistics (12–18 weeks)

Objective: transform survival groups into visibly different villages through physical construction and society-dependent projects.

Deliverables: parcels/rights; 16 building archetypes; modular 3D kits; construction/damage/repair; paths/roads/bridges; workplaces; 18 commodities; labor; markets; reservations; shipments; institutions; government presets; player housing/road orders; settlement inspectors and overlays.

Gate: physical-city-growth benchmark: at least two society presets create different layouts; neighborhood destruction produces displacement and rebuilding; Large simulation remains within budget.

75.6 Phase 5 — Knowledge, culture, politics and civilization formation (12–18 weeks)

Objective: create long-horizon divergence and multi-settlement authority.

Deliverables: capability graph; invention/adoption/loss/reverse engineering; culture/religion; patriotism; factions; legitimacy; policy; leadership/succession; civilization records; strategic bounded plans; direct decision control; cultural/state/population extinction; speech template thin implementation.

Gate: paired seed suites demonstrate causal technology/urban/political divergence, peaceful reform and coercive breakdown. Decision catalog and regression suite remain green.

75.7 Phase 6 — Diplomacy, physical war, ecosystems and unrestricted powers (14–20 weeks)

Objective: connect civilizations through trade, treaties, conflict, ecological pressure and complete player agency.

Deliverables: contact/belief, twelve diplomatic actions, executable treaties, espionage thin layer, militia/squads, sector combat, supply, occupation, refugees, damage, hybrid ecosystem, extinction/recovery, 24 powers, Force Outcome reconciliation, intervention history, Tier 3 ruins.

Gate: three-actor diplomacy, large-war stress, forced peace, forced city creation, extinction/restoration and landmark persistence scenarios pass.

75.8 Phase 7 — Vertical-slice stabilization (12–16 weeks)

Objective: turn the integrated build into a coherent, legible, indefinitely runnable first playable.

Deliverables: UI polish, overlays, causal summaries, speech library expansion, art-direction proof, performance optimization, save migrations, accessibility baseline, 24-hour/500-year soaks, balance seed suites, content validation, release process and creator acceptance.

Gate: all Section 74 acceptance criteria pass. The build is tagged `first-playable-1.0` only after no unresolved critical risks remain.

75.9 Calendar reality

The phase estimates assume sustained expert coding-agent work, disciplined review and limited final art. Sequential ranges total approximately 70–102 working weeks. Some content, UI and tooling can overlap after architecture stabilizes, but the critical path remains approximately **18–30 months** for a robust first playable under AI-heavy solo direction. A small experienced human team could reduce calendar time; unmanaged parallel agents can increase it.

The complete north-star game requires several additional release families over multiple years. Modern cities, advanced warfare, nuclear effects, dense industrial networks, digital systems and future technology are not one post-launch patch.

75.10 First month

Week 1: repository, bootstrap, protected branches, source-of-truth docs and CI.

Week 2: fixed clock, stable IDs, keyed RNG and canonical checksum.

Week 3: semantic persistence skeleton and headless runner.

Week 4: representative synthetic benchmark, presentation bridge and AI Decision Laboratory scaffolding.

At month end the project should have almost no gameplay but strong evidence that it can safely begin.

75.11 What not to build in the first six months

Do not build nuclear weapons, skyscrapers, modern cars, aircraft, detailed navies, advanced constitutions, full religions, large speech libraries, final character art, interior rooms, multiplayer, workshop scripting, procedural planets, realistic gore, cinematic campaigns or a universal technology tree. Do not ask an agent to “finish the AI.”

76 Later-age expansion strategy

76.1 Expansion rule

A feature is content expansion when it adds validated definitions over stable runtime behavior: building modules, occupations, recipes, policies, doctrine, speech, species, disasters, technologies or visual kits. It is architectural expansion when it changes spatial scale, authority, save semantics, route topology, simulation cadence or fundamental capability dimensions.

Content does not bypass performance, trace or save contracts. Every new definition has stable ID, schema version, localization keys, validation scenarios, presentation mapping and compatibility metadata.

76.2 Mostly content-driven expansion

After the first playable, early-modern depth can add crops, livestock, crafts, weapons, diseases, doctrine modules, government combinations, building styles, speech, occupations and disasters using existing systems. Content packs can emphasize different climates, cultures or settlement origins without fixed real-world nations.

76.3 Architectural expansion triggers

Industrialization introduces powered machinery, factories, rail, utilities, dense pollution, mass education, labor organization and capital equipment. Electrification introduces power grids, telecommunications and urban utilities. Modern warfare introduces vehicles, air/naval layers and large logistics. Nuclear/information eras add contamination, deterrence, computing and automation. Multi-planet systems require new topology and logistics.

Each trigger begins with an ADR, benchmark and migration plan. No expansion silently overloads first-playable abstractions.

76.4 Content quality gates

- Definition validates and uses stable IDs.
- Every capability has visible effect and blockers.
- Every building has complete construction/damage/LOD states.
- Every speech line is context-safe and localizable.
- Every policy has enforcement semantics.
- Every species has habitat, pressure and extinction behavior.
- Performance and save growth stay within allocated budget.
- AI behavior scenarios cover positive and negative use.

76.5 Release family A — Foundation depth

Deepen early-modern worlds: origins, crops, animals, crafts, disease, doctrine, government combinations, espionage, refugee integration, laws, building modules, speech and data modding. Improve UI, accessibility and content authoring. Most work is bounded content and system depth.

76.6 Release family B — Industrial transition

Add steam, mining pumps, machine tools, factories, rail, canals, telegraph, mass education, labor organization, industrial pollution, utilities, artillery and larger cities. This is the first major architectural expansion because powered capital equipment and dense networks become central.

76.7 Release family C — Electrified and combustion societies

Add electric grids, oil, internal-combustion vehicles, modern sanitation, hospitals, mass media, modern finance foundations, highways, aviation, mechanized armies and high-rise construction. City pathfinding and power/utilities receive new gates.

76.8 Release family D — Modern conflict and global order

Add mass politics, international institutions, ideological blocs, decolonization, naval/air warfare, logistics at continental scale, intelligence services, refugee crises and advanced public administration.

76.9 Release family E — Nuclear and information age

Add nuclear energy and weapons, layered effects, deterrence, fallout, contamination, computers, digital communication, automation, satellites and information warfare. Nuclear use must create persistent metropolitan scars and cultural consequences.

76.10 Release family F — Divergent futures

Add biotechnology, advanced robotics, novel energy, climate engineering, space industry and society-specific future paths. Future content remains a capability network, not one linear “future era.”

76.11 Distant expansion — Multi-planet civilization

A new topology abstraction adds planets, orbital layers, interplanetary routes, communication delay and planetary ecosystems. It is not a flag on the flat-world renderer. The original world remains a valid bounded planet-scale map or becomes one region under explicit migration.

76.12 Staged feasibility verdict

Foundation through industrial transition is plausible if first-playable gates pass. Modern and future eras are plausible as subsequent studio-scale programs. Multi-planet simulation is a distant research/expansion project. The architecture preserves the possibility without pretending the first implementation solves it.

77 Risk register

The register ranks probability and impact for the first-playable program. Owners refer to repository roles, not named individuals. Triggers are measurable; mitigations are expanded in Section 51.

Table 15 — Risk register

ID	Risk	Probability	Impact	Trigger / evidence	Primary mitigation	Owner
R01	Interaction complexity exceeds tick budget	High	Critical	p95 tick/backlog grows as systems integrate	Per-system budgets, bounded candidates, cadence, representative benchmarks	Technical architecture
R02	City pathfinding dominates CPU	High	Critical	Route queue grows; cache hit below 70%	Hierarchical graphs, corridor reuse, invalidation locality, shared fields	World/navigation
R03	Physical city rendering is unreadable or slow	High	High	<30 FPS stress or people/buildings merge visually	Billboard/building proof, LOD, occlusion, density budgets	Presentation
R04	Camera-facing billboards obscure direction/action	Medium	High	Combat/work user tests cannot identify facing	Equipment orientation, projectiles, shadows, poses, indicators	Character presentation
R05	Modular buildings look incoherent	Medium	High	Invalid silhouettes, entrances or style combinations	Authored module grammar, validators, limited palettes	Building art/tooling

ID	Risk	Probability	Impact	Trigger / evidence	Primary mitigation	Owner
R06	Procedural cities look generic across societies	Medium	Critical	Same-seed presets converge in layout metrics	Property/planning tests, society modifiers, paired scenarios	Settlement design
R07	No-manual-placement loop feels passive	Medium	High	Players cannot form meaningful intent	Strong decision-level orders, progress UI, Force Outcome distinction	Game design/UI
R08	Unrestricted powers trivialize observation	Medium	Medium	Players use only instant outcomes	Default remains unrestricted; UI foregrounds consequences/history, optional modes later	Game design
R09	Force Outcome creates broken states	High	Critical	Invalid references, negative inventory, orphaned households	Reconciliation layer, fuzz tests, transactional mutation	Player powers/persistence
R10	Omniscient UI overwhelms players	High	High	Players cannot find key cause or perspective	Search, progressive disclosure, overlay composition, presets	Information design
R11	Decision traces consume memory/performance	Medium	High	Trace volume exceeds budget or changes tick	Three trace modes, typed buffers, promotion/compaction	AI/telemetry
R12	Decision traces are technically complete but unreadable	Medium	High	Player cannot answer "why" despite trace	Ranked summaries, localization keys, trace UX tests	AI/UI
R13	Coding agents invent parallel AI frameworks	High	Critical	Duplicate scoring/RNG/planners appear	Canonical primitives, dependency/analyzer checks, ADR default rejection	AI architecture
R14	AI exhibits hidden omniscience	High	Critical	Decision uses authoritative hidden deposit/force	Observation contexts, adversarial fixtures, API isolation	AI architecture
R15	AI thrashes or loops	High	High	Repeated action switches/cancel-reserve cycles	Hysteresis, cool-down, invalidation, loop metrics	AI runtime
R16	Traits become stereotypes or decorative	Medium	High	No measurable behavior or deterministic outcomes by label	Contract modifiers, contextual tests, trace terms	People design
R17	Trait interactions become untunable	High	High	Score terms explode; unrelated behavior drifts	Modifier caps, factor groups, scenario dashboards	AI tuning
R18	Statistical tests are flaky	Medium	High	CI changes due to small random sample	Fixed seed suites, confidence thresholds, larger scheduled runs	Verification
R19	Speech templates repeat or reveal hidden state	High	Medium	Repetition metrics or knowledge violations	Tag/precondition validation, cooldowns, rumor semantics	Narrative systems
R20	Overlay composition becomes visually overloaded	High	Medium	Legends conflict; players misread map	Base/primary/secondary limits, patterns, presets	UI
R21	Economy oscillates or deadlocks	Medium	High	Prices/wages alternate extremes; reservations freeze	Damping, emergency allocators, convergence tests	Economy
R22	Resource and shipment state grows excessively	Medium	High	Shipment count/save bytes rise superlinearly	Batch manifests, route aggregation, lifecycle cleanup	Logistics
R23	Government presets remain cosmetic	Medium	High	Land, project, tax and succession outcomes converge	Shared primitive matrix tests, causal metrics	Politics
R24	Technology converges on one path	Medium	Critical	Portfolio diversity below threshold	Alternative recipes, institution/resource gates, lineage	Knowledge
R25	Physical warfare cannot scale	High	Critical	Active sectors, VFX or pathing exceed budget	Squad authority, task-group routes, sector cap, LOD	Warfare
R26	Stylized violence misses target rating/ tone	Medium	Medium	Creator/ratings review rejects presentation	Low-detail effects, configurable blood, decision by content-rating milestone	Creative direction

ID	Risk	Probability	Impact	Trigger / evidence	Primary mitigation	Owner
R27	Refugee/displacement systems create stranded populations	High	High	Households remain impossible/idle	Emergency shelter, migration plans, aid and recovery contracts	People/settlements
R28	Ecosystem causes runaway collapse or irrelevance	Medium	High	Most seeds extinct or no effect on societies	Carrying calibration, regional model, seed suites	Environment
R29	Extinction/recovery logic becomes incoherent	Medium	Critical	"Extinct" species/culture still has carriers or revives magically	Explicit state machine, viable remnant checks, fixtures	Environment/history
R30	Persistent ruins make saves grow without bound	High	High	Tier 3 geometry/history grows faster than budget	Significance tiers, landmark manifests, geometry compaction	Persistence
R31	Compaction deletes causal evidence	Medium	Critical	History links break or landmark vanishes	Retention references, protected IDs, compaction invariants	Persistence
R32	Save migration fails after broad agent changes	High	Critical	Old fixtures cannot load	Semantic DTOs, migration gates, protected schema review	Persistence
R33	Long-running worlds overflow or leak	Medium	Critical	Memory/event/ID slope or numerical instability	500-year/24h soak, fixed units, tombstone policy	Core simulation
R34	Unity/DOTS complexity creates mixed architecture	High	High	Hot simulation appears in MonoBehaviours	Assembly rules, examples, analyzer, code review	Technical architecture
R35	Unity package or licensing changes disrupt production	Medium	High	Required package/support/terms change	Pin LTS, review windows, Bevy fallback benchmark	Production/technical
R36	Bevy fallback costs more than it proves	Low	Medium	Spike expands into tool rewrite	Port headless benchmark only, four-week cap	Technical direction
R37	Agent-generated code is locally correct but globally contradictory	High	Critical	Duplicate concepts, inconsistent units, hidden state	Source-of-truth routing, task contracts, entropy audit	Production architecture
R38	Review agents rubber-stamp implementers	Medium	High	No new negative scenarios or findings	Explicit adversarial mandate, independent worktree, review metrics	Production
R39	Content volume outpaces validation	High	High	Broken references, missing states, balance noise	Content schemas, preview tools, batch validators	Content production
R40	First playable becomes a disconnected technical demo	Medium	Critical	Systems exist but do not form long-session loop	Vertical acceptance, integrated seed suites, creator playtests	Product direction
R41	Scope expands into modern/future eras prematurely	High	High	Advanced assets/features before gates	Deferred list, milestone protection, task approval	Product owner
R42	Heavy art is commissioned before architecture proof	Medium	High	Asset spend before city/engine gates	Placeholder/pixel prototype, explicit art gate	Production
R43	Accessibility is deferred until UI locks	Medium	Medium	Overlay/color/input assumptions harden	Baseline requirements from first UI, automated checks	UI/accessibility
R44	Localization becomes impossible due to templates	Medium	High	Concatenated strings and grammar failures	Typed templates, localization keys, pseudo-localization	Speech/UI
R45	No experienced human technical review is available	High	Critical	Critical architecture decisions only agent-reviewed	Seek periodic external code/architecture audit; stronger automated gates	Creator/production
R46	Story Director fabricates or overrules outcomes	Medium	Critical	Candidate lacks real resolution owner or checksum differs by camera	Authority boundary tests, Pure Simulation control, trace review	Story systems
R47	Incident pacing feels artificial	Medium	High	Severe events cluster without causal relation or quiet worlds receive forced drama	Recovery debt, regional pacing, profile comparison suites	Story systems

ID	Risk	Probability	Impact	Trigger / evidence	Primary mitigation	Owner
R48	Incidents become repetitive	High	Medium	Repetition keys exceed threshold in long runs	Cooldowns, semantic variants, chain diversity metrics	Narrative content
R49	Generated chronicle prose contradicts facts	Medium	Critical	Sentence lacks supporting node/edge or names wrong actor	Typed fact slots, provenance validation, no live LLM	History/UI
R50	Event graph or historical queries exceed budget	Medium	High	Query latency/save slope breaches budget	Significance tiers, chunked indexes, depth/result caps	History/persistence
R51	Negative-event overload exhausts players	Medium	High	Notification load and severity remain above recovery target	Chronicler recovery windows, aggregation, profile controls	Story/UI
R52	Perspective history is confused with authoritative truth	Medium	High	Players cannot distinguish belief from fact	Explicit perspective labels, confidence and correction links	History/UI
R53	Historical summaries do not localize	Medium	High	Grammar failures in pseudo-locales	Typed templates, complete clauses, localization tests	Narrative/localization
R54	Long-world chronicles become unreadable	High	High	Search/filter use rises while useful retrieval falls	Significance hierarchy, eras, filters, bookmarks and summaries	History/UI
R55	Historical maps inflate save size	Medium	High	Keyframes/deltas exceed bytes-per-century budget	Coarser aging cadence, chunk deduplication, event-driven keyframes	Persistence

78 Mitigations

78.1 Mitigation system

Risks are mitigated by recurring controls rather than isolated fixes:

- **Phase gates** stop breadth when core proof fails.
- **Behavior contracts** prevent vague AI scope.
- **Representative benchmarks** measure integrated load early.
- **Stable IDs and semantic saves** contain long-term change.
- **Canonical primitives** reduce duplicated systems.
- **Adversarial review** searches for classes of failure.
- **Regression fixtures** convert mistakes into permanent knowledge.
- **Progressive disclosure and overlays** make complexity legible.
- **Architecture-reserved boundaries** preserve north-star scope while deferring content.

78.2 Triggered contingency plans

Tick or routing failure

Freeze feature additions; capture profiler and query-cardinality report; identify the top two systems; reduce candidate set, cadence, cache invalidation or data movement; rerun baseline. Only after two bounded iterations consider engine-specific fallback. Never solve by camera-based fabrication.

City unreadability

Run controlled visual tests at fixed density. Adjust billboard scale, shadows, lane spacing, building transparency, roof fade, palette contrast and clutter LOD. Do not replace physical people with statistics at close zoom.

AI incoherence

Create a deterministic fixture before editing weights. Inspect authority, knowledge, preconditions, reservations, trace and invalidation. If multiple contracts fail similarly, fix canonical primitives. Update regression ledger.

Force Outcome corruption

Disable the affected Force power in release builds while retaining unrestricted categories that pass. Preserve saves. Add transactional mutation and rollback. A broken power is not hidden behind balance cost; it is repaired or explicitly unavailable.

Save-size growth

Break down bytes by domain and significance. Compact routine transactions and geometry; never demote protected landmarks silently. If current authoritative state is genuinely too large, revise world scale or representation through ADR.

Speech repetition

Measure line-family frequency, context coverage and localization safety. Expand line pools only after selector logic and cooldowns are validated. Suppress bubbles rather than showing incorrect or repetitive text.

Scope drift

Reject tasks without milestone linkage and explicit non-goals. Advanced-era content waits in architecture documents and content backlog. A compelling demo idea does not bypass the current gate.

| 78.3 Human review mitigation

The most important nontechnical mitigation is periodic external expert review. Even with adversarial agents, a project relying almost entirely on generated code should budget for occasional senior Unity/DOTS, systems-design and persistence audits before irreversible milestones. This does not replace automated evidence; it challenges assumptions the automated system may share.

| 78.4 Creator-facing risk dashboard

A simple dashboard reports each risk's current probability, evidence, owner, next review, trigger and mitigation status. The creator sees technical risks in player-visible terms: "orders can create invalid worlds," "cities unreadable at 5,000 people," "old saves not loading," rather than only profiler jargon.

79 Deferred features

| 79.1 World and environment

Full oceans/naval navigation, global climate systems, tectonics, volcanoes, tsunamis, complex hydrology alteration, detailed food webs, many persistent wildlife species, advanced invasive ecology and multi-planet worlds are deferred. The world remains flat and bounded.

| 79.2 People and society

Detailed romance, exhaustive genealogy, large social graphs, full crime/justice depth, complex disability/mental-health simulation, extensive language evolution, thousands of speech lines, fully developed education levels and interior-room behavior are deferred.

79.3 Settlement and economy

Skyscrapers, modern apartment systems, electricity, sewage networks, mass transit, rail, highways, complex ports, banks, securities, central banking, consumer markets, advanced corporate structures and detailed utility maintenance are deferred. Schemas reserve physical footprints, route types, utilities, capital equipment and ownership.

79.4 Technology and eras

Steam industry, modern manufacturing, automobiles, aircraft, antibiotics, nuclear power/weapons, digital communication, computing, automation, rockets and future technologies are deferred. The capability graph and knowledge states support later addition.

79.5 Politics, diplomacy and war

Complex constitutions/courts, mass parties/campaigns, international organizations, sanctions regimes, colonial administration depth, naval and air warfare, artillery ballistics, strategic deterrence, nuclear diplomacy, detailed insurgency and tactical player battles are deferred.

79.6 Product features

Multiplayer, console certification, cloud saves, cross-platform profiles, executable workshop mods, achievements/challenges, commercial integrations, final localization breadth, voice acting, final soundtrack and full marketing telemetry are deferred.

79.7 Explicitly never core

Live LLM citizens, manual normal-city placement, rendered ordinary interiors, realistic gore, universal technology tree, spherical core world and black-box ML civilization intelligence remain non-goals rather than deferred promises.

80 Definition of done

80.1 Phase 0 done

Clean checkout reproduces build/test; dependencies pinned; protected branches active; source-of-truth and AI docs validated; risk/task/ADR process working; destructive-operation policy tested; no gameplay required.

80.2 Phase 1 done

Fixed clock, IDs, RNG, checksum, semantic save, headless runner, representative synthetic load, route spike, presentation bridge and telemetry meet the engine gate. Failure path and Bevy spike decision are documented.

80.3 Phase 2 done

AI Decision Laboratory passes all three initial decisions, deterministic/property/metamorphic/statistical tests, save/replay, full trace comparison, performance and adversarial review. Canonical primitives are frozen to versioned APIs.

80.4 Phase 3 done

Generated worlds support founding groups, people, households, inheritance, traits, needs, health, work and initial shelter/food. Omniscient inspection and trace UI explain survival and failure. 100-year demographic run passes.

80.5 Phase 4 done

Physical villages emerge through construction, roads, economy, logistics, institutions and government. Society presets create different layouts. Player project orders work without manual placement. Physical-city benchmark is GO.

80.6 Phase 5 done

Technology, culture, religion, politics, civilizations, direct decision control, speech thin layer and civilizational extinction work. Paired seed suites demonstrate divergence and traces. No major behavior lacks contract.

80.7 Phase 6 done

Diplomacy, trade, physical war, occupation, ecosystems, extinction/recovery, unrestricted powers, Force Out-come and landmark ruins function together. Three-actor, war, restoration and persistence scenarios pass.

80.8 Phase 7 done

All vertical-slice criteria pass; performance and long-run gates pass; UI and accessibility are usable; save migrations are supported; content is validated; creator accepts representative sessions; known risks are below release threshold.

80.9 Feature-level done

A feature is done only when player-visible behavior, authoritative logic, behavior contract where relevant, tests, traces, save impact, performance, documentation, evidence, review and creator acceptance are complete. Code-complete or content-complete alone is not done.

81 Recommended first 25 implementation tasks

The first tasks are small, reviewable contracts. Each produces commands, tests, evidence and rollback. No task is accepted solely because it compiles.

Task 1 - Establish repository governance and reproducible Windows bootstrap

Protected Git repository, pinned Unity revision/packages, PowerShell bootstrap/build/test/package commands, clean-account verification and recoverable baseline.

Task 2 - Create source-of-truth, story and AI document set

Add `AGENTS.md`, architecture map, unit catalog, save rules, history/event schema, storyteller authority, AI golden principles, decision hierarchy, contract template and regression ledger.

Task 3 - Add architecture and code-quality gates

Assembly dependency tests, analyzer rules, forbidden API boundaries, content-schema validation and no-authoritative-state-in-presentation checks.

Task 4 - Build command-line and CI harness

Headless build/test, artifact retention, performance baselines, test reports, deterministic seed suite and failure diagnostics.

| Task 5 - Implement authoritative world clock and scheduler shell

64-bit world time, fixed base loop, due-time queue, speed controls, backlog watchdog and deterministic order.

| Task 6 - Implement stable identity registry

Never-reused domain IDs, typed references, tombstones, lookup/index ownership and migration fixtures.

| Task 7 - Implement keyed deterministic random streams

Counter/key-based access, stream catalog, occurrence counters, bounded random semantics and tests.

| Task 8 - Implement historical event graph foundation

Stable event IDs, typed causal edges, participant/location references, intervention ancestry, significance and persistence before complex systems emit unrecordable history.

| Task 9 - Implement canonical checksum and state diff

Deterministic reduction, subsystem digests, divergence localization and replay reports.

| Task 10 - Implement semantic persistence skeleton

Versioned DTOs/chunks, manifest, atomic save, prior-good backup, migrations, corruption diagnostics and active-event/history storage.

| Task 11 - Build synthetic 10,000-person engine benchmark

Compact people, households, needs, traits, jobs and locations without full AI; measure memory, tick, save and load.

| Task 12 - Implement spatial chunks and bounded queries

Authoritative coordinates, region ownership, indexes, candidate caps and deterministic iteration.

| Task 13 - Implement hierarchical navigation prototype

Strategic/settlement/local graphs, corridor cache, bounded invalidation and route-disruption benchmark.

| Task 14 - Build 2.5D presentation proof

Terrain, elevation, water, camera, zoom, selection, camera-facing billboards, modular structures, occlusion and debug overlays.

| Task 15 - Create AI Behavior Contract schema and decision catalog

Machine-readable fields, Markdown template, versioning, authority validation and repository routing.

| Task 16 - Implement canonical AI primitives

Utility curves, preconditions, reservations, cooldown, hysteresis, tie-breaking, knowledge filtering and trace emission.

| Task 17 - Build AI Decision Laboratory runner

Fixture loader, scenario/property/metamorphic/multi-seed tests, before/after comparison and per-decision performance.

| Task 18 - Implement founding-group site-selection contract

Knowledge-limited candidates, survival utility, stable randomness, traces and site scenarios.

| Task 19 - Implement urgent individual-needs contract

Emergency food/shelter/care choices, reservations, unreachable-target handling and regressions.

| Task 20 - Implement settlement labor-allocation contract

Demand, skills, tools, travel, ordered priorities, emergency overrides and distribution tests.

| Task 21 - Implement procedural world generation and validation

Hydrology, terrain, climate, soil, visible/hidden resources, hazards and viable contrasting regions.

| Task 22 - Implement persistent people, households, inheritance and traits

Life state, dependents, pooled survival, inherited tendencies, slow personality and save continuity.

| Task 23 - Implement physical camp and building-construction proof

Projects, footprints, entrances, materials, hauling, progress, damage, ruins and building state.

| Task 24 - Implement paths, road edges, parcels and project-order boundary

Travel heat, road proposals, route candidates, land rights and decision-level player ordering without manual placement.

| Task 25 - Execute integrated settlement-formation and chronicle gate

Run three groups across contrasting regions; prove settlement formation, collapse explanation, event graph, basic chronicle, save/load, camera independence and performance.

| Tasks immediately after the first 25

Story Director candidate selection and Chronicler profile follow the integrated gate, then institutions, technology opportunity, trade, diplomacy, limited physical war, omniscient comparison UI and the divergence prototype. Later-age systems remain prohibited.

82 Glossary

Action definition — Versioned content record describing one candidate action, its owning layer, preconditions, evaluators, reservations, effects and trace keys.

Active tactical sector — Bounded physical area in which hostile squads are resolving combat at the encounter cadence.

Adversarial reviewer — Independent agent role tasked with finding hidden assumptions, invalid state, non-determinism, authority violations, exploits and missing tests rather than approving intent.

AI Behavior Contract — Structured authoritative specification for one decision, including owner, trigger, observations, candidates, preconditions, utility, randomness, traces, invariants, budgets and scenarios.

AI Decision Catalog — Repository index of all decision IDs, owners, versions, policies, tests, trace schemas and performance budgets.

AI Decision Laboratory — Headless deterministic environment and toolchain used to implement, compare, benchmark and regress bounded decision systems before broad game behavior.

Architecture Reserved — Current schemas and boundaries are required to permit later implementation without replacing authoritative state; current content is not required.

Authoritative simulation — Camera-independent state and rules that determine actual outcomes. Presentation and UI observe it but do not own or alter it directly.

Believed state — Information an actor or institution holds, including confidence and staleness; may differ from ground truth.

Binding Order — Player command that obligates the owning civilization decision layer but does not bypass capability, materials, law, transport or execution.

Bounded candidate set — Explicit maximum number of actions or targets a decision may evaluate.

Camera-facing billboard — Pixel-art plane or sprite that rotates toward the camera while direction and action are conveyed through motion, equipment, pose and effects.

Canonical checksum — Deterministic digest over ordered authoritative fields used to detect replay divergence; excludes presentation-only state.

Canonical AI primitive — One approved runtime implementation of utility, curves, preconditions, reservation, hysteresis, randomness, scheduling, tracing or knowledge filtering.

Capability — A society's ability or knowledge related to a problem, principle, procedure, component, tool, institution or application.

Capability hypergraph — Directed network in which capability requirements may contain multiple simultaneous or alternative prerequisites.

Candidate action — One feasible or potentially feasible option considered by a decision contract.

Causal ancestry — Links from an event or state change to contributing events, decisions and player interventions.

Civilization knowledge perspective — UI mode that shows the world as a selected civilization currently believes it to be.

Cold state — Authoritative data that remains persistent but is processed rarely because no due event or active interaction requires it; not fabricated summary state.

Commitment — Selected action or goal retained for a minimum period until completion, cancellation or invalidation.

Compaction — Controlled reduction of historical or visual detail while preserving current state, stable identity, canonical evidence and protected landmarks.

Confidence — Quantified reliability of believed information, separate from its value.

Construction work package — Bounded stage of a physical project with material, labor, tool and authority requirements.

Content definition — Immutable, namespaced, schema-validated data record referenced by stable content ID.

Controlled randomness — Keyed, bounded random contribution declared by a behavior contract; never a substitute for preconditions.

Counterfactual experiment — Player save branch or intervention used to compare how changed conditions alter history.

Decision cadence — Scheduled or event-driven frequency at which a contract may reevaluate.

Decision ID — Stable namespaced identifier for a specific behavior contract; retired IDs are not reused.

Decision layer — Authority level—person, household, workplace, institution, settlement, region, government, civilization, alliance or international system—that owns a decision.

Decision trace — Structured record of observations, candidates, preconditions, score terms, randomness, selection, policy, reservations and invalidation.

Deferred content — Excluded content that uses existing architecture later.

Derived cache — Rebuildable acceleration data such as spatial indexes, route corridors, HLOD or UI read models; never the sole source of truth.

Deterministic replay — Re-execution from seed/snapshot and command log that reproduces canonical authoritative checkpoints on the supported reference build.

Direct decision control — Player control over civilization priorities and decisions without lower-layer micromanagement.

Domain ID — Stable world identifier independent of engine object or ECS handle.

ECS — Entity Component System; data-oriented storage and processing model used for simulation state.

Emergent outcome — Result generated by interactions among declared systems rather than a prewritten outcome selector.

Event-driven update — Work scheduled because a relevant state changed or a due event occurred rather than polled every frame.

Evidence pack — Build, test, replay, save, trace, performance, documentation and visual artifacts required to accept a change.

Explicit non-goal — Capability intentionally excluded to protect product identity, performance or scope.

Fast state — Mood, immediate fear, anger, exhaustion, hunger and other rapidly changing conditions.

Fixed point — Integer-backed numeric representation used where accumulation and deterministic stability matter.

Fixed step — Simulation update at controlled boundaries independent of render frames.

Force Outcome — Player power that bypasses ordinary feasibility and directly mutates authoritative world state through validation and reconciliation.

Founding group — Persistent migration/expedition population attempting to establish a settlement.

Functional extinction — Population or culture remains only in numbers or records too small to sustain ordinary recovery.

Goal invalidation — Event that makes a commitment impossible, illegal, irrelevant or unsafe and schedules reevaluation.

Ground truth — Authoritative world state, fully visible to the default omniscient player but not necessarily to AI actors.

Headless runner — Simulation executable without rendering/UI, used for tests, benchmarks and accelerated worlds.

Hierarchical pathfinding — Route selection through coarse regions/corridors before local refinement.

Historical-significance tier — Retention class governing whether a ruin, event or geography can be summarized, simplified or must remain permanent.

Household — Lightweight co-residence, support and economic unit for housing, dependents, pooled resources, migration and inheritance.

Hysteresis — Preference for continuing an existing commitment unless an alternative exceeds it by a declared margin.

Indirect material intervention — Player alteration of physical, biological, informational or social conditions whose effects propagate through normal systems.

Institution — Durable organization providing education, faith, research, health, market, governance, administration or other social capacity.

Intervention history — Permanent record of player state changes, including bypassed rules and affected entities.

Invariant — Condition that must always hold, such as nonnegative inventory, valid references, legal authority or nonreused IDs.

Keyed random stream — Random sample derived from stable identifiers and occurrence count rather than shared mutable sequence order.

Knowledge carrier — Person, record, artifact, school, workshop or institution that preserves and transmits capability evidence.

Knowledge filter — Canonical boundary that restricts observations to what an actor is entitled to know.

Landmark manifest — Save record protecting the identity, graph, structures, damage and evidence of a historically significant place.

Layer-authority violation — Decision or code path in which an entity acts outside its defined decision layer.

Lightweight social tie — One of a small capped set of persistent relationships with direct gameplay value.

Local variant — Civilization-specific implementation of a solution using locally available materials, tooling, institutions and knowledge.

Metamorphic test — Test that changes an input in a controlled way and verifies the output changes—or does not change—in an expected relationship when an exact oracle is unavailable.

Mod-safe definition — Data schema with stable IDs, validation and compatibility metadata that requires no unsafe executable code.

Monotonic ID — Identifier sequence that never reuses a world-local value.

Observation context — Actor- and layer-scoped set of sensed facts, beliefs, reports, confidence and time used by a decision.

Omniscient overlay — Map visualization of ground-truth authoritative data.

Organization — Shared concept for households, workplaces, institutions, governments, military units and factions where membership/ownership primitives overlap.

Person persistence — Stable identity and causal state throughout a person's life regardless of camera or update frequency.

Physical batch — Integer quantity of fungible goods stored or transported at a real location.

Physical city — Settlement whose buildings, roads, infrastructure, damage and movement exist as authoritative world objects rather than threshold labels.

Policy data — Validated curves, weights, action sets and modifiers loaded by stable execution code.

Precondition — Hard requirement that must pass before an action can be selected or executed.

Presentation snapshot — Read-only changed-state record consumed by rendering and UI.

Property test — Automated test over generated inputs that verifies general invariants rather than one expected example.

Reconciliation — Transactional process that updates all related records after a complex or forced state mutation.

Regression fixture — Saved deterministic state reproducing a previously observed defect.

Regression ledger — Repository record of behavior defects, root causes, fixes, contracts and permanent tests.

Reservation — Temporary claim on inventory, capacity, land, worker, route or unique object that prevents double use.

Response curve — Canonical mapping from observed value to normalized consideration.

Retail bounded trace — Limited trace retained during ordinary play for recent and important decisions.

Route corridor — Shared long-range path over strategic and road graphs reused by people, shipments or forces.

Scenario fixture — Versioned controlled world state with commands, allowed/forbidden behavior, timing, trace and performance expectations.

Semantic save — Versioned serialization of domain meaning rather than engine memory layout.

Slow personality — Core behavioral tendencies that change only gradually through major repeated experience.

Society-dependent planning — Urban project and placement behavior shaped by law, ownership, culture, finance, government and institutions.

Stable content ID — Namespaced versioned identifier for a recipe, building, policy, action or other definition.

State extinction — End of a political entity or government while people or culture may persist.

Statistical seed test — Multi-seed evaluation of distributions and trends rather than one exact outcome.

Strategic raster — Coarse authoritative world grid for terrain, environment, resources, claims and movement cost.

Tombstone — Minimal retained record for a destroyed or dead entity still referenced by history or law.

Trace mode — Retail, debug-selective or laboratory-full decision logging configuration.

Trait modifier — Declared effect of a behavioral tendency on response curve, threshold, utility, cadence or commitment.

Unrestricted Sandbox — Default rule set in which every power implemented in the current build is immediately available without balance currency or progression.

Utility score — Normalized comparison value for a candidate action after preconditions, observations and modifiers.

Vertical slice — Integrated first playable that proves the defining product loop, not a collection of isolated prototypes.

Visible causality — Design requirement that physical state, UI and history expose how local actions produce larger outcomes.

World-defining ruin — Historically significant destroyed geography that remains physically recognizable for the world's life unless explicitly transformed.

Chronicle - A localized, template-generated historical view derived from structured event facts and perspective records.

Historical Event Graph - Stable event nodes and typed causal/entity edges used by chronicles, maps, landmarks and the Why interface.

Incident candidate - A simulation-valid opportunity with explicit preconditions and an authoritative resolution owner; not a prewritten outcome.

Pure Simulation profile - Storyteller control profile with minimal selection influence, used as behavioral and checksum reference.

Recovery debt - Bounded pacing state indicating that a region or civilization recently absorbed severe pressure and should receive reduced unrelated story selection under profiles that protect recovery.

Story Director - Deterministic policy layer that selects, paces, frames and highlights eligible incidents without owning authoritative outcomes.

Storyteller profile - Saved weighting and pacing configuration such as Pure Simulation, Chronicler, Adversarial, Chaotic or Custom.

Perspective record - What an observer or civilization believed about an event, separate from authoritative facts.

Prototype Charter - Approved document defining falsifiable questions, time boxes, pass/fail evidence and stop conditions for pre-production experiments.

83 Assumptions

The following assumptions are explicit. A disproved assumption triggers an ADR or blueprint revision rather than silent drift.

1. The creator remains the final product and preference authority but does not perform routine production programming.
2. Coding agents generate most production code and documentation under repository constraints.
3. Periodic external human technical review can be obtained before critical irreversible milestones; if not, risk R45 remains high.
4. Development is Windows-first; Linux player builds are required; other platforms are deferred.
5. Unity 6.3 LTS and required DOTS packages remain available and commercially acceptable through the initial benchmark period.
6. Exact Unity and package revisions are pinned; this blueprint does not authorize automatic upgrades.
7. The authoritative simulation targets 10,000 persistent people for the first-playable Large benchmark.
8. A 20 Hz base loop and the declared game-time mapping provide acceptable readability and speed.
9. The first playable starts in an early-modern capability envelope.
10. Standard world size near 8 km square provides enough geographic differentiation for the first playable.
11. Flat bounded topology is sufficient for the core game identity.
12. A camera-facing billboard approach can remain readable with supplemental direction cues.
13. Modular low-poly buildings with pixel textures can produce coherent city identity under constrained grammars.
14. Exterior-only building simulation provides enough physicality without rendered rooms.
15. People can remain meaningful with lightweight households and capped social ties.
16. Controlled baseline inheritance plus environmental development provides useful variation without detailed genetics.
17. Culture and political attitudes can be represented as continuous carried tendencies rather than fixed labels.

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18. Bounded utility, state machines, policies and limited planning are sufficient for the intended coded intelligence.
 19. Full player omniscience is compatible with AI knowledge limits through separate query contexts.
 20. Unrestricted powers do not require balanced costs; player trust depends instead on clear consequences and history.
 21. Binding decisions plus autonomous implementation provide sufficient direct control without manual construction.
 22. Force Outcome can be implemented safely through transactional reconciliation.
 23. Physical commodity batches preserve the necessary logistics causality.
 24. Hierarchical pathfinding and shared corridors can support the target world and city scale.
 25. A capability hypergraph can express divergent technology without unmanageable authoring complexity.
 26. Region-level ecosystem populations are sufficient for common wildlife while selected animals remain persistent.
 27. Significant ruins can be preserved within save budgets through geometry simplification and significance tiers.
 28. MessagePack-CSharp/LZ4 or an equivalent semantic chunk format remains suitable after benchmark.
 29. A representative 100-year Large save can remain below 100 MB under planned retention.
 30. Full decision traces can be restricted to tests/debug while bounded retail traces support player explainability.
 31. Property, metamorphic and statistical tests can detect meaningful classes of AI regression.
 32. Multi-seed acceptance thresholds can be tuned without scripting identical outcomes.
 33. Final art, audio, commercial model and platform expansion occur after technical gates.
 34. The north-star product is a multi-year program; the first playable is not expected within a few months.
 35. Modern, nuclear, information and future eras can extend rather than replace core people, authority, route, building, capability and save models.
 36. Multi-planet play will require architecture extension and is not promised by the first-playable implementation.
 37. No proprietary internal architecture of comparable games is assumed.
 38. Research links and current product/version facts are reverified at implementation milestones where they may have changed.
 39. Performance budgets are provisional engineering targets until measured on committed reference hardware.
 40. Accessibility and localization architecture begin early even though final priorities remain creator decisions.
 41. The default content rating and exact violence ceiling are not yet selected; visual design remains stylized and configurable.
 42. The project can postpone unrestricted executable mods until schemas stabilize.
 43. The creator accepts that a staged GO/NO-GO process may require architecture redesign or reduced first-playable density while preserving the north-star identity.
 44. No simulation can retain infinite history or geometry; compaction is acceptable only when current causes and protected landmarks survive.
 45. "Indefinite simulation" means no intentional terminal date and bounded stable operation, not unlimited state growth on fixed hardware.

Additional Version 2.0 assumptions:

1. Version 1.0 can be a complete early-modern game without later-age novelty.
 2. Structured authored templates can produce sufficient chronicle readability without live LLM prose.
 3. Story selection can remain deterministic and subordinate to simulation truth.
 4. Historical maps can use keyframes/deltas rather than retaining exact positions of every entity at every time.
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5.

The event graph can compact ambient detail while preserving world-defining causal evidence.
6.

Prototype gates may stop or redesign the project before full Version 1.0 production.
7.

Storyteller profiles are world rules and must be included in replay/save state.
8.

A no-compaction debug mode is acceptable for verification even when it is not a supported retail save mode.

84 Remaining creator decisions

All routine engineering choices are resolved in this blueprint. The remaining decisions require personal, artistic, commercial or release preference. Recommended temporary defaults permit production to continue.

Table 16 — Remaining creator-preference decisions

ID	Decision	Why personal preference is required	Temporary default	Deadline	Affected systems
C01	Final title and brand	Identity and market positioning cannot be derived technically	"Civilization Sandbox" working title	Before public announcement	UI, saves, store, legal, marketing
C02	Tonal balance	Degree of seriousness, humor, tragedy and absurdity determines writing/art direction	Serious systemic history with restrained dry humor	Before large speech/content production	Speech, events, art, rating
C03	Audio and music direction	Cultural palette, diegetic density and emotional style are aesthetic	Layered ambient world audio; adaptive restrained score	Before vertical-slice audio pass	Audio runtime, content budget
C04	Commercial model and price	Revenue strategy depends on personal/business goals	Premium single-purchase first; no gameplay micro-transactions	Before store planning	Scope, platform, live operations
C05	Final platform scope	Support cost versus audience is commercial preference	Windows first; Linux player supported; macOS/console deferred	Before public system requirements	Input, build, QA, UI
C06	Modding and UGC policy	Openness, moderation and support tolerance are preference choices	Validated data mods after schema stabilization; no executable mods early	Before early access	Content schemas, security, save support
C07	Localization priorities	Markets, budget and language preference drive order	English source with localization-safe architecture and pseudo-localization	Before speech-library expansion	UI, speech, fonts, QA
C08	Accessibility ambition beyond baseline	Additional accommodations require product prioritization	Meet Section 42 baseline; budget a formal audit before release	Before UI lock	UI, input, art, audio
C09	Content rating and final violence ceiling	Acceptable blood, bodies and themes are ethical/market preferences	Stylized low-detail visible violence; configurable blood; no realistic gore	Before final combat art	Art, effects, marketing, platform certification
C10	Release, community and marketing strategy	Early access, closed testing, public roadmap and community involvement are business preferences	Private milestone builds until vertical-slice stability; then closed external test	Before first public build	Telemetry, support, roadmap, store

These decisions must not reopen locked technical choices. For example, selecting a lower content rating can reduce blood effects but does not make war abstract or remove physical casualties. Selecting console targets later may alter input/UI but does not justify camera-dependent simulation.

85 Revision change log

Version 2.0 is a complete replacement rather than an errata sheet. The table summarizes the formal change map; detailed integration appears throughout the main sections.

Table 17 — Formal revision change log

Baseline area	Change made	Reason / locked decision	New status	Affected systems	Scope/performance/ first-playable impact
Executive summary	Reframed around physical cities, unrestricted powers, omniscience, decision control and AI verification	5.1–5.8, Section 9	Confirmed/Recommended	Whole product	First playable expanded in control/UI/testing
Player-mode recommendation	Replaced Limited Influence default with Unrestricted Sandbox	5.1	Superseded → Confirmed	Rules, UI, powers, progression	Removes costs/unlocks; optional modes deferred
Direct control	Defined binding settlement-to-alliance decision orders	5.2–5.3	Confirmed	Authority, AI, UI, projects	Adds first-playable order system
Creative overrides	Split Force Outcome from normal orders and added reconciliation	5.3	Confirmed	Commands, persistence, history	Adds state-mutation tests and power scope
Indirect intervention	Required physical/social propagation rather than aggregate edits	5.4	Confirmed	Economy, people, environment, politics	Adds destruction/displacement scenarios
Information model	Made player fully omniscient by default; retained civilization knowledge separately	5.5	Confirmed	UI, overlays, AI knowledge	Adds full inspectors and compare perspective
Physical representation	Strengthened requirement that cities, war, goods and infrastructure visibly exist	5.6	Confirmed	World, settlement, war, render	Raises physical-city benchmark and rendering risk
Violence	Specified visible stylized pixel violence without realistic gore	5.7	Confirmed	Combat presentation, rating	Adds visual direction; no nuclear first-playable requirement
World format	Locked flat bounded 2.5D; removed globe ambiguity	5.8	Confirmed	World, camera, routing	Planetary systems become future architecture
Character facing	Replaced four-direction close sprites with camera-facing billboards	5.9	Confirmed	Assets, animation, combat readability	Reduces asset scope; adds direction-readability benchmark
Art direction	Locked approximate 32-bit detailed pixel aesthetic	5.10	Confirmed	All presentation	Replaces restylable ambiguity with production direction
Appearance	Added simulation-linked procedural sprite layers	5.11	Confirmed	People, content, UI	Adds descriptors and validation pipeline
Buildings	Locked modular low-poly 3D with pixel textures and full physical states	5.12	Confirmed	Settlement, render, save	Adds building grammar and skyline architecture
Interiors	Locked exterior-only authoritative interior state	5.13	Confirmed	Buildings, movement, render	Prevents room/floor scope; reserves functional modules
City development	Required society-dependent hybrid organic/planned growth	5.14	Confirmed	Settlements, politics, economy	Adds paired layout acceptance tests
Manual placement	Explicitly prohibited ordinary building/road placement	5.15	Confirmed	Player control, UI	Direct orders replace city-builder controls
Speech	Added contextual authored/template bubbles and bounded social effects; no live LLM	5.16	Confirmed	People, UI, localization	Adds thin first-playable library and repetition budget
Households	Clarified lightweight shared survival/economic model	5.17	Confirmed	People, economy, migration	Split into dedicated section and acceptance tests
Inheritance	Added controlled parental baselines with environment; excluded cultural genetics	5.18	Confirmed	People, health, traits	Adds first-playable inherited tendencies
Traits	Required direct coded decision effects and contextual tradeoffs	5.19	Confirmed	AI, people, politics, war	Adds contract modifiers and multi-seed tests
Personality	Added fast/medium/slow change rates	5.20	Confirmed	People, AI, history	Adds rate-limited state model

Baseline area	Change made	Reason / locked decision	New status	Affected systems	Scope/performance/first-playable impact
Social ties	Added capped meaningful persistent ties	5.21	Confirmed	People, politics, military	Adds small state budget; preserves non-goal of social sim
World density	Specified moderate zoom/region/era-dependent physical detail	5.22	Confirmed	Presentation, performance	Adds clutter and LOD budgets
Overlays	Expanded to comprehensive omniscient overlay set	5.23	Confirmed	UI, query, render	Adds composition architecture and performance budget
Resources	Added hybrid physical surface/overlay underground representation and flow inspection	5.24	Confirmed	Resources, logistics, UI	Adds extraction visuals and overlay requirements
Ecosystems	Replaced thin habitat concept with hybrid individual/regional model	5.25	Confirmed	Environment, people, economy	Adds first-playable thin ecosystem and tests
Extinction	Added local/functional/global species and state/cultural/population civilization distinctions	5.26	Confirmed	Environment, history, politics	Adds first-playable controlled extinction scenarios
Recovery	Added remnant recovery and unrestricted restoration	5.27	Confirmed	Environment, civilization, powers	Adds restoration contracts and history
Ruins	Added permanent world-defining city scars and significance tiers	5.28	Confirmed	Save, world, render, history	Increases save/geometry risk and budgets
Game AI	Replaced broad utility guidance with layered authority and behavior contracts	Section 6.1–6.2	Confirmed/Recommended	All behavior	Major repository and testing expansion
AI milestone	Inserted AI Decision Laboratory before broad behavior	6.3	Confirmed	Roadmap, tasks, CI	Adds 5–8 week phase and GO/NO-GO
Behavioral tests	Added deterministic, invariant/property, metamorphic and statistical categories	6.4	Confirmed	Testing	Expands CI and acceptance
Tracing	Made structured trace mandatory for important decisions	6.5	Confirmed	AI, UI, telemetry	Adds trace schema and performance modes
AI primitives	Required one canonical implementation set	6.6	Confirmed	Core AI	Prevents parallel frameworks
Rules/execution	Moved policy variation into validated data	6.7	Recommended	Content/AI	Improves extensibility and review
AI review	Added implementer/adversarial reviewer separation	6.8	Confirmed	Production	Changes approval workflow
Regression ledger	Added permanent behavior fixture process	6.9	Confirmed	Testing/docs	Creates project-specific behavioral memory
Machine learning	Restricted ML to bounded experimental uses with strict gates	6.10	Confirmed	AI architecture	Prevents black-box core intelligence
Repository docs	Added nine required docs/ai files and new package/test layout	6.11	Confirmed	Repository	Expands first tasks and documentation
First playable	Preserved defensible quantities; added households, inherited traits, direct orders, Force Outcome, omniscience, traces and deterministic AI tests	Section 11	Confirmed	Whole vertical slice	Scope increases but remains early-modern
Performance	Added billboards, buildings, overlays, speech, traces, ecosystems and ruins to budgets	Section 17	Recommended	Runtime/save	New benchmarks and degradation policy
Roadmap	Expanded from seven phases to eight with AI Lab gate	Section 10/48	Recommended	Production	Calendar estimate revised to 18–30 months

Baseline area	Change made	Reason / locked decision	New status	Affected systems	Scope/performance/first-playable impact
First 25 tasks	Reordered around governance, deterministic core, save, headless, AI Lab, traits, 2.5D proof and settlement founding	Section 10/55	Recommended	Execution	Nuclear/advanced-era tasks excluded
Risk register	Added 20+ new risks covering physical cities, billboards, speech, omniscience, Force Outcome, traces, ruins and extinction	Section 10/50	Recommended	Production	Mitigations and owners updated
Final evaluation	Added explicit engine, AI, settlement, city and divergence go/no-go benchmarks and staged feasibility verdict	Section 25	Recommended	Governance	Prevents monolithic implementation

85.1 Removed assumptions

The following baseline assumptions are removed: Limited Influence as the default; discovery unlocks as required engagement; four-direction close character sprites; art style remaining unspecified/restylable; a potential globe topology as a core-world option; player direct control being temporary/budget-limited; and the idea that ordinary old city ruins may be compacted without explicit landmark significance.

85.2 Unchanged high-value baseline decisions

Version 2.0 retains the core game identity, persistent people, nonlinear capability network, physical batch economy, hierarchical routing, semantic saves, stable IDs, camera-independent authority, Unity provisional selection, 8 km Standard/16 km Large presets, 40-person founding groups, baseline content counts, 20 Hz scheduling, 10,000-person benchmark, 6 GB memory target, save/load budgets, headless workflow, protected branches, task contracts and staged release families.

85.3 Version 2.0 additions

- Promoted Version 1.0 to a complete early-modern/1700s product boundary and prohibited simultaneous all-age development.
- Added sequential major-version age strategy and local capability-state model.
- Added Prototype Charter, deterministic/scale/AI/visual/rough-settlement/divergence/vertical-slice/full-production distinctions.
- Added simulation-grounded storyteller, candidate-selection authority, profiles, event graph, chronicles, historical maps and retention tiers.
- Reordered foundational work so event history exists before complex systems produce irrecoverable events.
- Expanded performance, risks, glossary, assumptions, acceptance criteria, diagrams and implementation tasks for storytelling and prototype gates.
- Superseded Version 2.0 in full while retaining all compatible locked decisions, architecture, citations, tables and appendices.

86 Bibliography

The bibliography separates verified product/runtime facts, official development guidance, comparable-game disclosures and primary technical sources. Access dates correspond to the research verification date unless a publication date is shown. Project-specific recommendations remain in the main text; a citation does not imply that a source endorses this exact architecture.

86.1 Engine, runtime, editor, licensing and CI sources

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- [S-ENG-15] Epic Games. "Unreal Engine 5.0 Release Notes — MassEntity." Public description of MassEntity and tens-of-thousands-agent use cases; original feature status was experimental. https://dev.epicgames.com/documentation/en-us/unreal-engine/unreal-engine-5.0-release-notes?application_version=5.0
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86.5 Research-status notes

- Unity support, pricing, package and licensing information may change; verify before dependency and commercial milestones.
- Bevy's warning reflects its repository at the verification date and may improve later.
- Unreal MassEntity status and APIs are version-specific; verify against the selected engine release if reconsidered.
- Comparable-game citations establish public design positions, not proprietary implementation.
- Academic algorithms are adapted to project constraints; no source is treated as a complete production solution.

87 Technical appendices

87.1 Appendix A — Detailed simulation schedule and ownership matrix

The matrix is the default scheduling contract. Individual behavior contracts may schedule earlier work through explicit invalidation events. "Saved" means authoritative state or future due work must survive save/load; derived caches rebuild.

System	Owning package/layer	Reads	Writes	Cadence	Camera dependent	Saved
World clock/scheduler	simulation.core	speed commands, due queues	time, due execution order	20 Hz wall loop	No	Yes
Weather/hazards	world/environment	climate normals, anomalies	regional weather, hazard events	game hour/day/event	No	Yes
Ecology/resources	ecosystems/world	climate, harvest, pollution	populations, yield, reserves	daily/seasonal/event	No	Yes
Spatial index	world	location/graph changes	bounded query caches	event/tick	No	Derived
Local movement	people/logistics	route, speed, hazards, occupancy	position, route progress	10–20 Hz	No	Yes
Immediate reactions	ai.runtime/person	hazard/attack/event context	reaction state, goal invalidation	event/2–10 Hz	No	Yes
Individual goals	people/AI person layer	needs, knowledge, opportunities, policy	commitment, reservation, trace	1–5 game min/event	No	Yes
Health	people	nutrition, exposure, care, age	vitality, condition, death events	daily + acute	No	Yes
Household	households	members, home, income, needs	budget, consumption, migration	hourly/daily/event	No	Yes
Labor matching	economy/workplace	jobs, skill, law, travel	contracts, assignments	daily/event	No	Yes
Workplace production	economy	labor, inputs, equipment, policy	output, wear, pollution	5 game min/event	No	Yes
Markets/prices	economy	stock, demand, shipments, policy	prices, orders, clearing	6 game hours	No	Yes
Construction	settlements	projects, labor, deliveries, weather	progress, building state	5 game min/event	No	Yes
Paths/roads/parcels	settlements/world	travel heat, projects, law, damage	graph edges, rights, condition	daily/event	No	Yes
Institutions	institutions	members, funding, capacity, policy	services, education, doctrine	daily/weekly	No	Yes
Technology projects	knowledge	pressure, people, evidence, inputs	evidence, prototype, capability	daily/event	No	Yes
Culture/attitudes	culture/people	rolling conditions, institutions, events	distributions, medium-rate attitudes	monthly/event	No	Yes
Government/politics	politics	attitudes, factions, law, capacity	policy, budget, legitimacy, succession	weekly/monthly/event	No	Yes
Civilization strategy	politics/civilization	regional conditions, orders, diplomacy	goals, programs, priorities	weekly/event	No	Yes
Diplomacy	diplomacywar	believed state, goals, treaties	proposals, incidents, clauses	weekly/event	No	Yes
Military operations	diplomacywar	goals, routes, supply, intelligence	task groups, objectives	hourly/event	No	Yes

System	Owning package/layer	Reads	Writes	Cadence	Camera dependent	Saved
Combat encounters	diplomacywar	squads, equipment, terrain, morale	wounds, losses, movement, damage	10 Hz/event	No	Yes
Player command validation	playerpowers	mode, target, parameters, authority	validated command/rejection	command boundary	No	Yes
Force reconciliation	playerpowers/persistence	mutation plan, references, invariants	transactional state delta	command boundary	No	Yes
History/significance	persistence/history	events, references, landmarks	canonical events, tiers, summaries	event/daily	No	Yes
Compaction	persistence/history	retention policy, references, age	summaries, simplified geometry	incremental	No	Yes
Presentation bridge	presentation	changed authoritative state	read-only snapshots	tick/frame boundary	Indirectly	No
Billboards/buildings/effects	presentation	snapshot, camera, assets	GPU/visual state	frame	Yes	No
Overlay query	UI/query	authoritative/believed read models	tiled visualization data	on demand/cached	Yes	No
Speech selection	UI/social info	speaker state, tags, view	bubble event/display	seconds/event	Display yes; social effect no	Effect event yes
Decision tracing	AI/telemetry	decision evaluation	trace buffers/history promotion	decision event	No	Bounded/selected
Benchmarks/tests	Tests	fixtures, commands	reports, diffs	explicit run	No	Artifacts only

87.1.1 Scheduling invariants

1. Presentation cannot cause earlier or later authoritative due work.
2. The same due-time set sorts identically by time, priority, stable ID and sequence.
3. A system that defers work records backlog and does not lose the due item.
4. Continuous rates integrate the same total across legal speed steps within numerical tolerance.
5. Full tracing may increase wall duration but not selected action or state.
6. A Force Outcome executes at a declared tick boundary and completes reconciliation transactionally.
7. Save barriers complete scheduled mutations before snapshot; no half-applied command is serialized.

87.2 Appendix B — Core data-model sketches

These semantic sketches guide final C# and save DTOs. Exact storage may differ after benchmarks, but fields and ownership semantics may not disappear without an ADR.

87.2.1 Stable identity

```

StableEntityId
  WorldGuid world
  UInt64 local_id           # monotonic, never reused

StableContentId
  String namespace
  String name
  UInt32 major_version

StableDecisionId
  StableContentId contract

StableEventId
  StableEntityId-like monotonic identity

```

87.2.2 Person record

```

PersonRecord
  StableEntityId id
  StableContentId name_culture
  LocalizedName canonical_name

```



```

WorldTime birth_time
LifeState life_state
BodyState body
NutritionState nutrition
ConditionRef[] acute_conditions
FixedTraitBaselines inherited_baselines
BehavioralTraitState personality
FastState fast_state
MediumAttitudeState attitudes
SkillSet skills
EducationState education
OccupationRef occupation
StableEntityId household_id
StableEntityId? home_building_id
StableEntityId? workplace_id
MembershipRef[] institutions
CultureExposure[] cultures
BeliefState belief
PoliticalState politics
MaterialState possessions_and_wealth
NeedState needs
LocatedState location
GoalCommitment current_goal
ExperienceSummary experience
SocialTieRef[CAP] persistent_ties
KnowledgeCarrierRef[] carried_knowledge
AppearanceDescriptor appearance
DecisionTraceRing recent_decisions
StableEntityId? parent_a
StableEntityId? parent_b

```

87.2.3 Household record

```

HouseholdRecord
StableEntityId id
StableEntityId[] members
StableEntityId[] guardians
StableEntityId[] dependents
StableEntityId? home_building_id
TenureAndRights tenure
Inventory shared_inventory
MoneyBalance balance
DebtRef[] obligations
BudgetPolicy budget_policy
MigrationCommitment migration
InheritanceRule inheritance
WellbeingSummary shared_conditions
ReservationSet reservations

```

87.2.4 Building and project

```

BuildingRecord
StableEntityId id
StableContentId archetype_id
Footprint footprint
Fixed height
EntranceRef[] entrances
ModuleState[] functional_modules
MaterialLayer[] materials
ConstructionState construction
DamageState damage
RepairState repair
AbandonmentState abandonment
OwnershipRights ownership
CapacitySet capacity
InventoryRef[] inventories
UtilityConnectionRef[] utilities
StyleDescriptor style
HistoricalSignificance significance
StableEntityId? predecessor
StableEntityId[] replacement_children

ProjectRecord

```

```

StableEntityId id
StableContentId project_type
DecisionLayer owner_layer
StableEntityId owner
PlayerOrderRef? binding_order
ProjectStatus status
CandidateSiteRef selected_site
WorkPackage[] work_packages
ReservationSet reservations
FundingPlan funding
RequiredCapabilityExpr requirements
BlockerRef[] blockers
WorldTime next_review

```

87.2.5 Route and shipment

```

RouteEdge
StableEntityId id
RouteType type
GeometryRef geometry
StableNodeId from
StableNodeId to
Fixed capacity
Fixed speed_modifier
Condition condition
AccessPolicy access
StableEntityId owner
StableEntityId maintainer
DamageState damage
GraphVersion version
HistoricalSignificance significance

```

```

Shipment
StableEntityId id
StableEntityId owner
InventoryManifest cargo
InventoryRef origin
InventoryRef destination
RouteCorridor corridor
TransportManifest transport
RiskState risk
WorldTime departed
WorldTime eta
SegmentProgress progress
EscortRef? escort

```

87.2.6 Capability and local knowledge

```

CapabilityDefinition
StableContentId id
StableContentId domain_id
ProblemTag[] solves
RequirementExpression requirements
AlternativeRecipe[] recipes
EvidenceType[] accepted_evidence
CapabilityEffect[] effects
AdoptionSector[] sectors
BehaviorContractRef[] decision_contracts
LocalizationKeys explanation

```

```

CapabilityHolderState
StableEntityId holder
StableContentId capability
Fixed awareness
Fixed understanding
Fixed practical_knowledge
bool has_plans
Fixed prototype_reliability
Fixed operate
Fixed repair
Fixed component_manufacture
Fixed full_manufacture
Fixed mass_production

```

```

AdoptionBySector adoption
KnowledgeCarrierRef[] carriers
StableEntityId[] variant_lineage
WorldTime last_practiced

```

87.2.7 Historical event and landmark

```

HistoricalEvent
  StableEventId id
  StableContentId event_type
  WorldTime start
  WorldTime? end
  StableEntityId[] actors
  StableEntityId[] affected_entities
  RegionRef location
  CauseRef[] causes
  OutcomeRef[] outcomes
  InterventionRef[] player_ancestry
  ImportanceTier tier
  LocalizationArgs display
  EvidenceRef[] retained_evidence

LandmarkManifest
  StableEntityId landmark_id
  StableEntityId? original_settlement_id
  RegionPolygon protected_extent
  GraphChecksum street_graph
  StableEntityId[] protected_structures
  DamageLayer[] damage_history
  EnvironmentalScarRef[] scars
  ImportanceReason[] reasons
  StableEventId[] canonical_references
  CompactionPolicy policy

```

87.2.8 Player order and Force Outcome

```

PlayerCommand
  StableCommandId id
  WorldTime issued_at
  PlayerRuleSet rule_set
  PlayerCommandClass class      # Observe, Order, Material, Force
  StableContentId power_id
  StableEntityId[] targets
  TypedParameter[] parameters
  ValidationResult validation
  ReconciliationPlan? reconciliation
  StateDeltaSummary result
  CommandId? caused_by

BindingOrder
  StableCommandId source_command
  StableEntityId target_authority
  StableDecisionId owned_decision
  Objective objective
  Fixed priority
  OrderStatus status
  RequirementState feasibility
  StableEntityId[] active_projects
  BlockerRef[] blockers
  WorldTime next_review

ForceOutcomeRecord
  StableCommandId source_command
  InvariantCheck[] pre_checks
  StateMutation[] mutations
  InvariantCheck[] post_checks
  StableEntityId[] created_ids
  StableEntityId[] destroyed_ids
  DeclaredSourceSink[] conservation_adjustments

```

87.3 Appendix C — Canonical AI Behavior Contract template

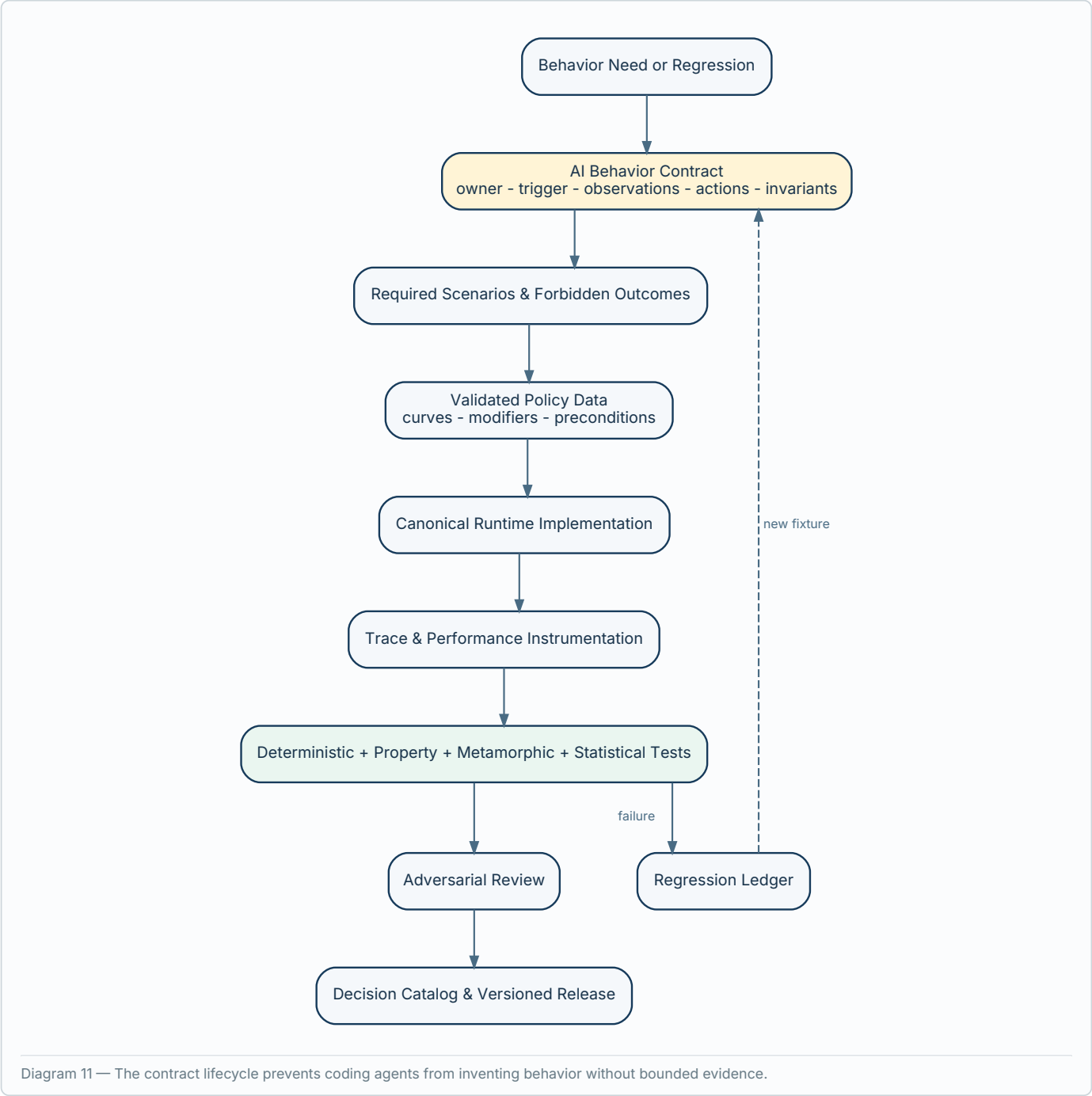


Table 12 — Required AI Behavior Contract fields

Field	Required content	Validation
Decision ID	Stable namespaced ID and version	Unique; retired IDs reserved
Owning layer	Person through international system	Matches authority matrix
Purpose	One bounded outcome	No vague “intelligence” objective
Trigger	Event and/or cadence	Declared invalidation path
Observations	Typed fields and source	Knowledge-filtered; bounded
Hidden information	Explicit inaccessible facts	Adversarial tests present

Field	Required content	Validation
Candidate actions	Stable action IDs	Maximum count declared
Preconditions	Hard legal/material/authority checks	Reason codes and tests
Utility terms	Curves, units, weights, aggregation	Canonical evaluators only
Controlled randomness	Stream, amplitude and selection semantics	Separate trace term
Tie-breaking	Stable deterministic order	Tested with equal scores
Reservations	Inventory/capacity/land/worker claims	Release and expiry defined
Cooldowns/hysteresis	Switch and repeat control	Loop tests present
Invalidation events	Target loss, hazard, order, time, policy	Replan semantics declared
State changes	Commitment, project, event or output	Package ownership valid
Required trace	Candidate and summary fields	Schema version specified
Invariants	Conditions that always hold	Property tests listed
Performance budget	Candidate/query/time/allocation cap	Benchmark fixture listed
Required scenarios	Positive, negative, boundary, save/replay	IDs exist in repository
Metamorphic relations	Directional expectations	Exceptions require trace cause
Statistical expectations	Seed suite and distribution threshold	Only where randomness matters
Explicit non-goals	Decisions and systems not owned	Scope gate
Save/version impact	Active commitment and occurrence state	Migration policy

87.3.1 Markdown contract template

```
# AI-BEHAVIOR-NNN – Decision title

Status: Proposed / Implemented / Superseded
Contract version:
Owning layer:
Owner package:

## Purpose
One bounded decision outcome.

## Trigger and cadence
Scheduled and event-driven triggers; next-review semantics.

## Observation context
Typed observations, knowledge source, confidence/staleness, candidate-query limits.

## Hidden information
Facts the actor must not access.

## Candidate actions
Stable action IDs and maximum count.

## Preconditions
Hard gates with reason codes.

## Utility model
Terms, units, response curves, weights, aggregation and clamps.

## Traits, attitudes and policy modifiers
Declared modifier groups and caps.

## Randomness and tie-breaking
Stream family, key, amplitude, weighted-choice semantics, stable tie-break.

## Commitment controls
Reservations, minimum duration, cooldown, hysteresis, change penalty.
```

```

## Invalidation
Events and resulting release/replan behavior.

## State changes
Owned records, commands, events and expected effects.

## Trace schema
Required candidate, selected-action, policy and invalidation fields.

## Invariants
Numbered conditions that always hold.

## Performance budget
Candidate/query/allocation/p95 targets and benchmark scenario.

## Required scenarios
Positive, negative, boundary, save/replay, camera-independent.

## Metamorphic relations
Input transformations and expected output relations.

## Statistical evaluation
Seed suite, metrics, thresholds and confidence.

## Explicit non-goals
Work the decision must not perform.

## Save compatibility
Fields/counters/commitments and migration notes.

```

87.3.2 Example contract summary — `Person.SelectUrgentGoal.v1`

Owner: Person. **Trigger:** 1–5 game minutes, critical need, hazard, target invalidation or binding emergency policy. **Observations:** current needs, reachable known food/water/shelter/care, household dependents, immediate hazards, current commitment and local law. **Candidates:** continue, eat, drink, rest, seek shelter, seek care, flee, assist dependent, idle. **Preconditions:** target exists and is known; route reachable; inventory/capacity reservable; actor physically able; authority not required beyond personal action. **Utility:** urgency × accessibility × expected relief × preference, modified by traits, dependents, fear and commitment. **Randomness:** ≤0.03 keyed adjustment among noncritical near-ties. **Invariants:** selected preconditions true; no negative inventory; critical flee outranks ordinary work when lethal hazard threshold passes. **Budget:** ≤12 candidates; ≤3 bounded opportunity queries; p95 ≤50 μs.

87.4 Appendix D — AI trace schema and explainability mapping

87.4.1 Trace term taxonomy

Every score term has stable key, raw unit, normalized value, weight, modifier source and contribution. Terms are grouped:

- need/urgency;
- access/distance/cost;
- expected effect;
- risk/safety;
- law/authority;
- material feasibility;
- household/social obligation;
- trait/personality;
- medium-rate attitude;
- culture/institution policy;
- higher-layer priority;
- commitment/inertia;

- random adjustment.

A player-facing summary maps technical keys to localization templates. “Food accessibility 0.82” becomes “Food is nearby and reachable.” Technical depth remains accessible.

87.4.2 Trace retention

A person retains the latest 32 compact traces. Major organizations retain 256. Selected entities can temporarily retain more. Decisions promoted to canonical events preserve summary, selected candidate, top positive/negative terms, blockers, controlling policies and intervention ancestry. Full candidate arrays remain in laboratory artifacts rather than permanent retail saves unless bookmarked.

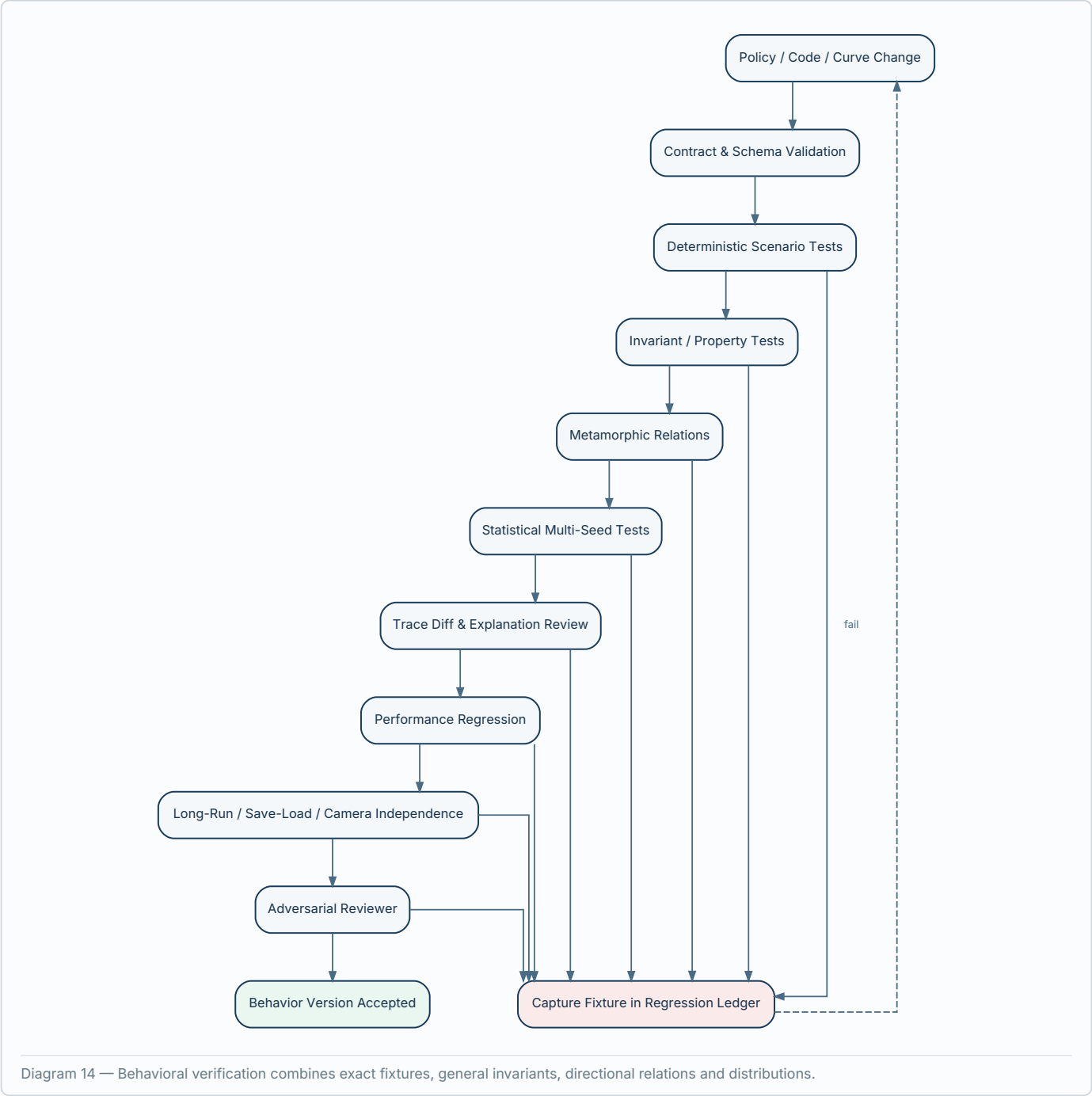
87.4.3 Trace invariants

- Contract and policy versions are present.
- Observation context identifies perspective.
- Every rejected candidate has precondition or selection reason.
- Selected candidate has all hard preconditions true.
- Random adjustment is explicit and within declared amplitude.
- Evaluation duration is diagnostic only and does not feed selection.
- Trace buffer overflow drops oldest permitted trace and increments telemetry; it cannot alter decisions.

87.4.4 Explainability UI mapping

Technical field	Player summary	Debug detail
precondition.route_unreachable	“No reachable route exists.”	Graph version, blocked edge IDs, observer knowledge
term.hunger_urgency	“Hunger is critical.”	Raw reserve, normalized curve, weight
policy.order.housing_priority	“The player ordered housing expansion.”	Command ID, authority, priority, issued tick
modifier.trait.caution	“Caution reduced the appeal of this risk.”	Baseline, expressed value, contribution
random_adjustment	“Chance modestly favored this option.”	RNG family/key/counter and numeric value
invalidation.target_destroyed	“The target was destroyed.”	Event ID and reservation releases

87.5 Appendix E — Behavioral testing catalogue



87.5.1 AI-SITE-001 — Reachable water

Initial state: founding group, three candidate sites, one with fresh water within 200 m, otherwise comparable.
Expected: water term and final rank for reachable site are not lower absent another declared cause.
Forbidden: use of hidden mineral deposit.
Trace: water, route, hazard, preference and random terms.
Metamorphic: removing the river lowers water/access terms.

87.5.2 AI-SITE-002 — Attractive floodplain with stale knowledge

Initial state: group believes flood risk low; authoritative risk high.

Expected: decision may select site using belief; trace identifies stale confidence.

Forbidden: hidden ground-truth avoidance.

After flood: memory and future site/policy decisions update.

87.5.3 AI-NEED-001 — Starving person beside food

Initial state: critical hunger, reachable unreserved food.

Expected: eat or move to food within timing limit.

Forbidden: start distant job, sleep or idle.

Property: inventory decreases only on consumption event.

87.5.4 AI-NEED-002 — Food behind impassable river

Initial state: critical hunger; visible/known food across destroyed bridge; no route.

Expected: food action rejected with route blocker; seek alternate, aid or migration.

Forbidden: crossing or consuming remotely.

Metamorphic: repairing bridge makes food action feasible after graph update.

87.5.5 AI-RES-001 — Unique tool reservation

Initial state: two workers, one required tool, two jobs.

Expected: one reservation succeeds; other job blocks/replans.

Forbidden: both jobs active with same tool.

Save test: reservation remains unique across save/load.

87.5.6 AI-LABOR-001 — Harvest versus ordered housing

Initial state: settlement has scarce workers, imminent harvest and binding housing order.

Expected: settlement allocates according to survival urgency and order priority; may delay housing with explicit blocker.

Forbidden: teleporting labor or ignoring order.

Trace: higher-order priority and harvest-loss risk.

87.5.7 AI-HOUSE-001 — Housing destruction

Initial state: 20 homes destroyed, 60 households, available public hall and neighboring vacancies.

Expected: displacement pressure, shelter assignments, overcrowding, migration/rebuild candidates.

Forbidden: households disappearing into aggregate score.

Metamorphic: adding reachable shelter cannot reduce available emergency capacity.

87.5.8 AI-POL-001 — Patriotic opposition

Initial state: high homeland patriotism, low government trust after documented corruption and military failure.

Expected: person can support reform/resistance while retaining patriotism.

Forbidden: patriotism forcing government loyalty.

Trace: separate polity/government attitudes.

87.5.9 AI-TECH-001 — Knowledge without manufacture

Initial state: inventor understands principle; no precision tools/material process.

Expected: research may produce plans or crude prototype; full manufacture blocked.

Forbidden: capability jump to mass production.

Metamorphic: adding tooling improves manufacturability but does not guarantee adoption.

87.5.10 AI-TECH-002 — Uranium and nuclear order

Initial state: known uranium deposit; government binding order for nuclear power; early-modern industry.

Expected: strategic order recorded; prerequisites identified; no reactor appears.

Forbidden: “uranium → nuclear power” shortcut.

Trace: scientific, institutional, industrial, safety and infrastructure blockers.

87.5.11 AI-DIP-001 — Misidentified convoy

Initial state: government believes construction convoy is army; confidence low.

Expected: threat response based on belief; verification/defense proposals plausible.

Forbidden: use of omniscient cargo truth.

Player UI: compare true versus believed.

87.5.12 AI-WAR-001 — Supply route severed

Initial state: task group in hostile region; bridge destroyed; three days rations.

Expected: readiness decline, reroute/forage/retreat; no supply teleport.

Metamorphic: restoring bridge improves reachable supply after travel delay.

87.5.13 AI-FORCE-001 — Instant factory

Initial state: rural settlement; Force Outcome creates factory.

Expected: valid building/owner/capacity records; zero output until workers, inputs, knowledge and power exist.

Forbidden: fabricated production.

History: permanent intervention and bypassed construction.

87.5.14 AI-FORCE-002 — Relocated population

Initial state: 100-person group moved to barren region.

Expected: households and memberships reconcile; shelter/food crisis, migration and mortality risk.

Forbidden: disappearance of dependents or possessions without declared loss.

87.5.15 ECO-EXT-001 — Local versus global extinction

Initial state: species populations in two isolated regions; one habitat destroyed.

Expected: local extinction in one region; global state remains extant.

Forbidden: global extinction flag.

Recovery: recolonization only with route/habitat or player action.

87.5.16 HIST-RUIN-001 — Metropolis compaction

Initial state: Tier 3 destroyed city, 200 years decay, aggressive save compaction.

Expected: protected street graph, major structures, craters and contamination remain; ordinary debris can merge.

Forbidden: city replaced by generic empty terrain.

Save test: landmark manifest references resolve.

87.5.17 UI-CAM-001 — Camera independence

Initial state: same seed and command log; one run follows city closely, one stays at world zoom, one headless.

Expected: canonical checkpoints identical.

Forbidden: AI cadence or random stream keyed to visibility.

87.5.18 AI-SPEECH-001 — Hidden deposit rumor

Initial state: speaker has no deposit knowledge; local rumor pool mentions “gold in the hill” as unverified claim.

Expected: bubble/event marked rumor with low confidence; AI does not treat as surveyed deposit.

Forbidden: speech revealing exact authoritative coordinates or grade.

87.6 Appendix F — Benchmark scenario catalogue

B01 Synthetic entity throughput

10,000 people with representative body, needs, traits, culture, politics, location, goal, household, workplace, appearance and trace-ring references; 2,000 active movement records; 1,500 workplaces; 3,000 shipments. Run headless and maximum presentation variants. **Pass:** Section 68 tick, memory, render and allocation budgets.

B02 Spatial churn

10,000 people move among dense chunks; 5% change chunk per game minute; construction changes local graph; 2,000 bounded neighborhood queries per tick. **Pass:** deterministic query order, candidate caps, no route backlog.

B03 Route disruption

Four settlements exchange food/tools. Destroy one bridge at a fixed tick. **Pass:** rerouting, shipment delay, shortage, price, policy and migration follow causal time; no duplication; explanation names bridge.

B04 Demographic century

Run 100 and 500 years with births, inherited baselines, aging, disease, migration and death; save/load every 10 years. **Pass:** stable IDs, valid households, bounded history and population behavior.

B05 Knowledge divergence

Two same-seed groups receive different river/coal/institution conditions and face one transport or drainage problem. **Pass:** statistically different solution portfolios; every outcome feasible and explained.

B06 Knowledge collapse and rediscovery

Destroy specialists, school, workshop and records; retain one damaged artifact. **Pass:** manufacture lost; awareness/evidence survives; successor may develop local derivative, not perfect unlock.

B07 Government/property matrix

Replay one settlement under four presets. **Pass:** authorization, land, revenue, project timing, inequality, succession and urban form differ through shared primitives.

B08 Three-actor diplomacy

Two weaker actors face a stronger third with asymmetric trade and beliefs. **Pass:** alliance, balancing, appeasement or opportunism emerges across seeds; no universal script.

B09 Large war stress

Ten task groups, 16 active sectors, 2,000 soldiers, damaged routes, refugees, fires and building destruction. **Pass:** performance floor, persistent individual outcomes, supply integrity and visible war.

B10 Save and migration torture

Repeated saves during construction, war, Force Outcome reconciliation, extinction and compaction across multiple schema versions; inject interruptions/corruption. **Pass:** recovery, migration, checksum and landmark preservation.

B11 Camera independence

Run observed, unobserved, world-zoom and headless variants. **Pass:** identical canonical checkpoints.

B12 10× scheduler stress

Large world with seasonal harvest, government review, war and disasters near due-time boundaries. **Pass:** no event loss, bounded backlog, clear CPU-limit reporting.

B13 Billboard/building stress

5,000 people, 1,000 high-detail buildings, 10,000 world structures, 40 speech bubbles, fire and overlay. **Pass:** 60 FPS ordinary, ≥30 FPS stress, directional readability.

B14 Omniscient overlay stress

Compose deposits, resource flows, population, military and civilization-knowledge difference over Large world. **Pass:** query and render budgets; legends remain readable.

B15 Landmark-century growth

Generate repeated town destructions and five Tier 3 metropolis ruins over 500 years. **Pass:** save growth projection acceptable; no protected deletion; render LOD stable.

B16 Force Outcome fuzz

Generate valid and invalid combinations of create/destroy/relocate/restore commands. **Pass:** transactional reconciliation, no invalid references, diagnostics and rollback.

87.7 Appendix G — Agent task-contract template

```
# TASK-NNN — Title

Status: Draft / Approved / In Progress / Verification / Done / Rejected
Risk tier:
Milestone:
Owner role:

## Objective
One verifiable outcome.

## Player-visible behavior
What the creator/player can observe.

## Authoritative inputs
Commands, state, content, time, observations and randomness.

## Outputs and state changes
Exact domains, components, events and schemas affected.

## Source-of-truth references
Blueprint sections, repository docs, behavior contracts and ADRs.

## Constraints and invariants
Authority, knowledge, units, determinism, performance and safety.

## Acceptance criteria
Numbered pass/fail statements.

## Required tests
Static, unit, property, scenario, metamorphic, statistical, integration,
replay, persistence, performance, UI and soak as applicable.

## Save compatibility
None / compatible addition / migration required / breaking proposal.

## AI behavior impact
Decision IDs, contract versions, trace and regression ledger.

## UI and explainability
Panels, overlays, reason codes, localization and accessibility.

## Performance budget
Scenario, p50/p95, allocations, memory, query and trace limits.
```

```
## Explicit non-goals
What must not be built, generalized or refactored.

## Allowed files/packages
Scope guard and forbidden dependencies.

## Destructive-operation disclosure
Deletes, rewrites, dependencies, backups and approvals.

## Evidence required
Logs, reports, traces, screenshots/video, profiler data and checksums.

## Rollback
Commit/tag/save recovery and exact commands.
```

87.8 Appendix H — Architecture decision record template

```
# ADR-NNN – Decision title

Status: Proposed / Accepted / Superseded / Rejected
Date:
Owner:
Risk tier:

## Context
Forces, constraints, evidence, affected saves, behavior contracts and dependencies.

## Decision
Exact selected approach and scope.

## Alternatives considered
Options, measurements, credible tradeoffs and rejection reasons.

## Consequences
Positive, negative, migration, performance, operations and creator-facing impact.

## Validation
Benchmarks, fixtures, tests, traces and review required.

## Rollback or supersession
How to undo or replace without losing source-of-truth history.

## Creator disclosure
Plain-language effect and explicit approval where required.
```

87.9 Appendix I — Repository review checklists

87.9.1 Pull-request checklist

- Task contract approved; scope unchanged or amendment documented.
- Diff coherent; generated/binary changes isolated and explained.
- Authoritative behavior matches specification and decision layer.
- No UI/presentation mutation or forbidden dependency.
- Stable IDs, units, time and RNG conventions followed.
- Behavior contracts, traces and catalog updated where relevant.
- Tests reproduce intended behavior and failure from clean checkout.
- Property/metamorphic/statistical categories included where required.
- Replay/checksum changes are explained.
- Save compatibility/migration reviewed.
- Performance before/after supplied for protected paths.
- UI explanations and localization keys supplied.
- Documentation/ADR/regression ledger updated.

-
- Rollback works.
 - Independent reviewer and creator acceptance complete for risk tier.

87.9.2 Dependency-upgrade checklist

- Need and alternatives documented.
- Current revision tagged and restorable.
- License, terms, security and support dates reviewed.
- Release notes and breaking changes analyzed.
- Upgrade isolated on dedicated branch.
- Clean build, tests, AI scenarios, replay, saves and performance run.
- Linux player build checked.
- Package lock committed.
- Rollback rehearsed.
- Creator approves high-risk upgrade.

87.9.3 Save-migration checklist

- Input/output schema versions explicit.
- Original save backed up before mutation.
- Pure migration function has fixture.
- Removed numeric keys reserved.
- Stable references resolve or diagnose.
- Active decisions, occurrence counters and reservations migrate.
- History, interventions and landmarks preserved.
- Old, migrated and fresh worlds converge on invariants.
- Failure leaves original untouched.
- Release notes explain compatibility.

87.9.4 AI behavior review checklist

- Contract owner/layer correct.
- Observation context excludes hidden information.
- Candidate and query caps enforced.
- Preconditions precede scoring.
- Canonical curves/primitives reused.
- Randomness keyed, bounded and traced.
- Reservations release on every exit.
- Hysteresis/cooldown prevent loops.
- Invalidation handles target, policy, hazard and save/load.
- Positive, negative and boundary scenarios present.
- Metamorphic relations cover expected direction.
- Multi-seed distribution reviewed where randomness matters.
- Trace contains losing candidates and policy.
- Performance meets per-decision budget.
- Regression ledger updated.

87.10 Appendix J — Document quality-control checklist

Before delivery of this edition, the exported package must pass:

- cover, subtitle, version, date and authority present;
- clickable table of contents and internal section anchors;
- all 61 required sections present;
- final feasibility evaluation, bibliography and appendices present;
- 33 required diagrams present as vector SVG with editable DOT source;
- page headers, footers and page numbers consistent;
- selectable text and standard/embedded fonts;
- no clipped body text, tables, captions or diagrams;
- no table wider than the content box;
- no accidental blank or nearly blank pages;
- no orphaned main headings;
- sufficient contrast and readable minimum type;
- external hyperlinks and internal PDF links generated;
- PDF parser reports expected page count and no structural error;
- required headings verified in extracted text;
- source Markdown, HTML, CSS, SVG, DOT and build scripts retained;
- every rendered final page visually inspected through page images/contact sheets;
- every locked decision represented in main text and change log;
- no superseded Limited Influence/default progression or four-direction-sprite recommendation remains;
- first-playable numbers reconcile across summary, budgets, roadmap and tasks;
- risks have mitigations and phase gates;
- citations and bibliography keys resolve.

87.11 Final feasibility evaluation

87.11.1 Candid assessment

The complete north-star game is difficult but staged-feasible. The first playable is itself a substantial product, not a prototype weekend. The combination of persistent named people, physically built cities, long-horizon civilization systems, unrestricted player mutation, omniscient causal UI and AI-only production has no single proven reference implementation. Success depends on refusing premature breadth and using measurable gates.

The architecture is credible because it separates identity from update frequency, physical truth from visual LOD, decisions from execution, player orders from Force Outcome, ground truth from actor belief, policy data from runtime, and permanent landmarks from routine history. None of those separations removes the need for implementation evidence.

87.11.2 Largest technical risks

1. Hierarchical pathfinding and route invalidation inside dense evolving cities.
2. Integrated tick cost after economy, politics, warfare, ecosystems and traces interact.
3. Semantic save migration and long-term state growth.
4. Transactional Force Outcome reconciliation.
5. Rendering and selecting thousands of billboards among tall modular buildings.

87.11.3 Largest design risks

1. The world feels passive despite decision-level control.

2. Complex causes remain unreadable even with traces.
3. Society-dependent city growth converges into generic layouts.
4. Unrestricted powers make experimentation shallow if consequences are not legible.
5. Traits, culture and politics become stereotypes rather than contextual systems.

87.11.4 Largest physical-world simulation risks

1. Construction and goods are physical enough to create hauling/path cost without enough visible value.
2. Warfare destroys more geometry and population than persistence/save budgets can retain.
3. City entrances, bridges, slopes and occlusion create local navigation failures.
4. Landmark preservation grows save and rendering cost over centuries.
5. Remote physical outcomes tempt implementation shortcuts that violate camera independence.

87.11.5 Largest AI-only development risks

1. Architectural drift across agent sessions.
2. Self-certifying behavior that passes happy-path tests.
3. Hidden omniscience introduced by convenient world queries.
4. Inconsistent utility curves, RNG and reservations.
5. Save/version impact omitted during local feature work.
6. Review agents sharing the implementer's assumptions.
7. The creator lacking independent technical signals when evidence is misleading.

The AI Decision Laboratory, adversarial roles and regression ledger directly target these risks. Periodic human expert audits remain strongly recommended.

87.11.6 Largest production risks

1. Treating the 61-section north star as one milestone.
2. Heavy art and content before engine/city gates.
3. Advanced-era enthusiasm displacing early-modern platform work.
4. Underestimating UI and content-validation labor.
5. Tool/version/licensing change during a long solo production.
6. Burnout from maintaining a large repository and evaluating agent output.

87.11.7 What must be proven before heavy art investment

- Unity engine GO with representative integrated load.
- AI Decision Laboratory GO with defect detection.
- Autonomous founding and survival seed suite.
- Society-dependent physical village growth.
- Billboard combat/work/crowd readability.
- Building module grammar and skyline proof.
- Neighborhood destruction, displacement and rebuilding.
- Save/load and compaction of a 100-year physical world.
- Omniscient trace UI usable by external playtesters.

Until these pass, art should remain modular prototype-quality with pixel-density and silhouette tests rather than final production volume.

87.11.8 Engine go/no-go benchmark

Use Section 57 exactly. A failure after two architecture-aware optimization passes triggers the bounded Bevy headless comparison. Do not port presentation or content during the fallback test. The benchmark decision is documented in an ADR and creator-facing report.

87.11.9 AI Decision Laboratory go/no-go benchmark

Use Section 40. The decisive proof is not that three actions look correct. It is that the harness detects deliberately seeded defects, reproduces decisions across save/load and scheduling variation, explains every selection, measures cost and prevents hidden information.

87.11.10 Settlement-formation go/no-go benchmark

A 40-person group must select a plausible site from its knowledge, establish shelter, water, food and storage, allocate work, construct visible structures, form paths, survive or fail for traceable reasons, and preserve people/households through save/load. At least 80% of valid baseline seeds reach one village by year 15.

87.11.11 Physical-city-growth go/no-go benchmark

Two society presets on the same terrain must create statistically and visually different parcel, road, housing and public-project patterns. The system must render and navigate a dense stress town, destroy a neighborhood, displace households, preserve ruins and rebuild without manual placement or camera-dependent outcomes.

87.11.12 Civilization-divergence go/no-go benchmark

Paired seed suites with changed geography, institutions or resources must produce different technology, economy, politics, diplomacy and city form by year 50. Differences must be traceable to conditions, not scripted civilization identities or unconstrained randomness.

87.11.13 Recommended first month

Create governance and CI; pin Unity; implement world time, stable IDs, keyed RNG, semantic save skeleton, headless runner, checksum/state diff, representative synthetic benchmark, presentation bridge and initial AI contract schema. End the month with a GO/NO-GO report for continuing to the AI Decision Laboratory.

87.11.14 What not to build during the first six months

No nuclear systems, skyscrapers, modern vehicles, aircraft, detailed navies, rendered interiors, complex constitutions, giant technology catalog, final art, voice acting, multiplayer, executable mods, console certification, monetization, live LLM dialogue or multi-planet systems. Do not write a broad civilization planner before the laboratory.

87.11.15 Final verdict on staged feasibility

Verdict: GO for a disciplined preproduction and first-playable program; NO-GO for immediate full north-star production. The project should proceed through Phase 0, Phase 1 and the AI Decision Laboratory. Continued investment is conditional on the engine, AI and settlement gates. The north-star vision should remain intact, but content and era breadth must stay deferred until the causal physical foundation is measured.

87.11.16 Exact next action after blueprint approval

Create the repository, copy the Version 2.0 source package into `docs/blueprint/`, open **TASK-001 — Establish repository governance and reproducible Windows bootstrap**, and do not authorize gameplay implementation until its clean-checkout evidence is accepted.

87.11 Appendix K - Historical event and incident schemas

87.11.1 Historical event

```

schema_version: 1
event_id: EventId
kind: HarvestFailure
start_time: WorldTime
end_time: WorldTime
spatial_bounds: RegionBounds
participants: [StableEntityRef]
authoritative_facts:
  yield_delta_milliunits: -420000
  affected_farms: 17
causes: [EventEdgeRef]
consequences: [EventEdgeRef]
player_intervention_ancestry: [InterventionId]
significance_tier: 3
presentation_key: history.harvest_failure
perspectives: [PerspectiveRecordRef]
physical_remnants: [LandmarkRef]

```

87.11.2 Incident candidate

```

incident_id: Incident.MerchantReliefMission.v1
owner_system: TradeDiplomacy
required_actors: [Settlement, MerchantInstitution]
preconditions:
  - shortage_severity >= 0.55
  - reachable_supplier_count > 0
  - available_merchant_capacity > 0
exclusions:
  - active_same_chain
cooldown: 180_game_days
resolution_owner: TradeMissionSystem
story_profiles: [Chronicler, Adversarial, Chaotic, Custom]
required_trace: true
historical_event_on_accept: ReliefMissionProposed
performance_budget_us: 80

```

87.12 Appendix L - Storytelling verification catalogue

- **STORY-AUTH-001:** camera movement cannot change candidate eligibility or outcome.
- **STORY-AUTH-002:** director cannot select an incident lacking required actors/resources.
- **STORY-REP-001:** repetition key caps equivalent incidents within the configured window.
- **STORY-REC-001:** Chronicler reduces unrelated severe selection during recovery debt.
- **STORY-TEXT-001:** every chronicle fact slot resolves to an event fact or perspective record.
- **STORY-Persp-001:** a mistaken civilization belief is labeled and does not mutate authoritative history.
- **HIST-GRAPH-001:** compaction preserves all edges required by a landmark and technology lineage.
- **HIST-MAP-001:** historical map view does not alter live authoritative state.
- **HIST-SAVE-001:** migrated saves preserve event IDs, significance and intervention ancestry.
- **HIST-L10N-001:** pseudo-localization produces no concatenation, overflow or missing grammar forms.